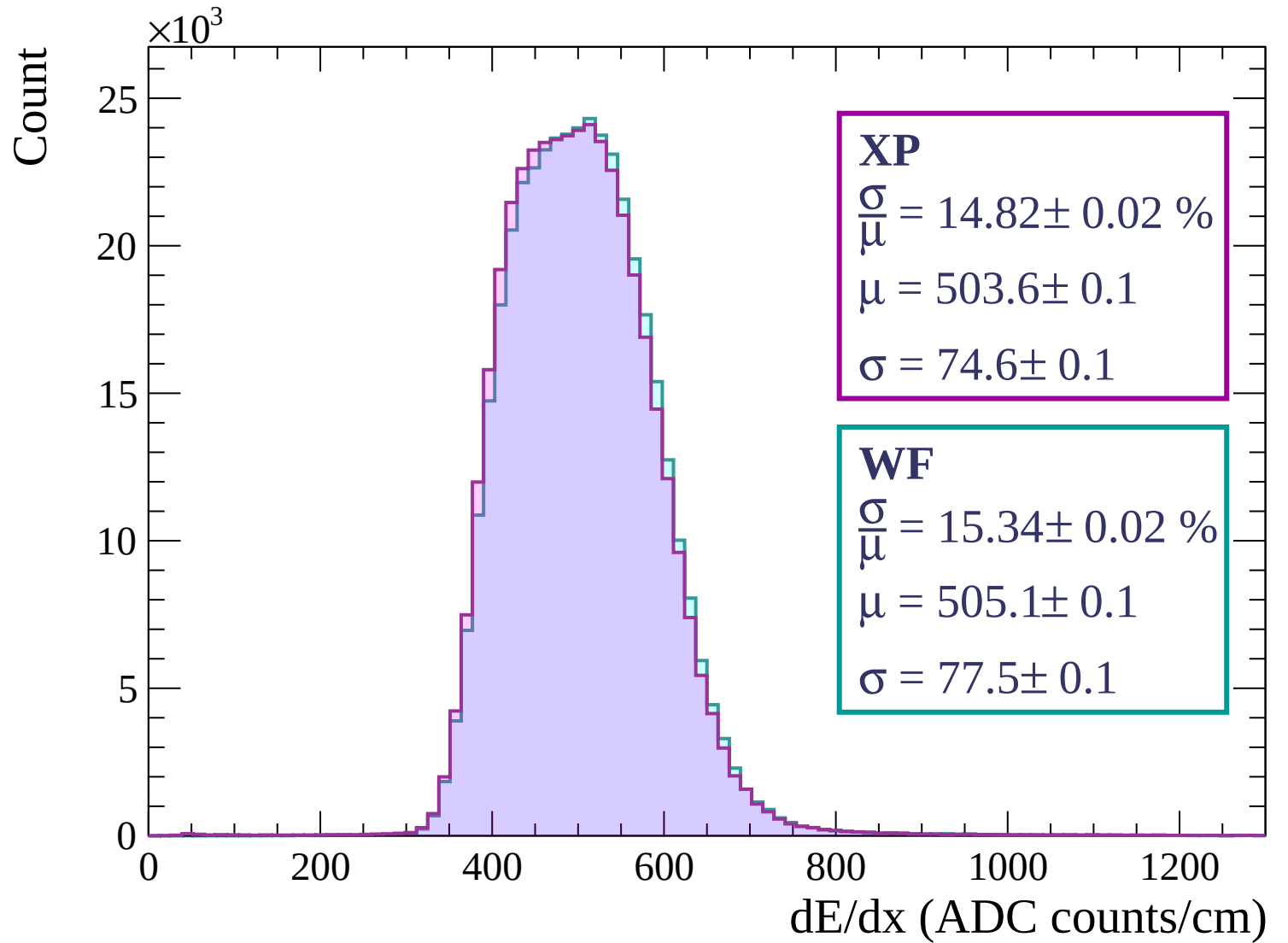
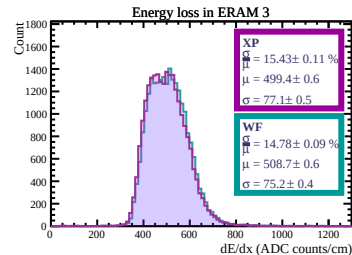
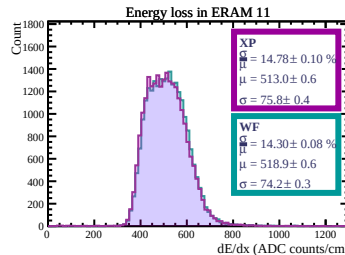
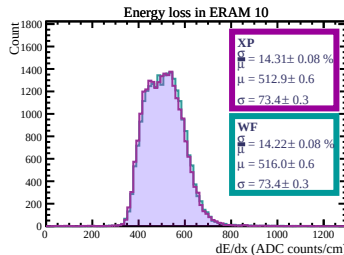
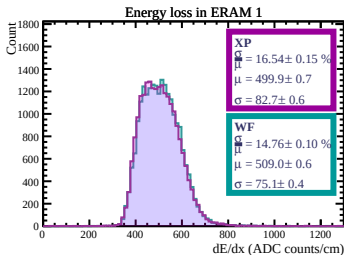
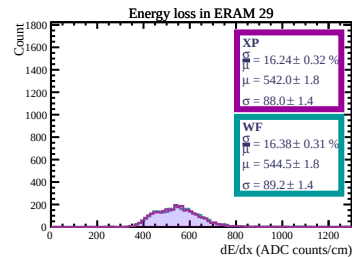
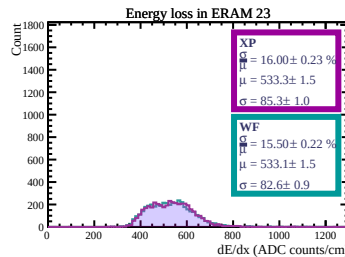
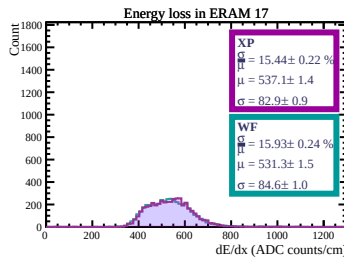
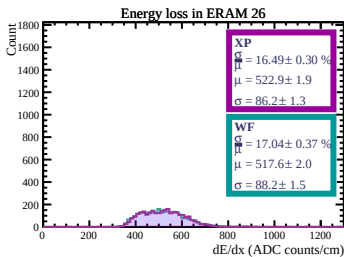
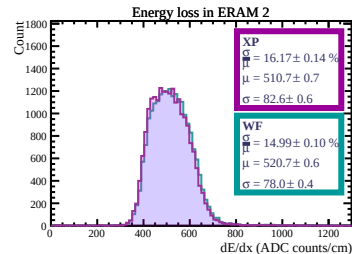
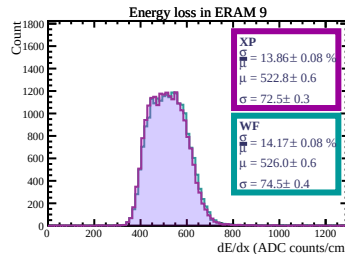
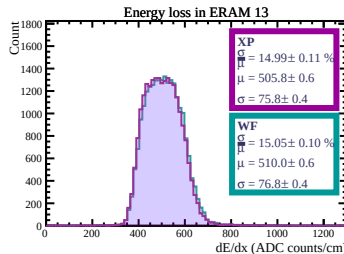
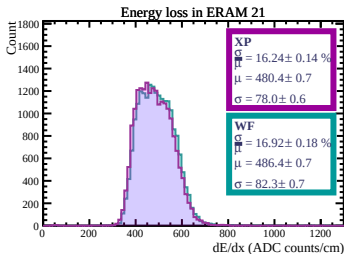
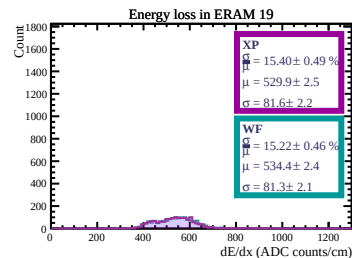
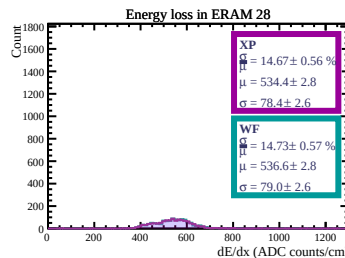
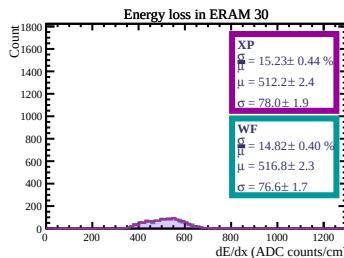
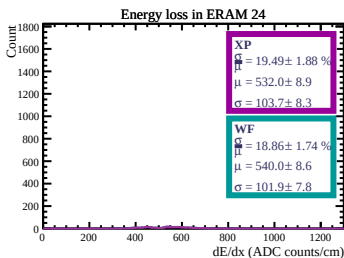
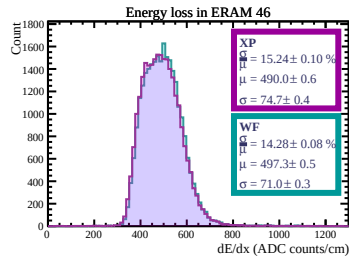
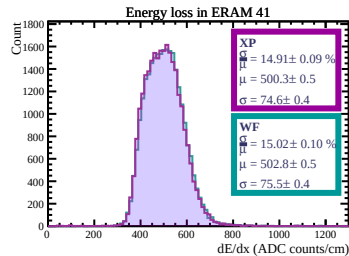
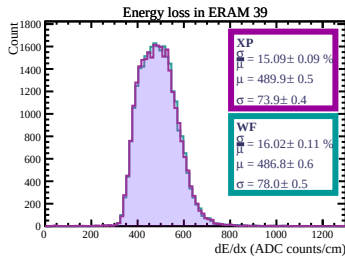
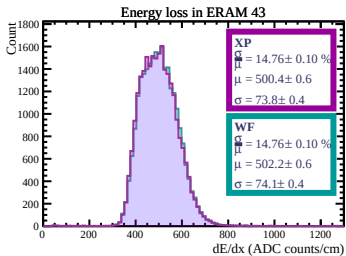
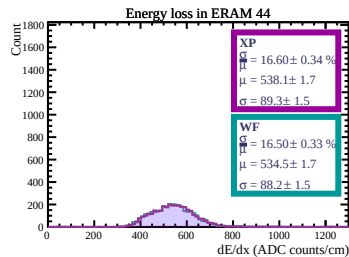
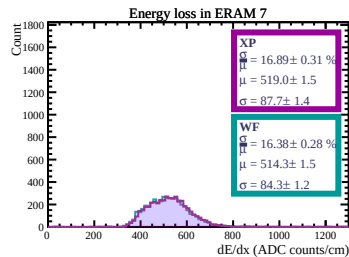
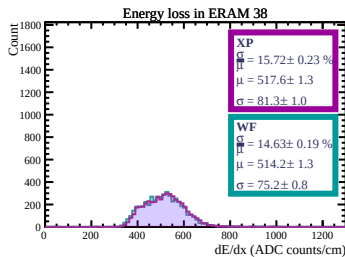
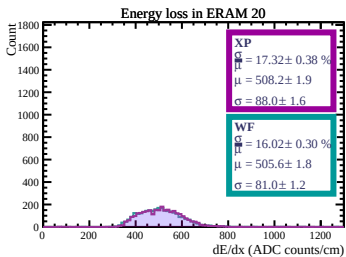
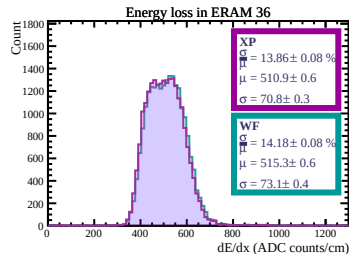
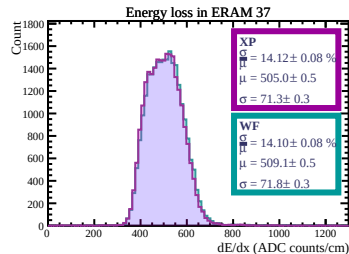
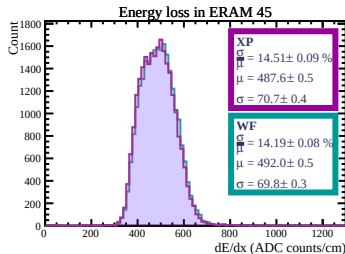
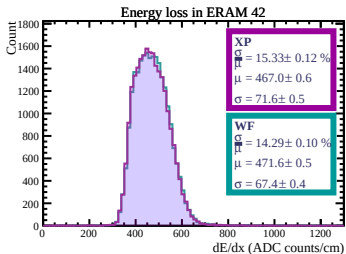
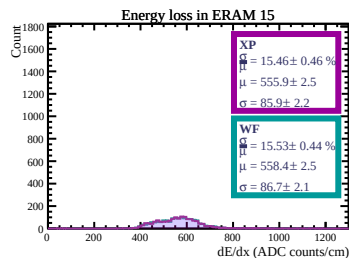
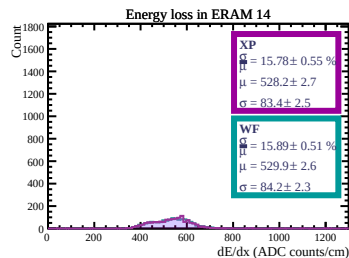
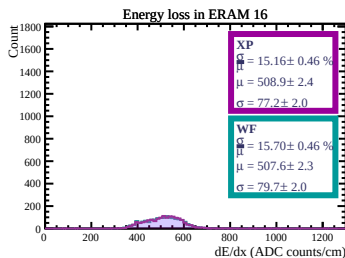
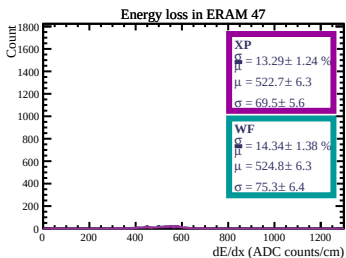
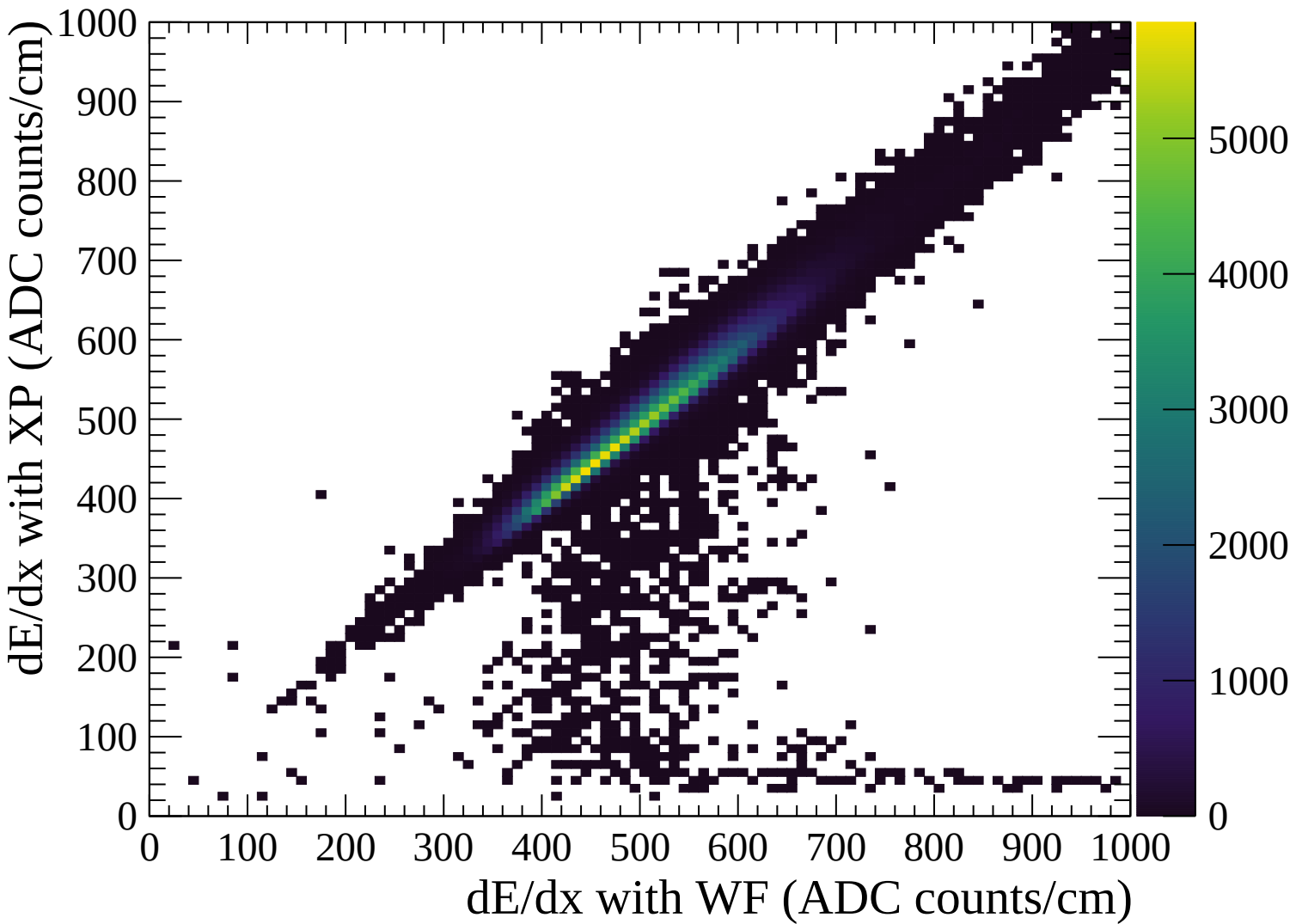
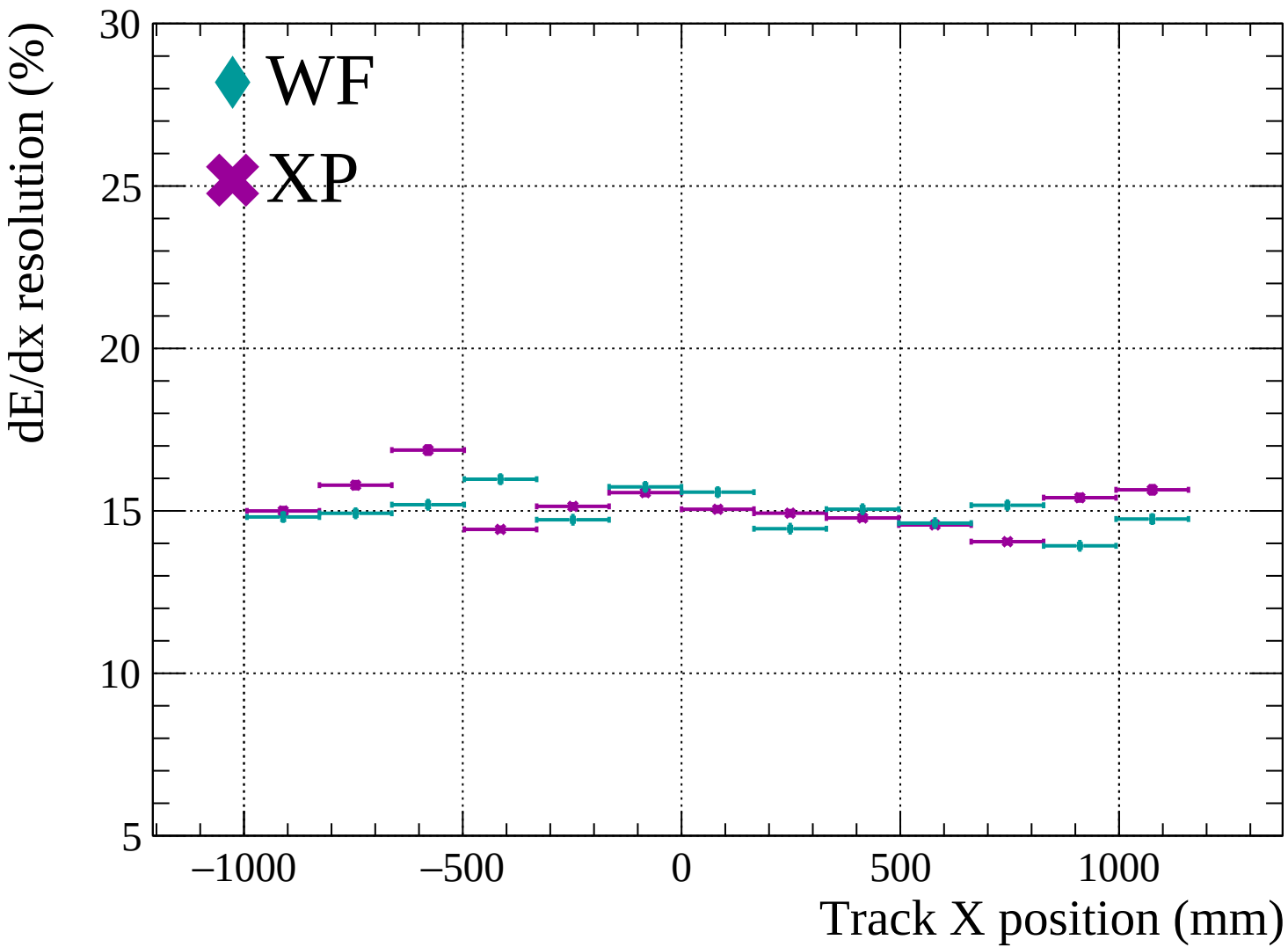


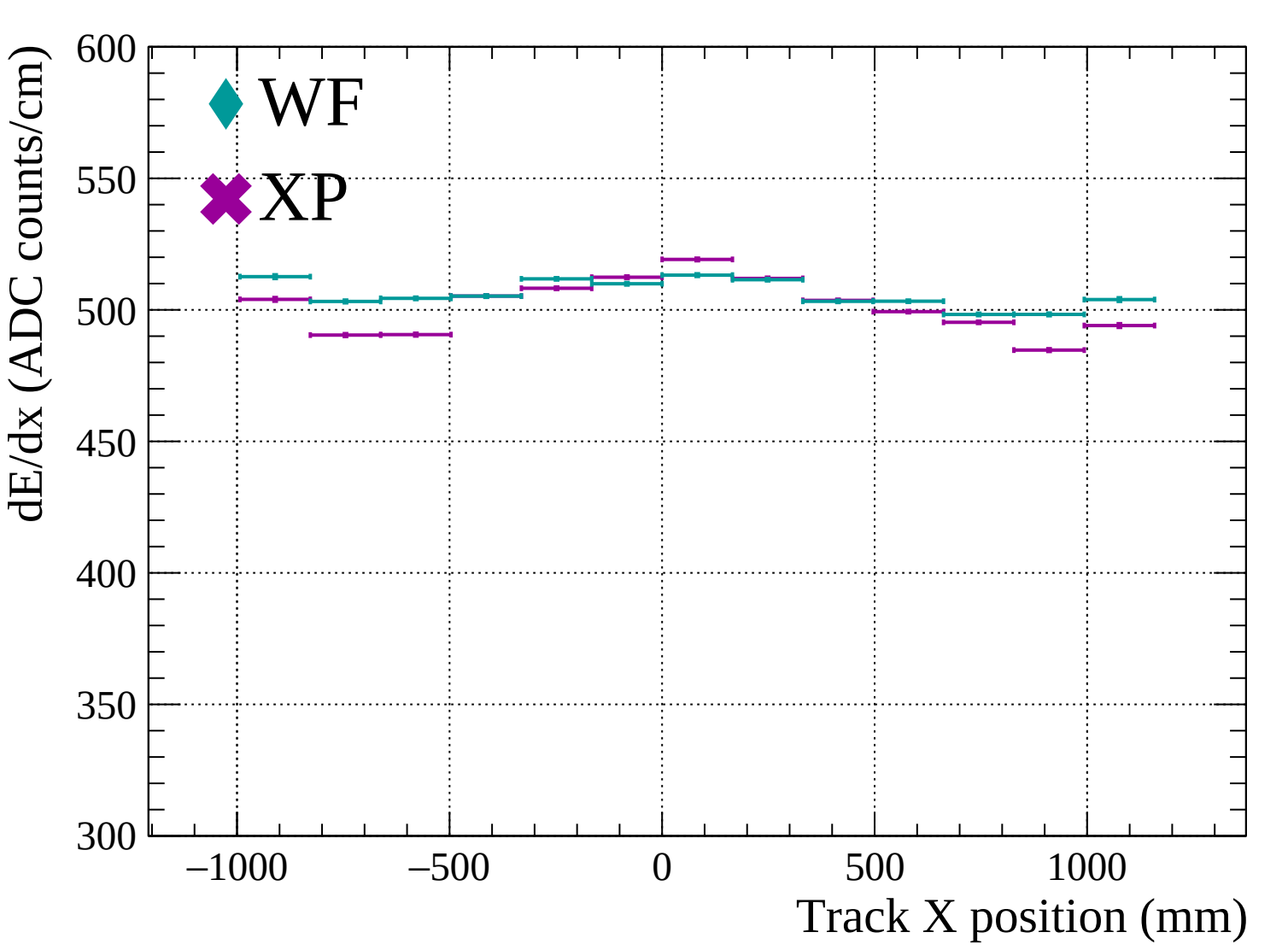


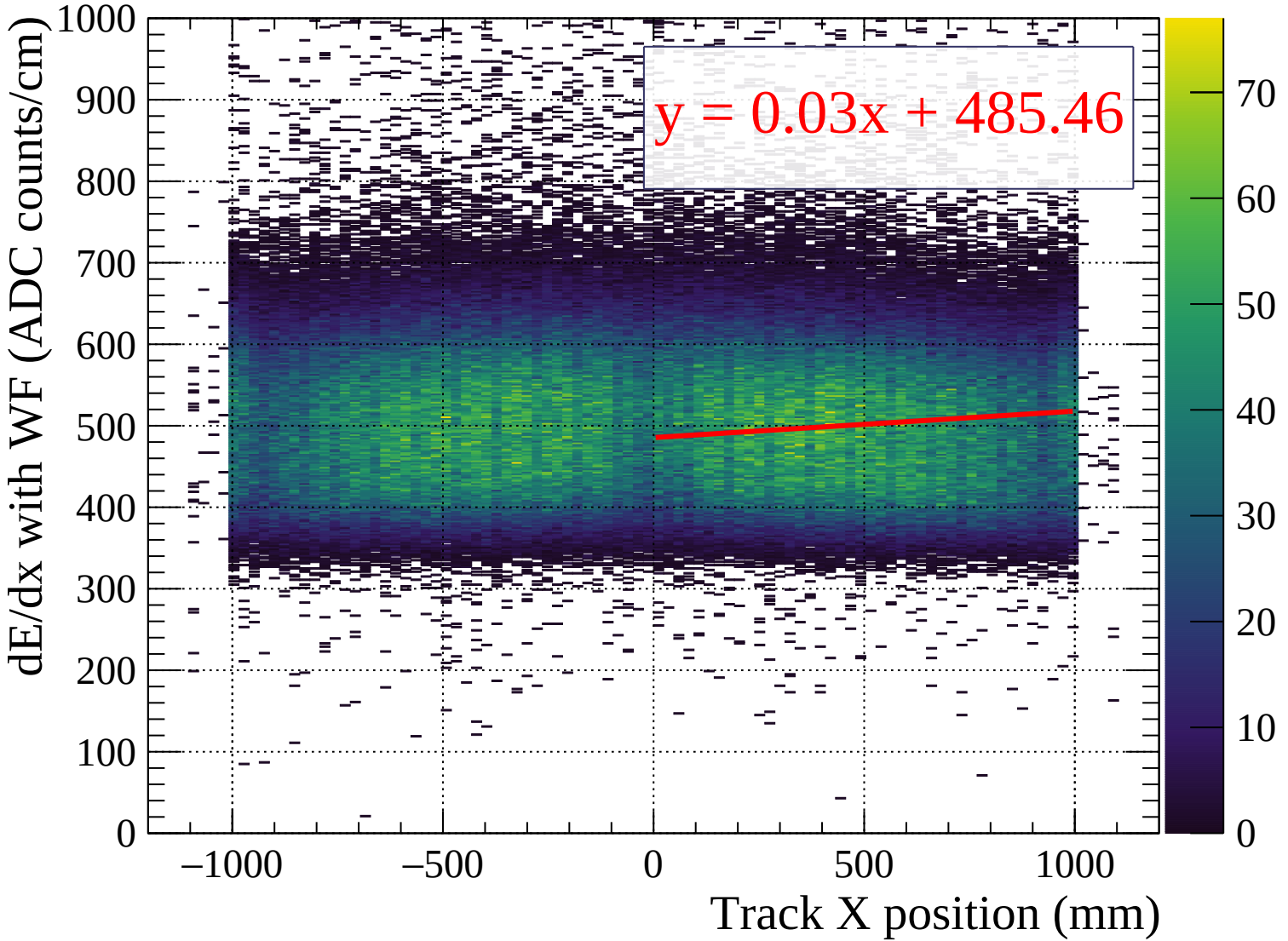


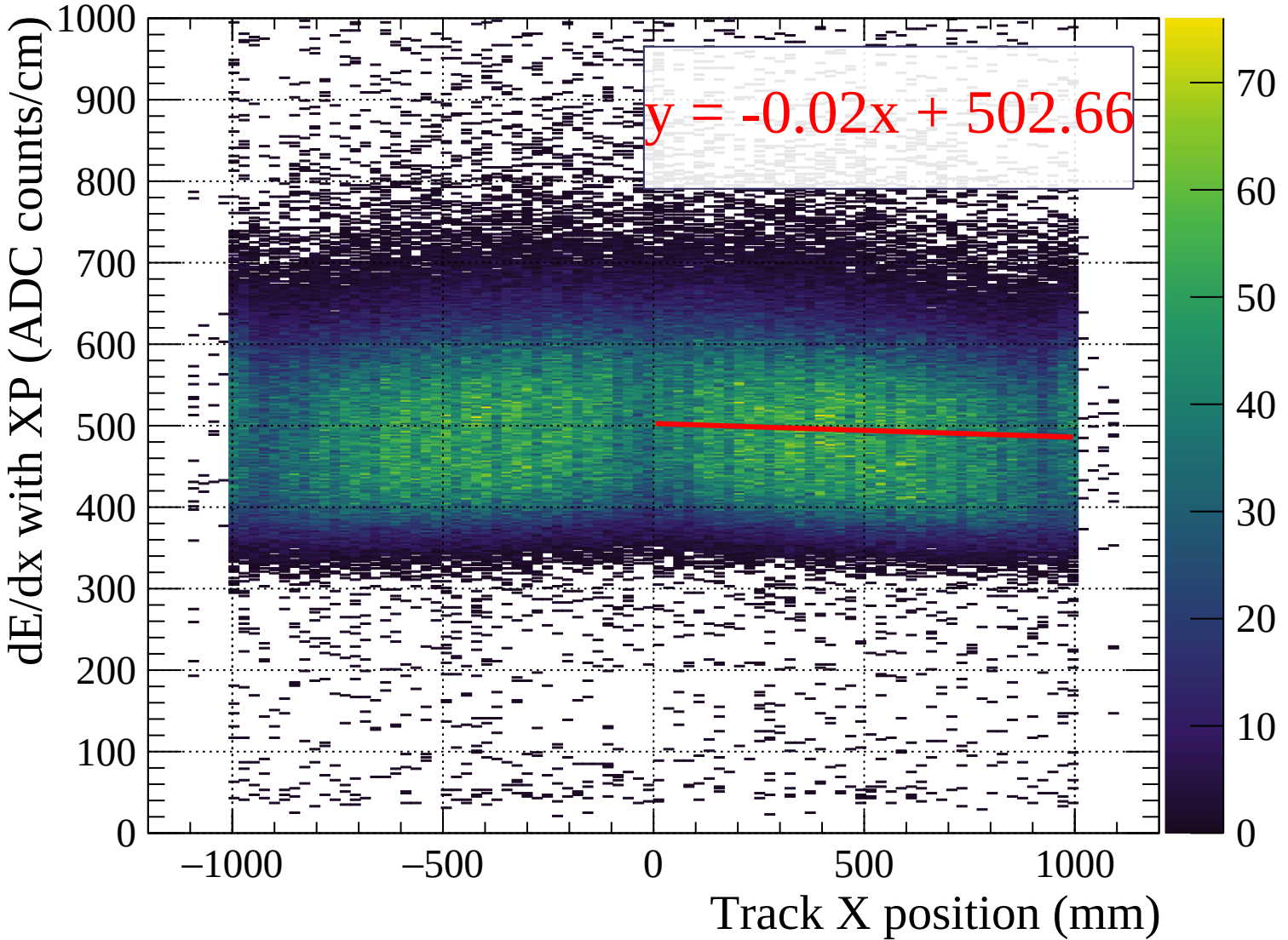


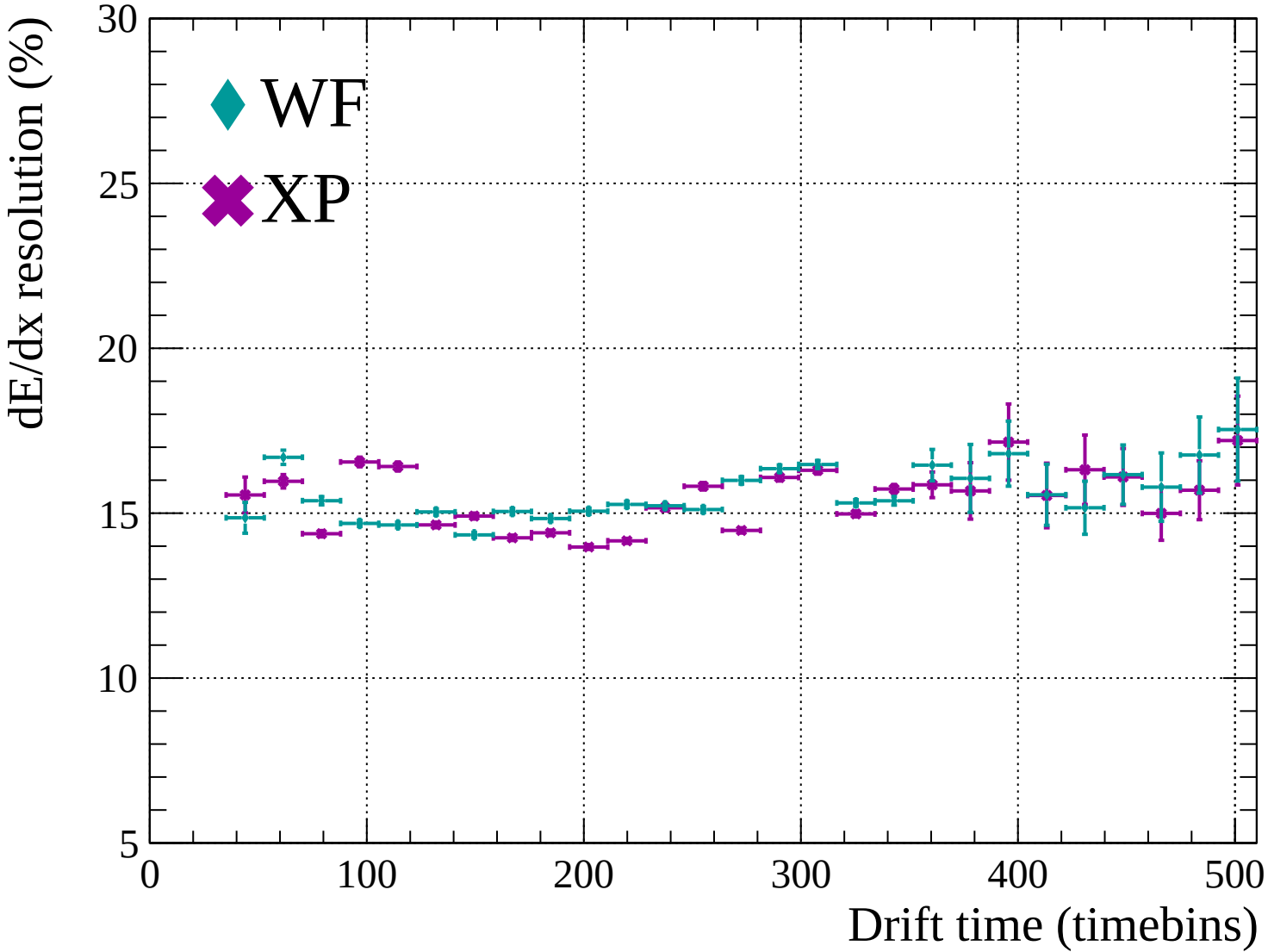


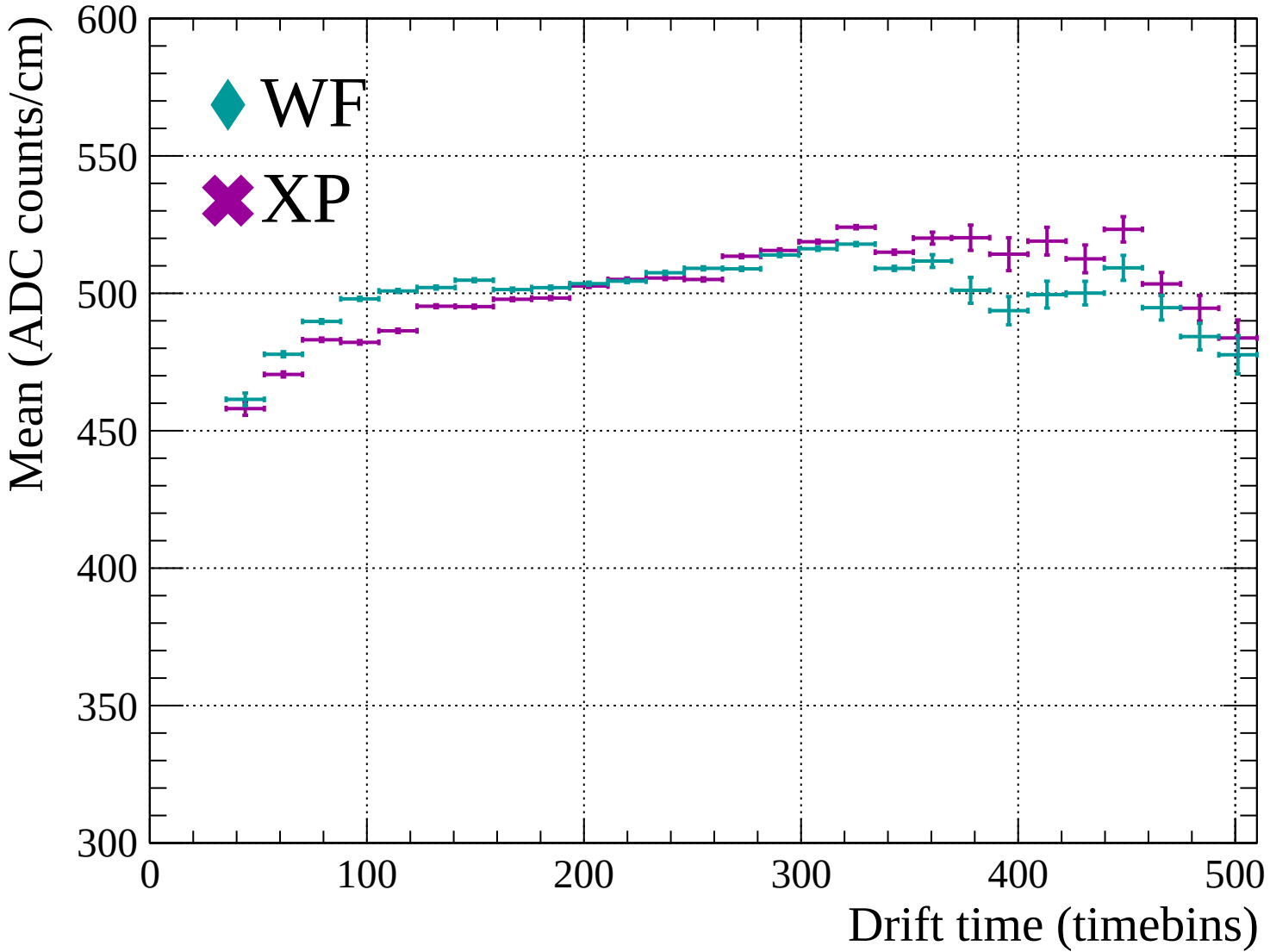


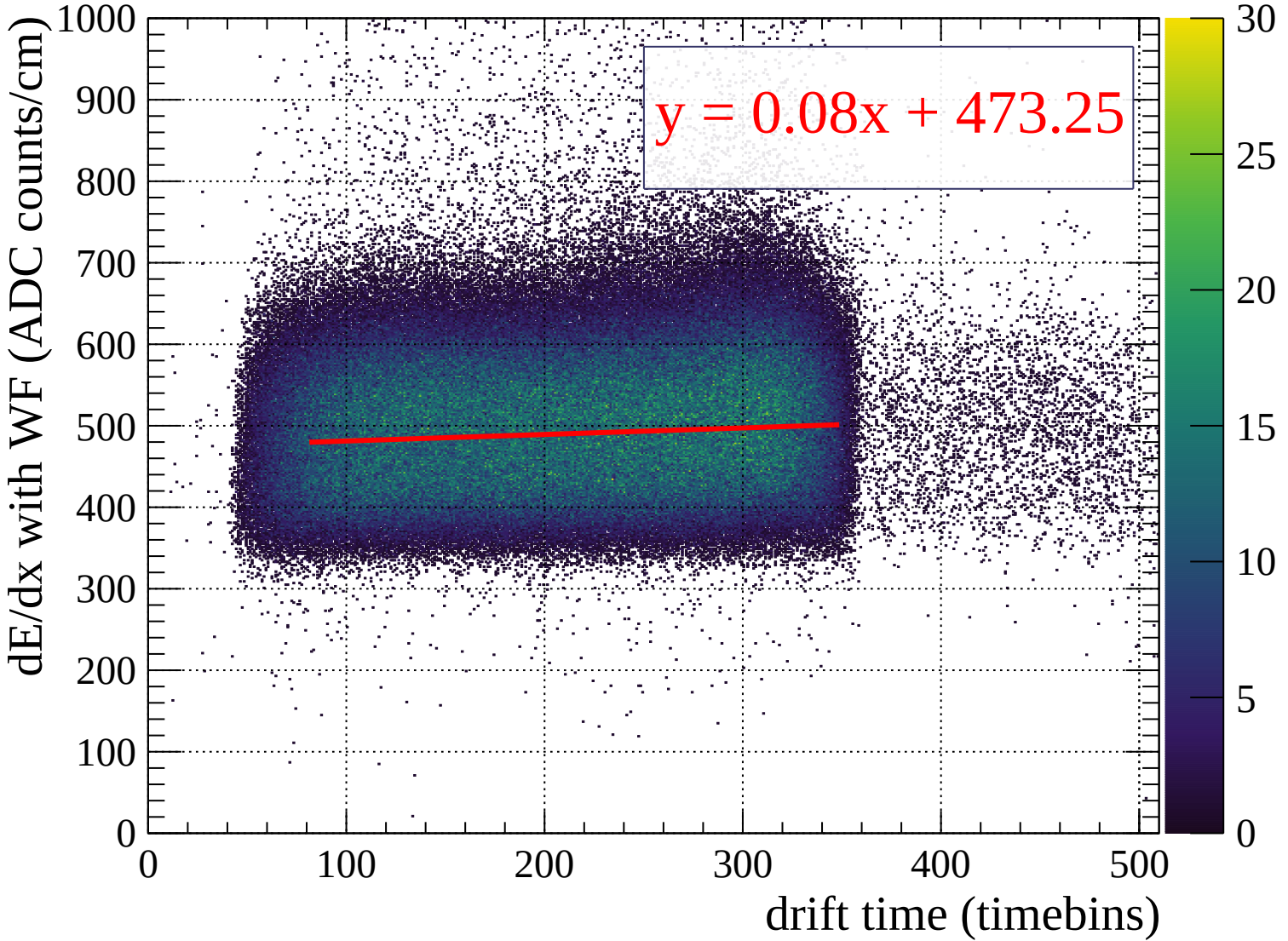


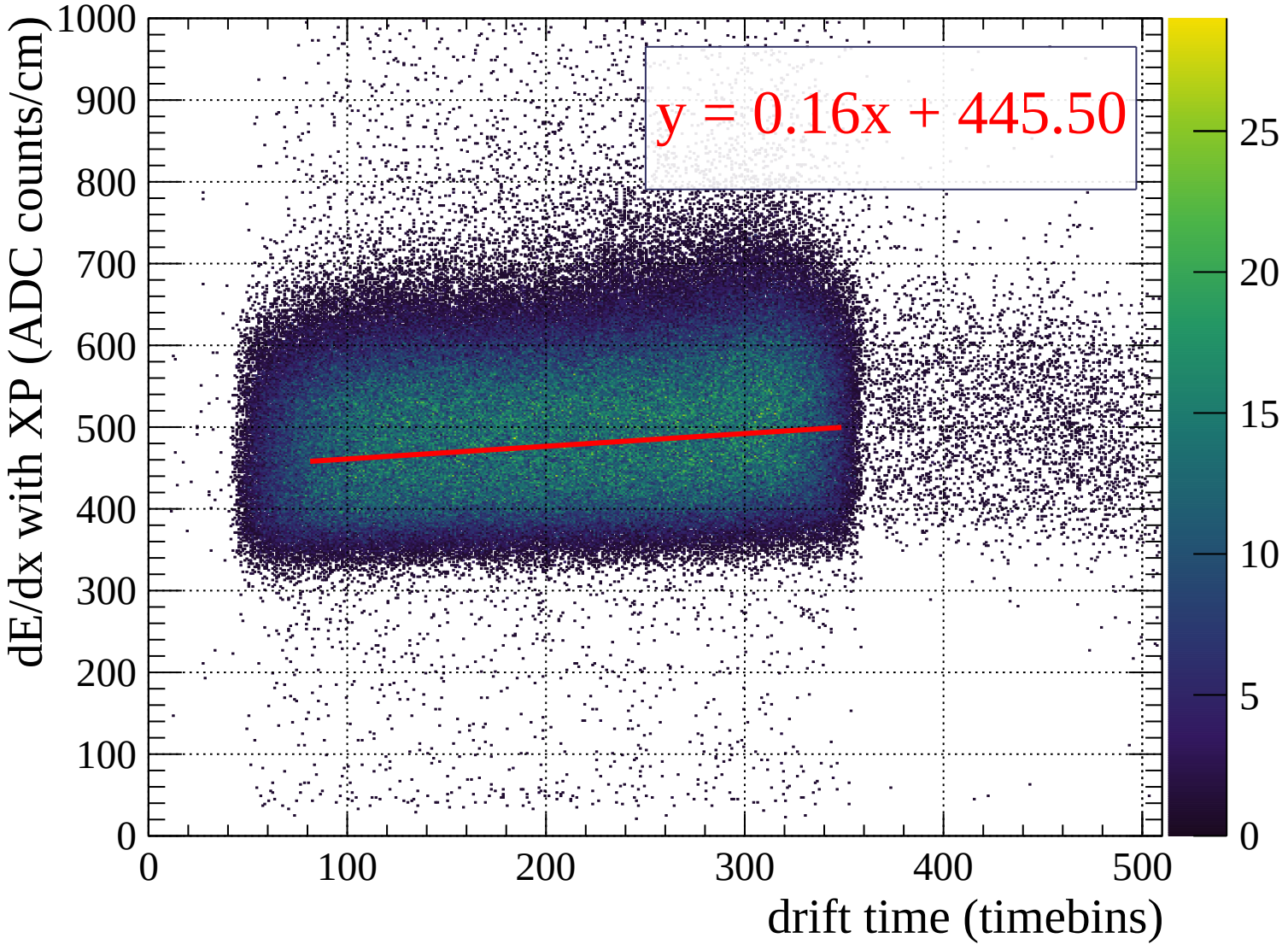


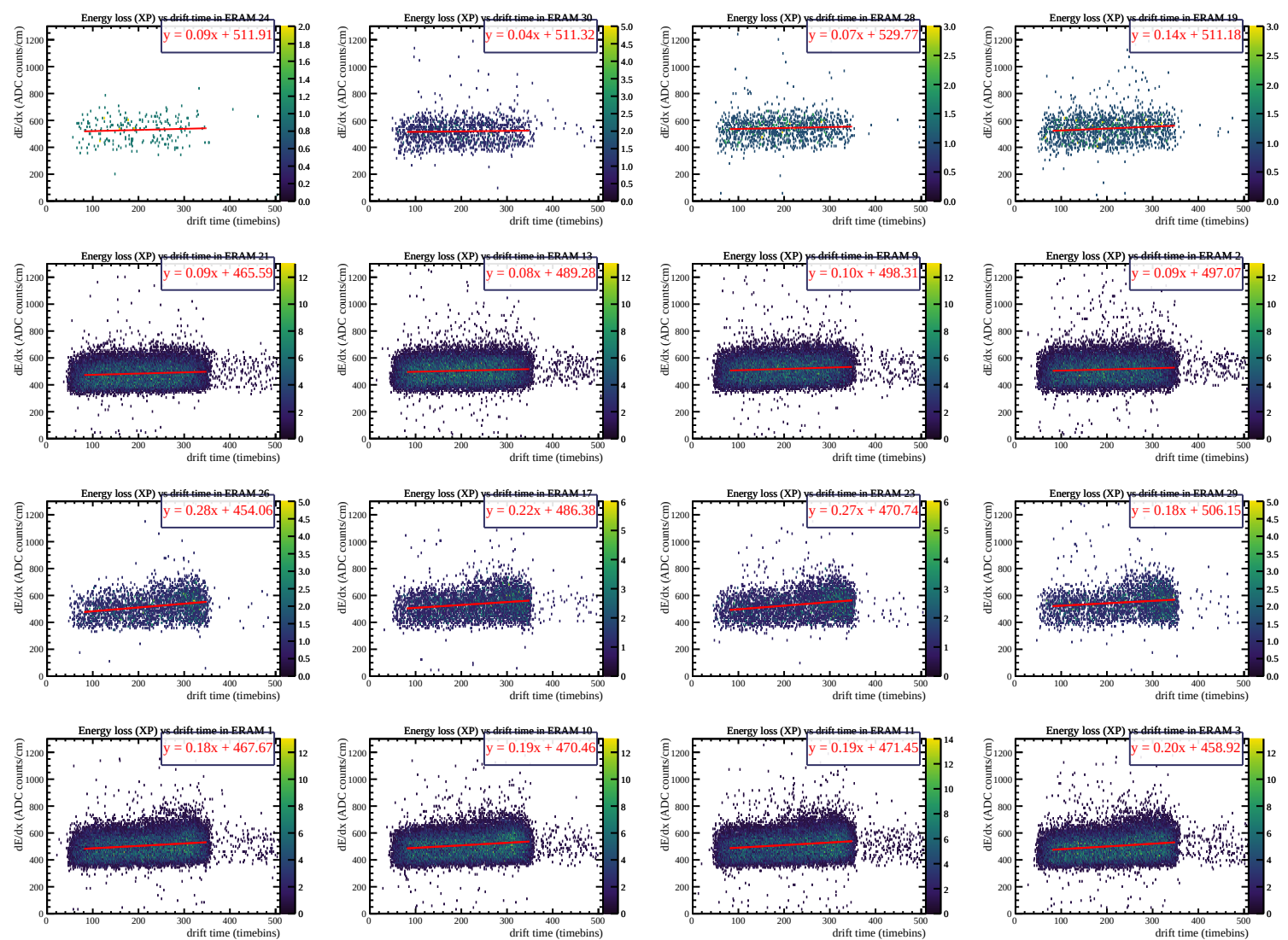


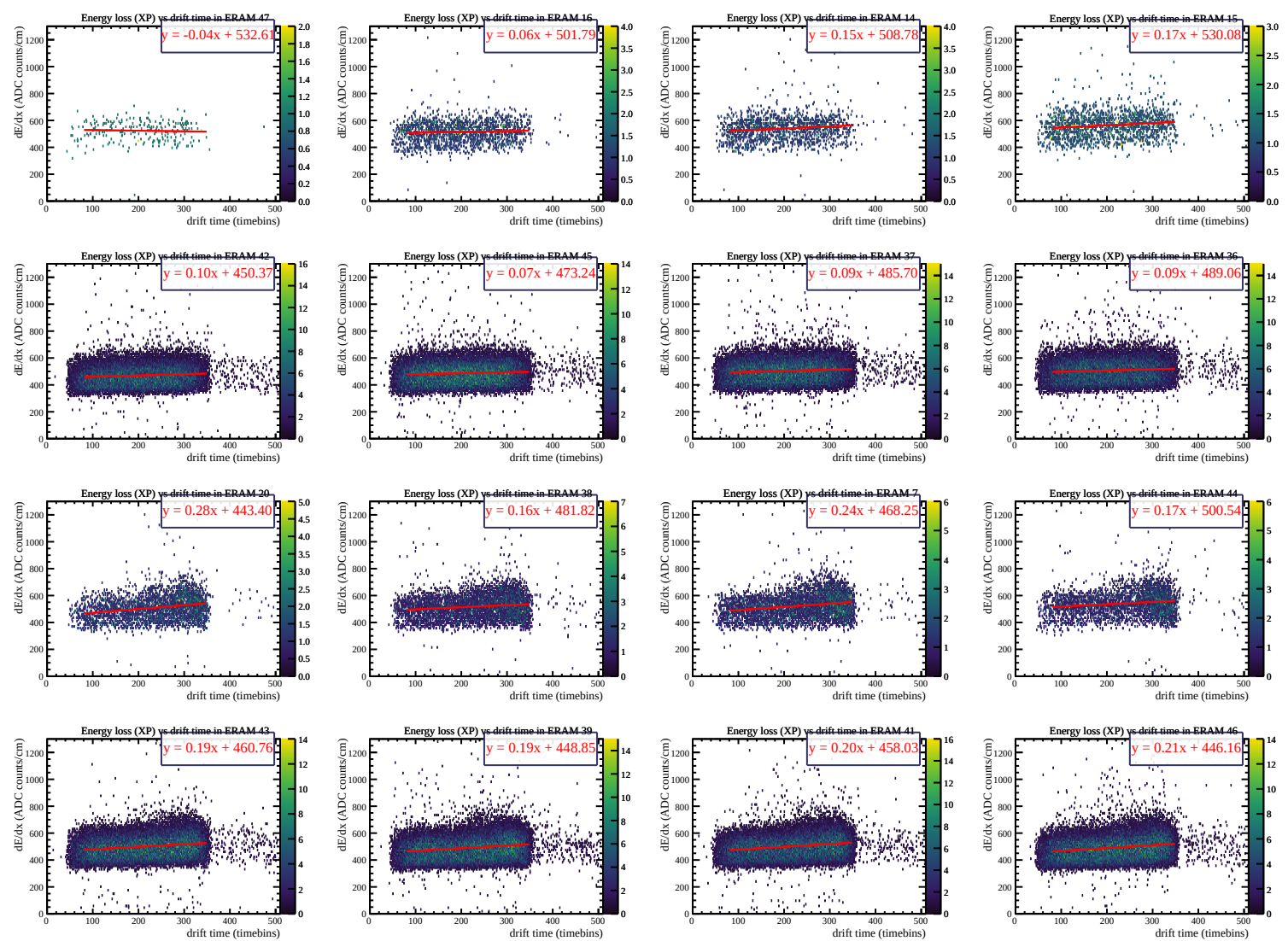


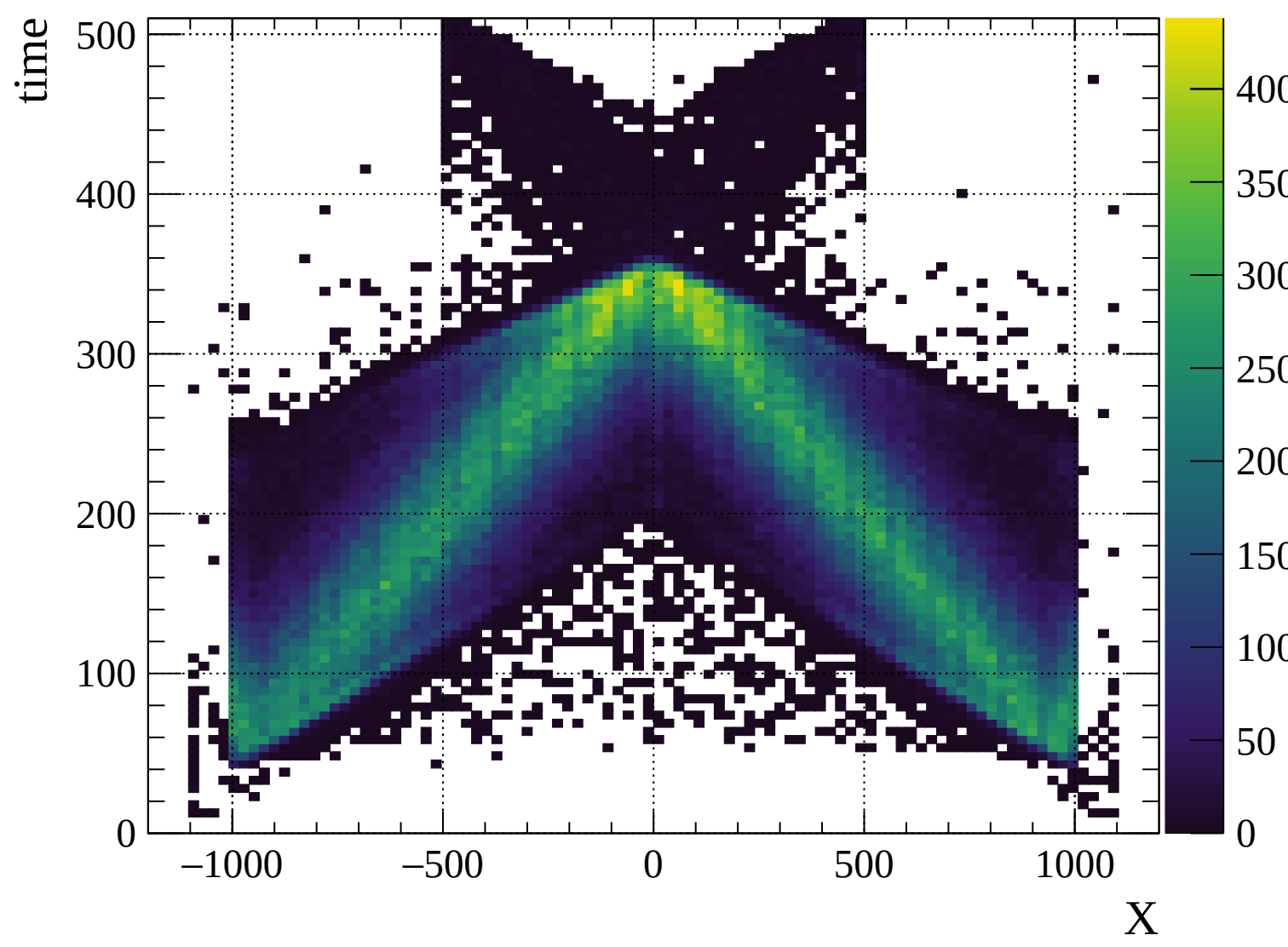


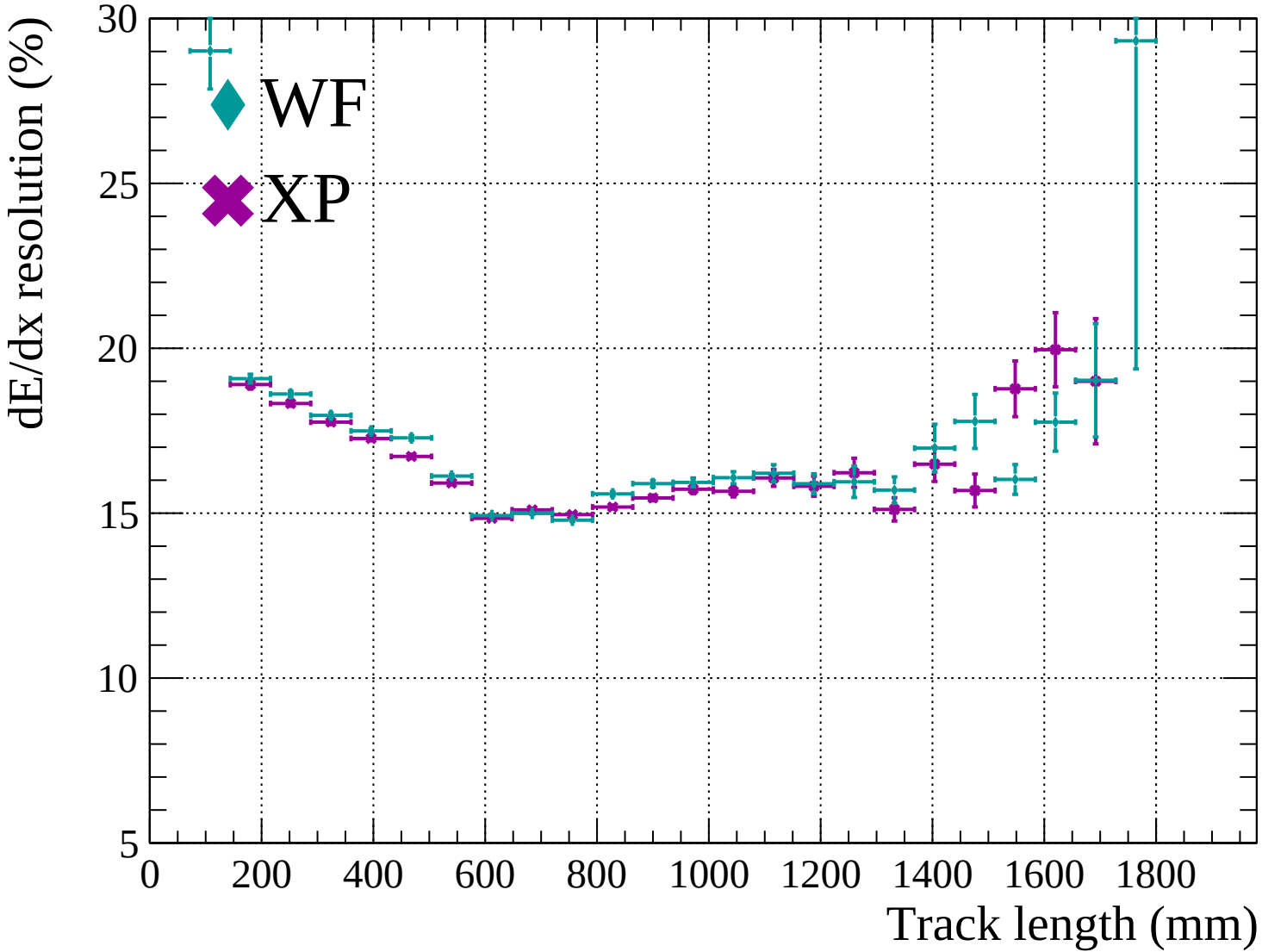


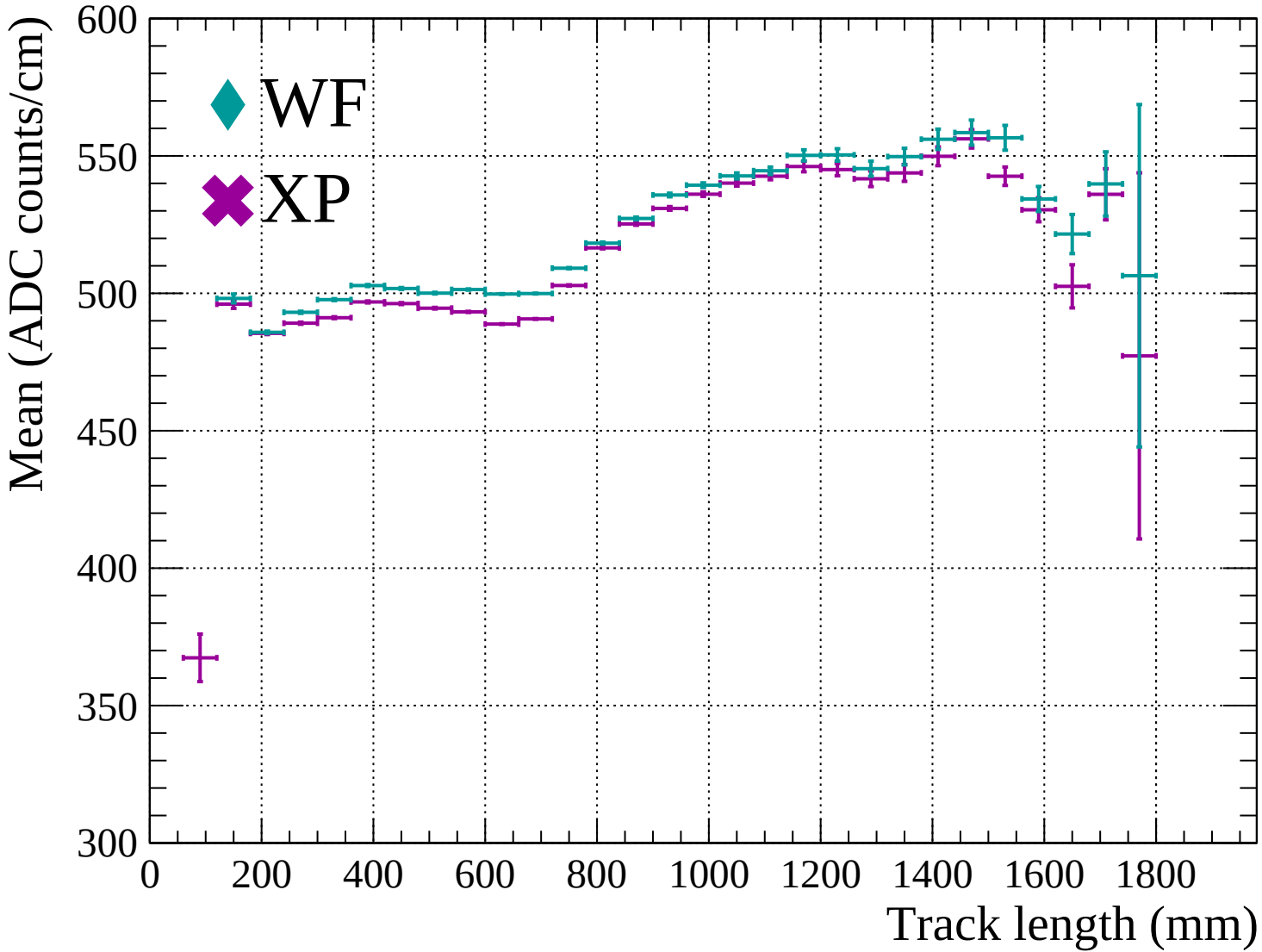


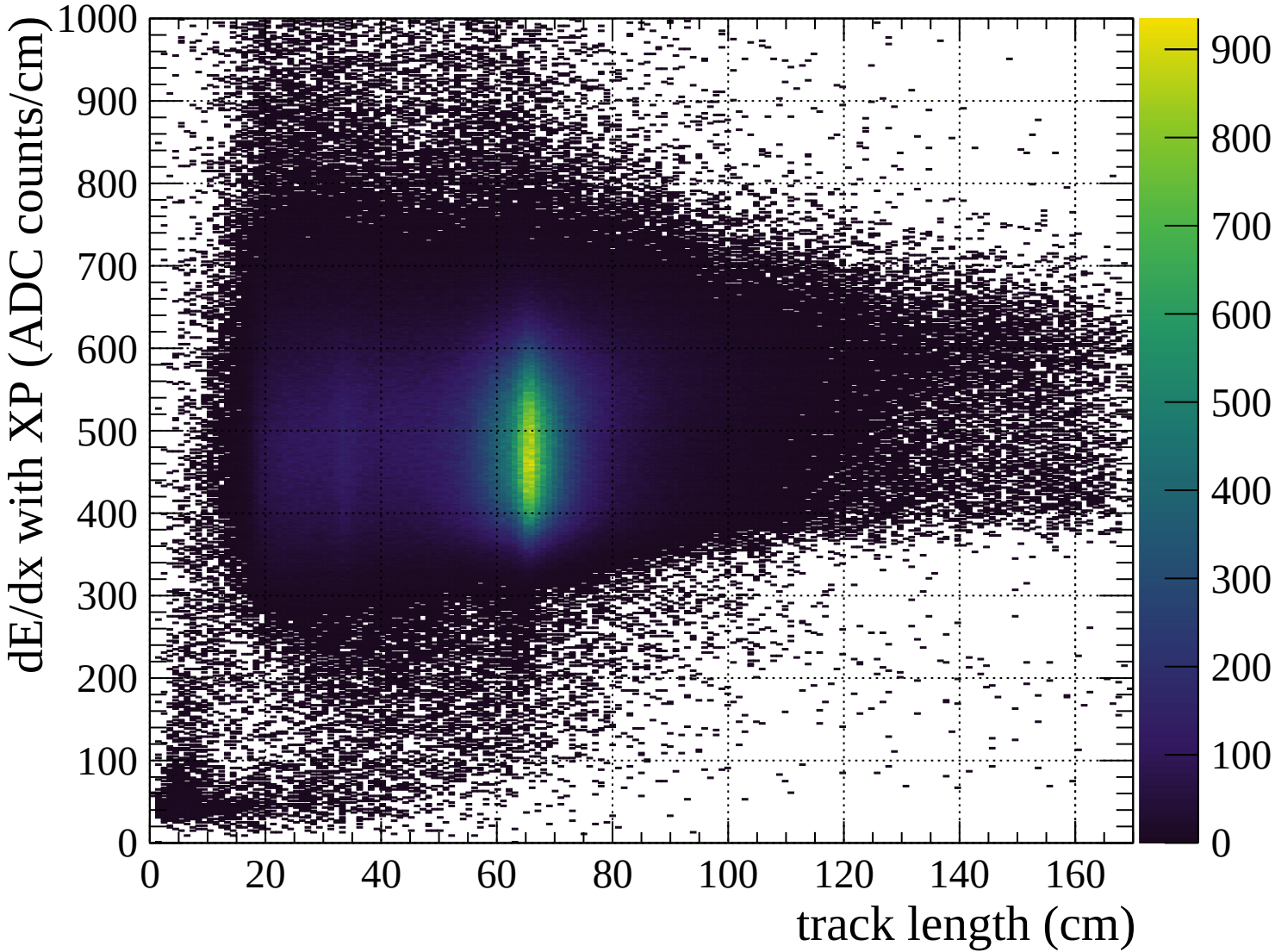


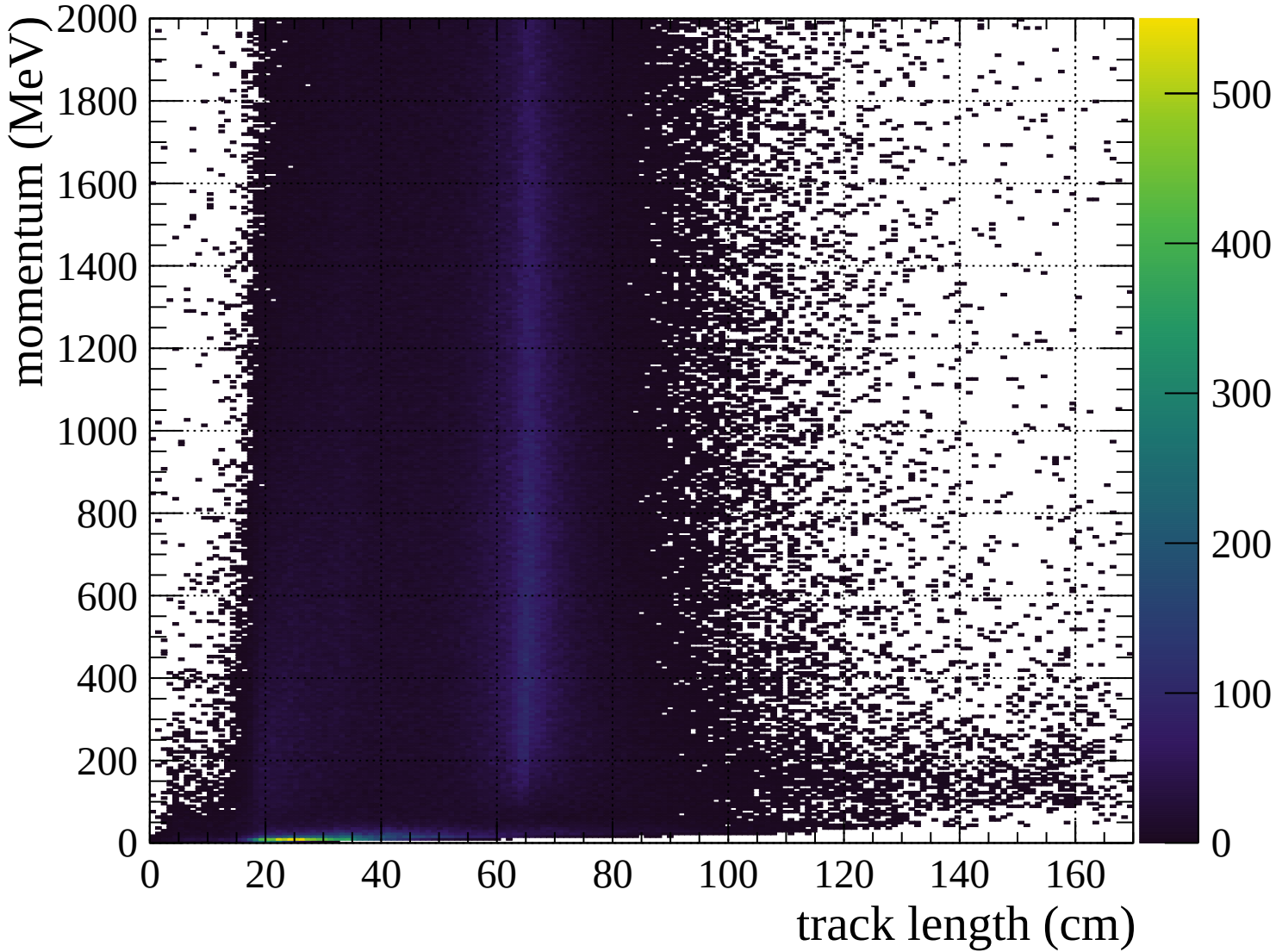


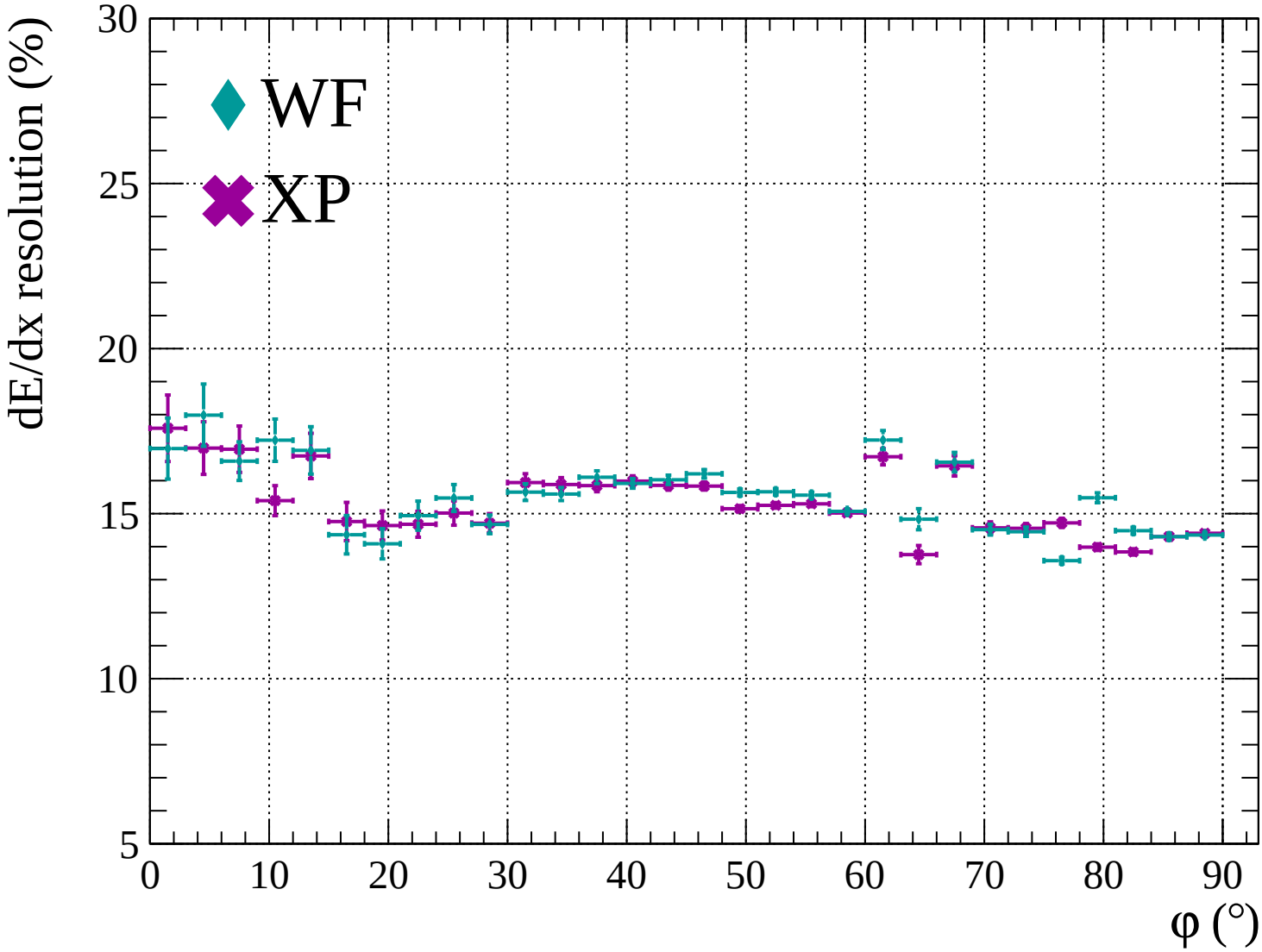


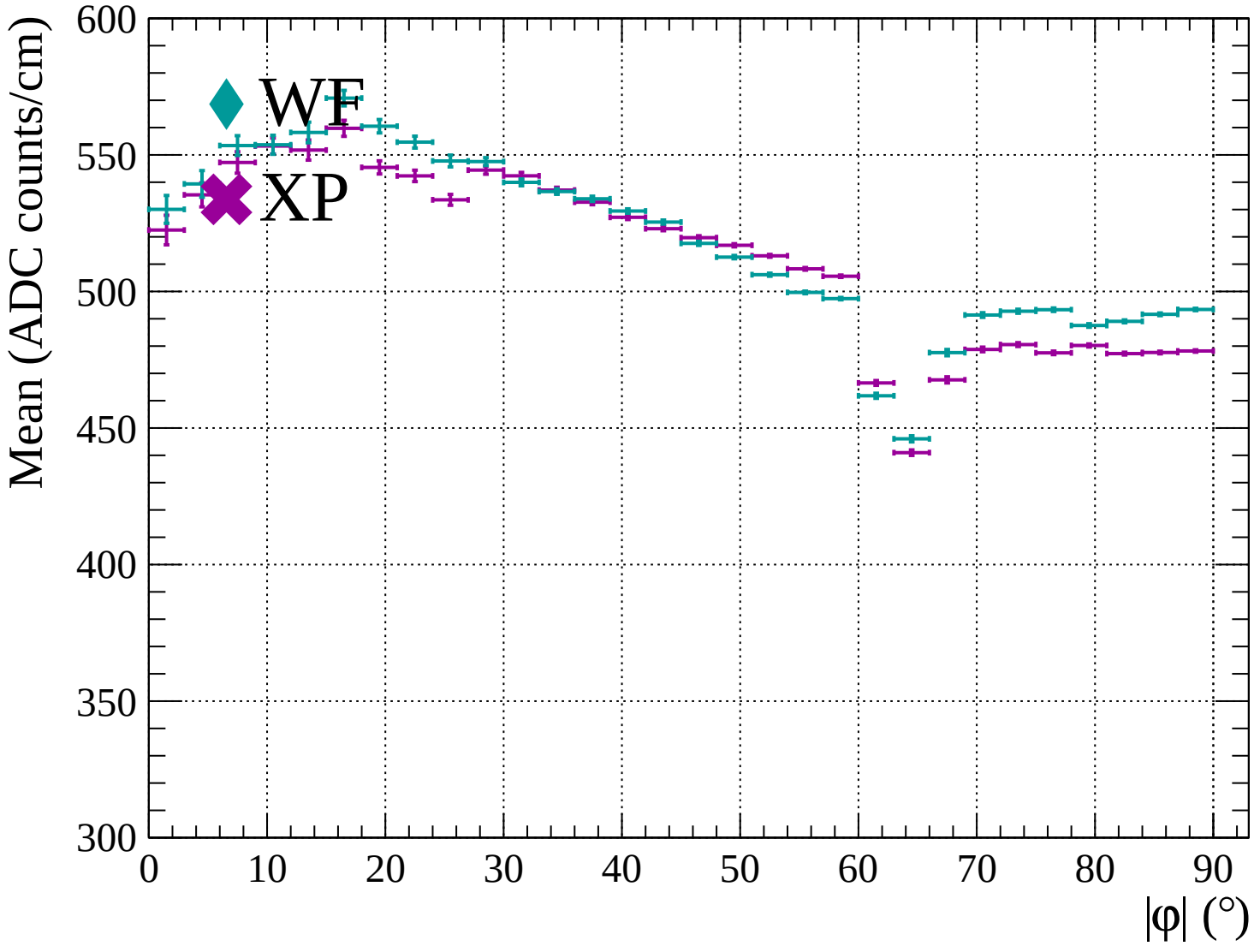


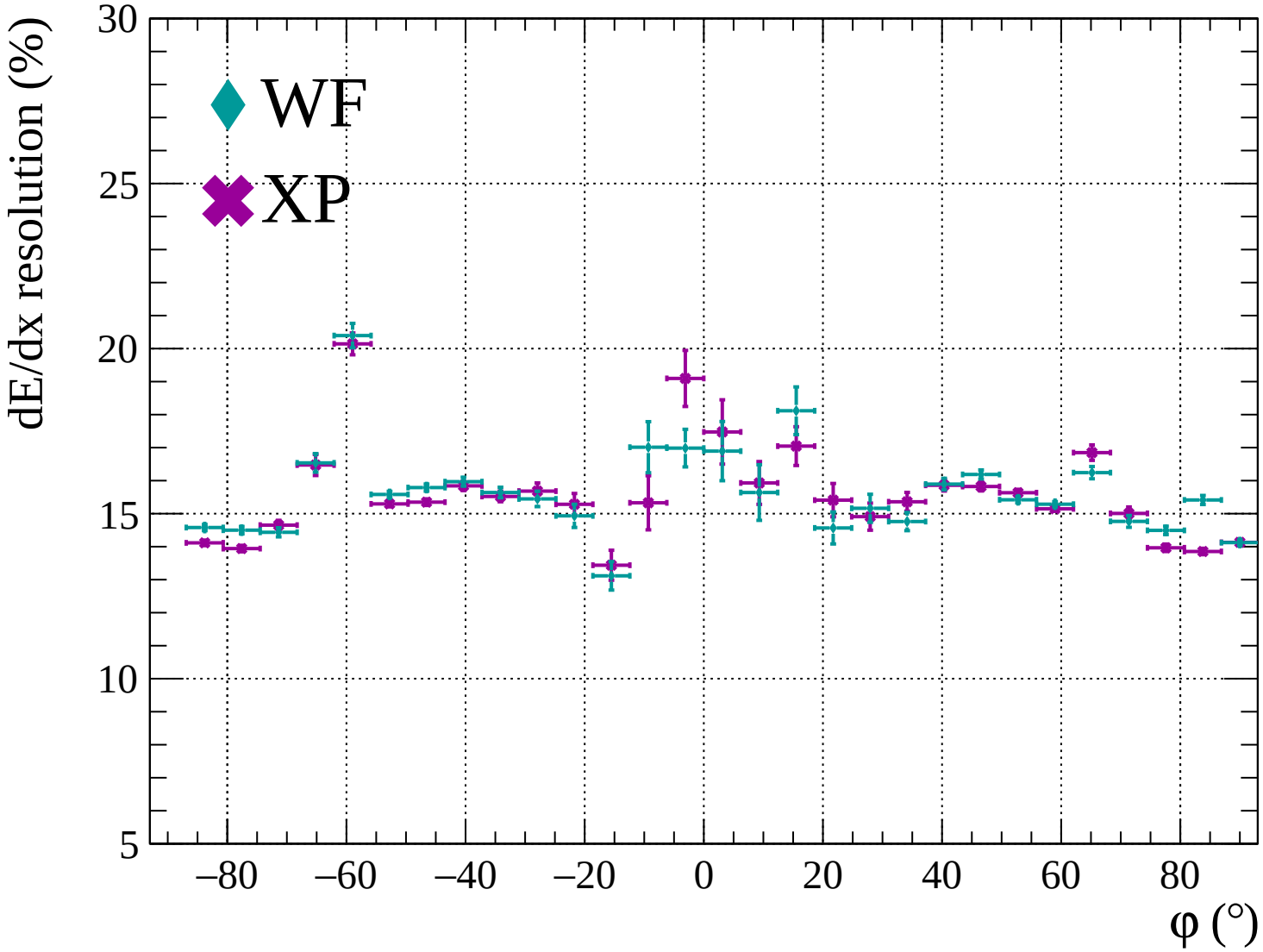


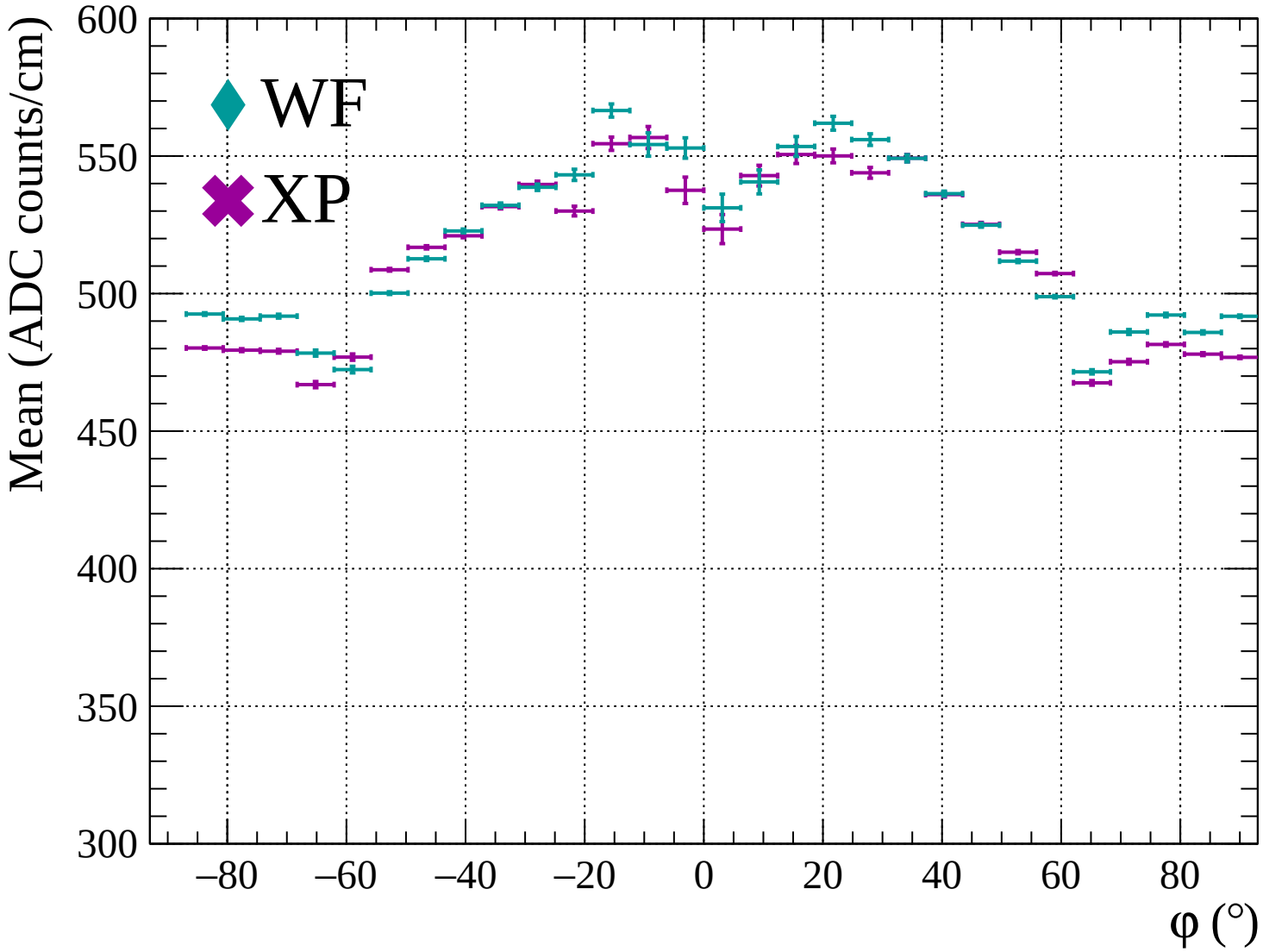


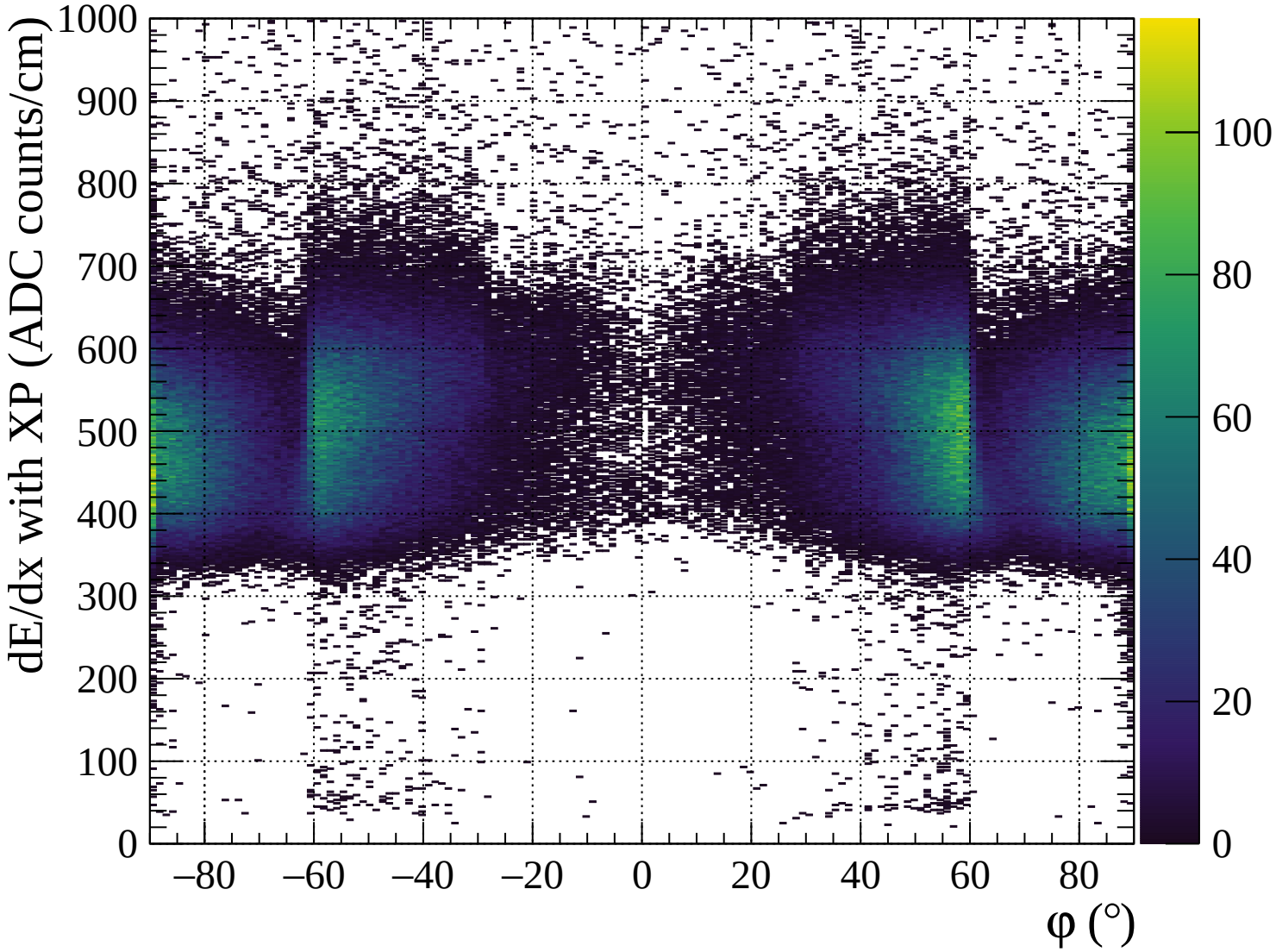


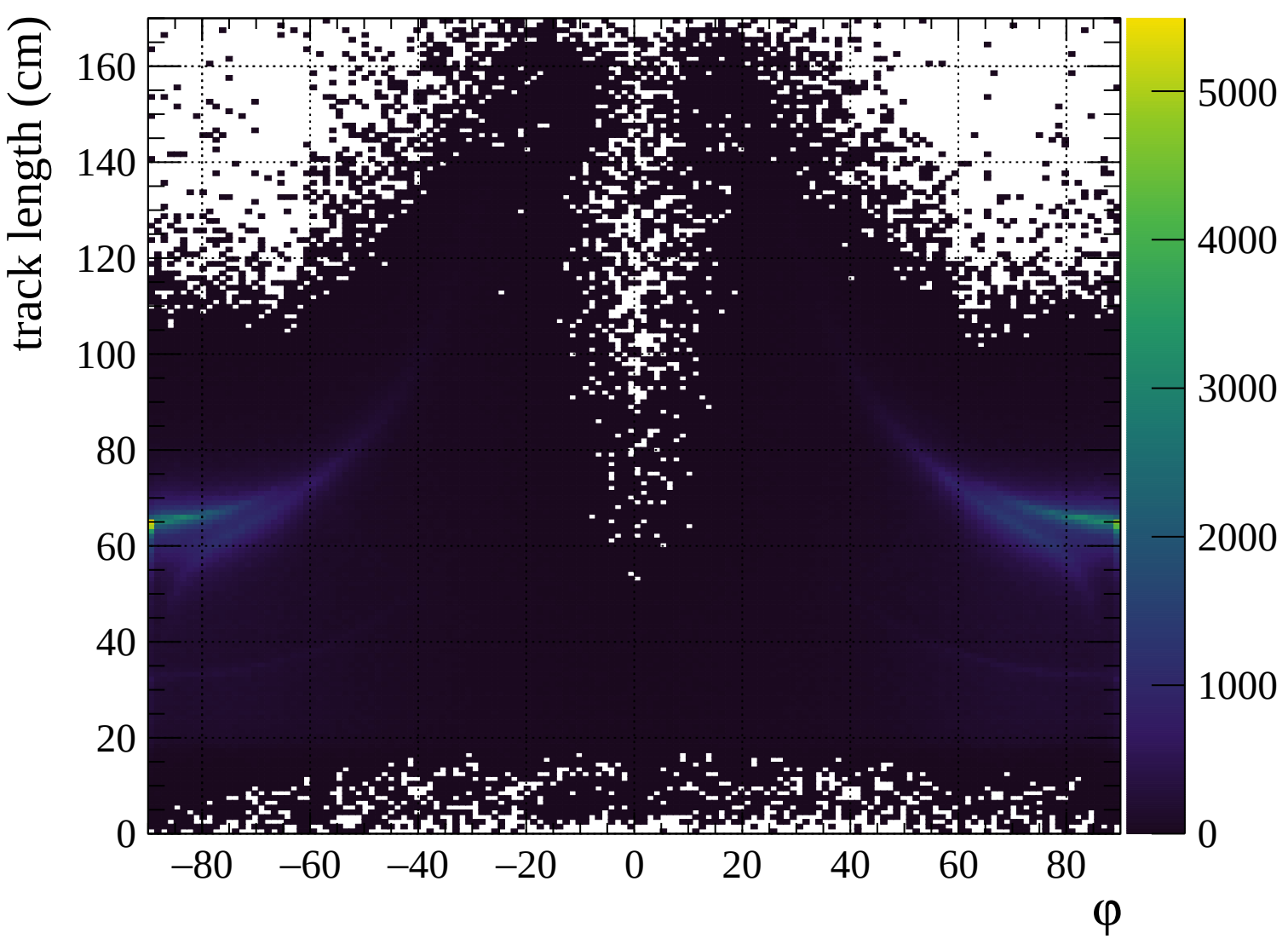


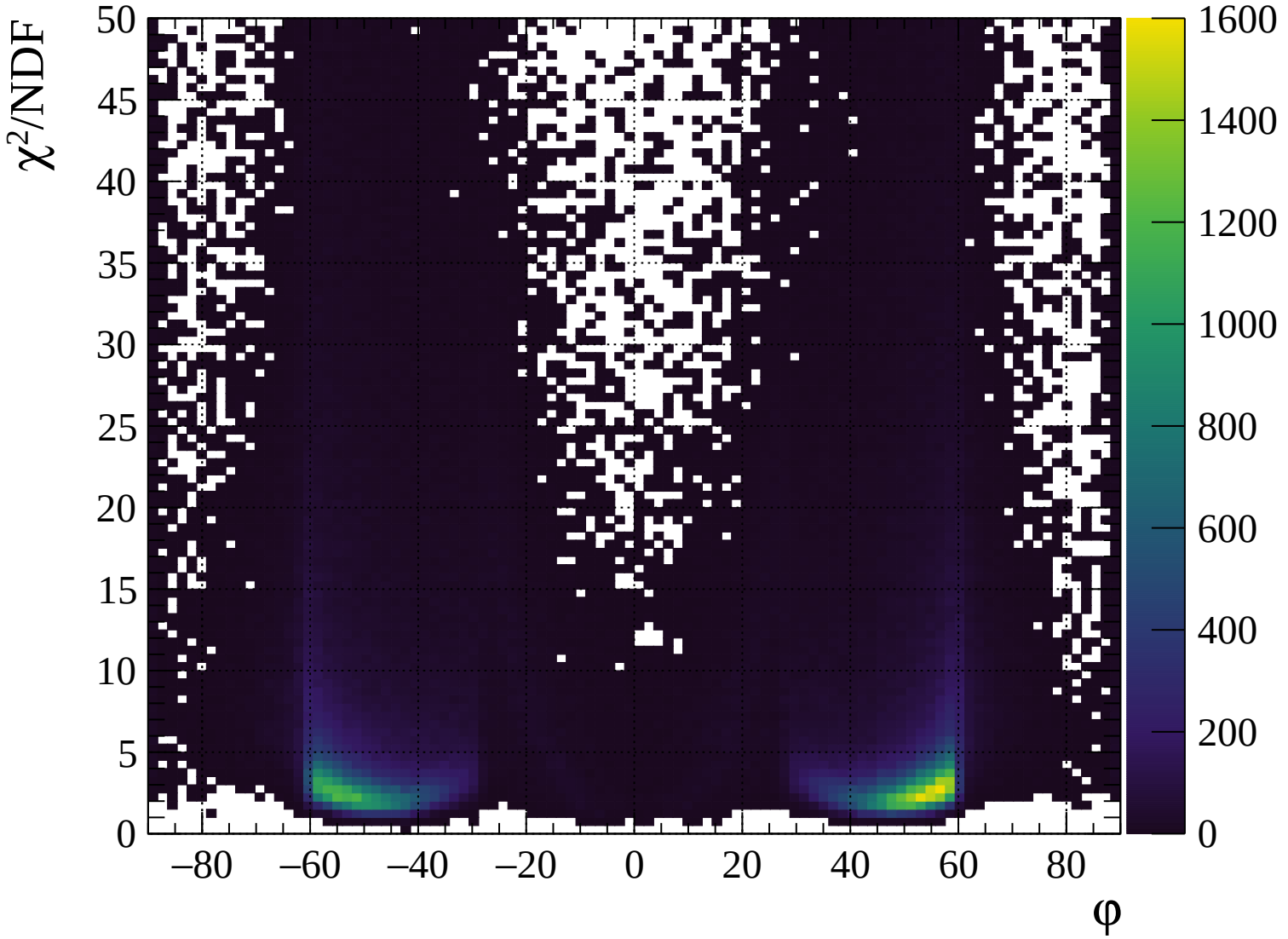


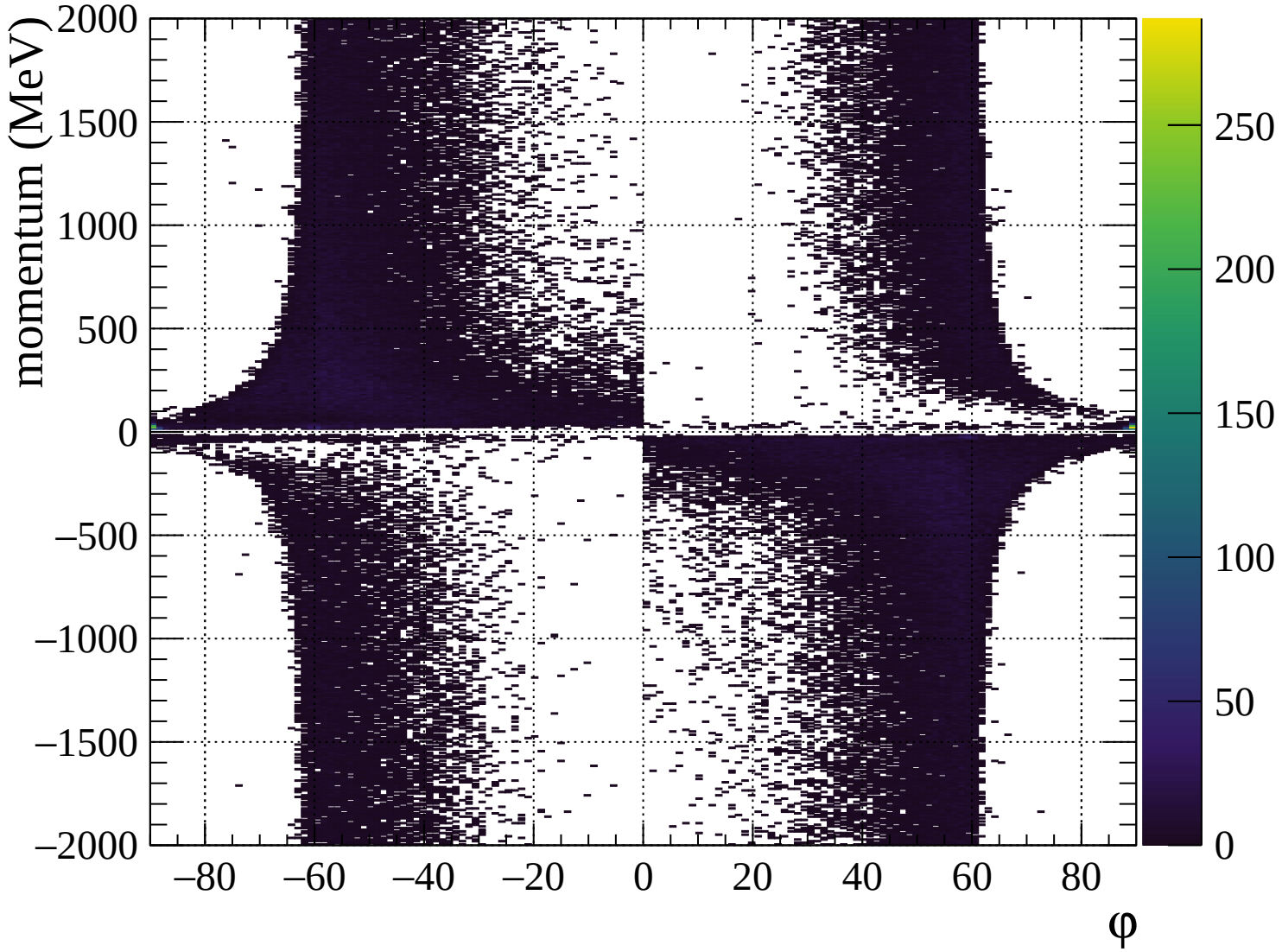


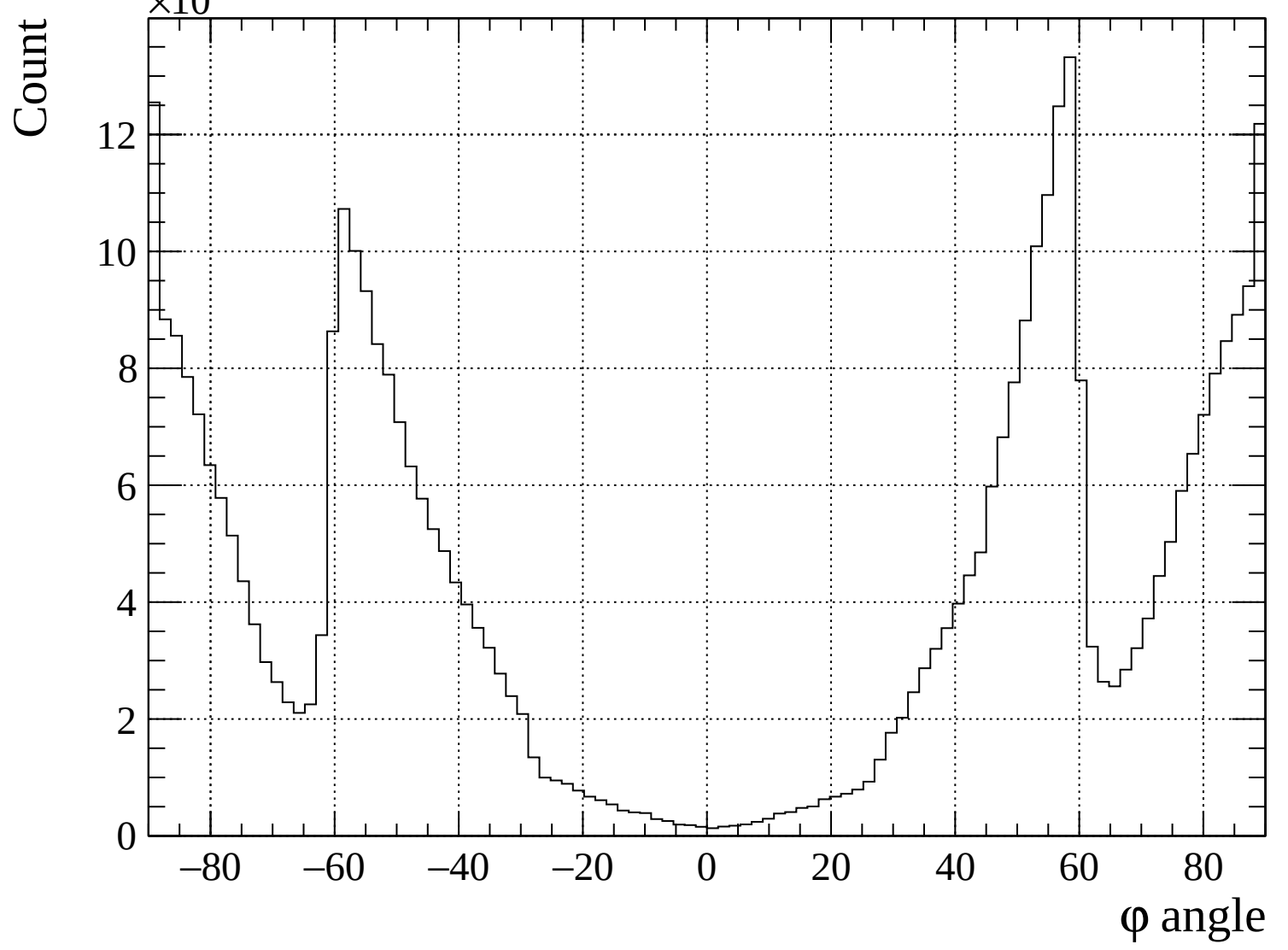


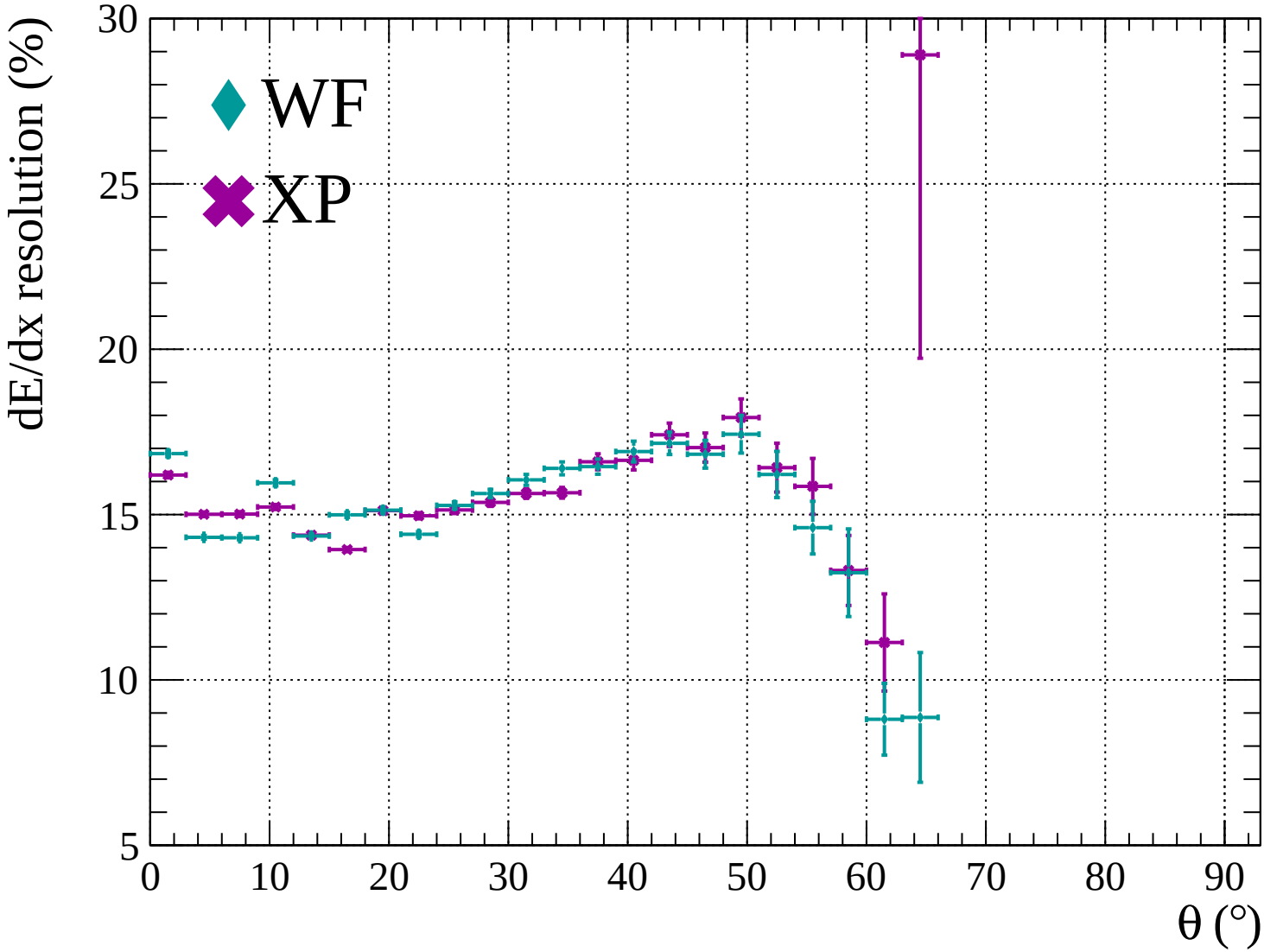


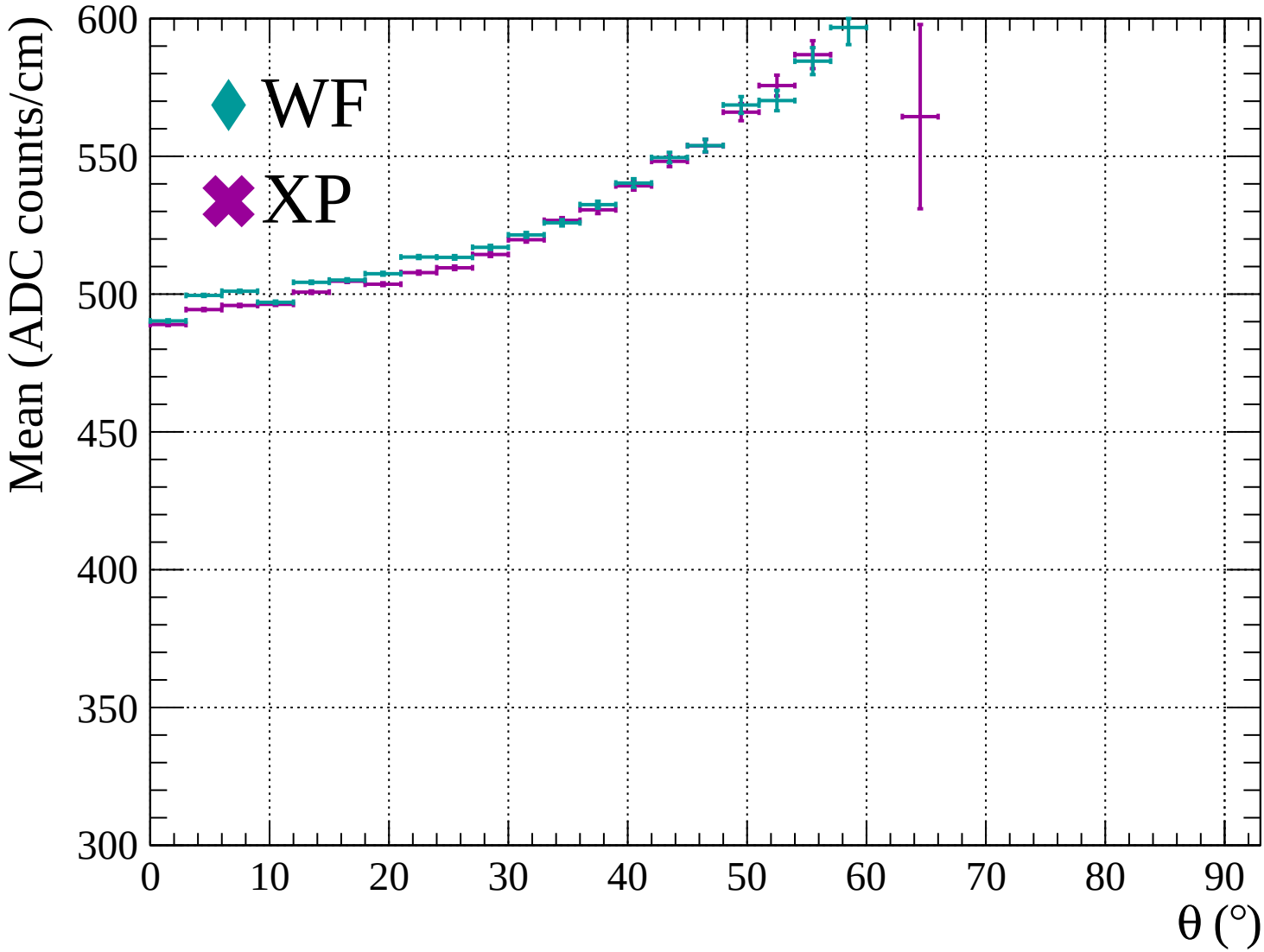


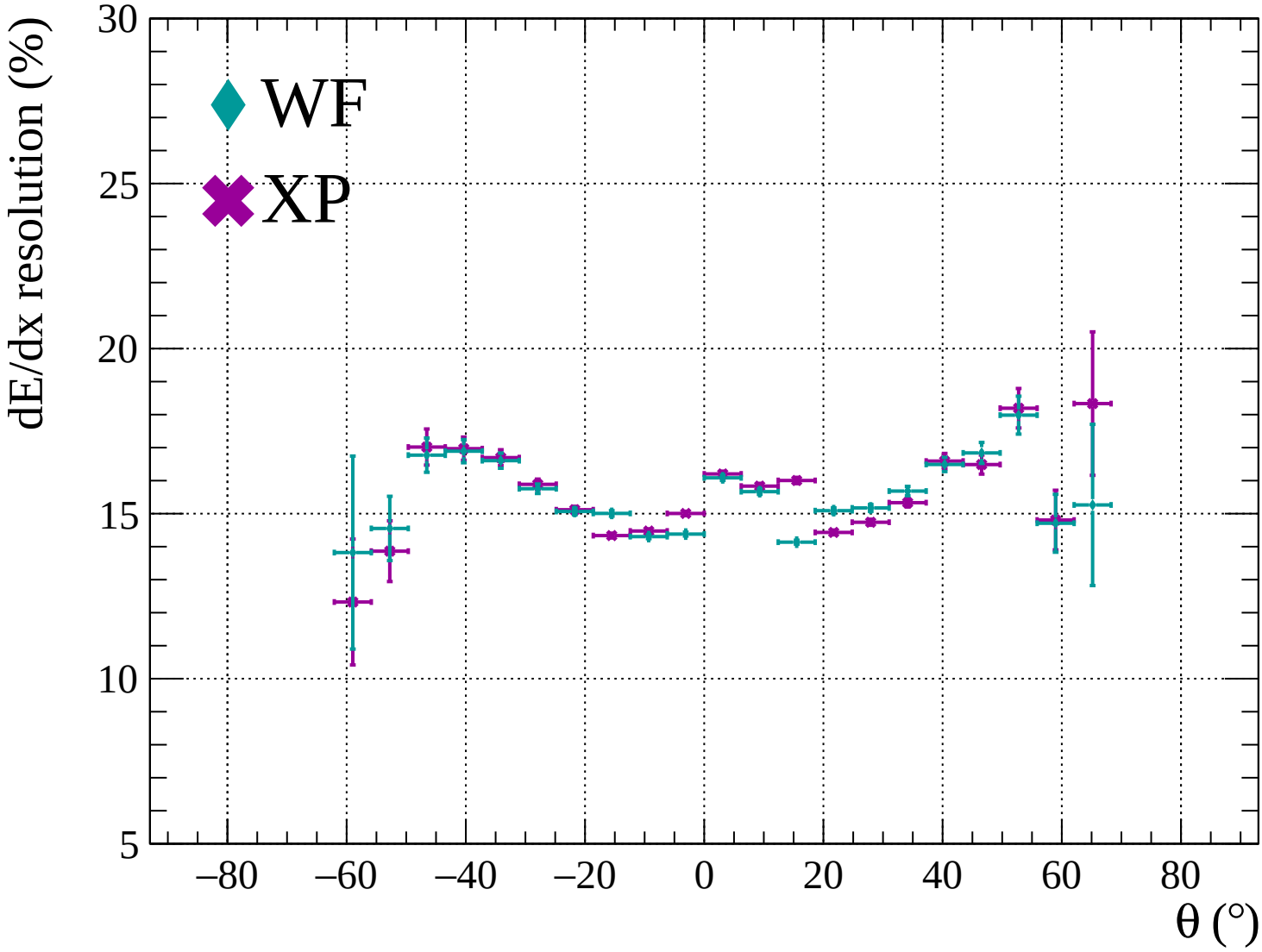


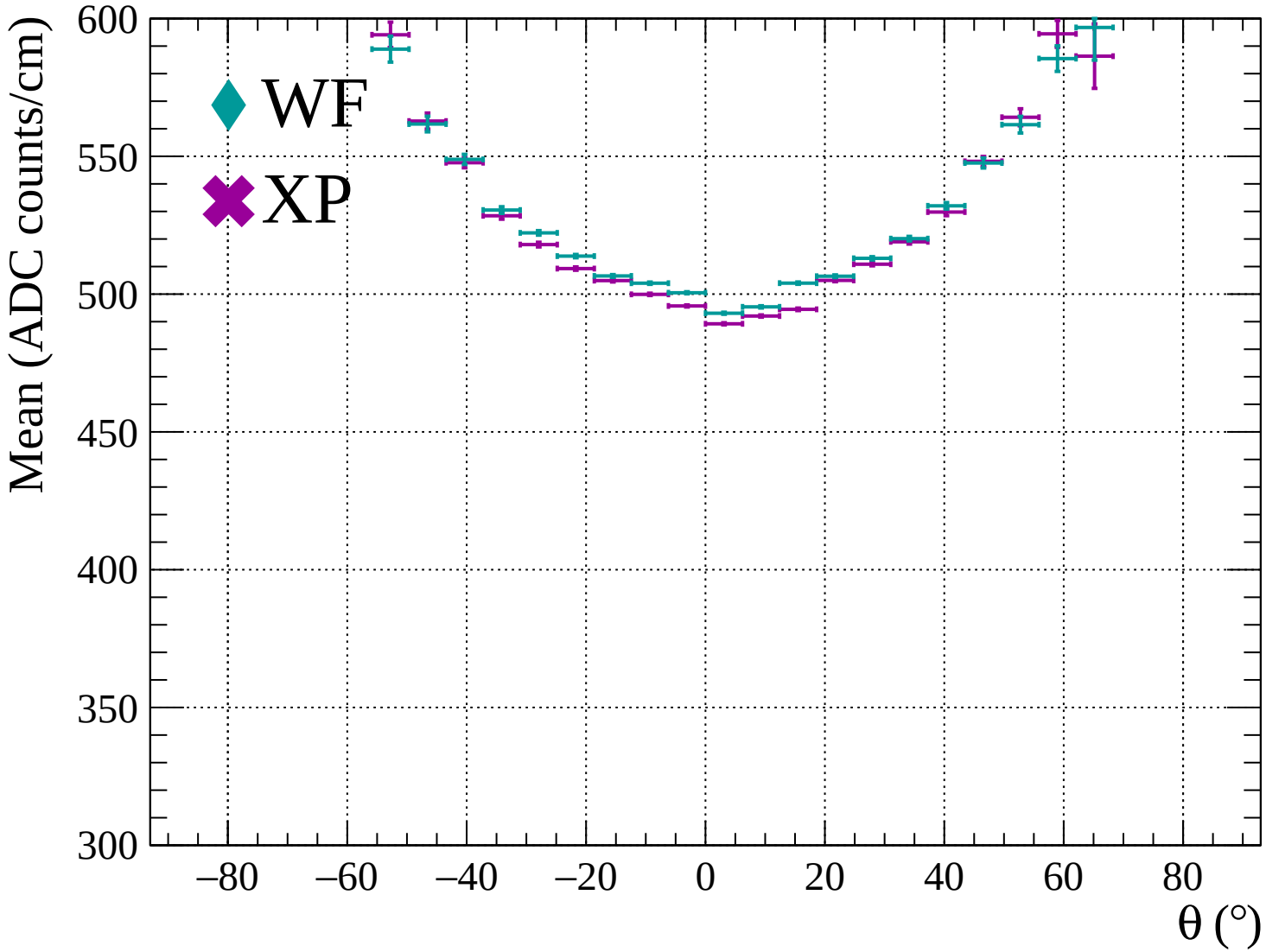


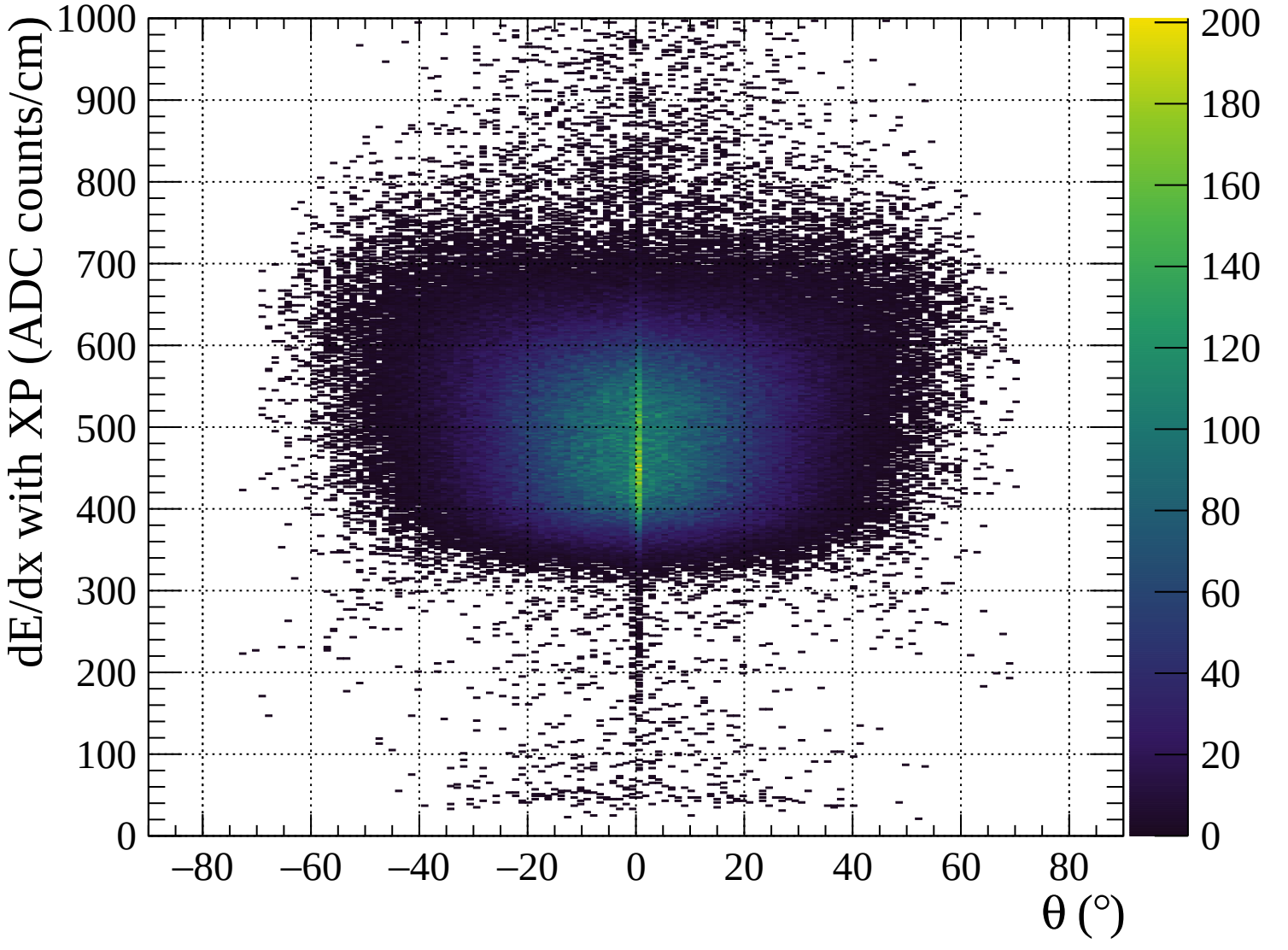


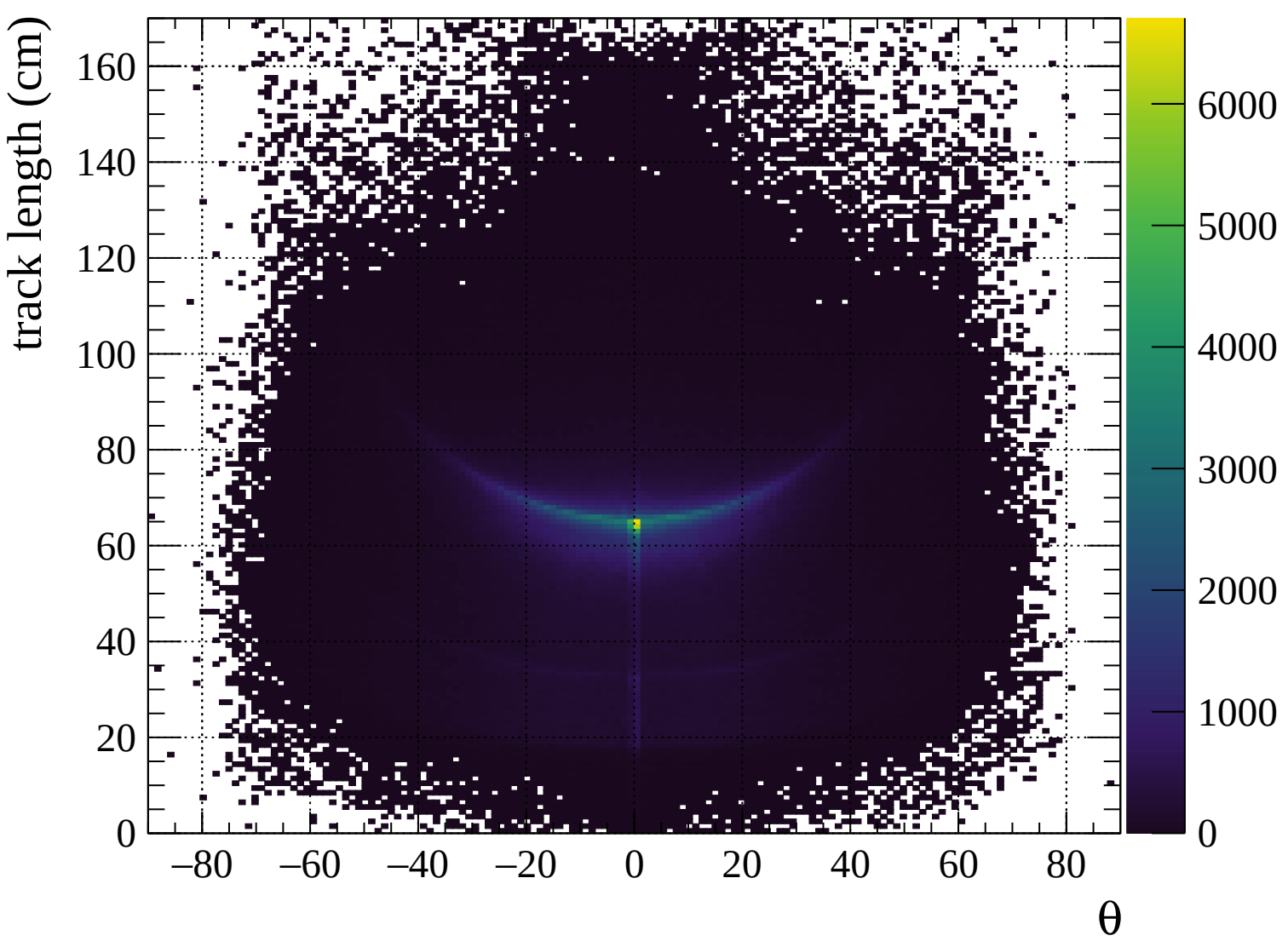


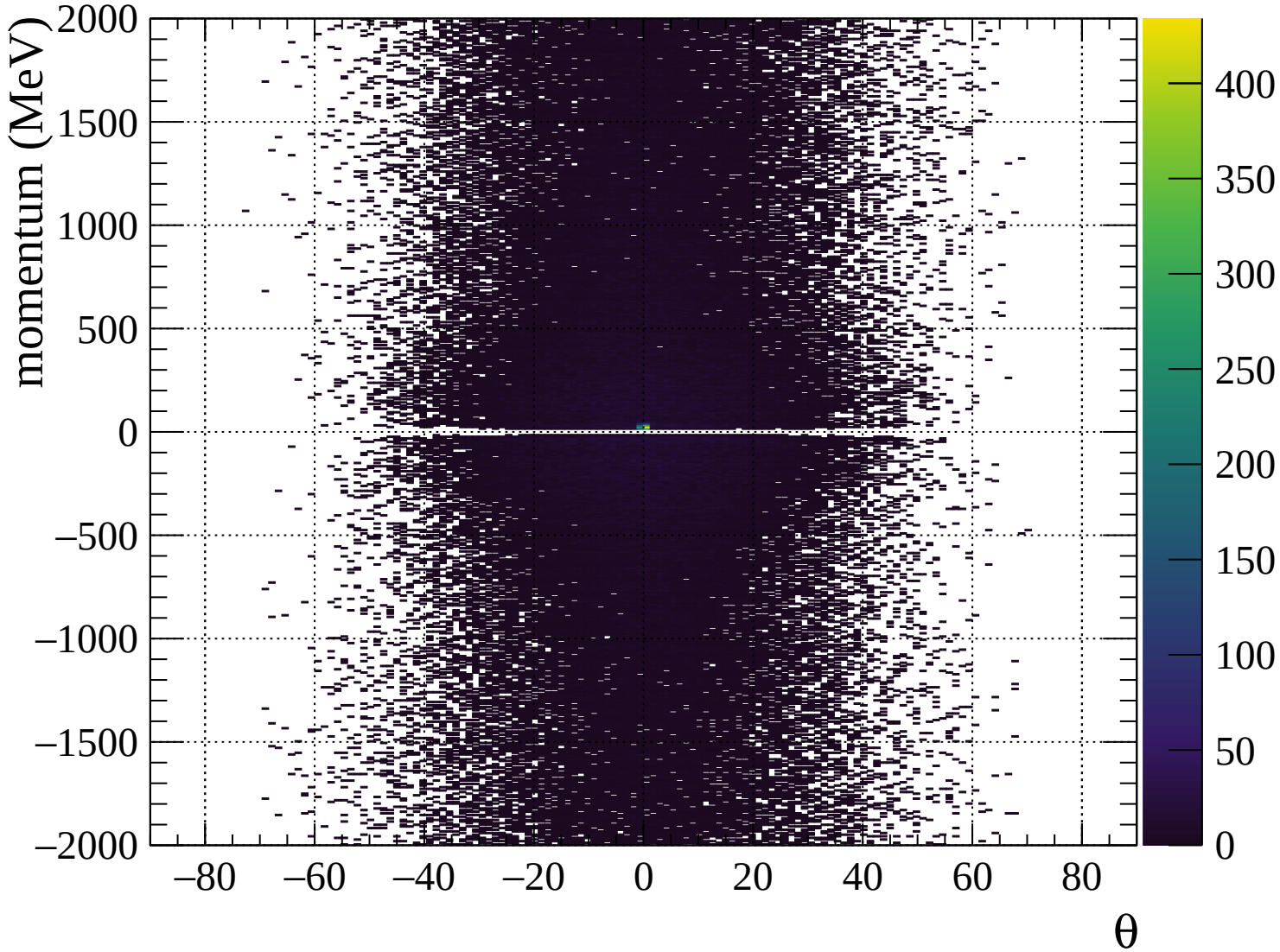


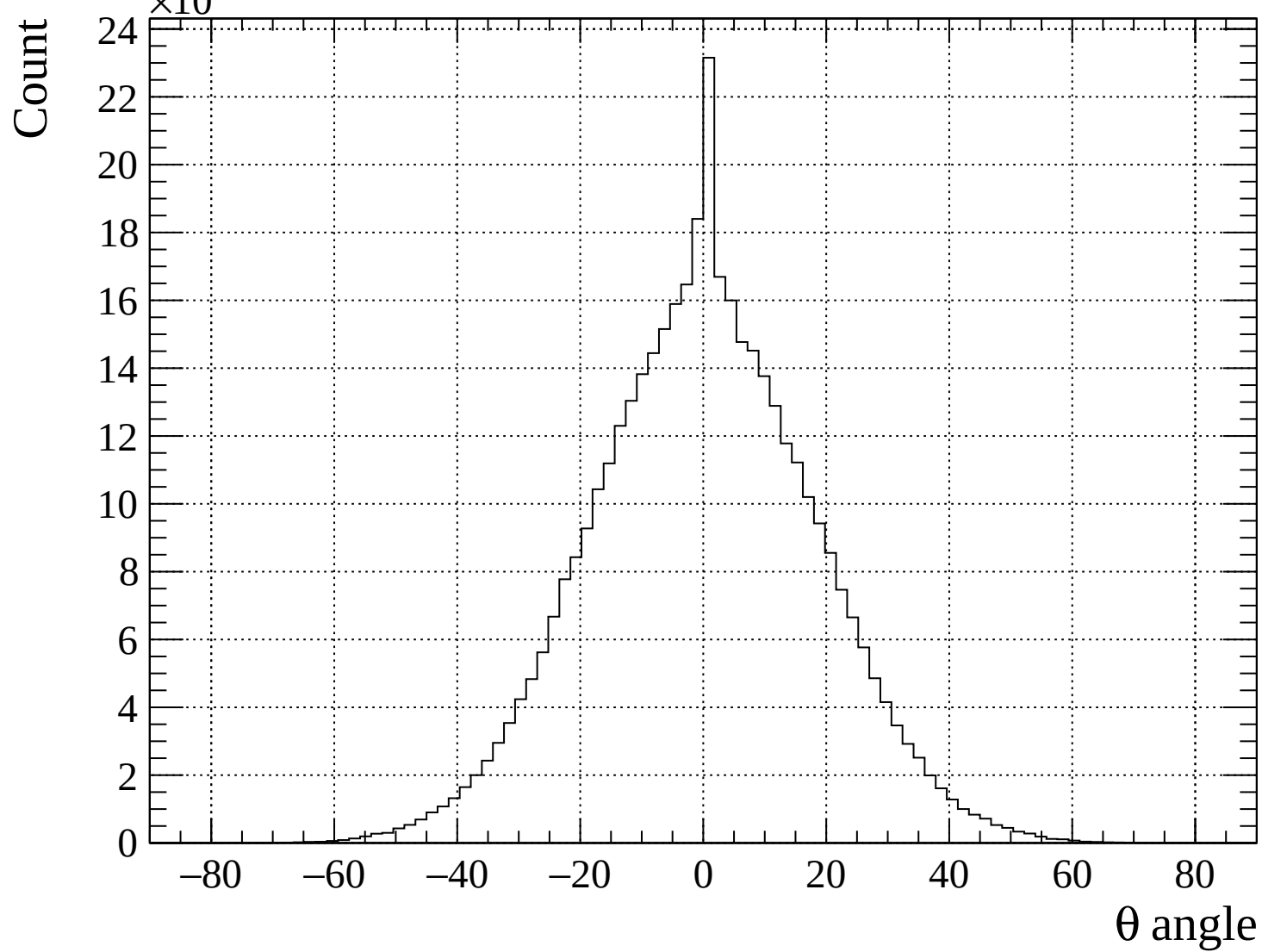


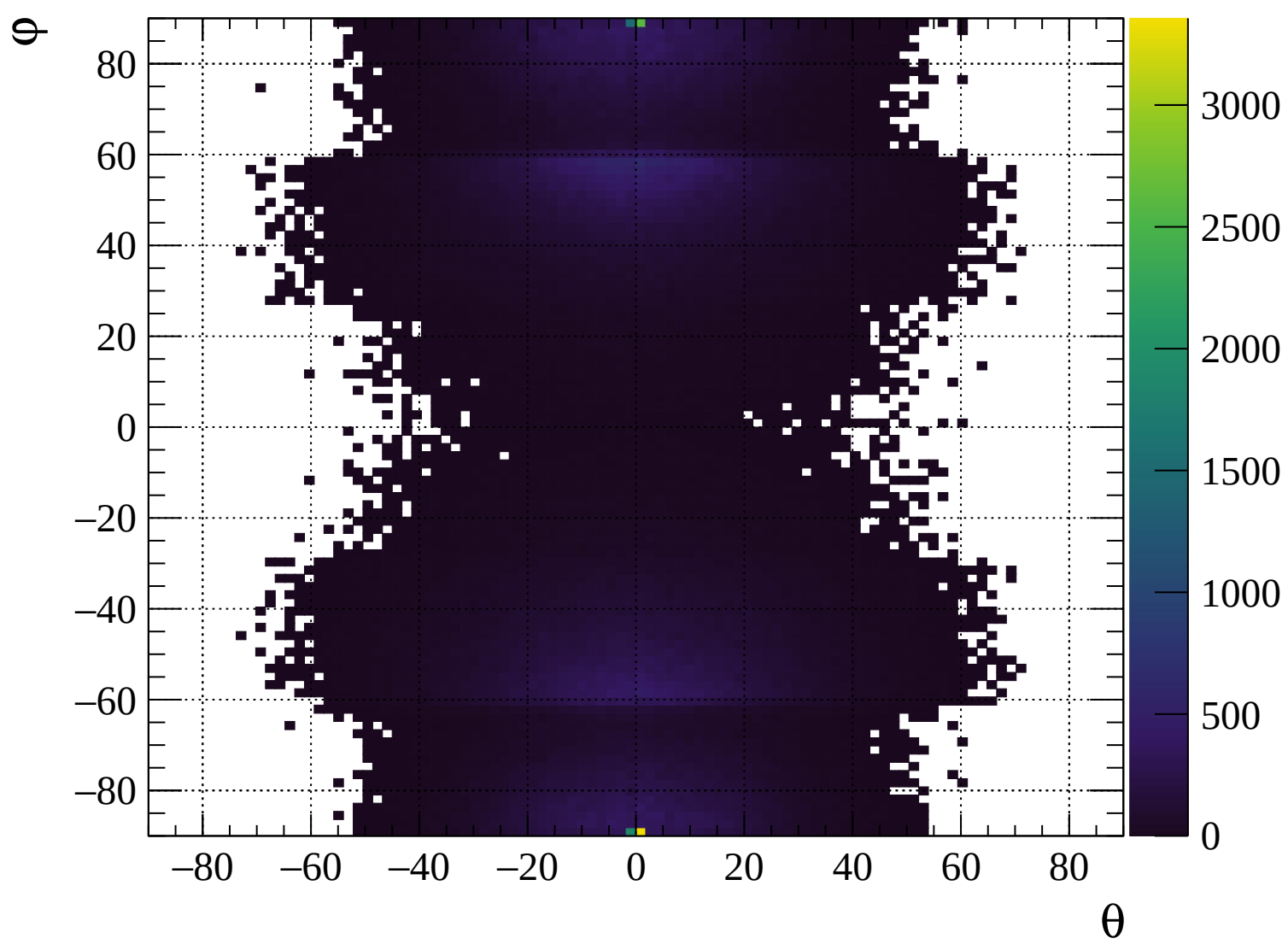


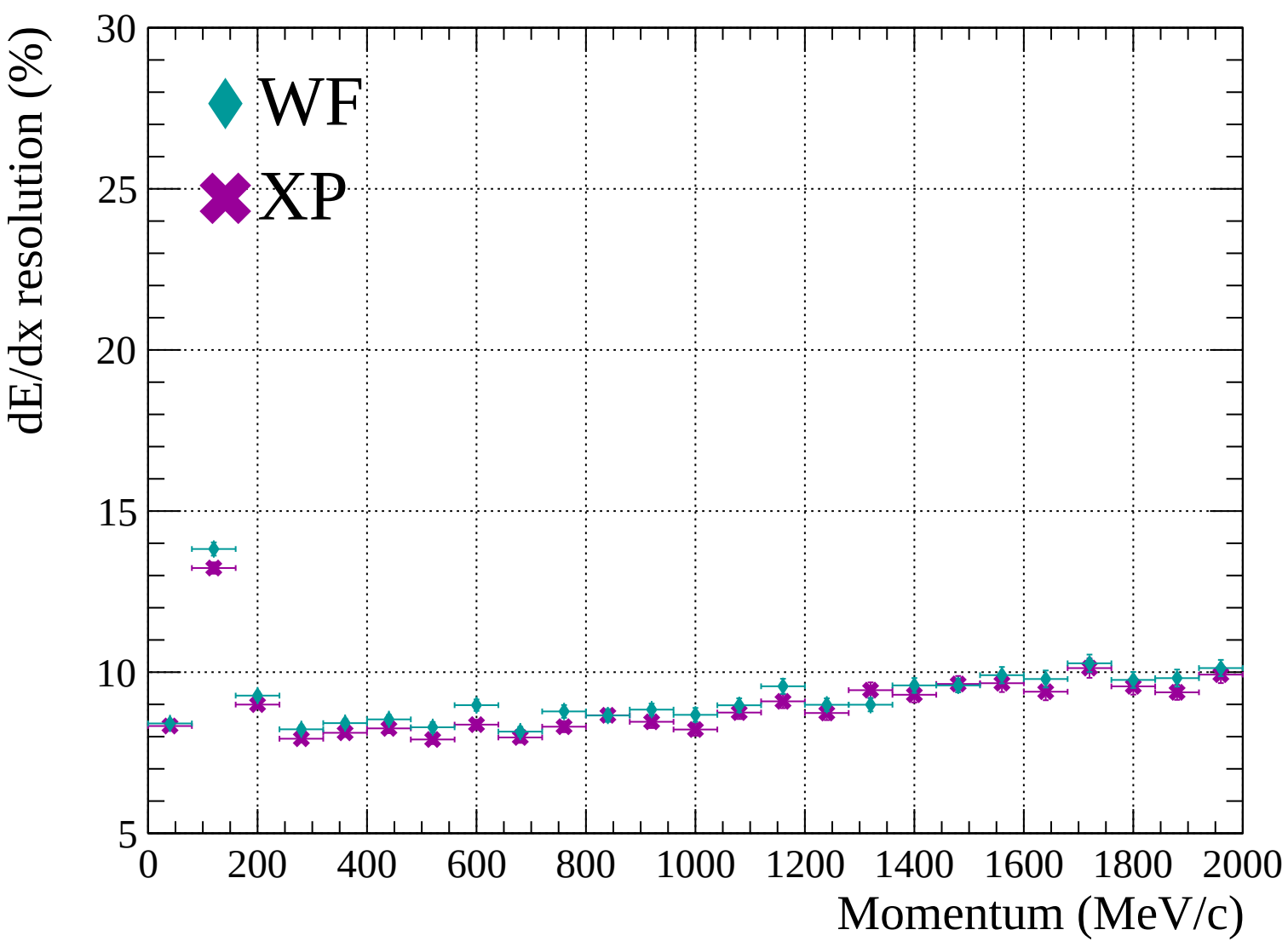


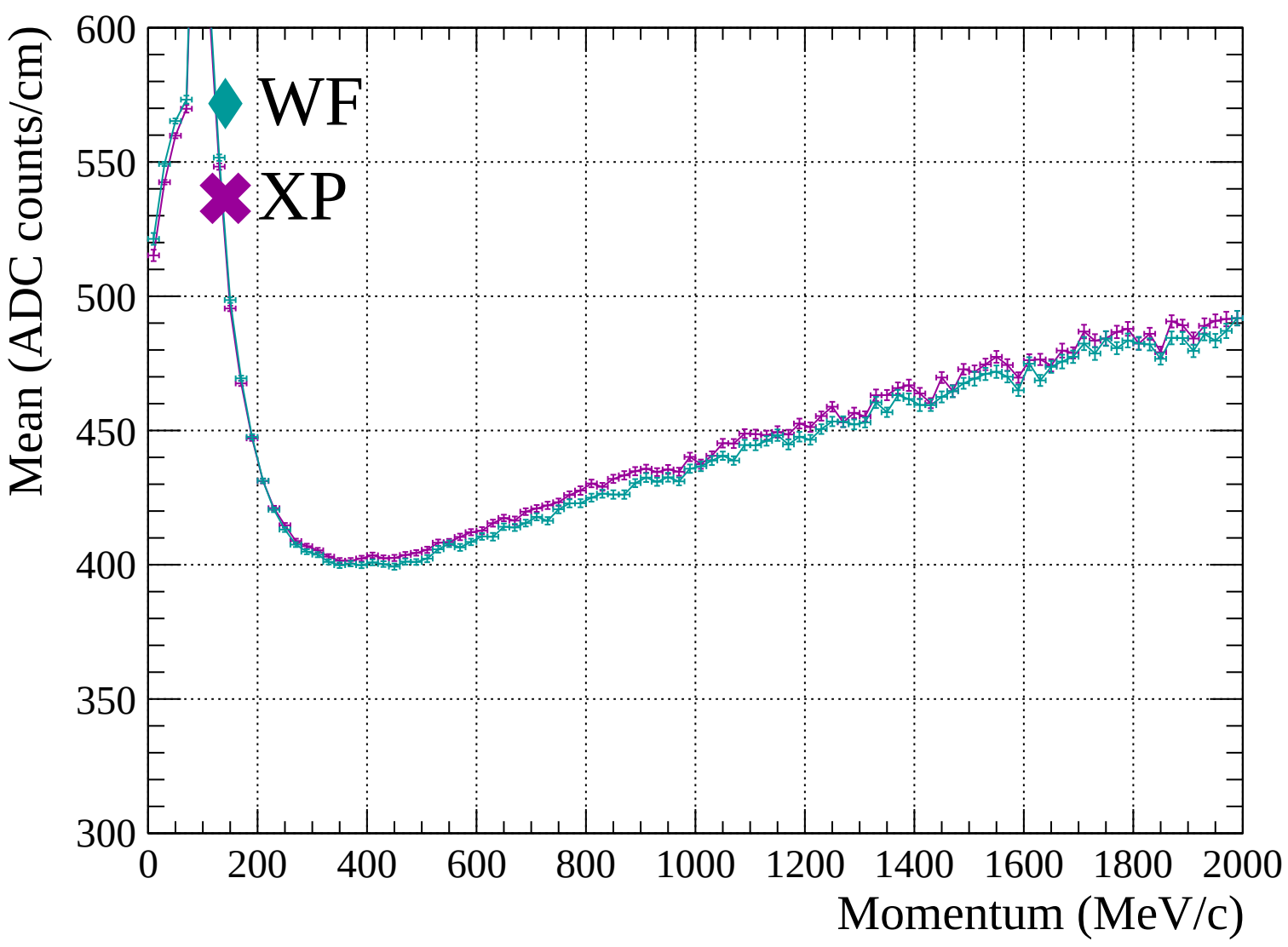


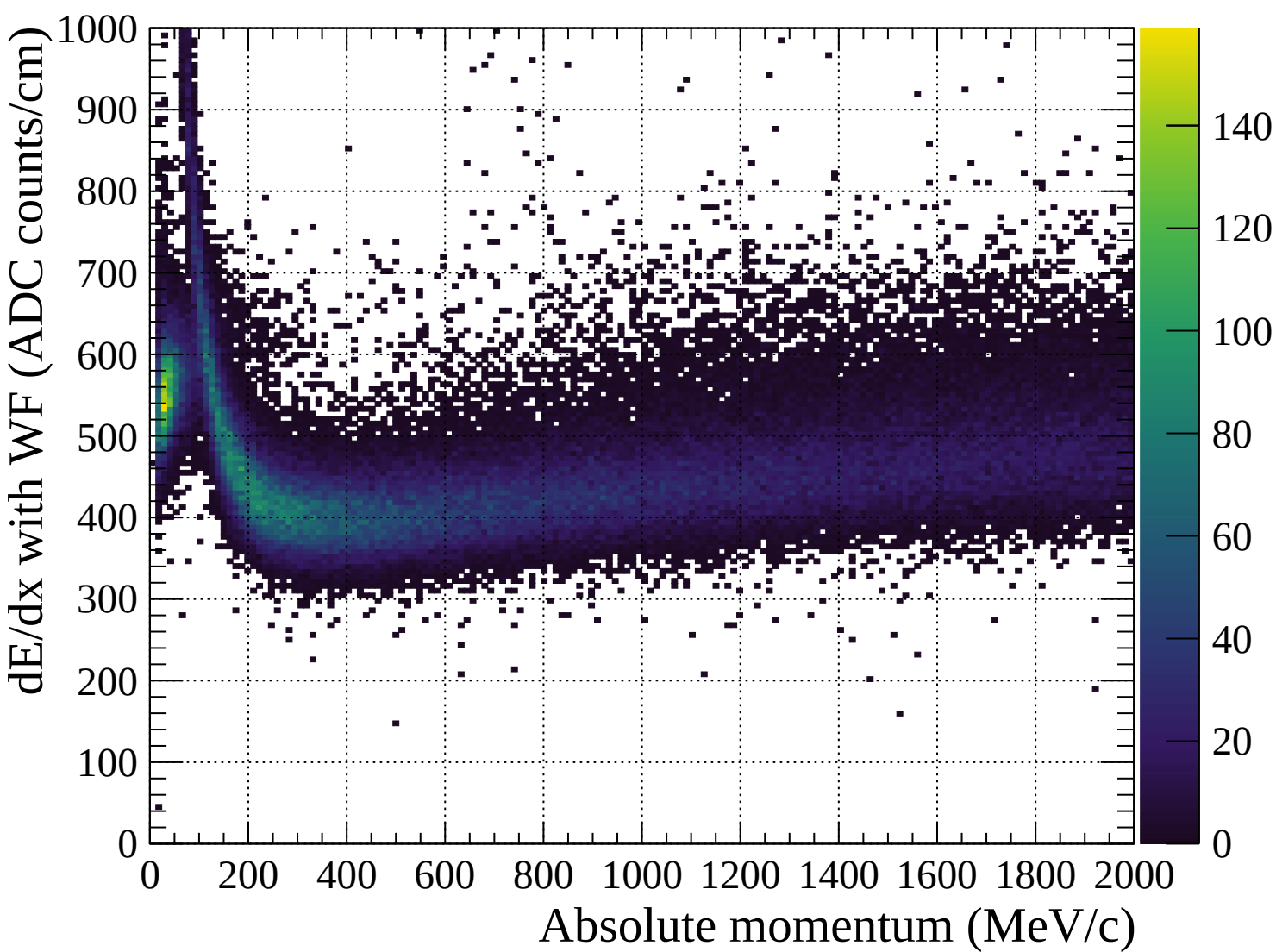


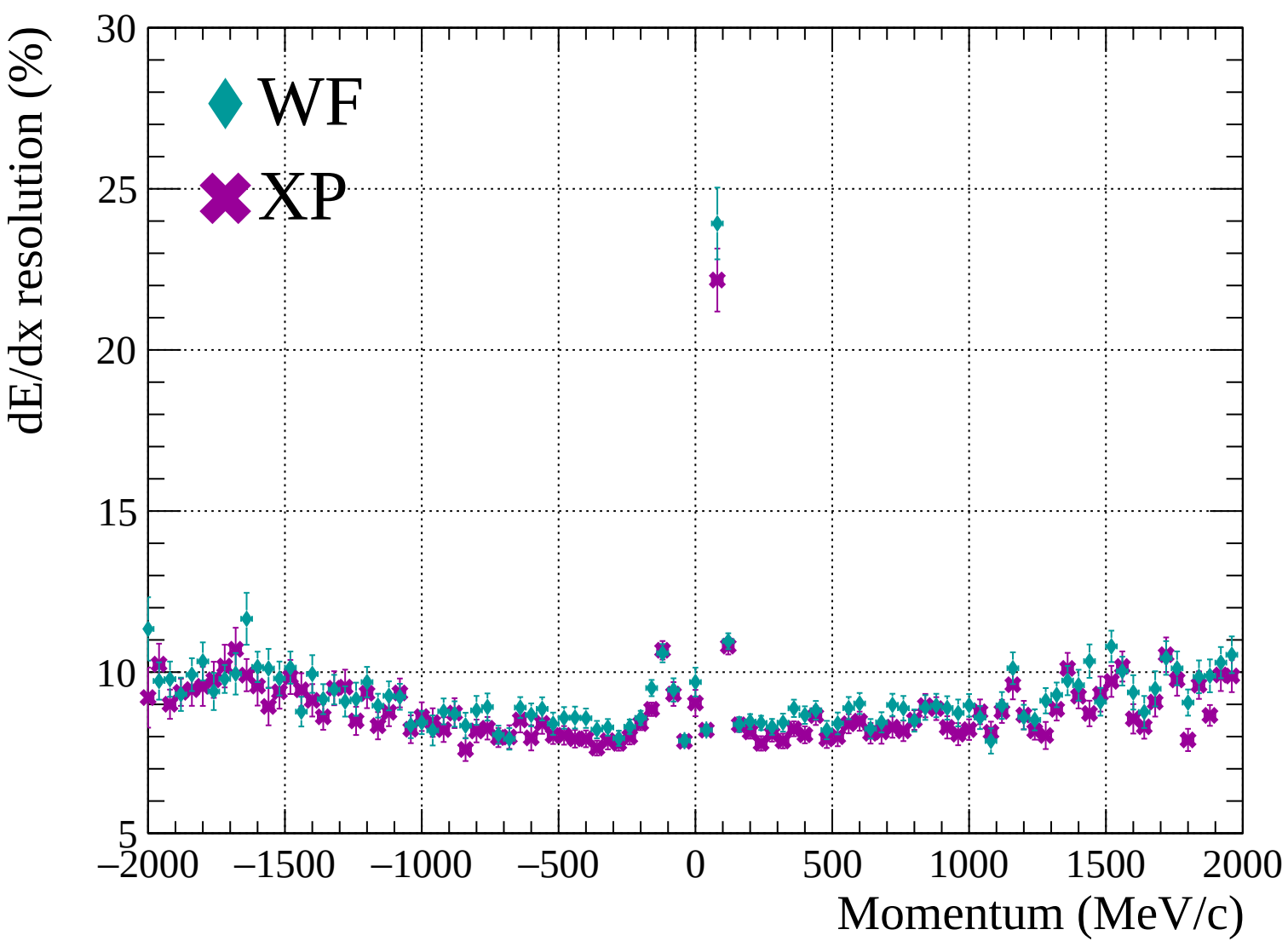


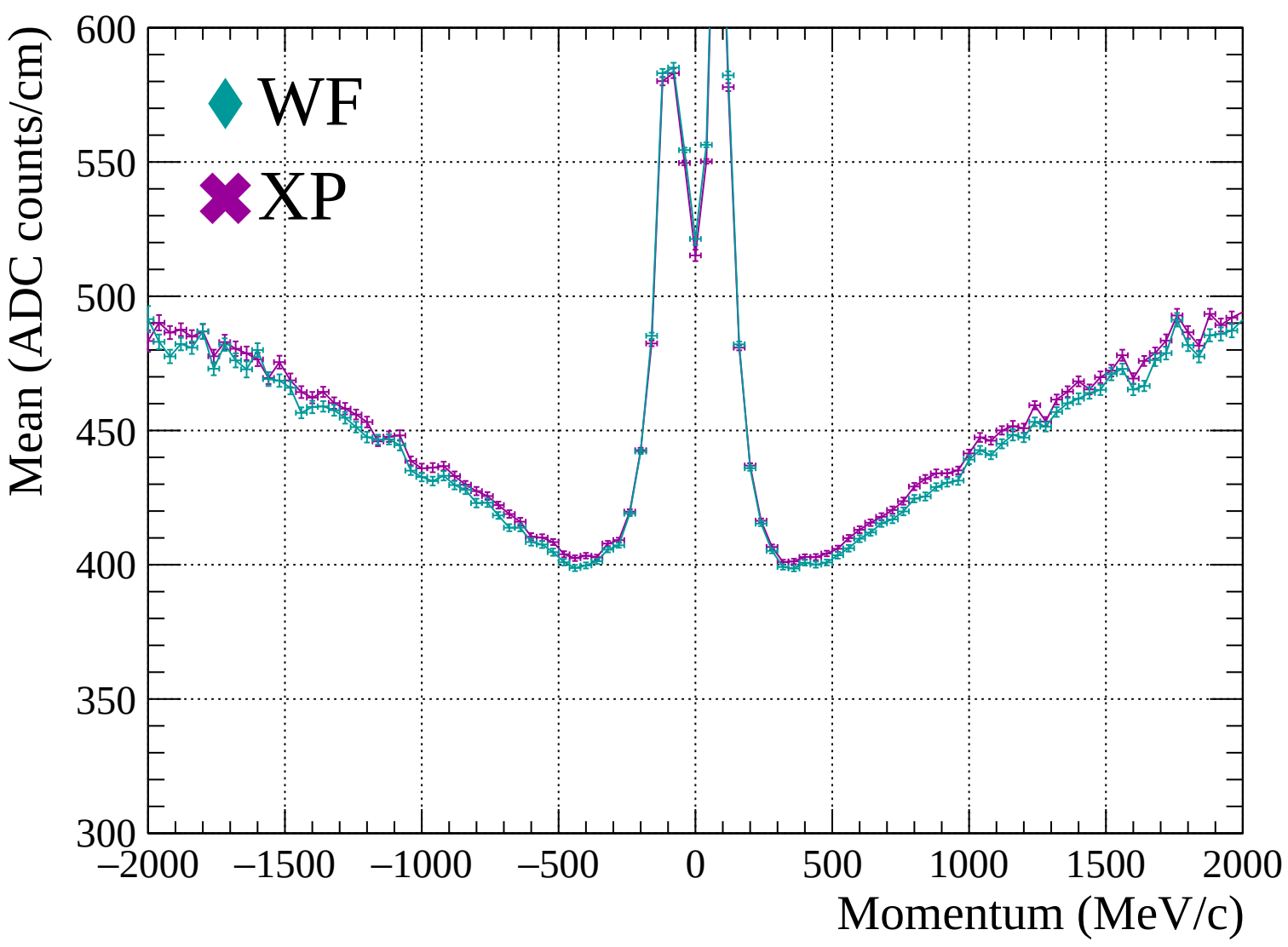


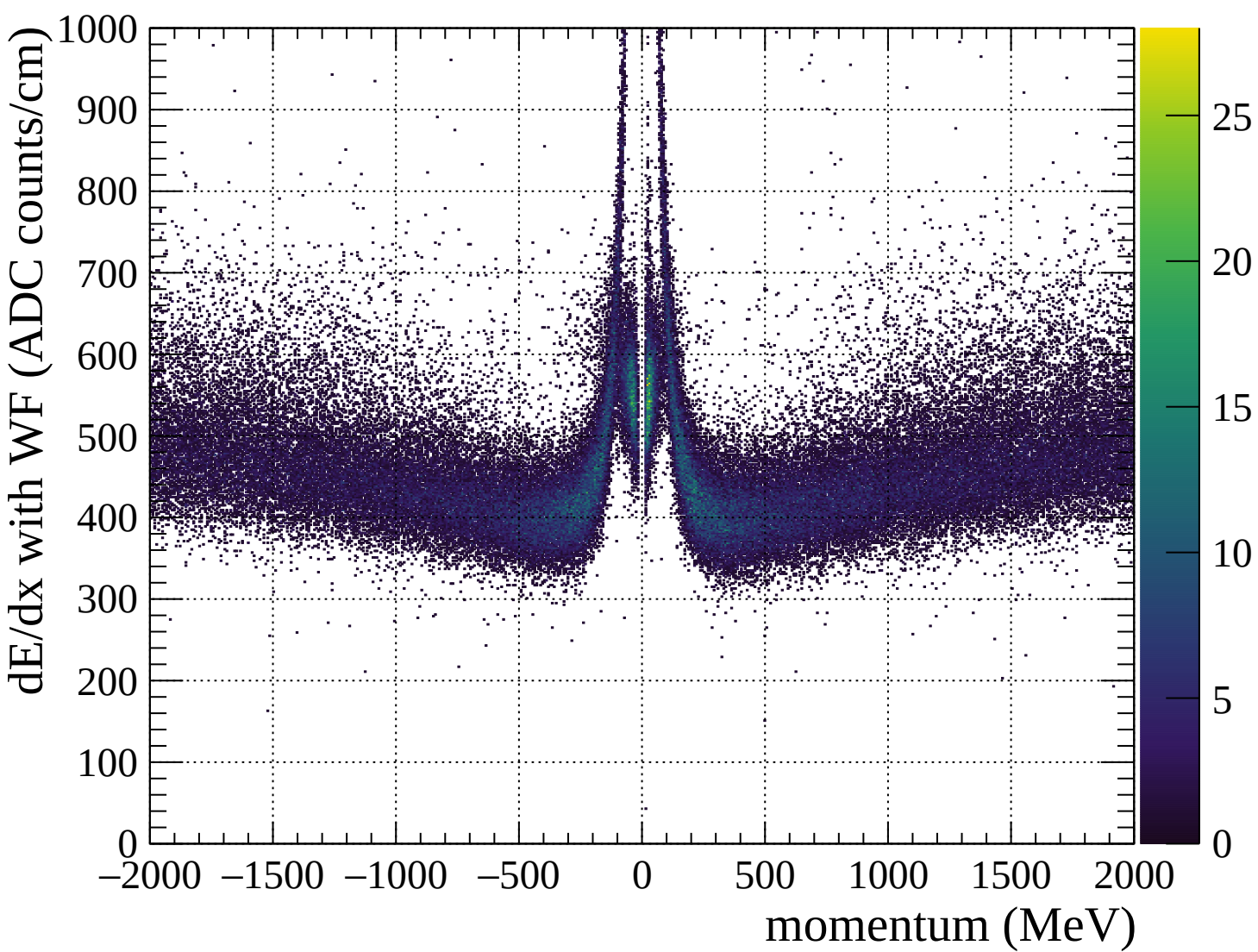


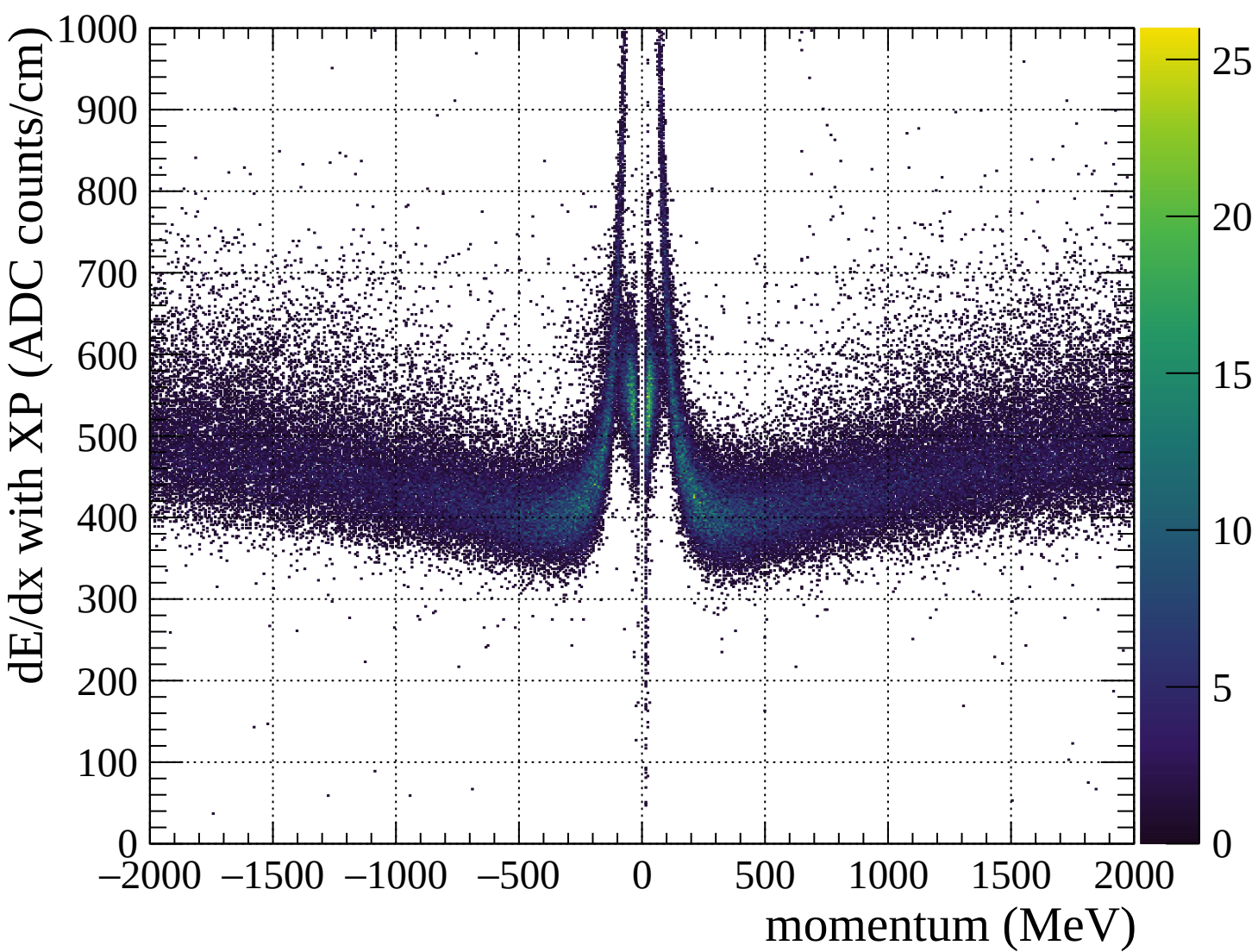


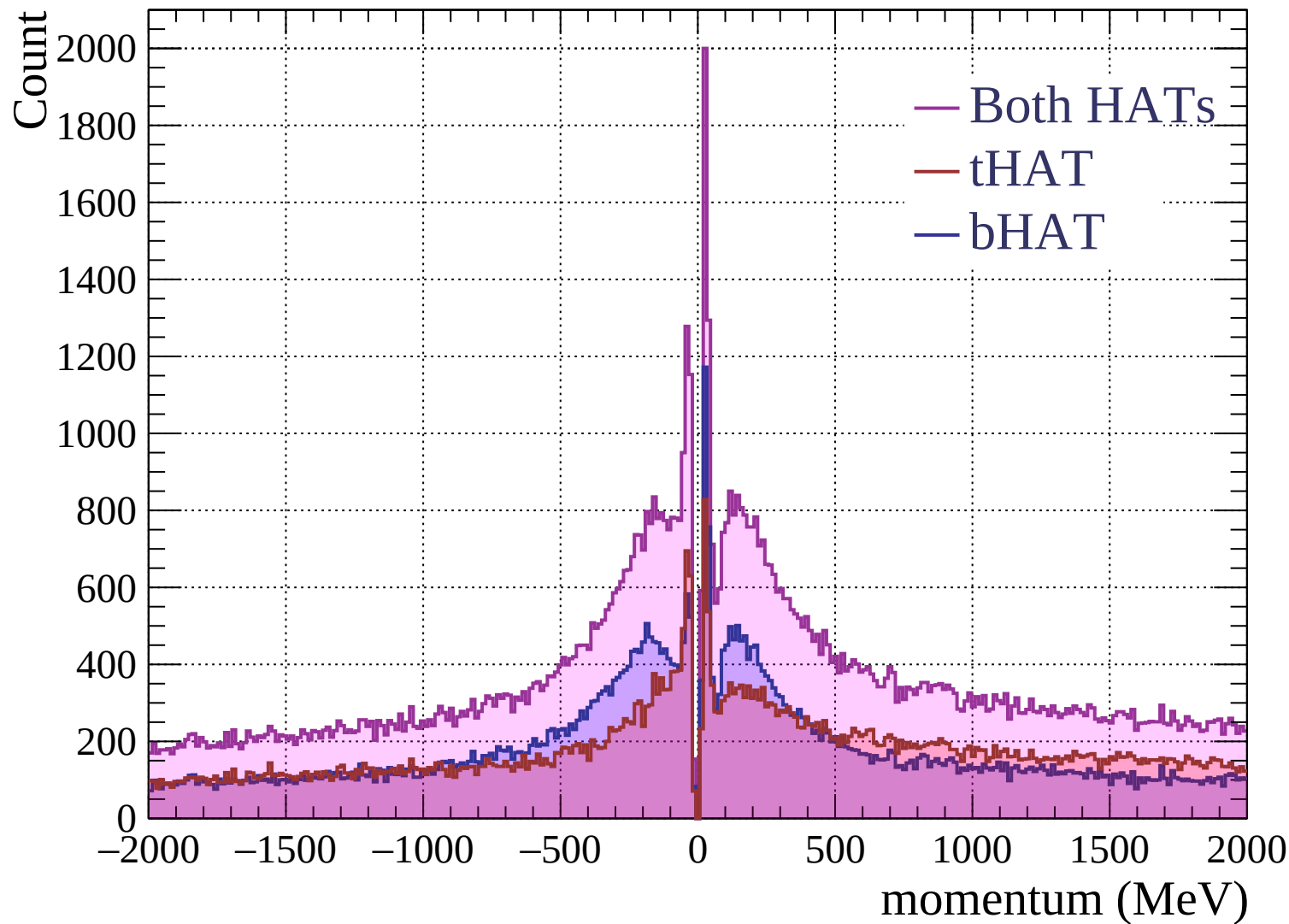


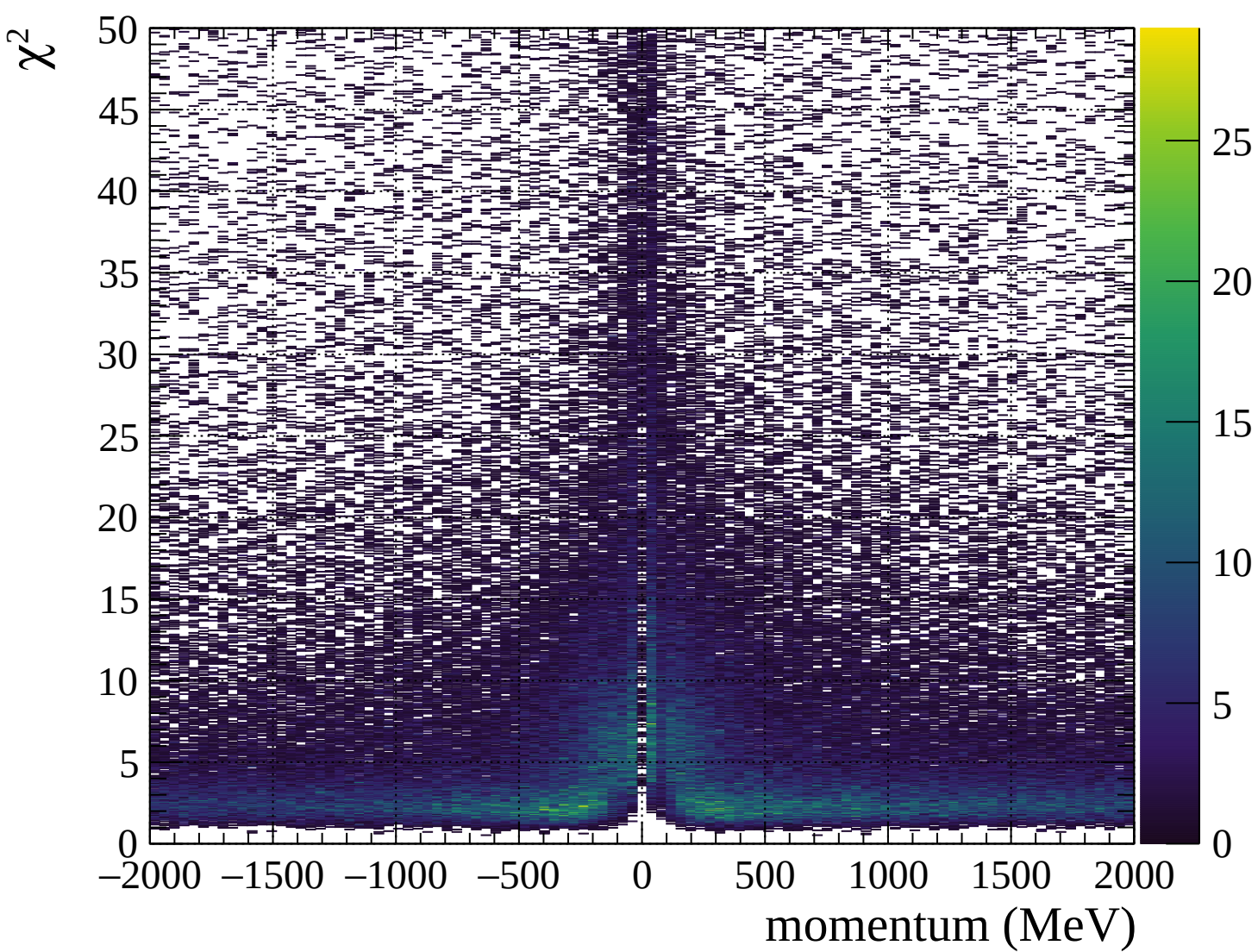




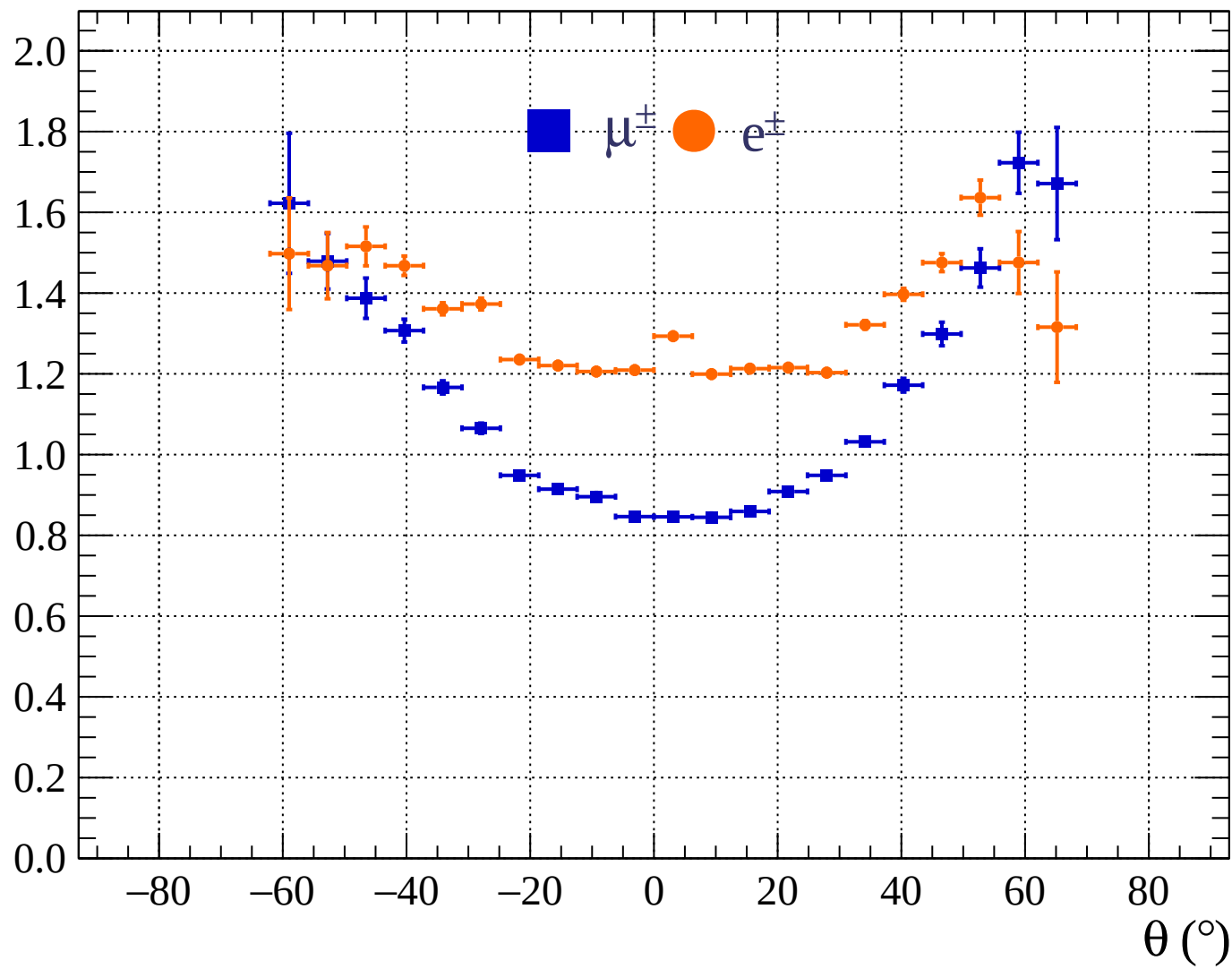




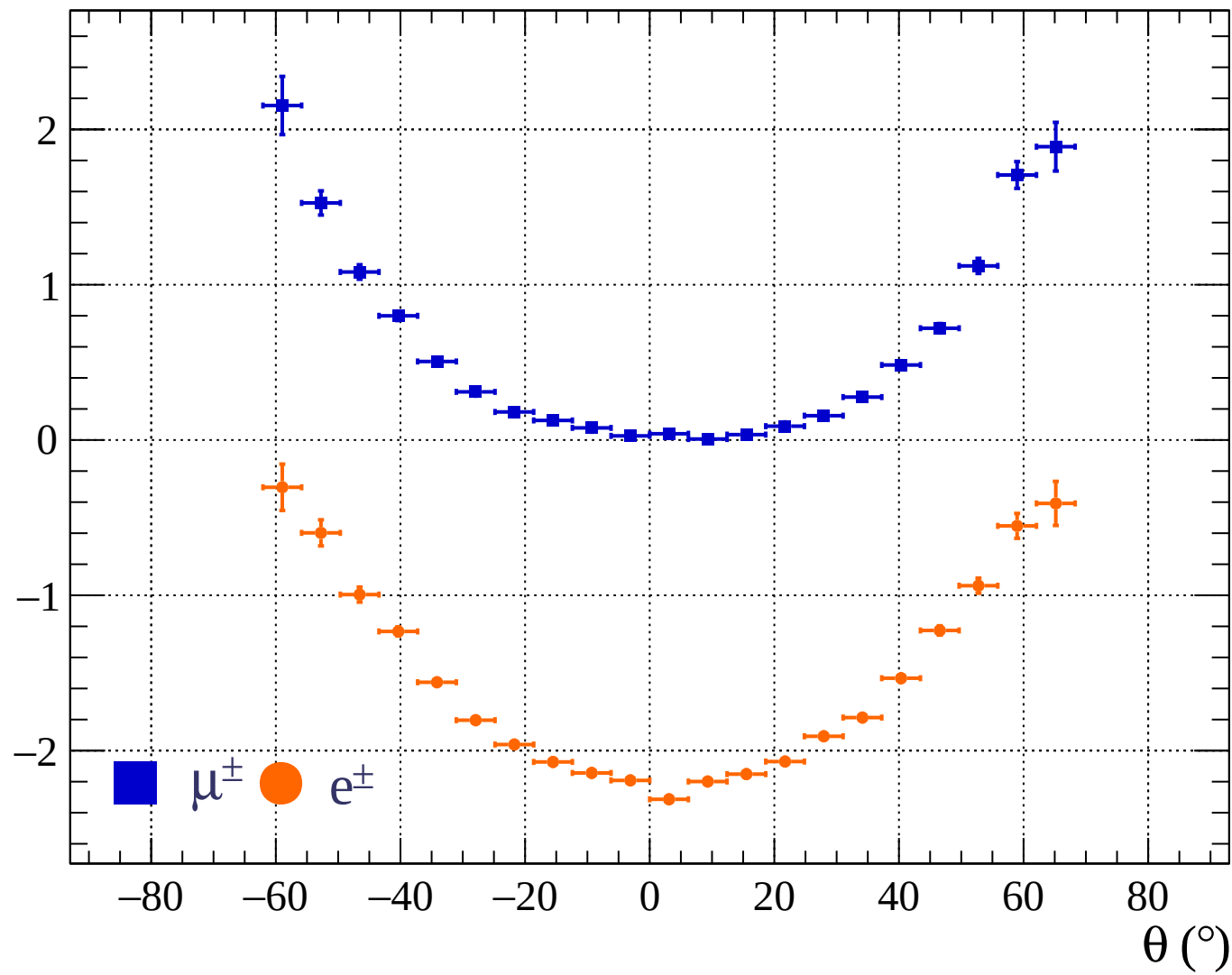




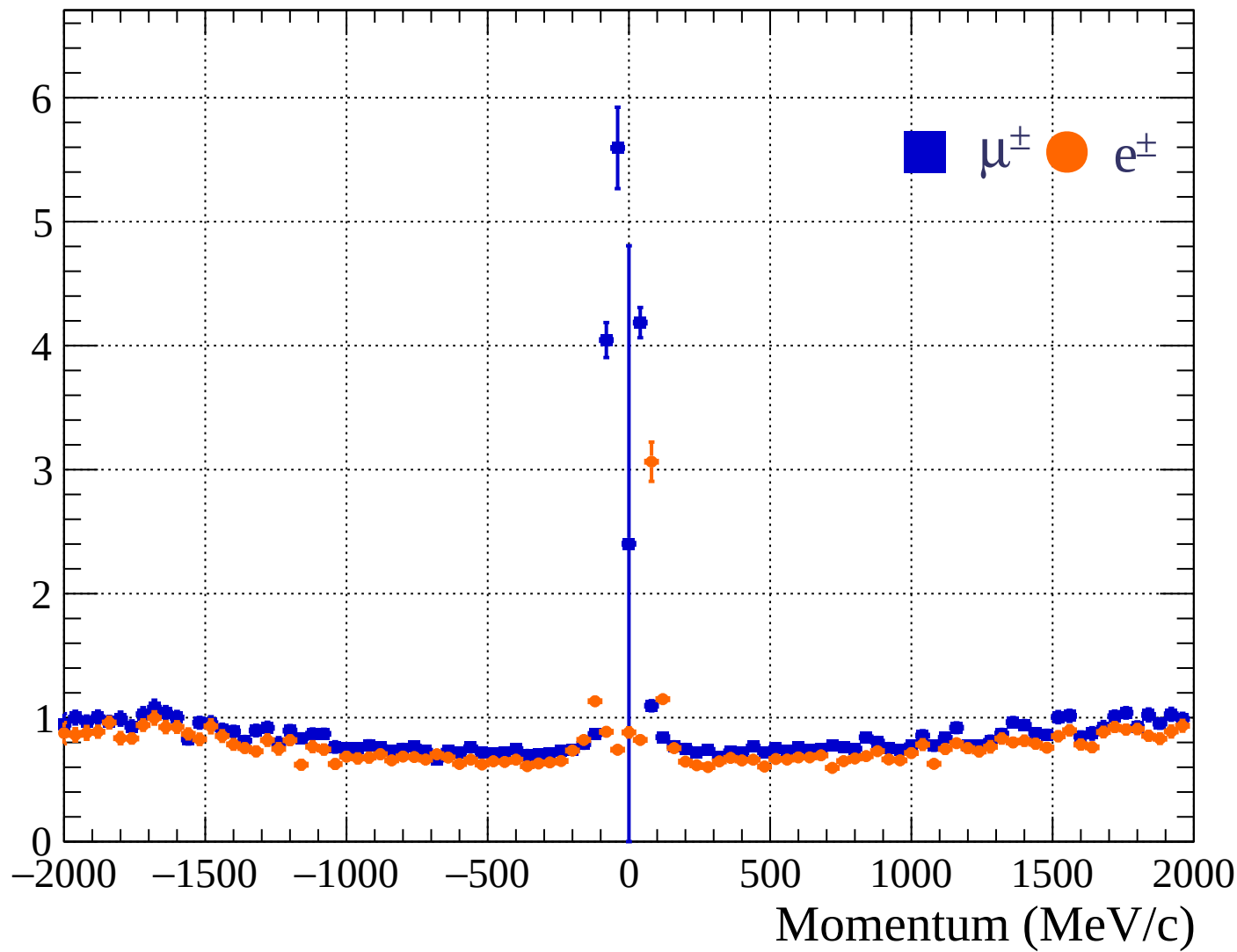
Pulls standard deviation



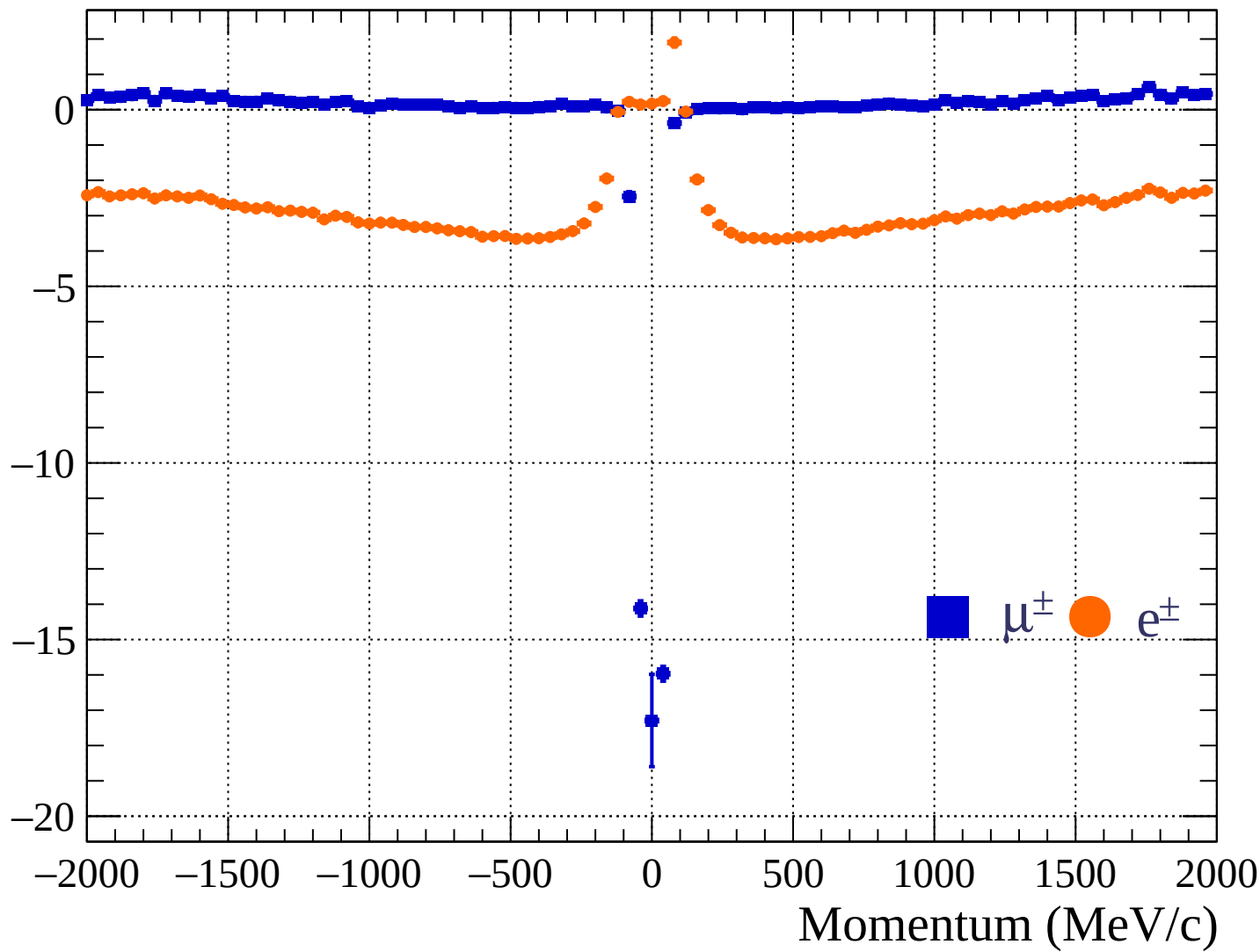
Pulls mean

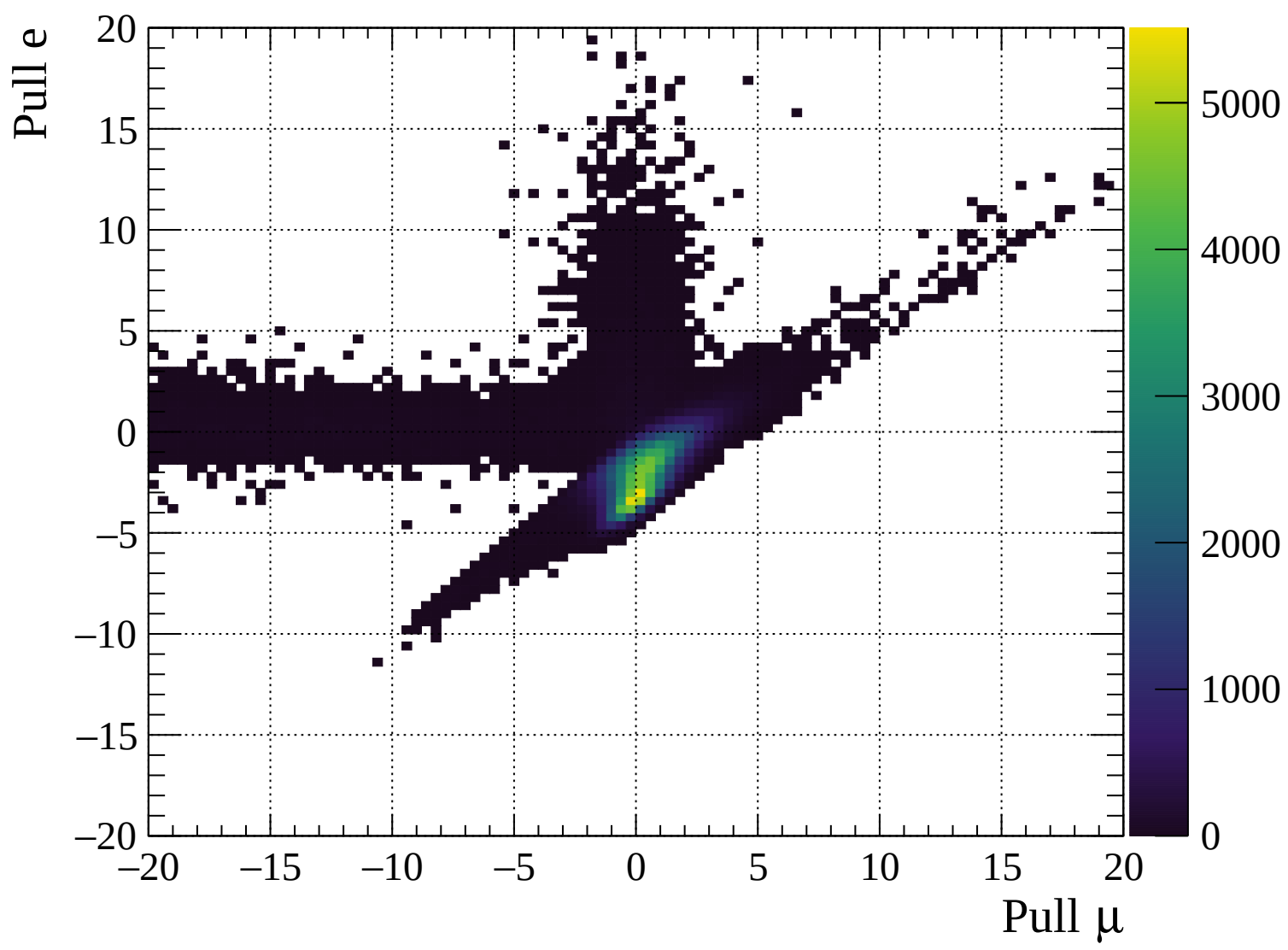


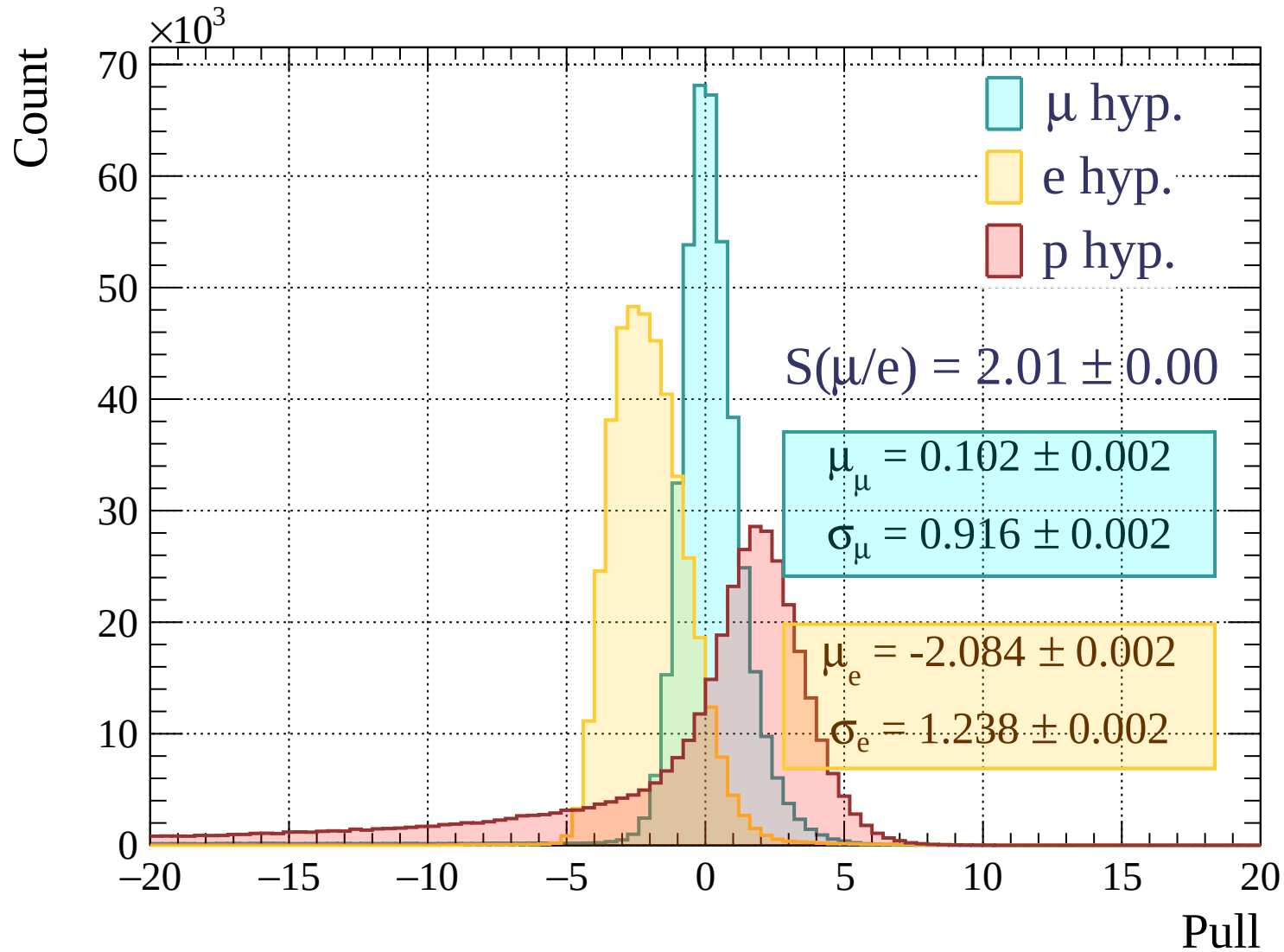
Pulls standard deviation

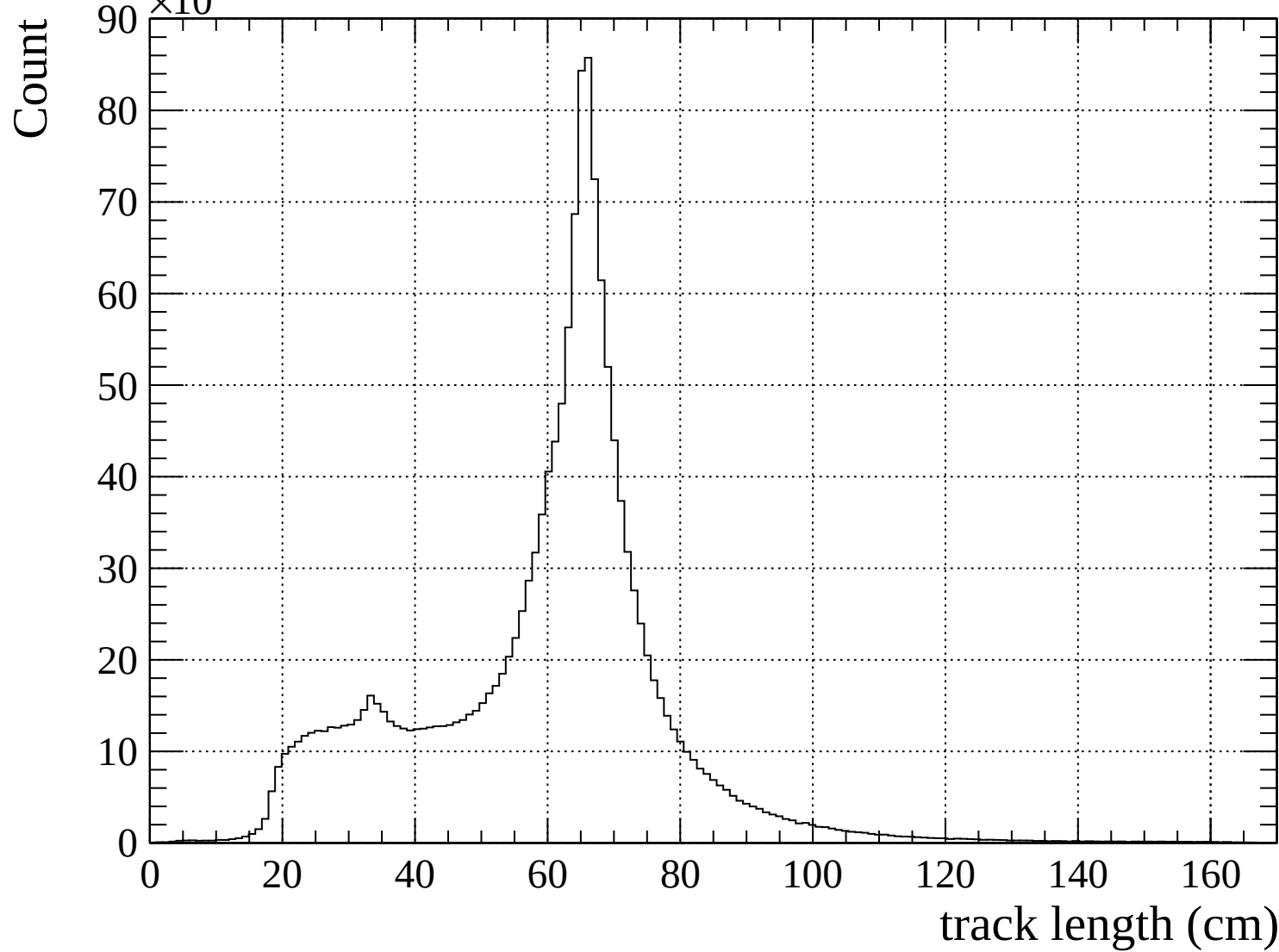


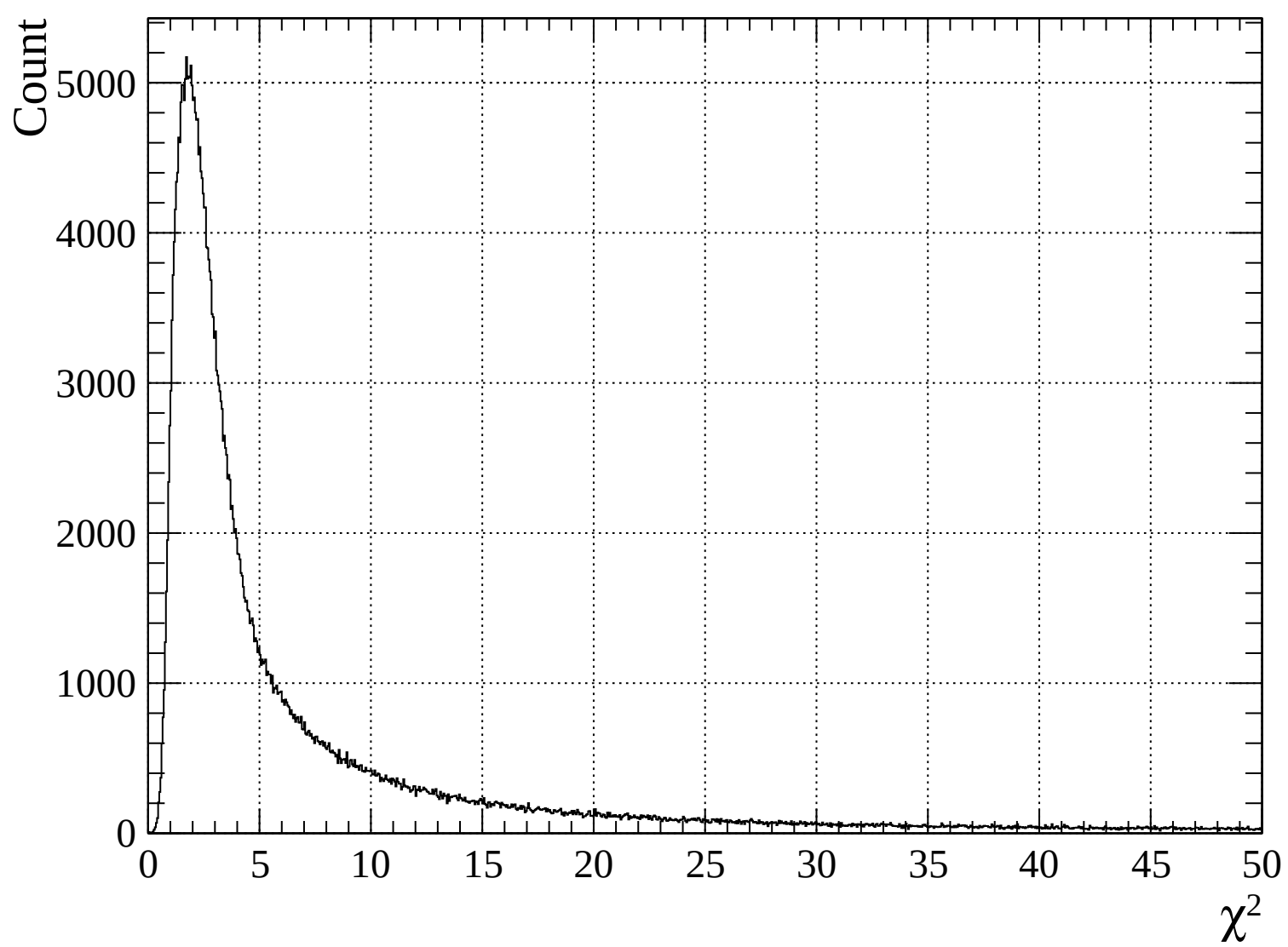
Pulls mean

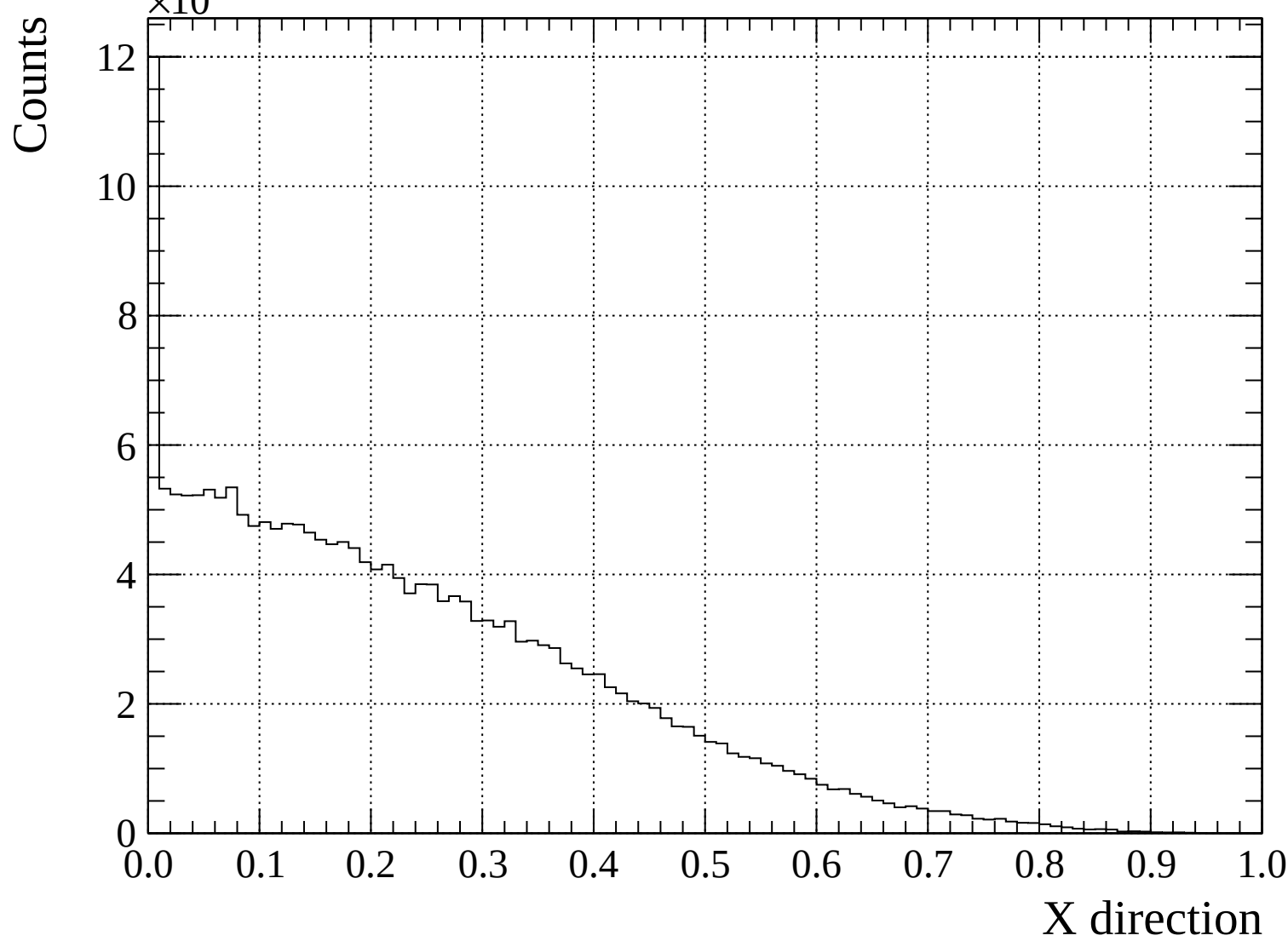


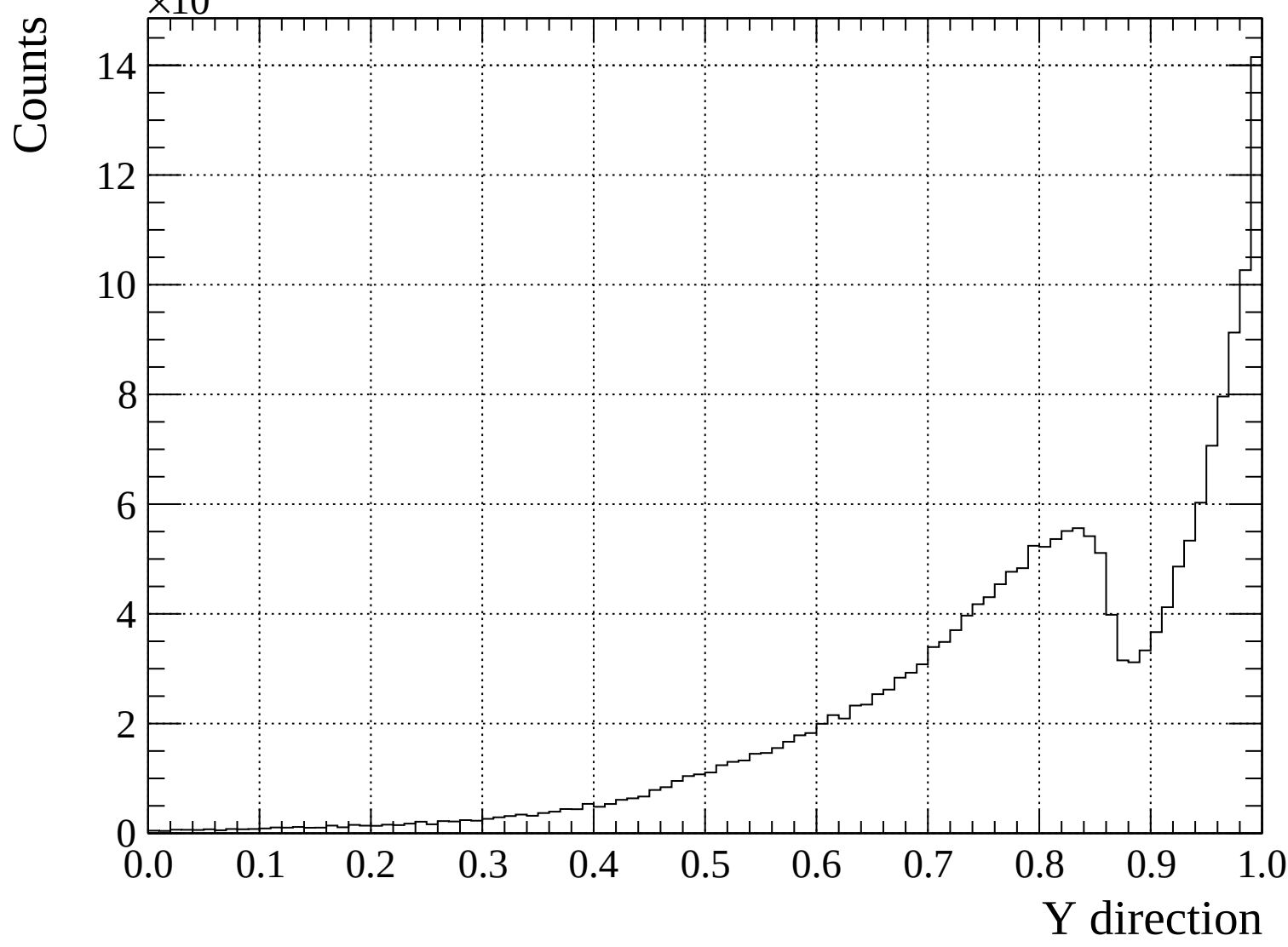


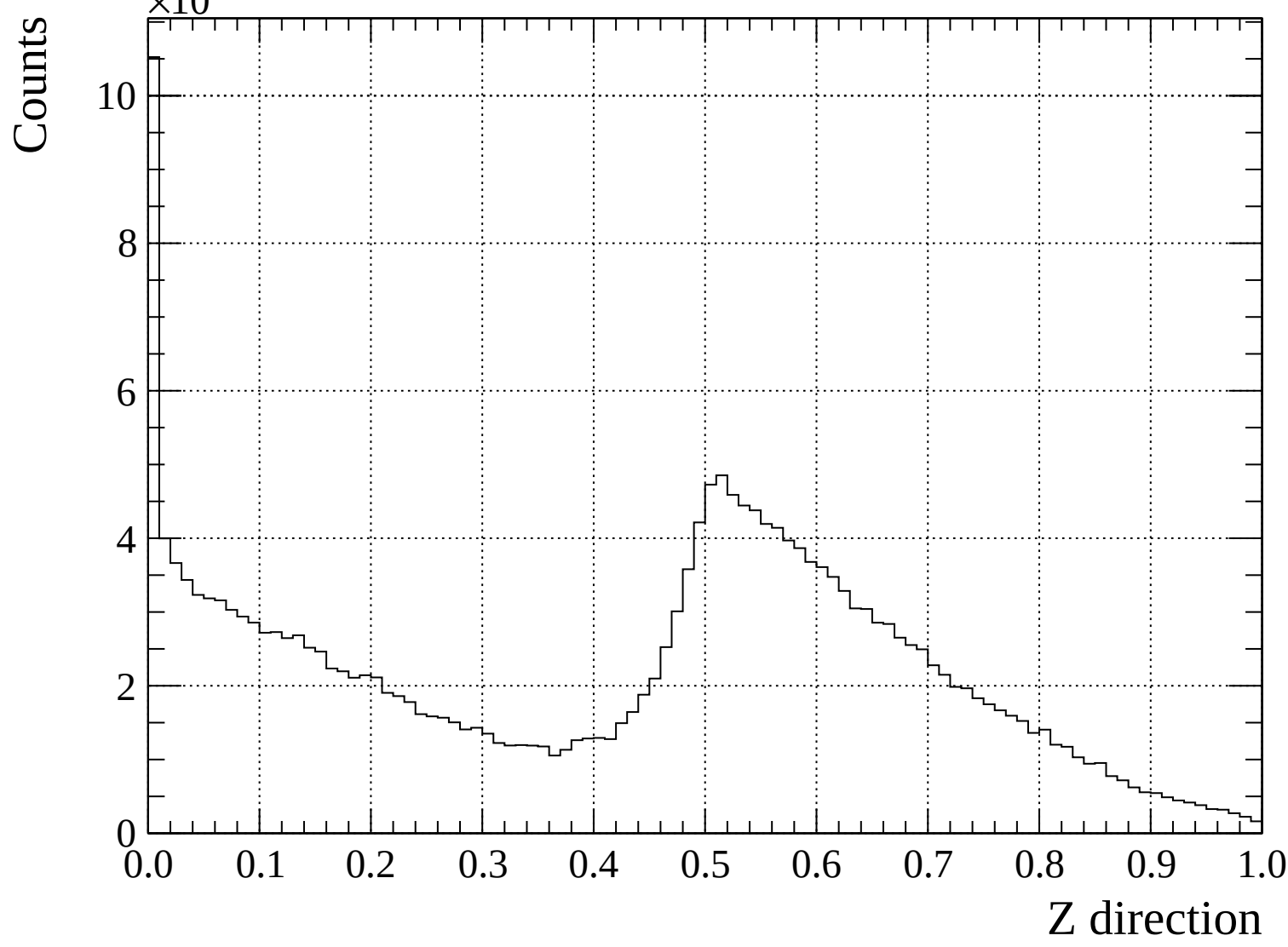


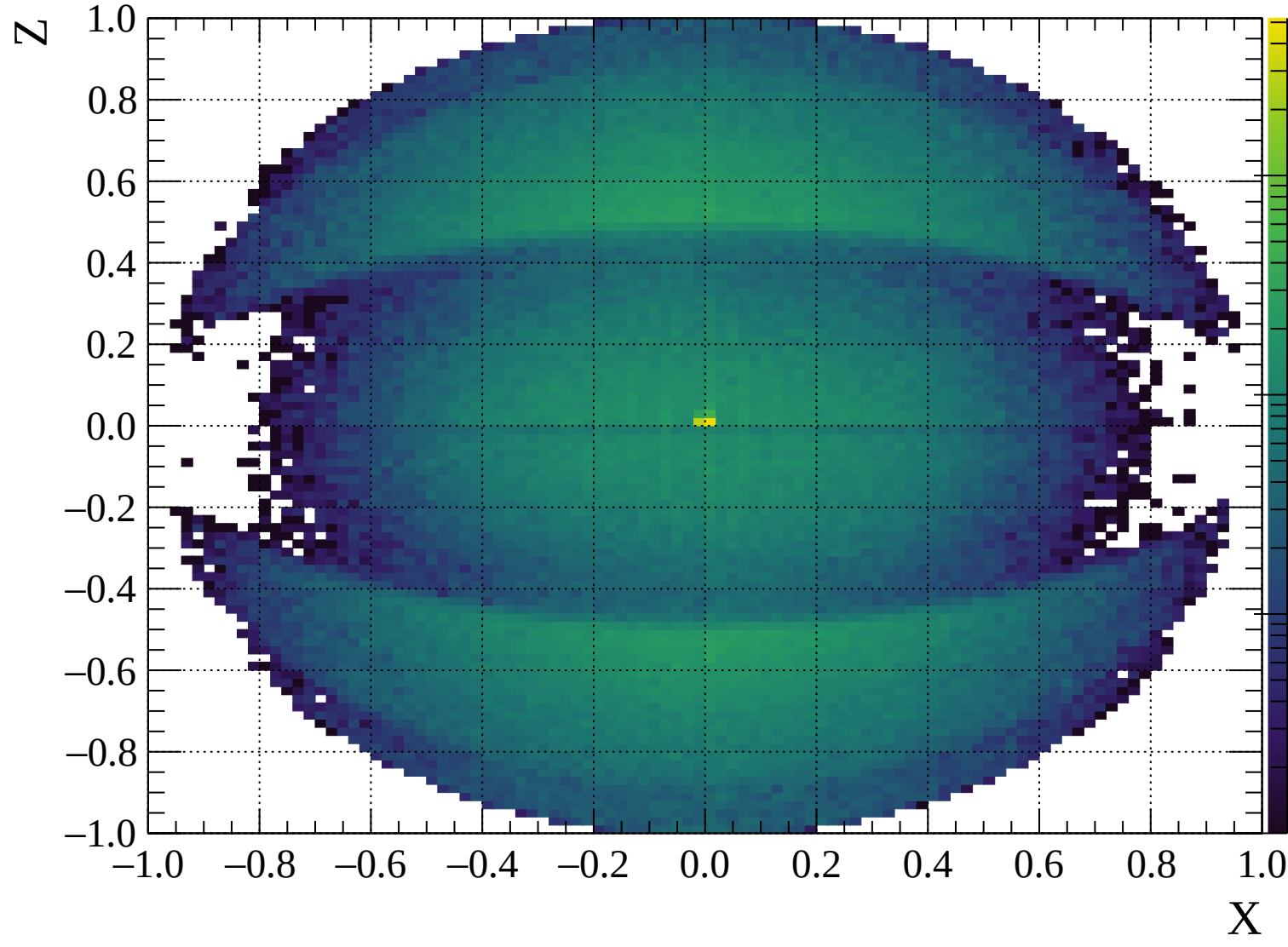


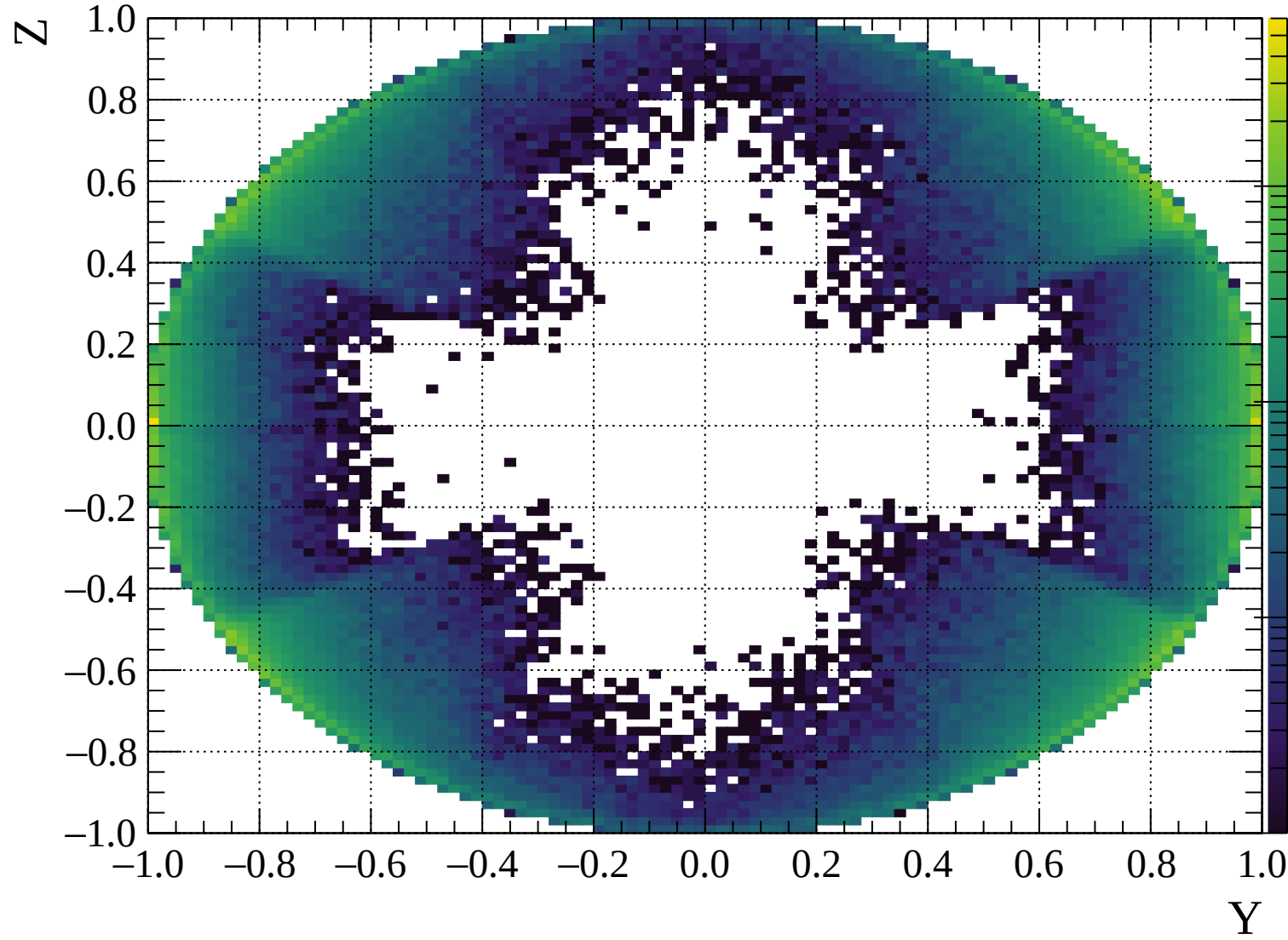


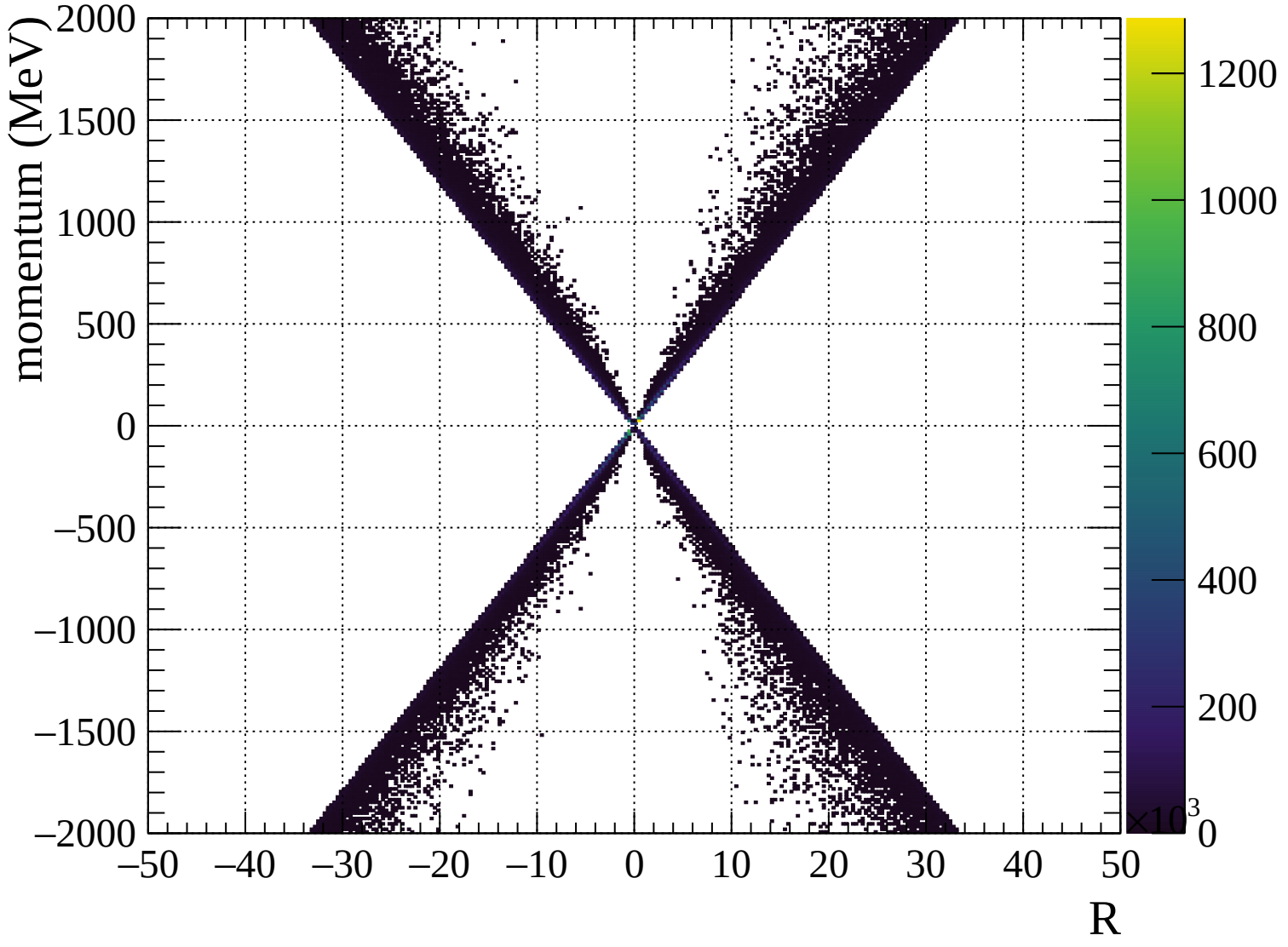


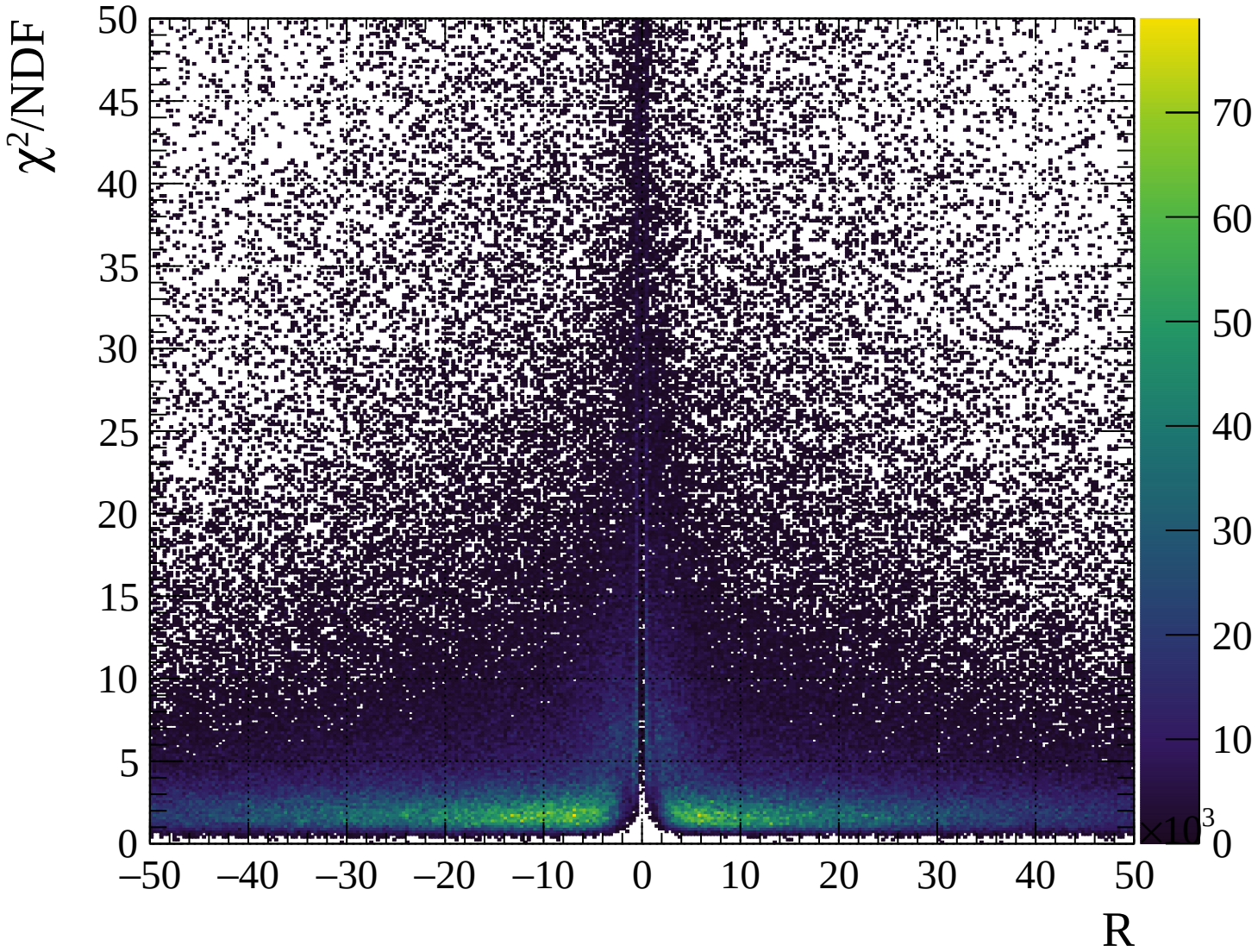




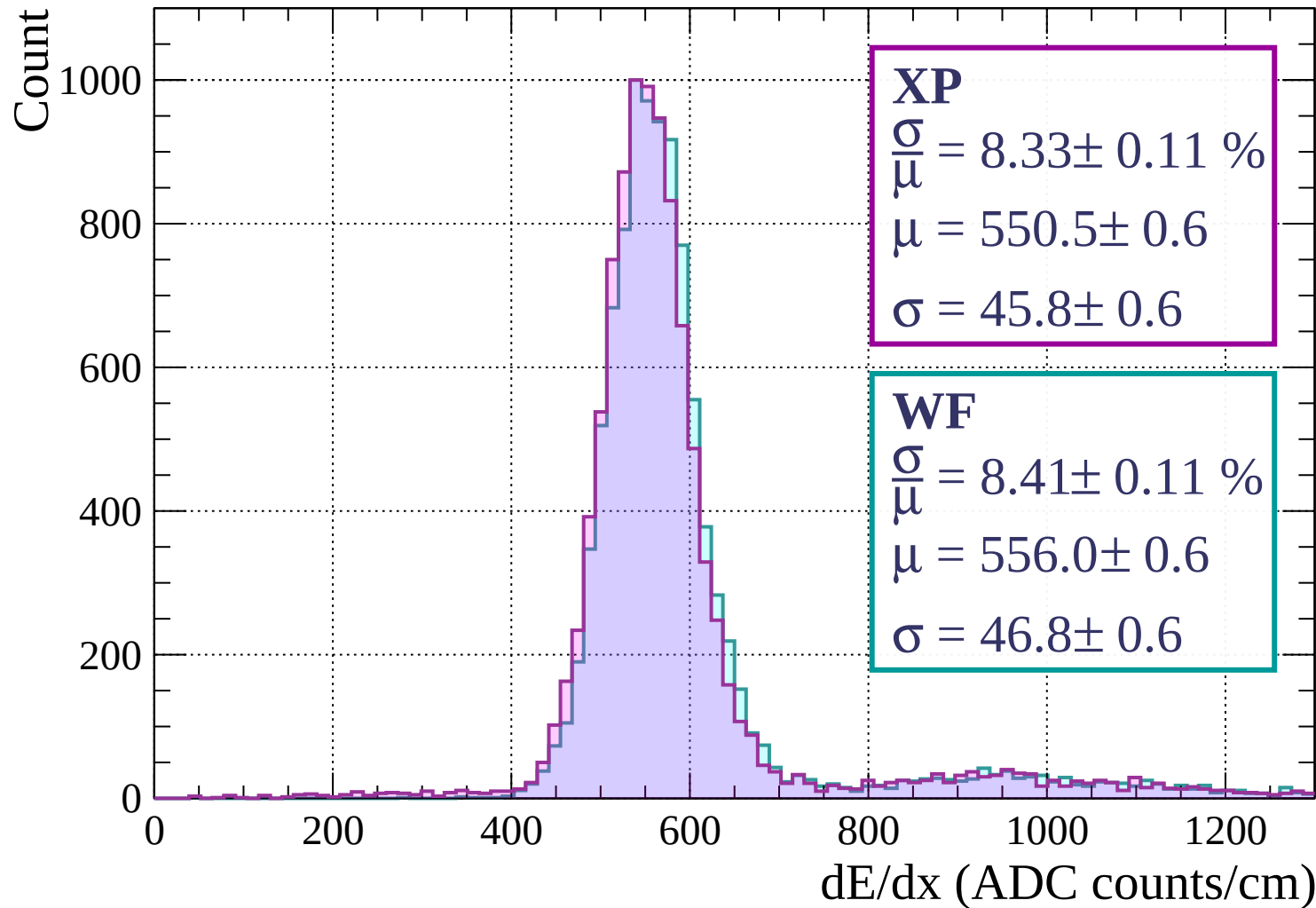




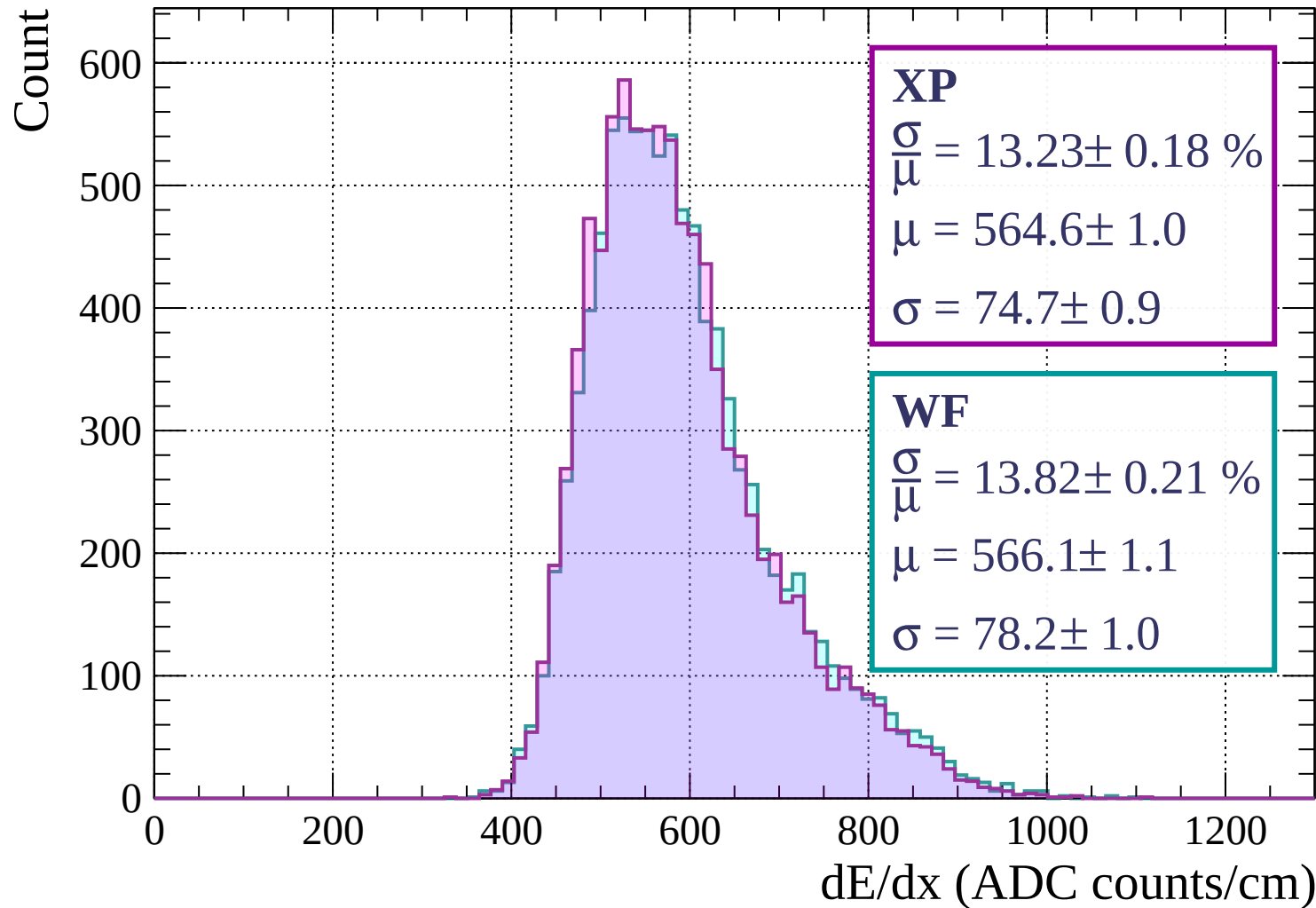




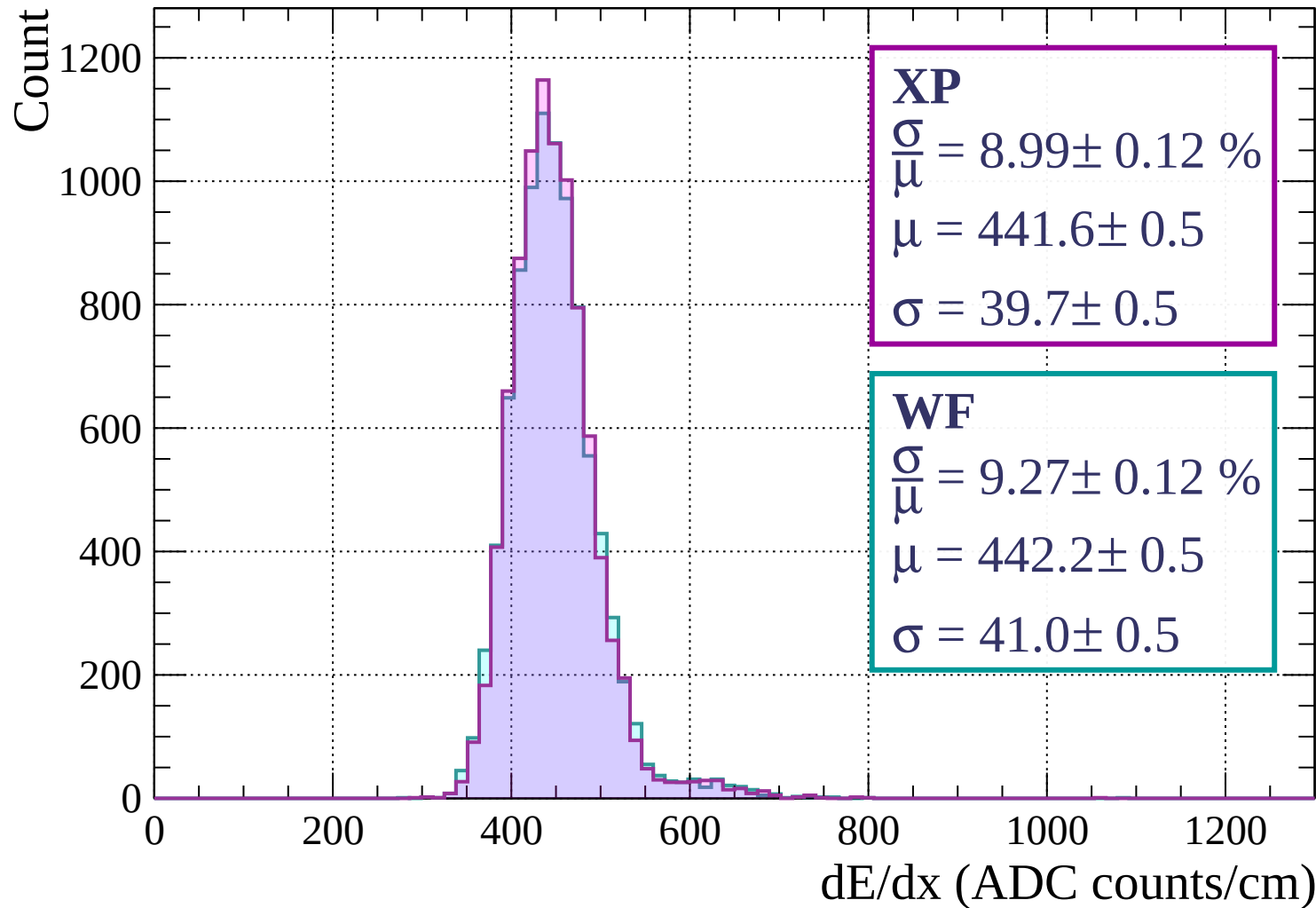
Energy loss | $0 < p < 80$



Energy loss | $80 < p < 160$



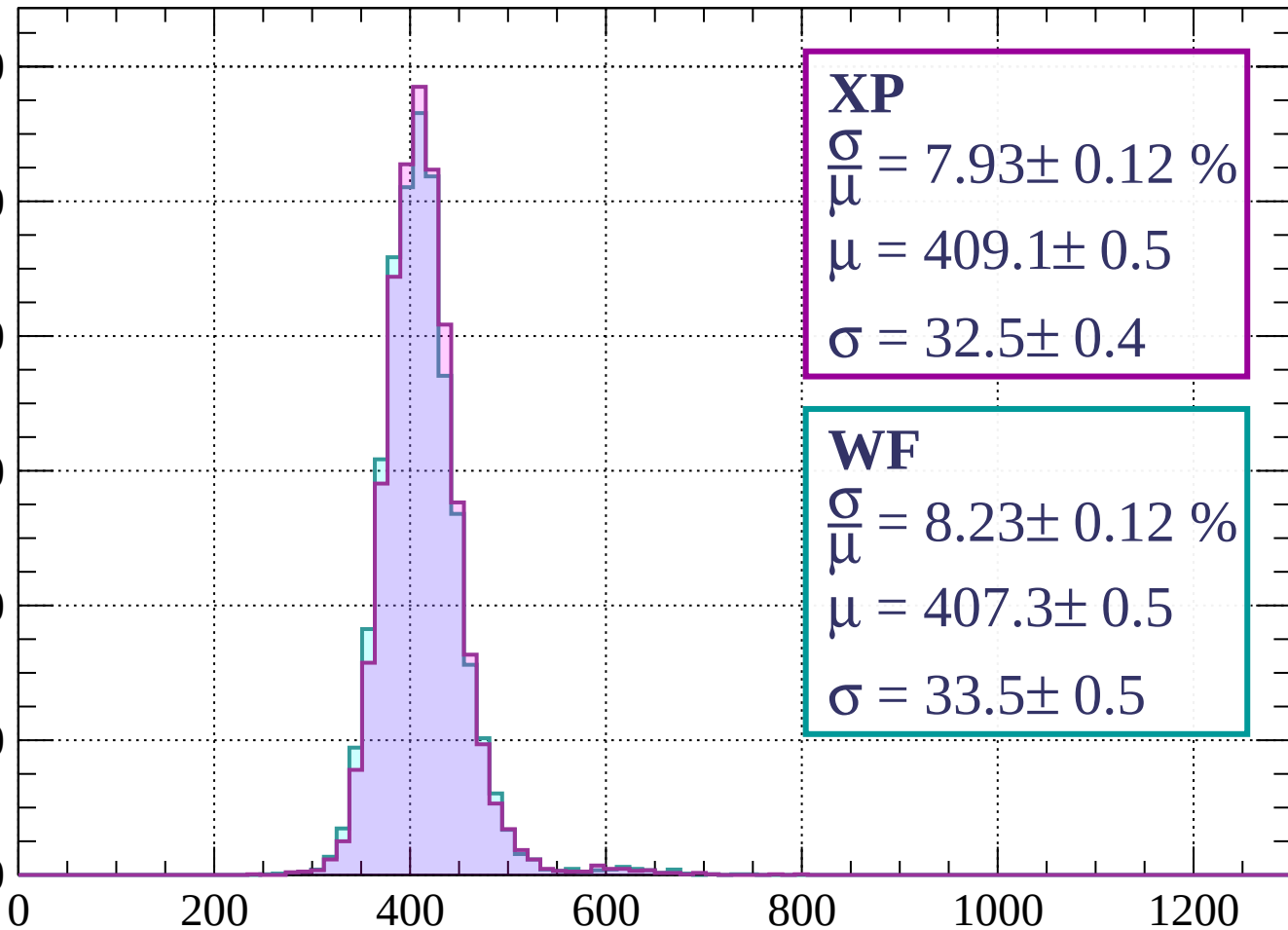
Energy loss | $160 < p < 240$



Energy loss | $240 < p < 320$

Count

1200
1000
800
600
400
200
0



XP

$$\frac{\sigma}{\mu} = 7.93 \pm 0.12 \%$$

$$\mu = 409.1 \pm 0.5$$

$$\sigma = 32.5 \pm 0.4$$

WF

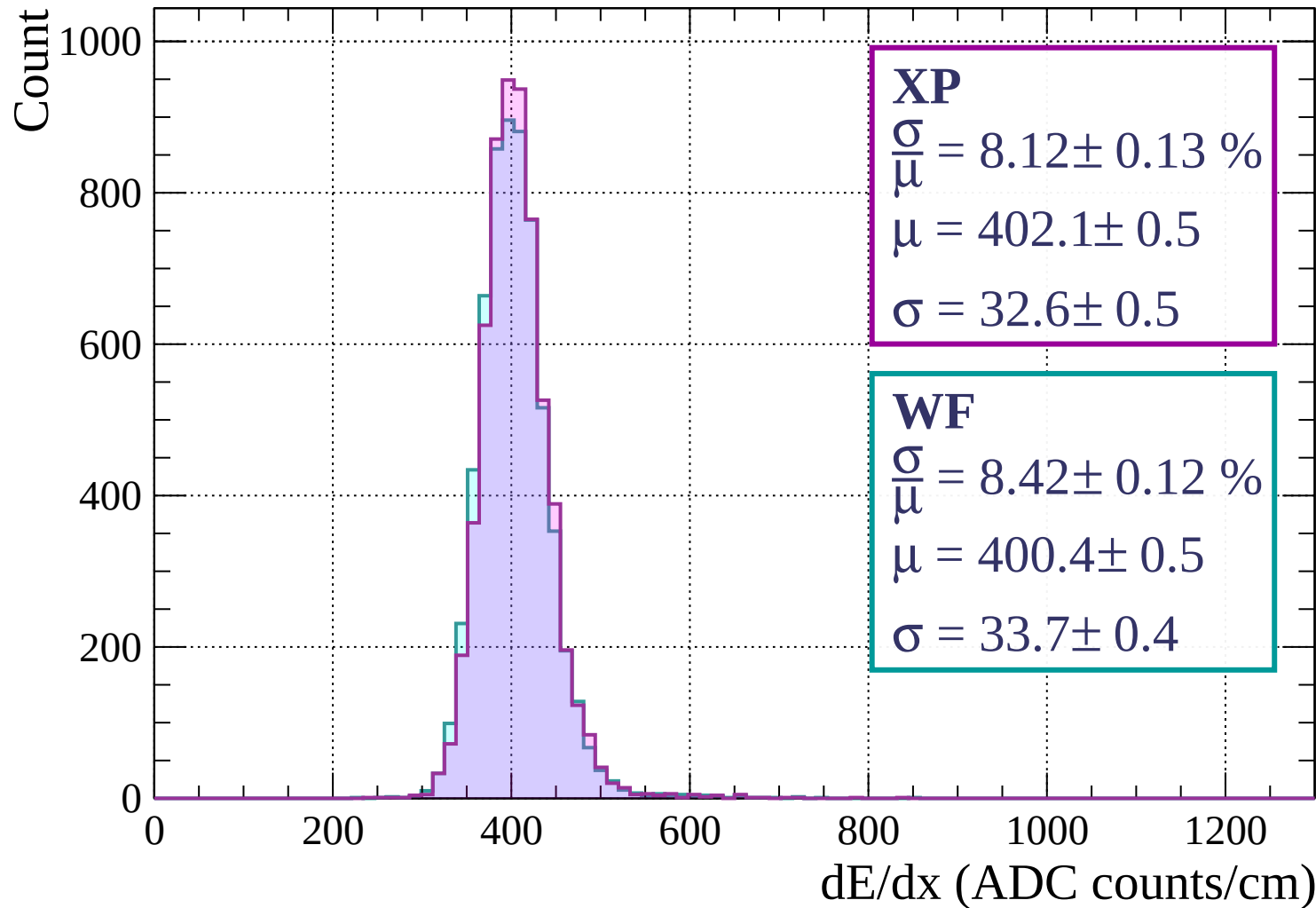
$$\frac{\sigma}{\mu} = 8.23 \pm 0.12 \%$$

$$\mu = 407.3 \pm 0.5$$

$$\sigma = 33.5 \pm 0.5$$

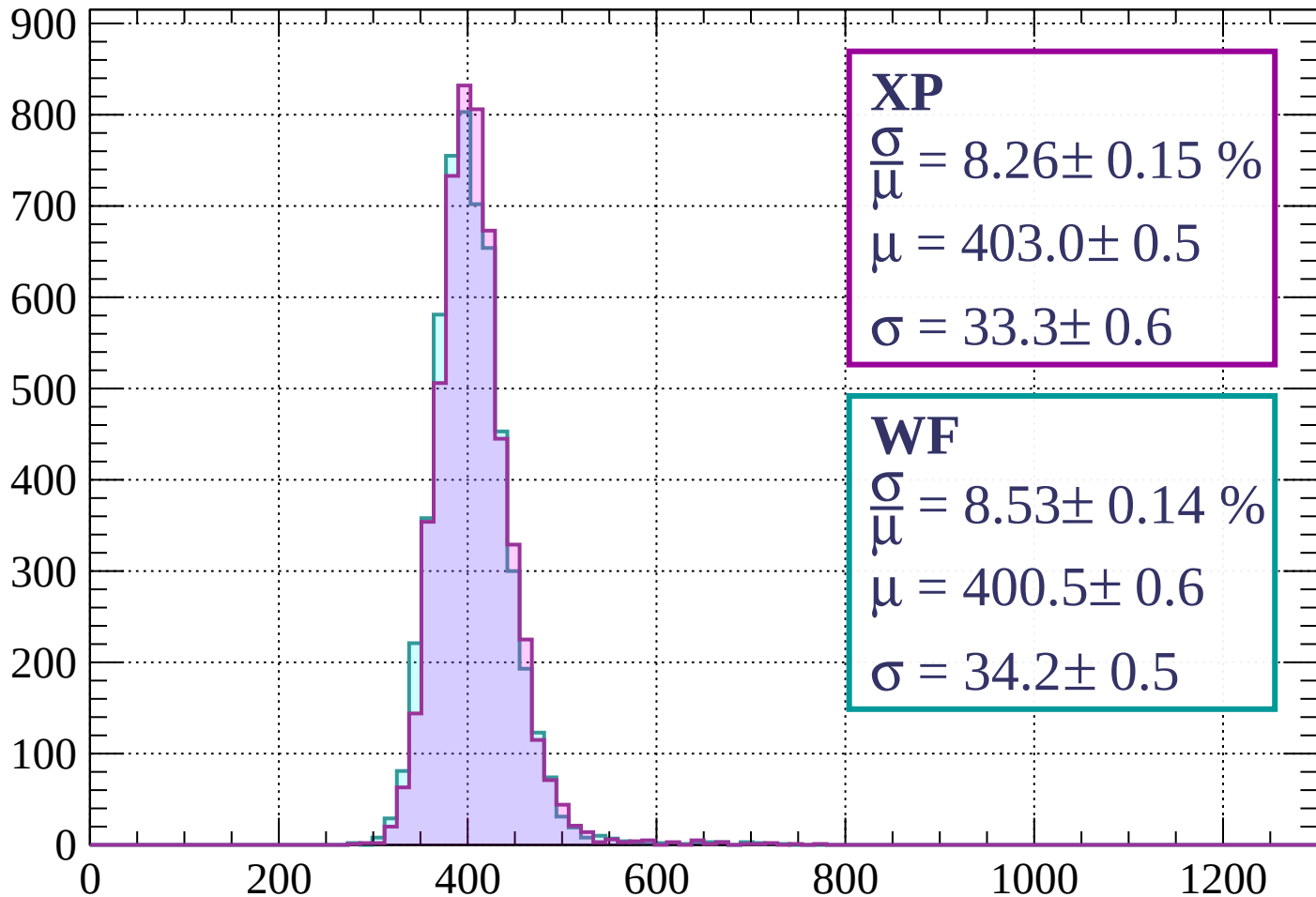
dE/dx (ADC counts/cm)

Energy loss | $320 < p < 400$



Energy loss | $400 < p < 480$

Count



XP

$$\frac{\sigma}{\mu} = 8.26 \pm 0.15 \%$$

$$\mu = 403.0 \pm 0.5$$

$$\sigma = 33.3 \pm 0.6$$

WF

$$\frac{\sigma}{\mu} = 8.53 \pm 0.14 \%$$

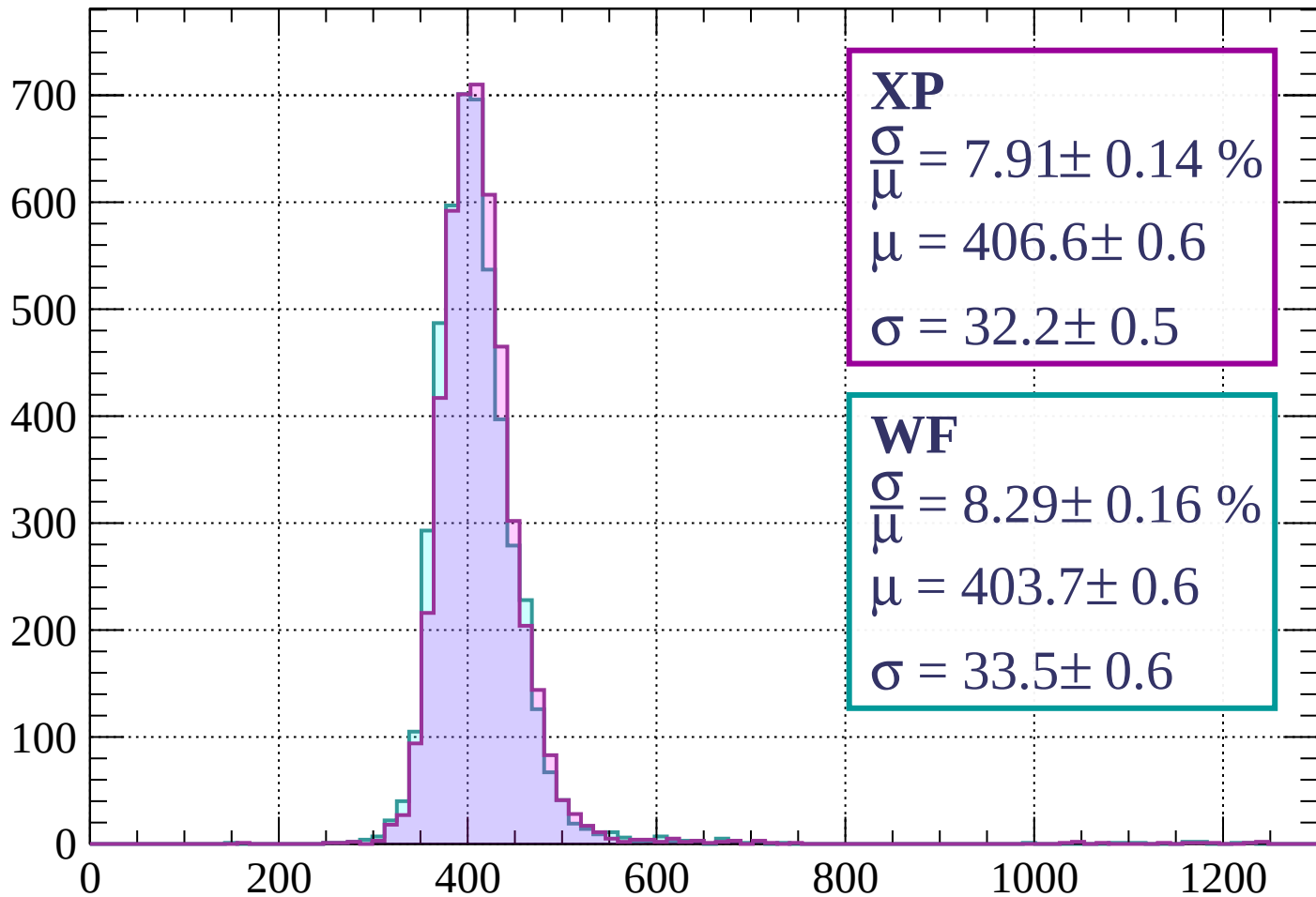
$$\mu = 400.5 \pm 0.6$$

$$\sigma = 34.2 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss | $480 < p < 560$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 7.91 \pm 0.14 \%$$

$$\mu = 406.6 \pm 0.6$$

$$\sigma = 32.2 \pm 0.5$$

WF

$$\frac{\sigma}{\bar{\mu}} = 8.29 \pm 0.16 \%$$

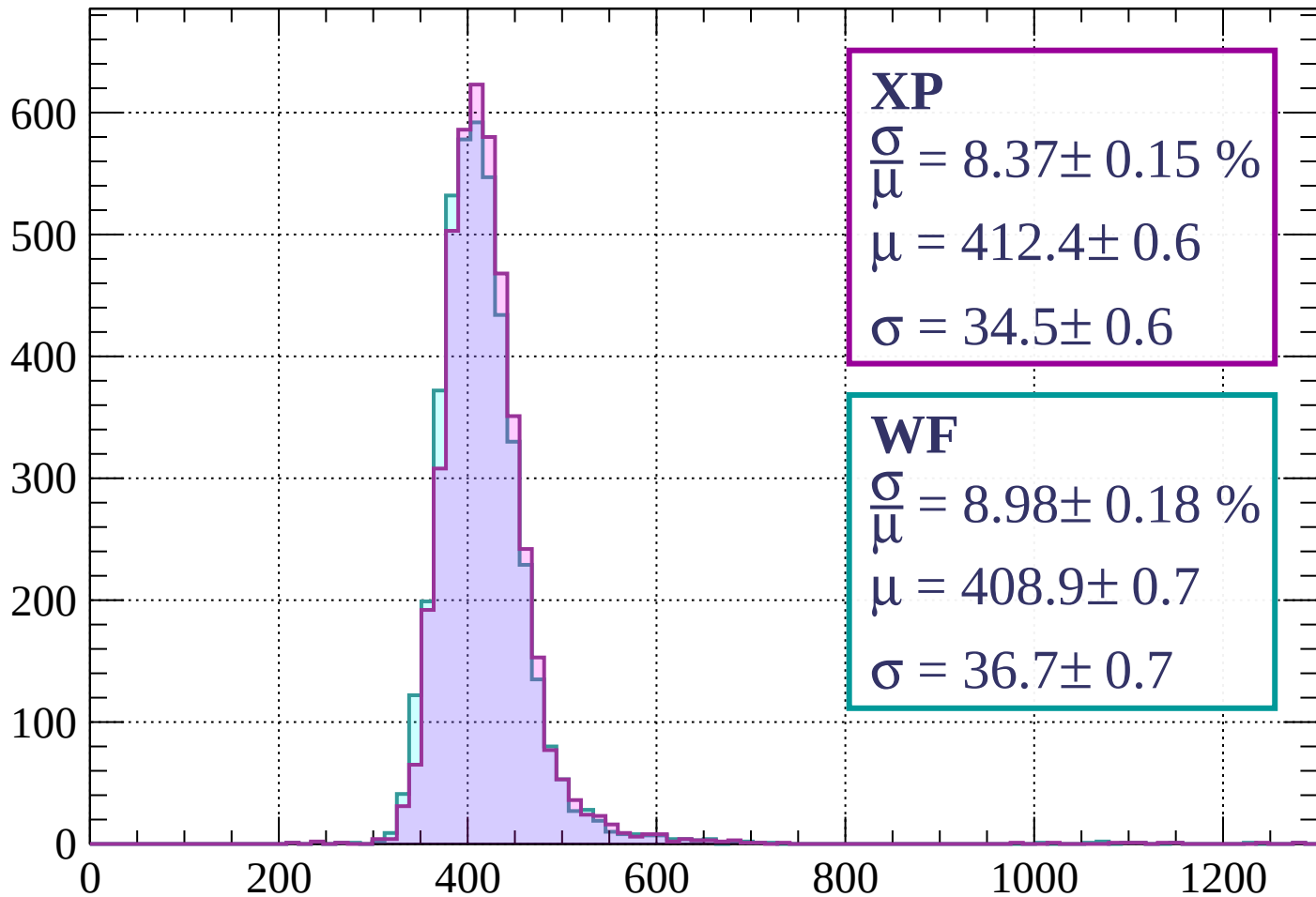
$$\mu = 403.7 \pm 0.6$$

$$\sigma = 33.5 \pm 0.6$$

dE/dx (ADC counts/cm)

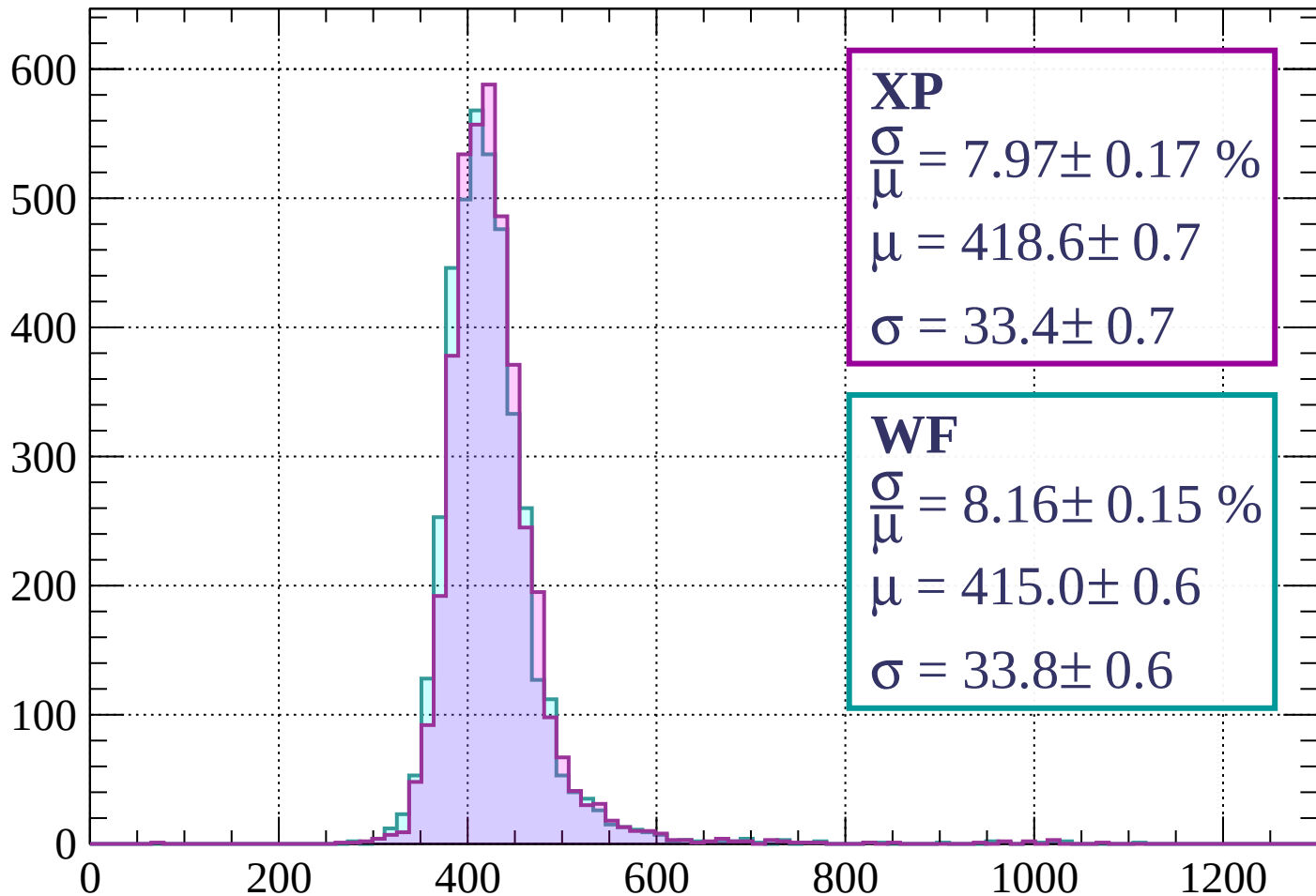
Energy loss | $560 < p < 640$

Count



Energy loss | $640 < p < 720$

Count



XP

$$\frac{\sigma}{\mu} = 7.97 \pm 0.17 \%$$

$$\mu = 418.6 \pm 0.7$$

$$\sigma = 33.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.16 \pm 0.15 \%$$

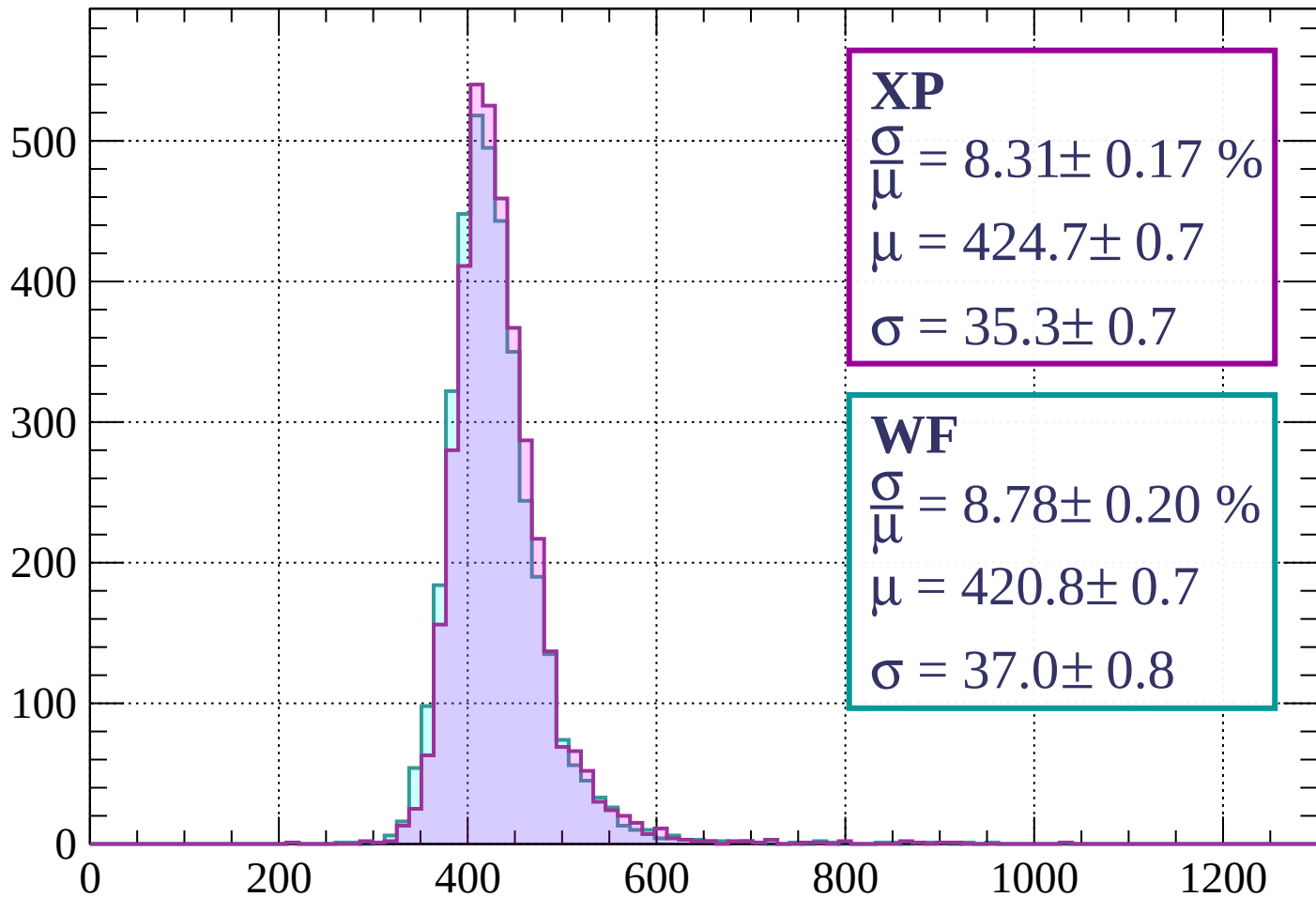
$$\mu = 415.0 \pm 0.6$$

$$\sigma = 33.8 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss | $720 < p < 800$

Count



XP

$$\frac{\sigma}{\mu} = 8.31 \pm 0.17 \%$$

$$\mu = 424.7 \pm 0.7$$

$$\sigma = 35.3 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.78 \pm 0.20 \%$$

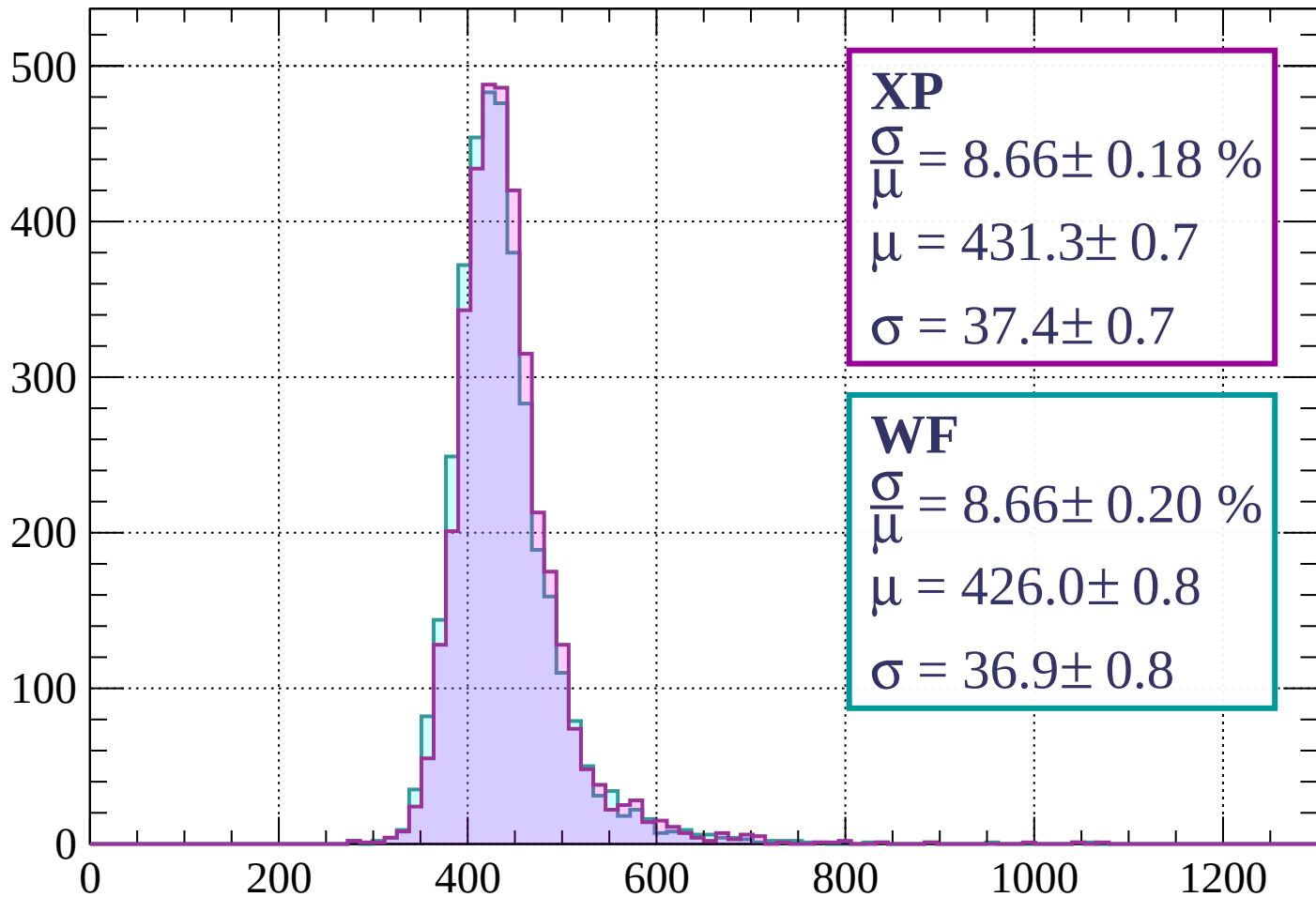
$$\mu = 420.8 \pm 0.7$$

$$\sigma = 37.0 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $800 < p < 880$

Count



XP

$$\frac{\sigma}{\mu} = 8.66 \pm 0.18 \%$$

$$\mu = 431.3 \pm 0.7$$

$$\sigma = 37.4 \pm 0.7$$

WF

$$\frac{\sigma}{\mu} = 8.66 \pm 0.20 \%$$

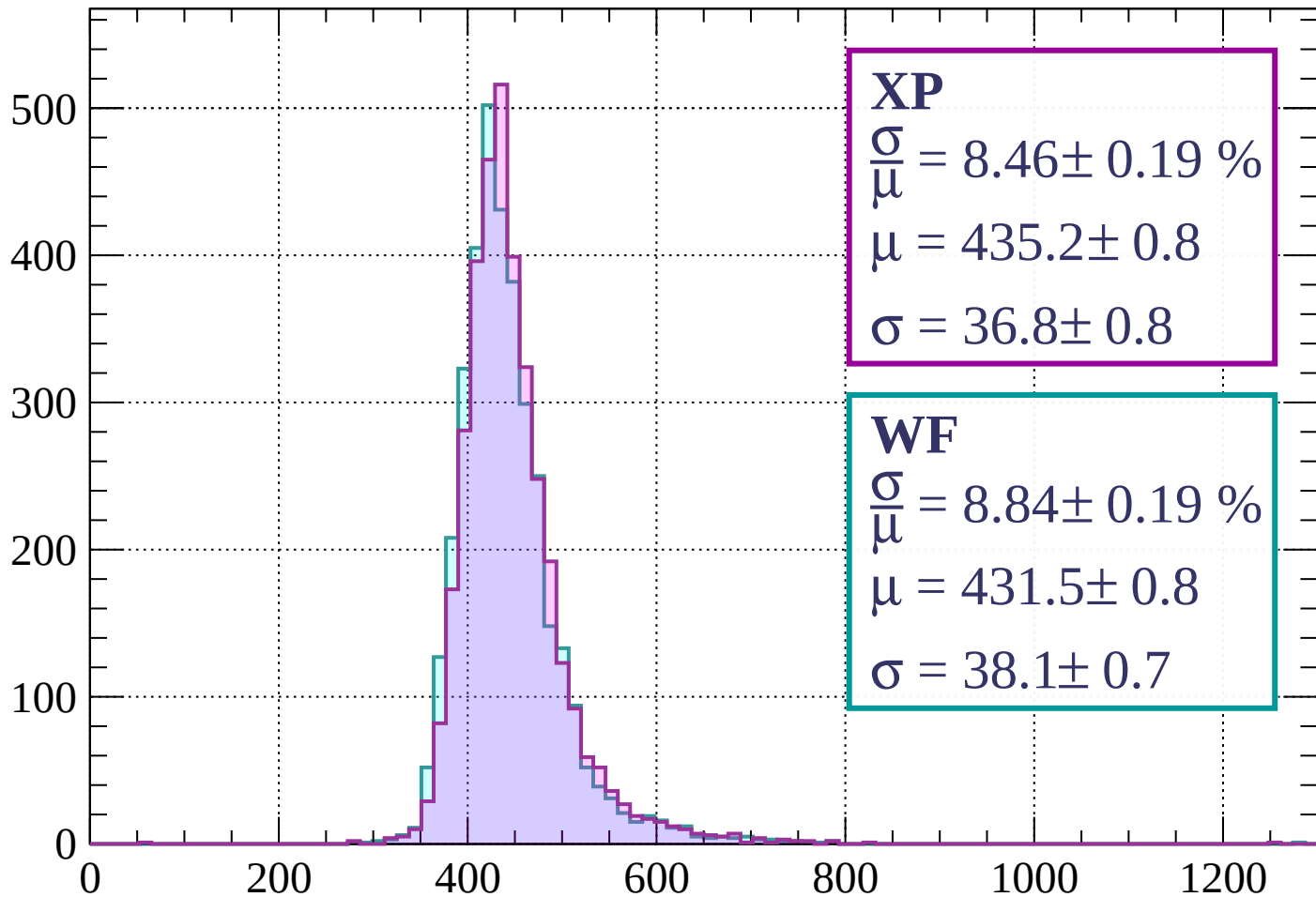
$$\mu = 426.0 \pm 0.8$$

$$\sigma = 36.9 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss | $880 < p < 960$

Count



XP

$$\frac{\sigma}{\mu} = 8.46 \pm 0.19 \%$$

$$\mu = 435.2 \pm 0.8$$

$$\sigma = 36.8 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.84 \pm 0.19 \%$$

$$\mu = 431.5 \pm 0.8$$

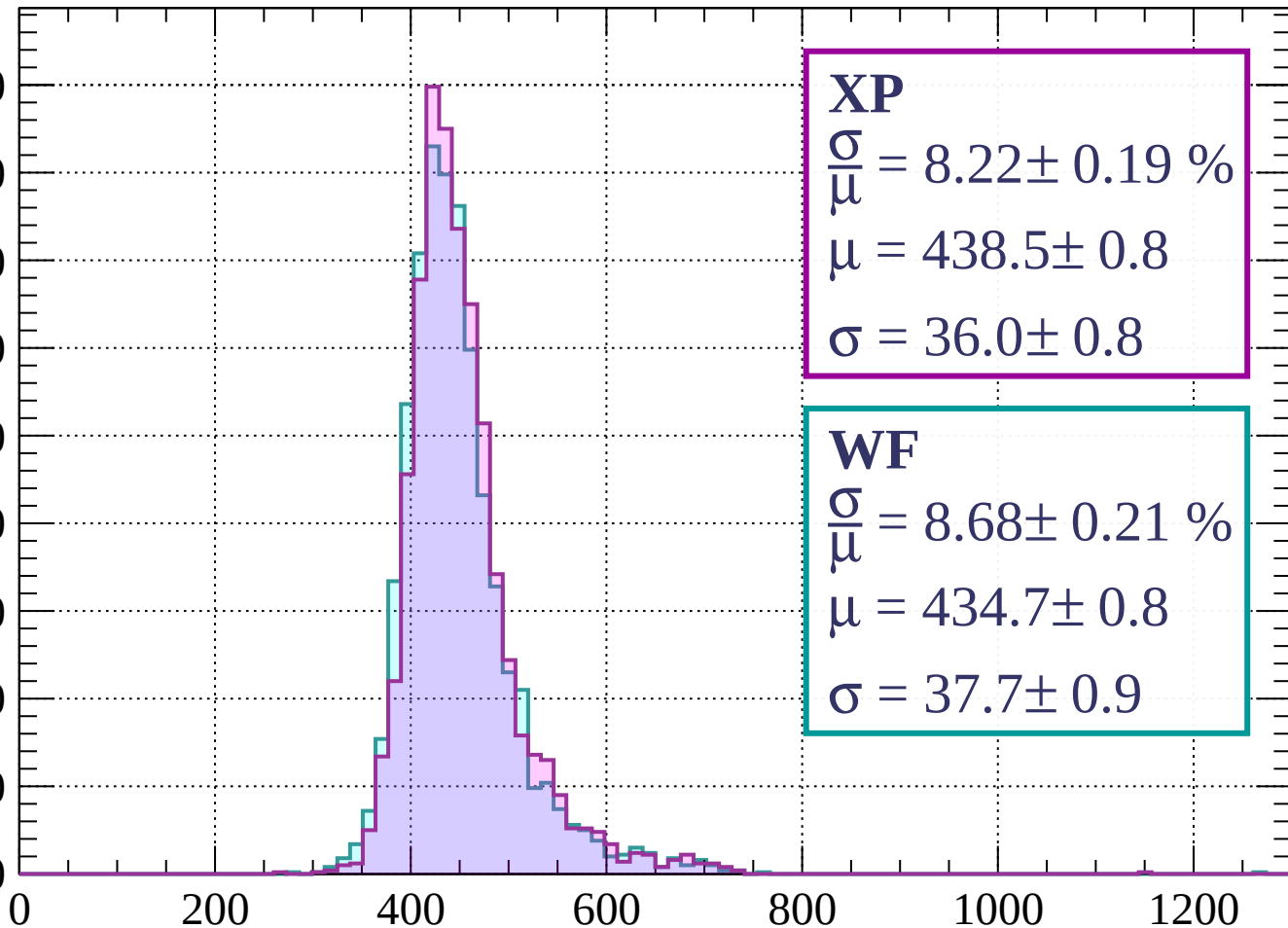
$$\sigma = 38.1 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss | $960 < p < 1040$

Count

450
400
350
300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 8.22 \pm 0.19 \%$$

$$\mu = 438.5 \pm 0.8$$

$$\sigma = 36.0 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.68 \pm 0.21 \%$$

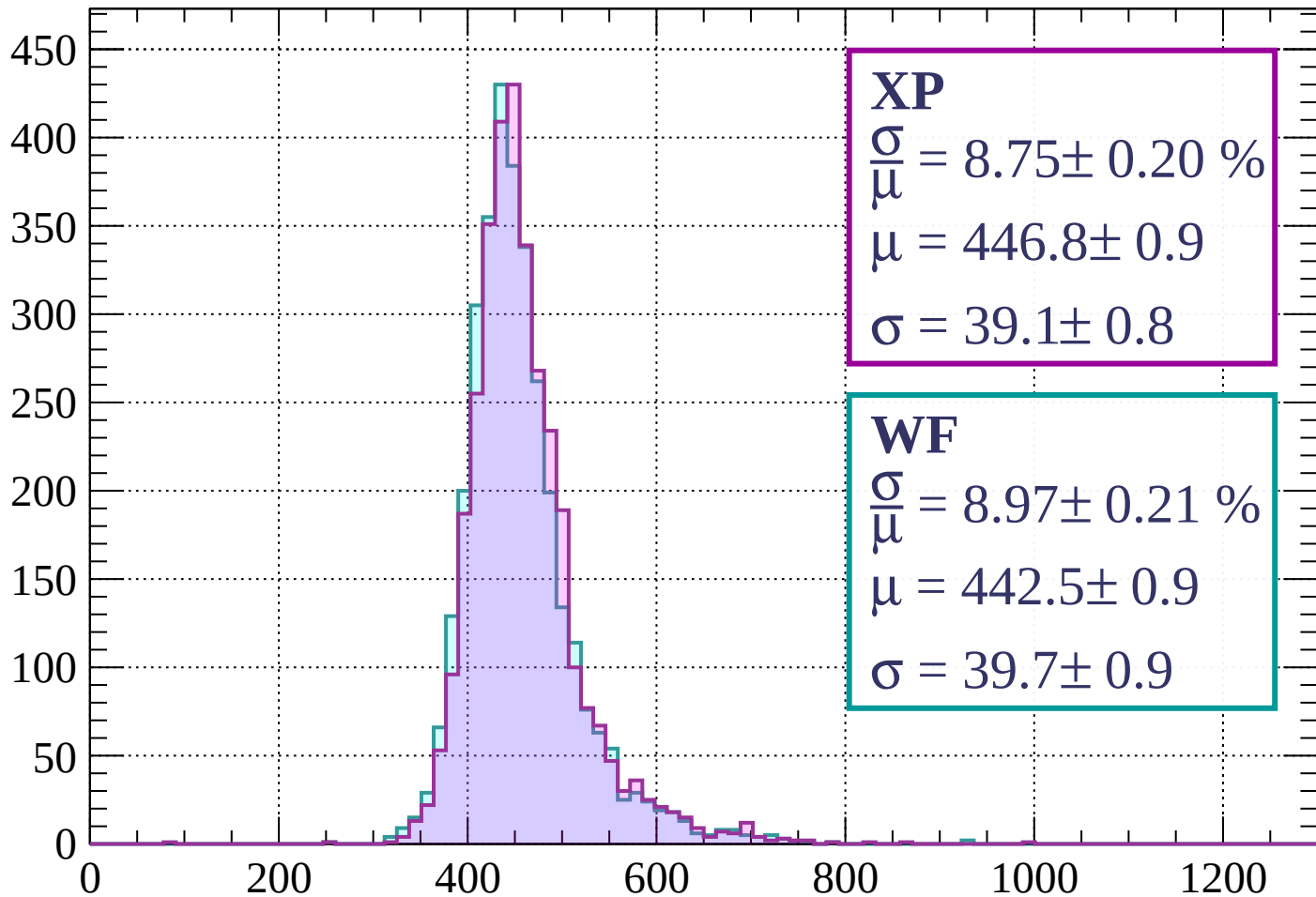
$$\mu = 434.7 \pm 0.8$$

$$\sigma = 37.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1040 < p < 1120$

Count



XP

$$\frac{\sigma}{\mu} = 8.75 \pm 0.20 \%$$

$$\mu = 446.8 \pm 0.9$$

$$\sigma = 39.1 \pm 0.8$$

WF

$$\frac{\sigma}{\mu} = 8.97 \pm 0.21 \%$$

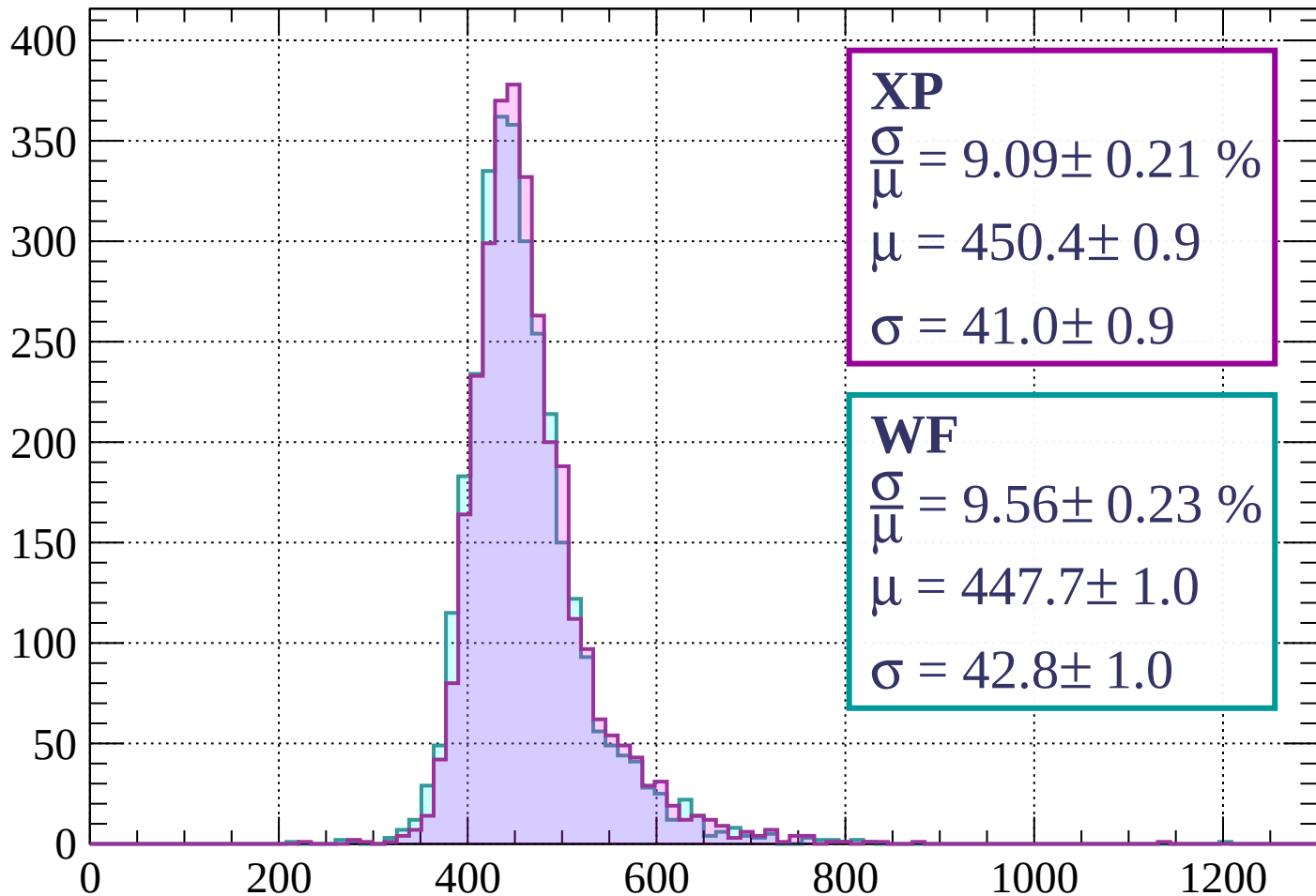
$$\mu = 442.5 \pm 0.9$$

$$\sigma = 39.7 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1120 < p < 1200$

Count



XP

$$\frac{\sigma}{\bar{\mu}} = 9.09 \pm 0.21 \%$$

$$\mu = 450.4 \pm 0.9$$

$$\sigma = 41.0 \pm 0.9$$

WF

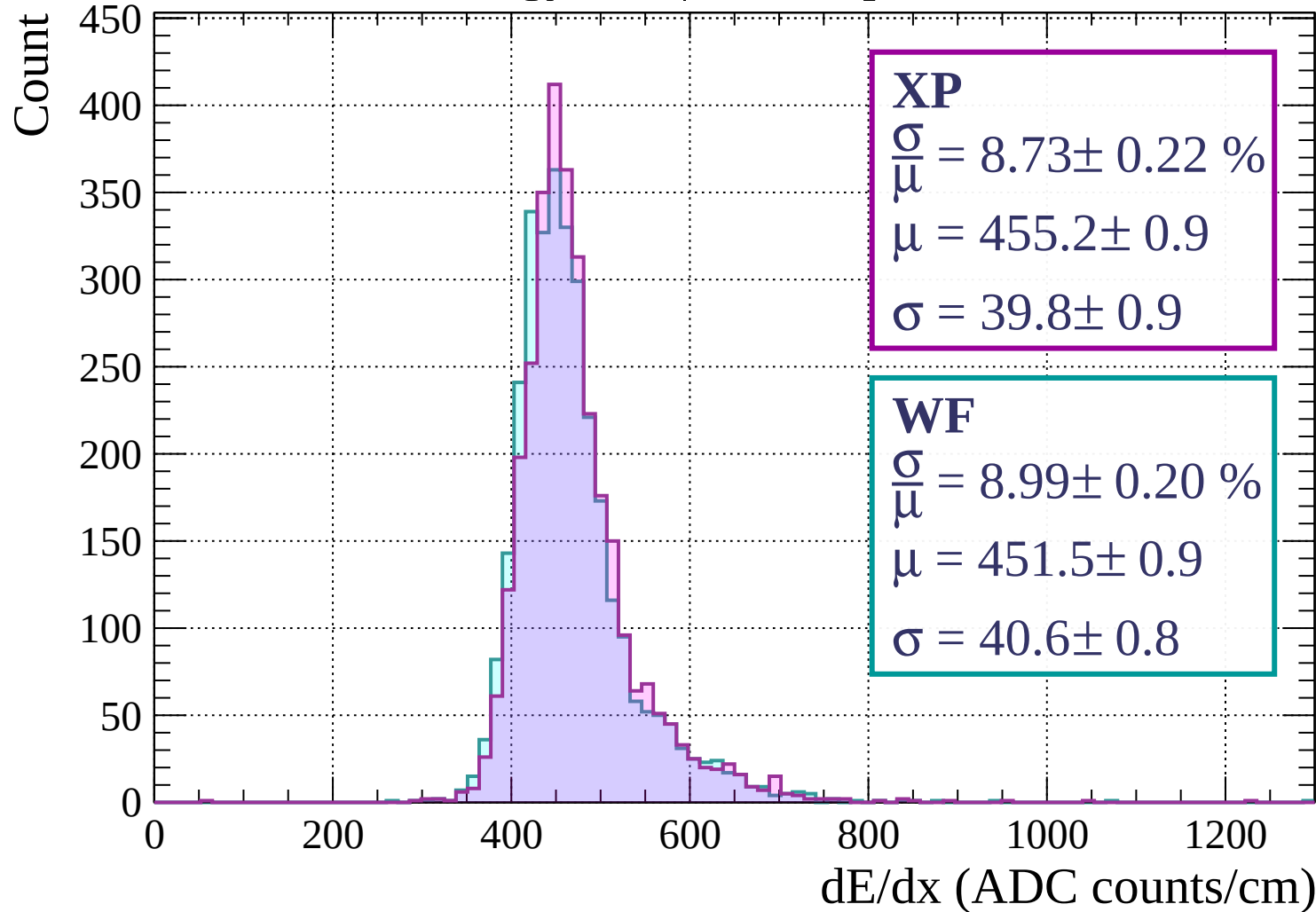
$$\frac{\sigma}{\bar{\mu}} = 9.56 \pm 0.23 \%$$

$$\mu = 447.7 \pm 1.0$$

$$\sigma = 42.8 \pm 1.0$$

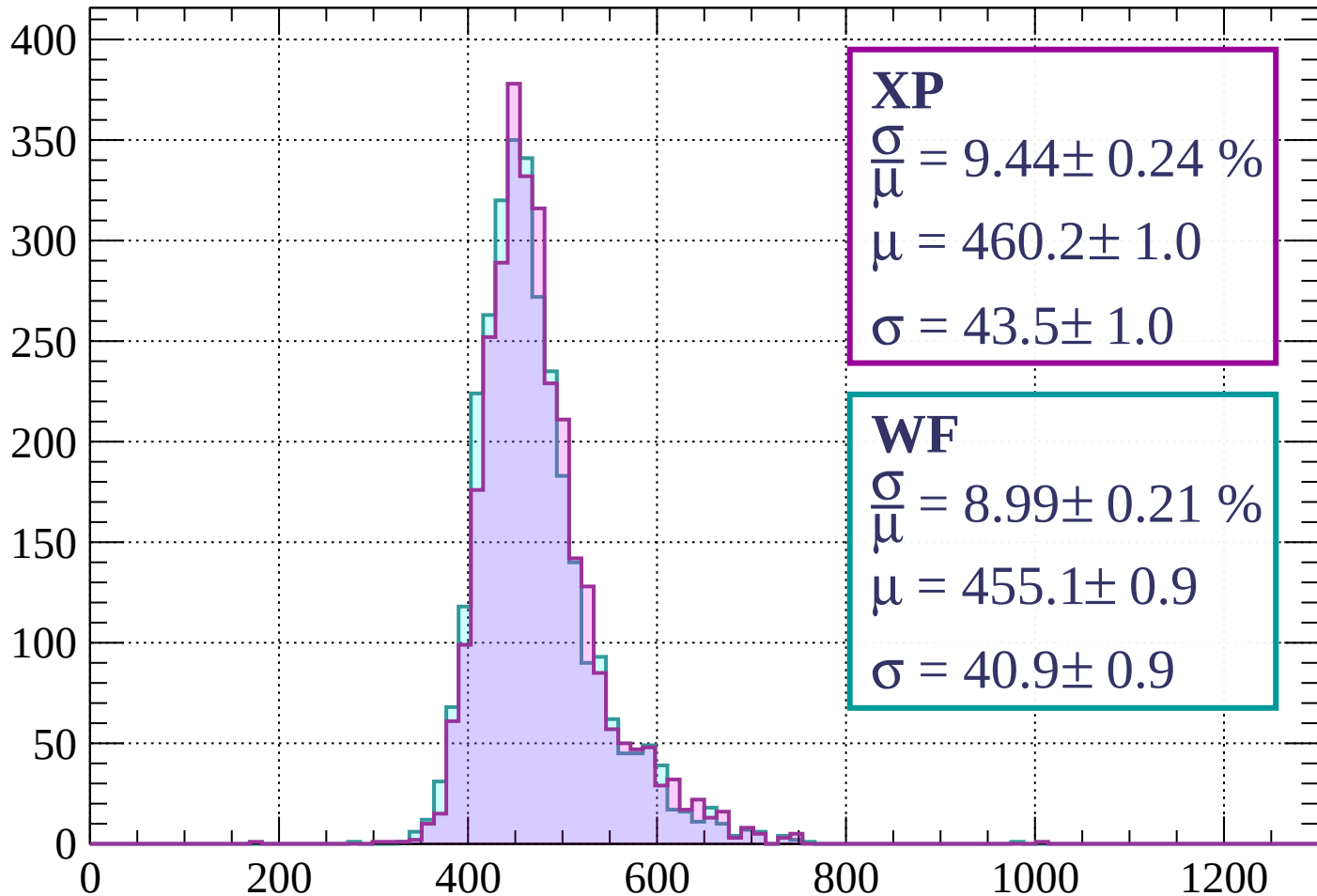
dE/dx (ADC counts/cm)

Energy loss | $1200 < p < 1280$



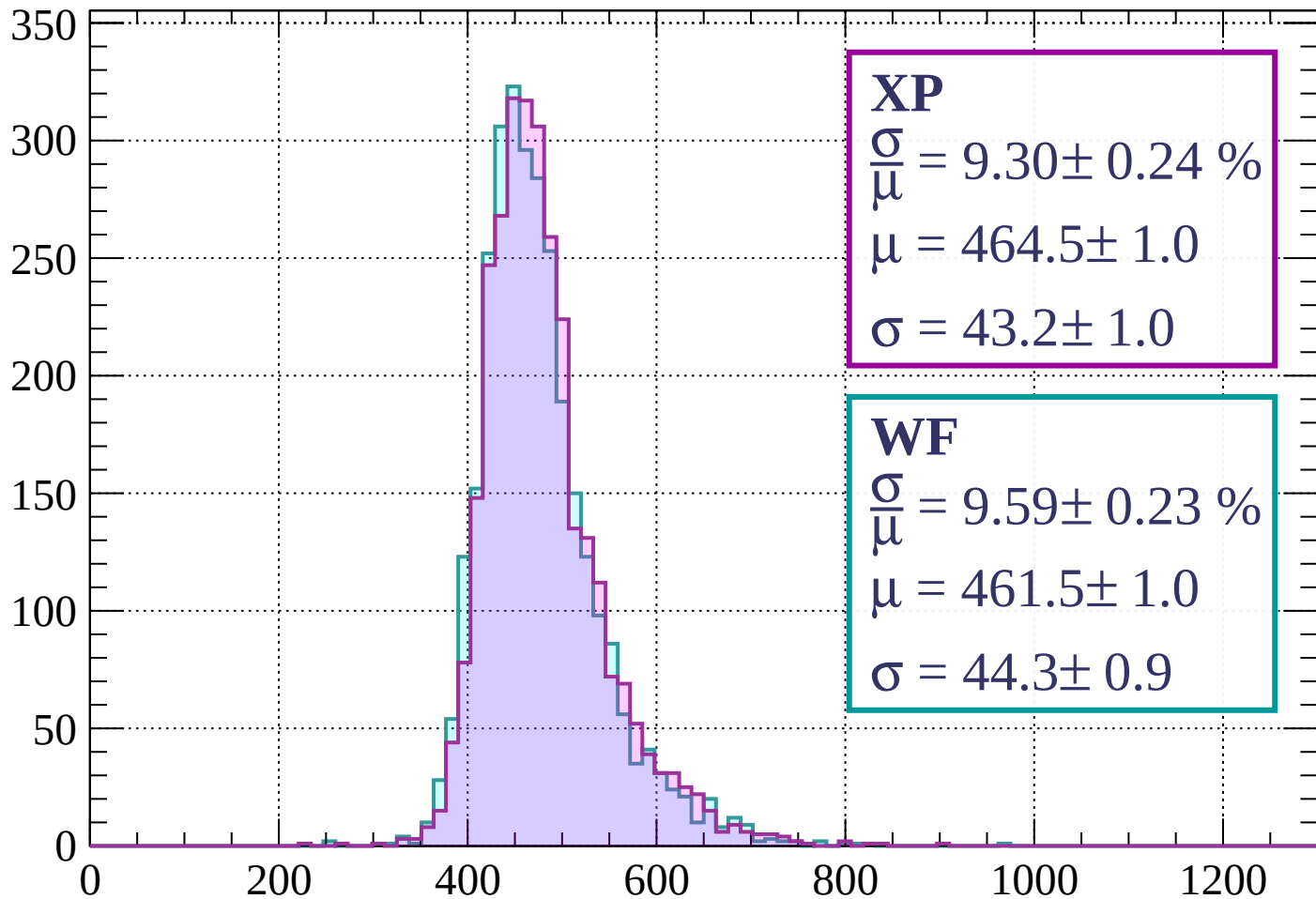
Energy loss | $1280 < p < 1360$

Count



Energy loss | $1360 < p < 1440$

Count



XP

$$\frac{\sigma}{\mu} = 9.30 \pm 0.24 \%$$

$$\mu = 464.5 \pm 1.0$$

$$\sigma = 43.2 \pm 1.0$$

WF

$$\frac{\sigma}{\mu} = 9.59 \pm 0.23 \%$$

$$\mu = 461.5 \pm 1.0$$

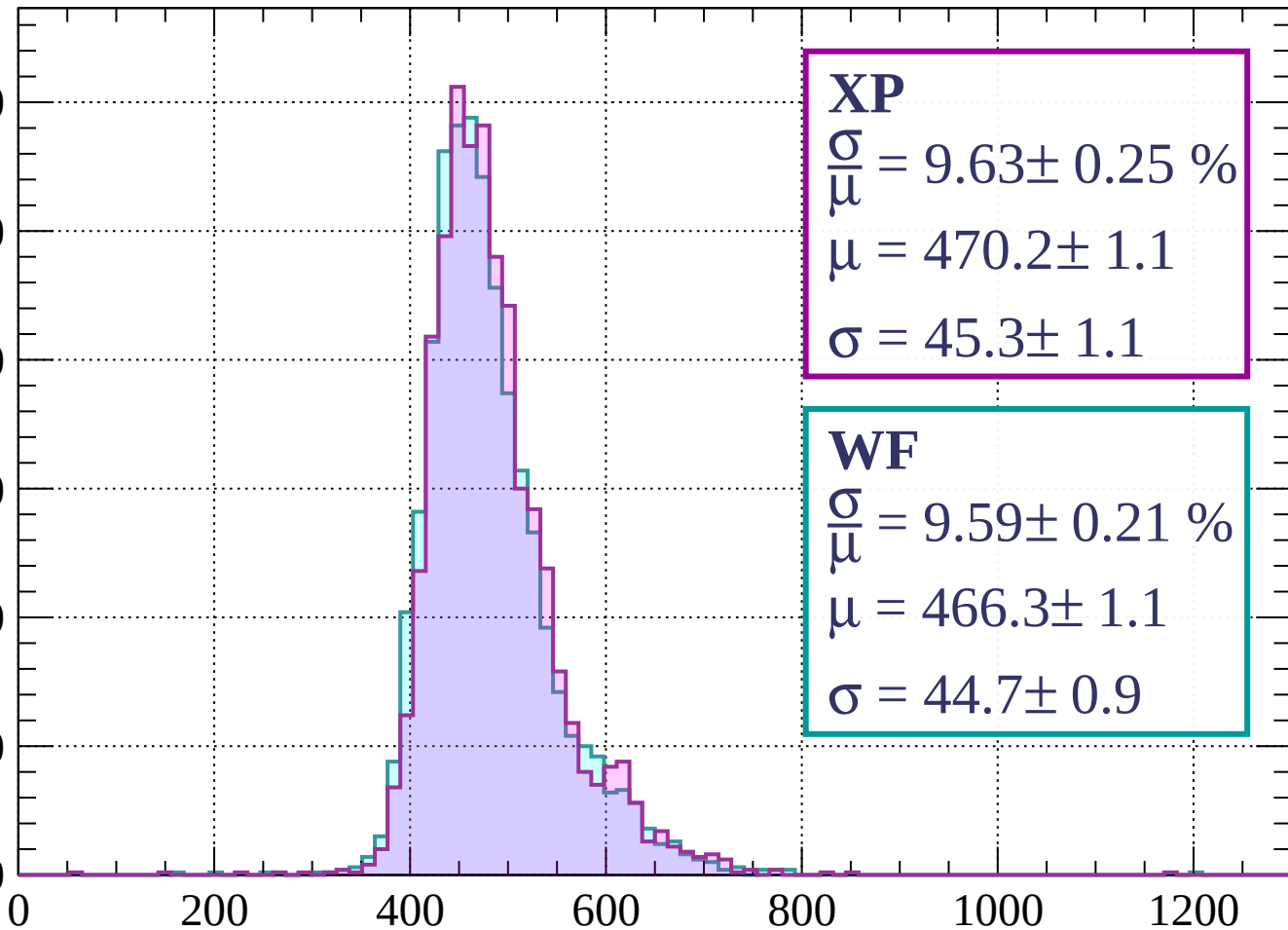
$$\sigma = 44.3 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss | $1440 < p < 1520$

Count

300
250
200
150
100
50
0



XP

$$\frac{\sigma}{\mu} = 9.63 \pm 0.25 \%$$

$$\mu = 470.2 \pm 1.1$$

$$\sigma = 45.3 \pm 1.1$$

WF

$$\frac{\sigma}{\mu} = 9.59 \pm 0.21 \%$$

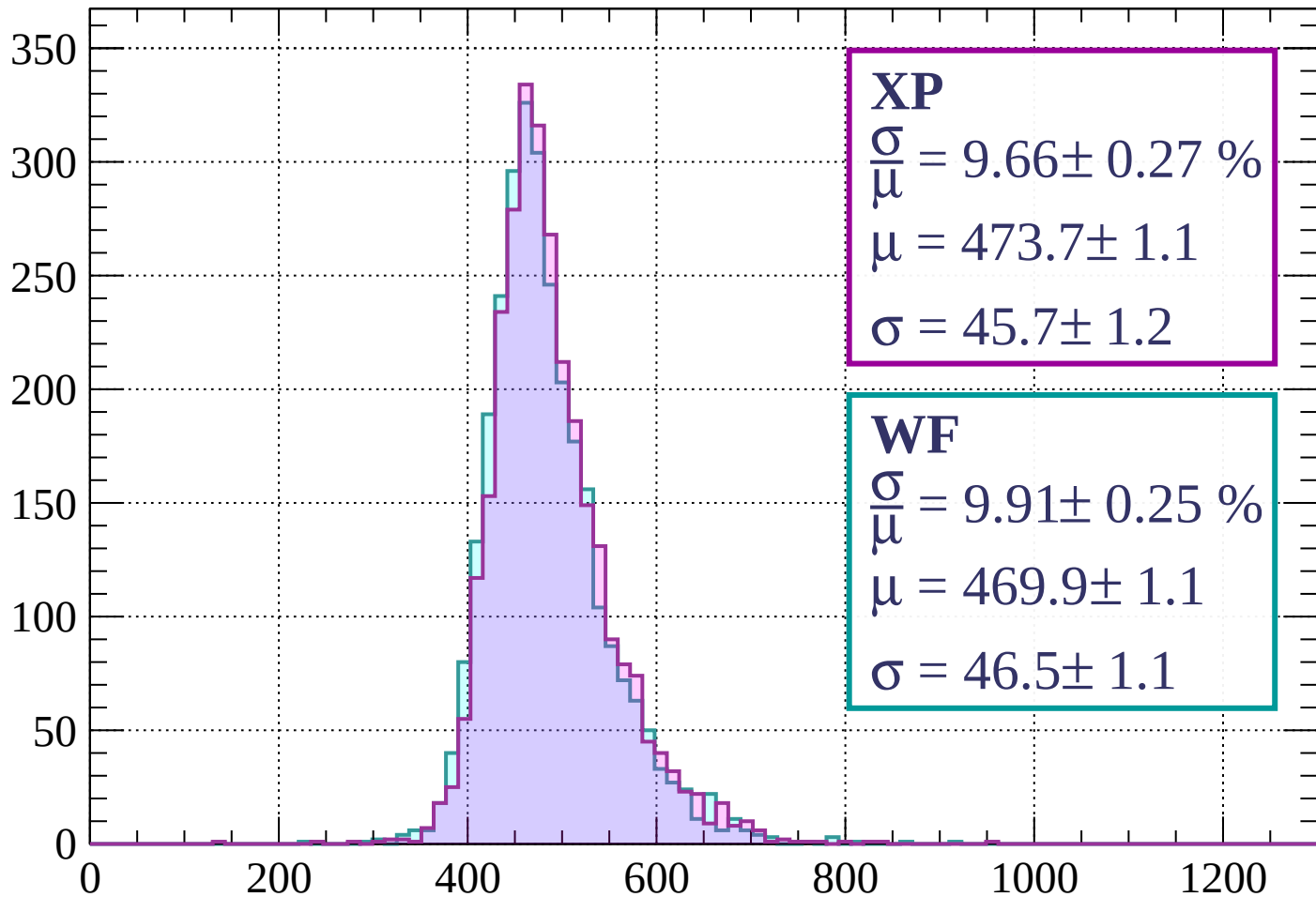
$$\mu = 466.3 \pm 1.1$$

$$\sigma = 44.7 \pm 0.9$$

dE/dx (ADC counts/cm)

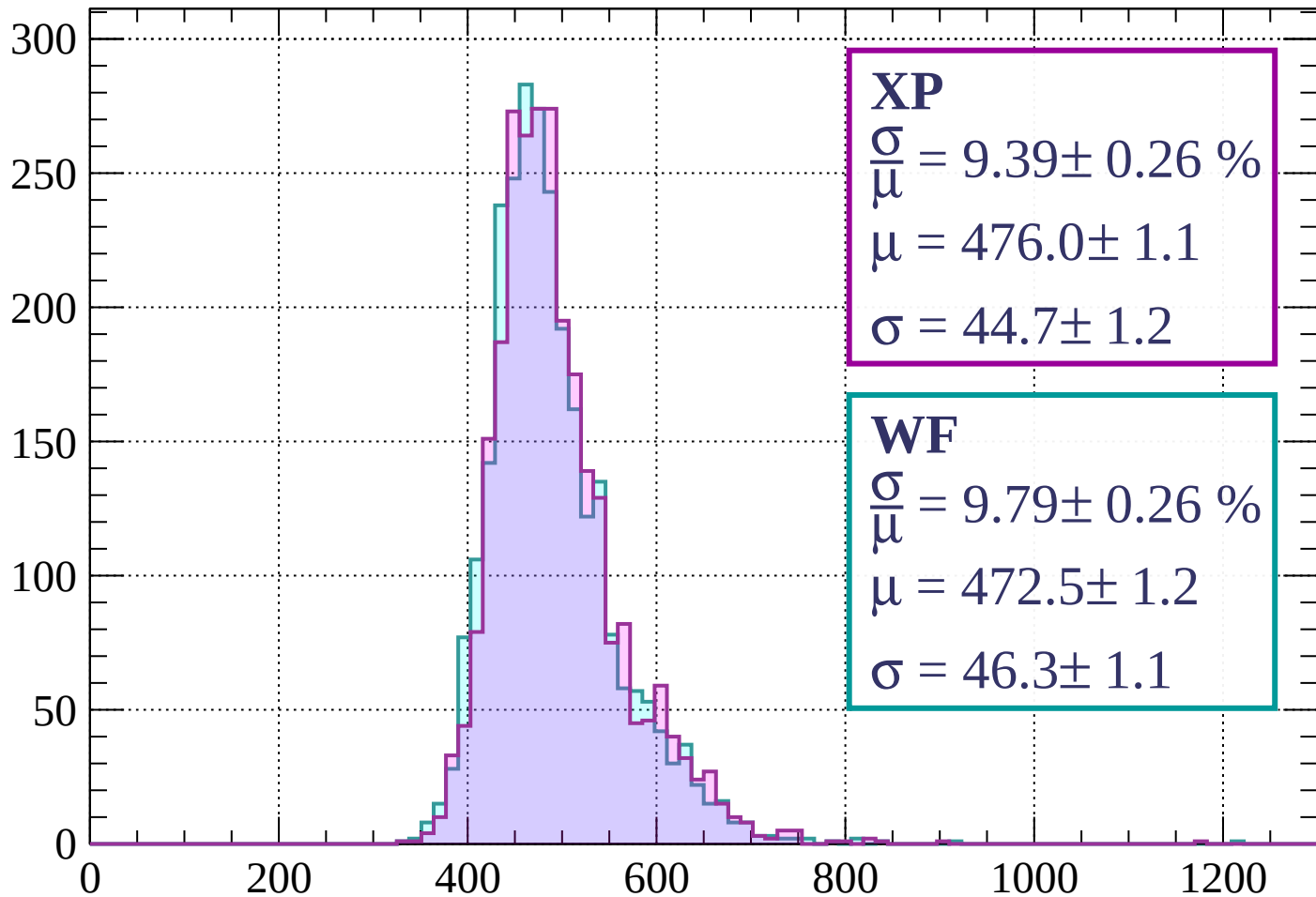
Energy loss | $1520 < p < 1600$

Count



Energy loss | $1600 < p < 1680$

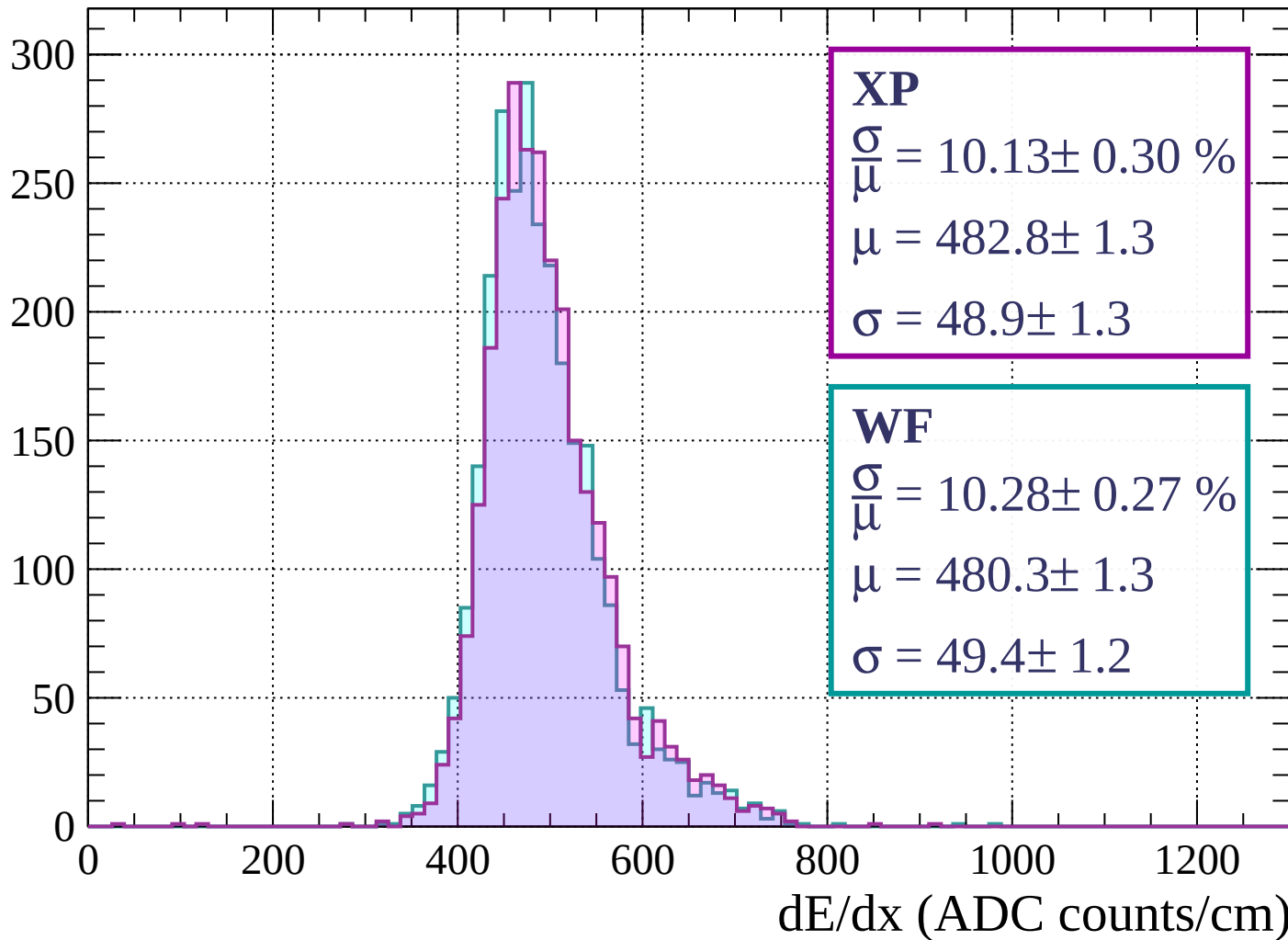
Count



dE/dx (ADC counts/cm)

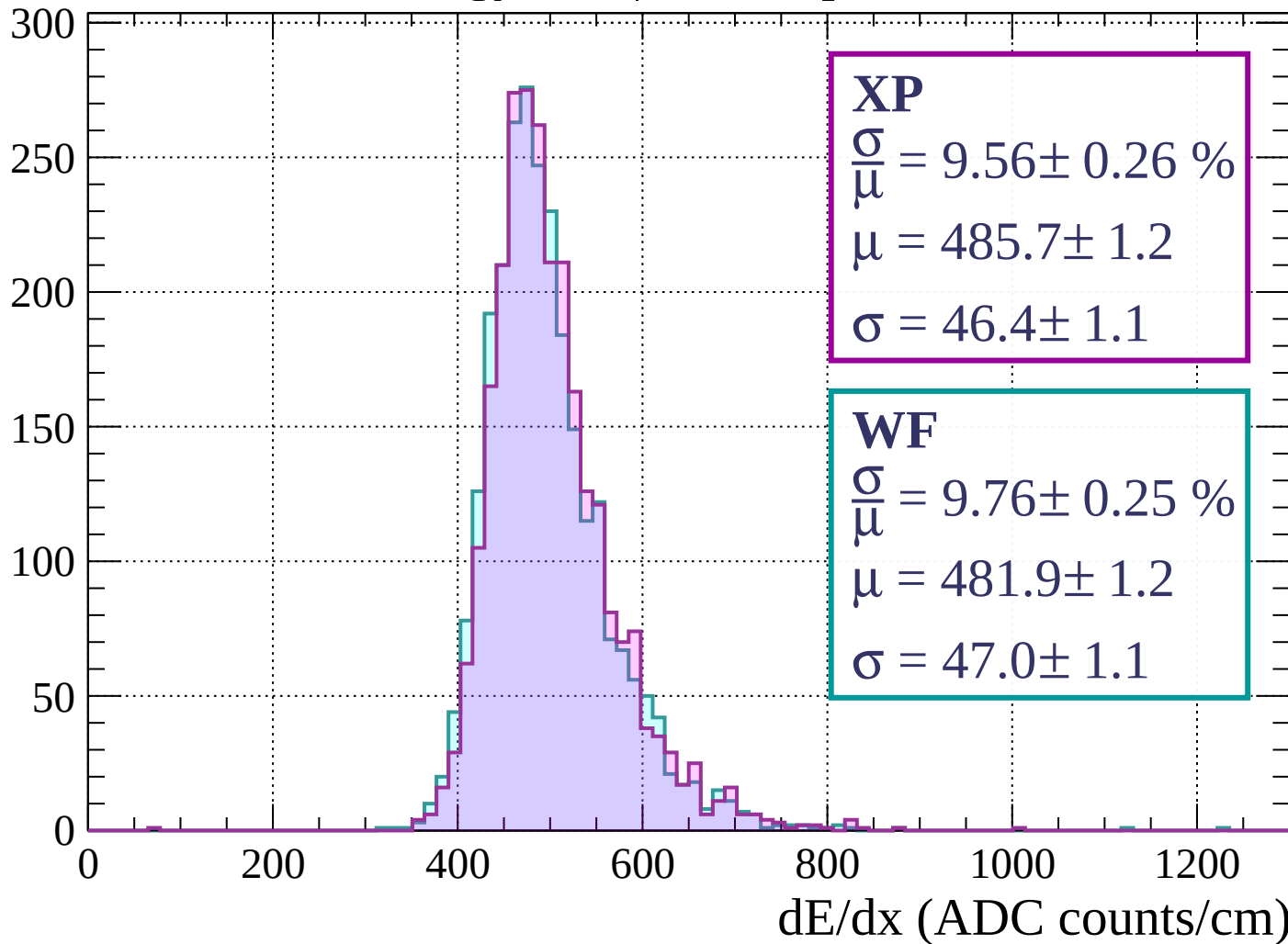
Energy loss | $1680 < p < 1760$

Count

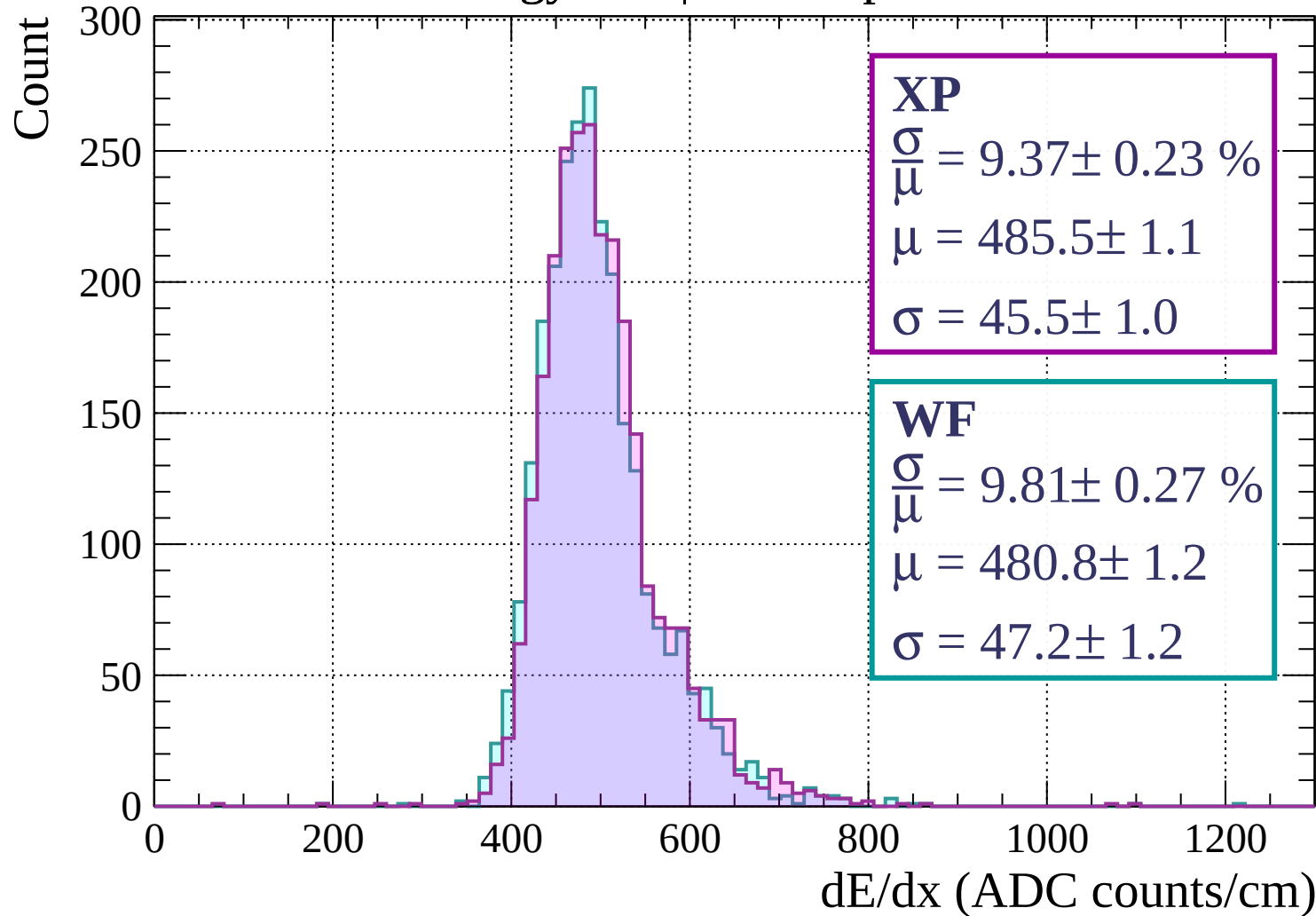


Energy loss | $1760 < p < 1840$

Count

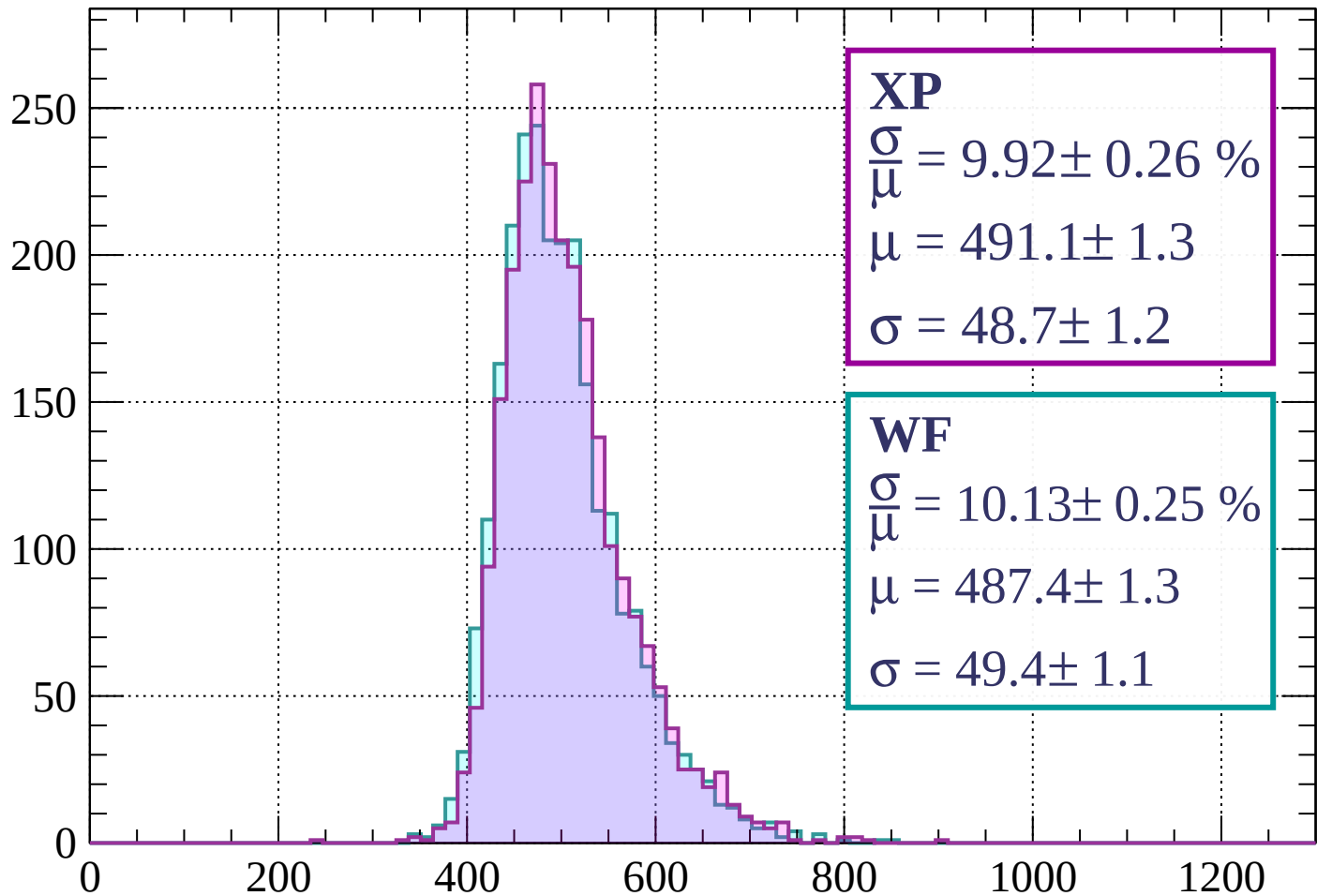


Energy loss | $1840 < p < 1920$



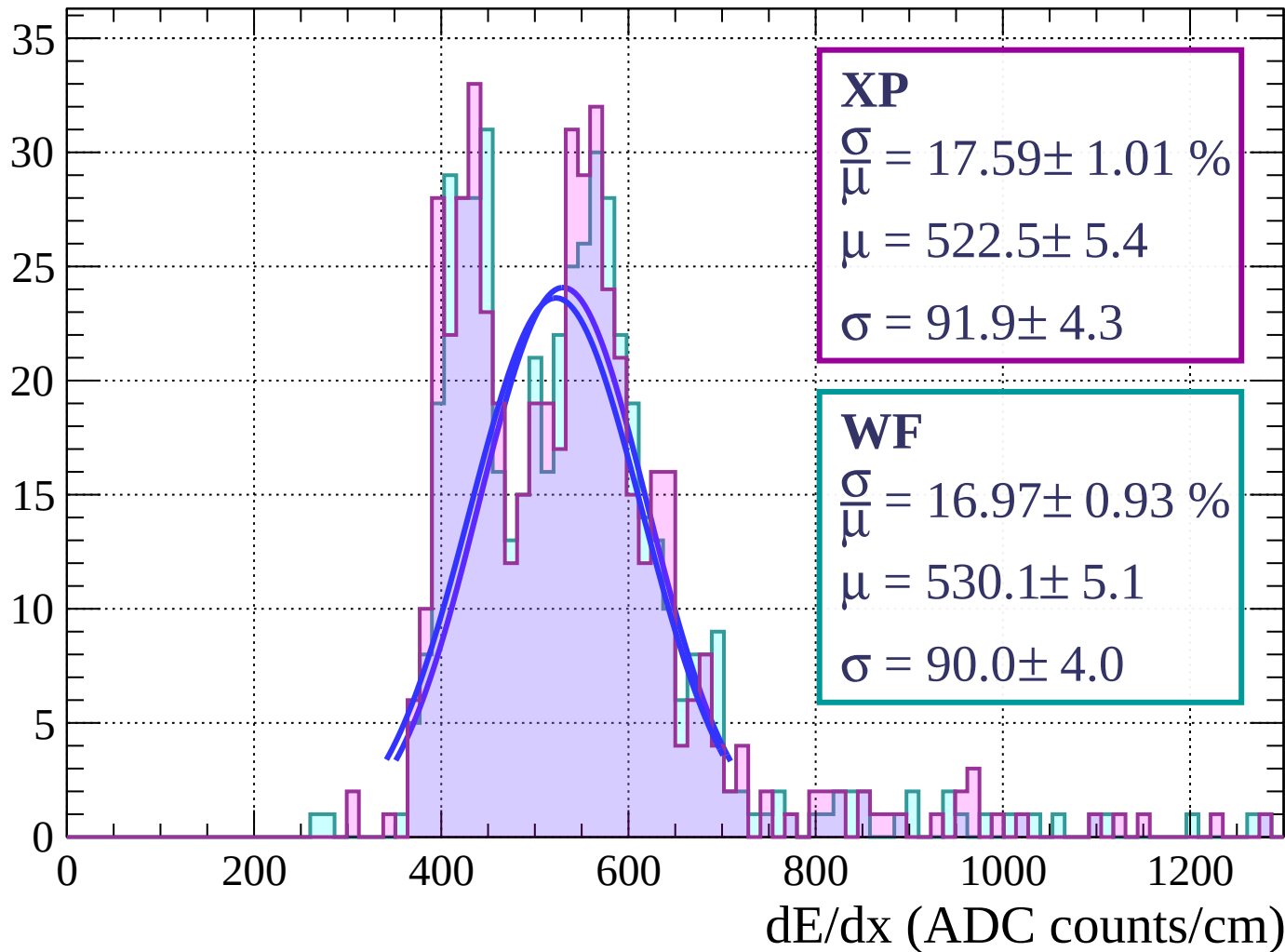
Energy loss | $1920 < p < 2000$

Count



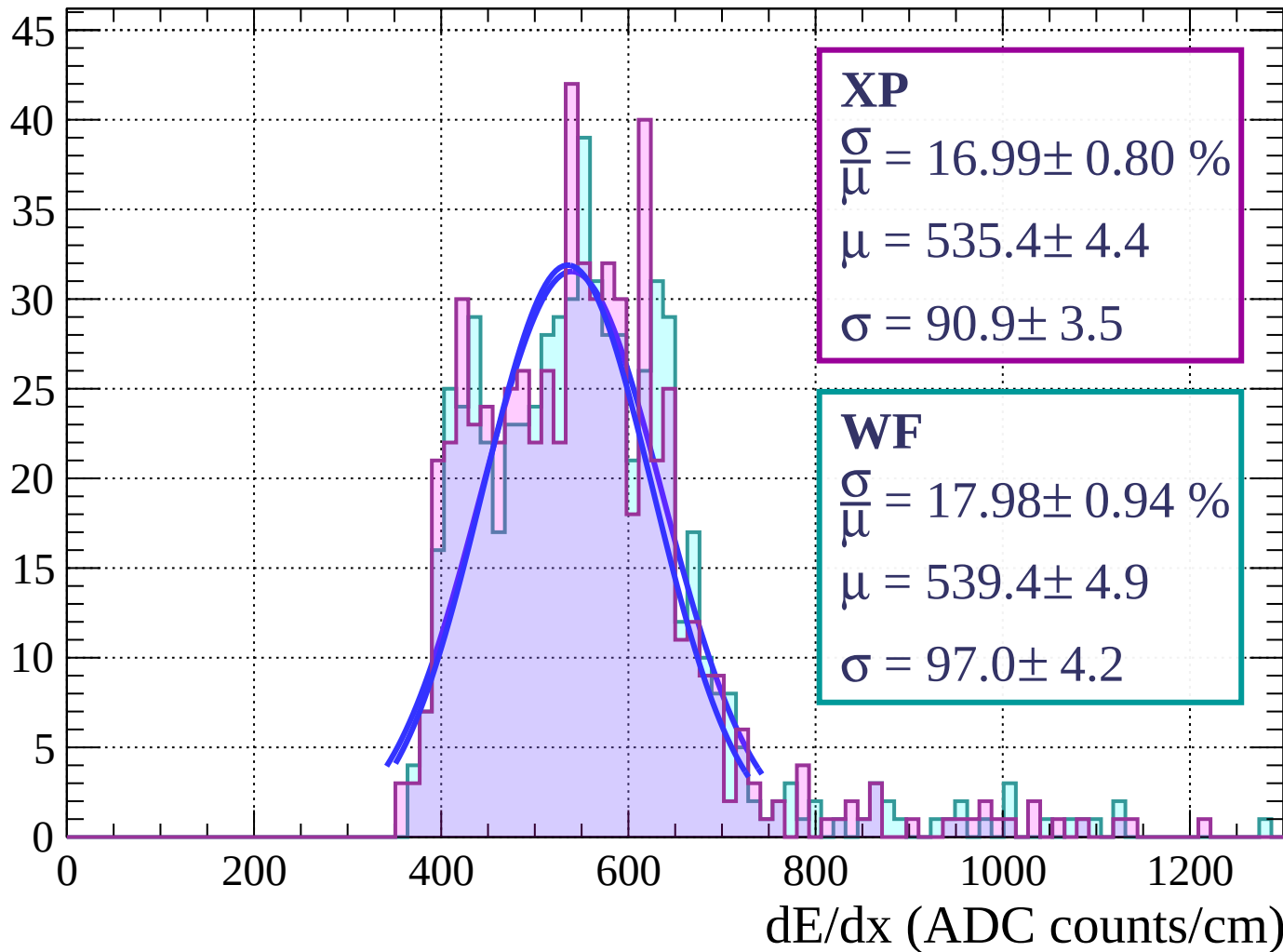
Energy loss | $0 < \phi < 3$

Count



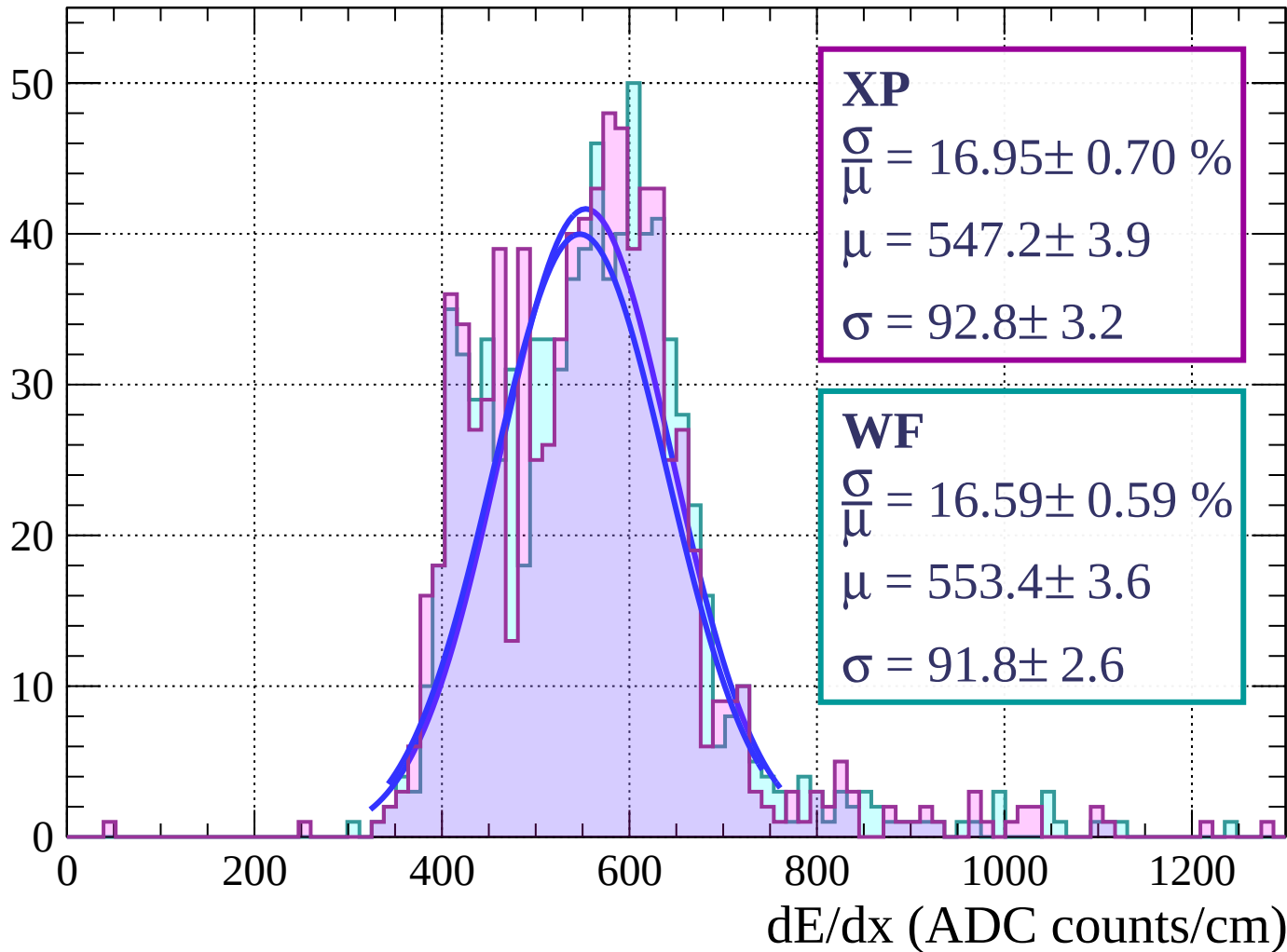
Energy loss | $3 < \phi < 6$

Count



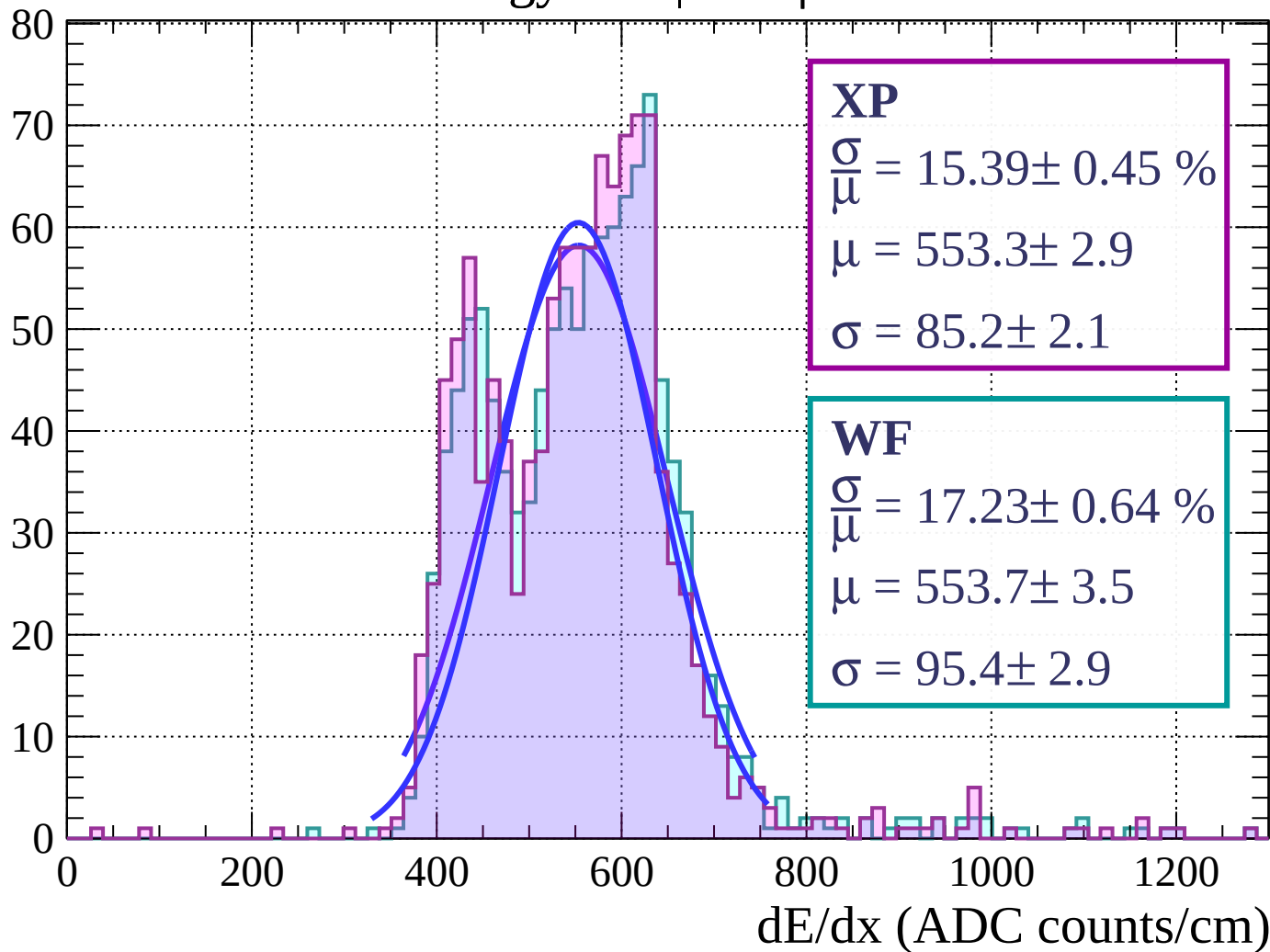
Energy loss | $6 < \phi < 9$

Count



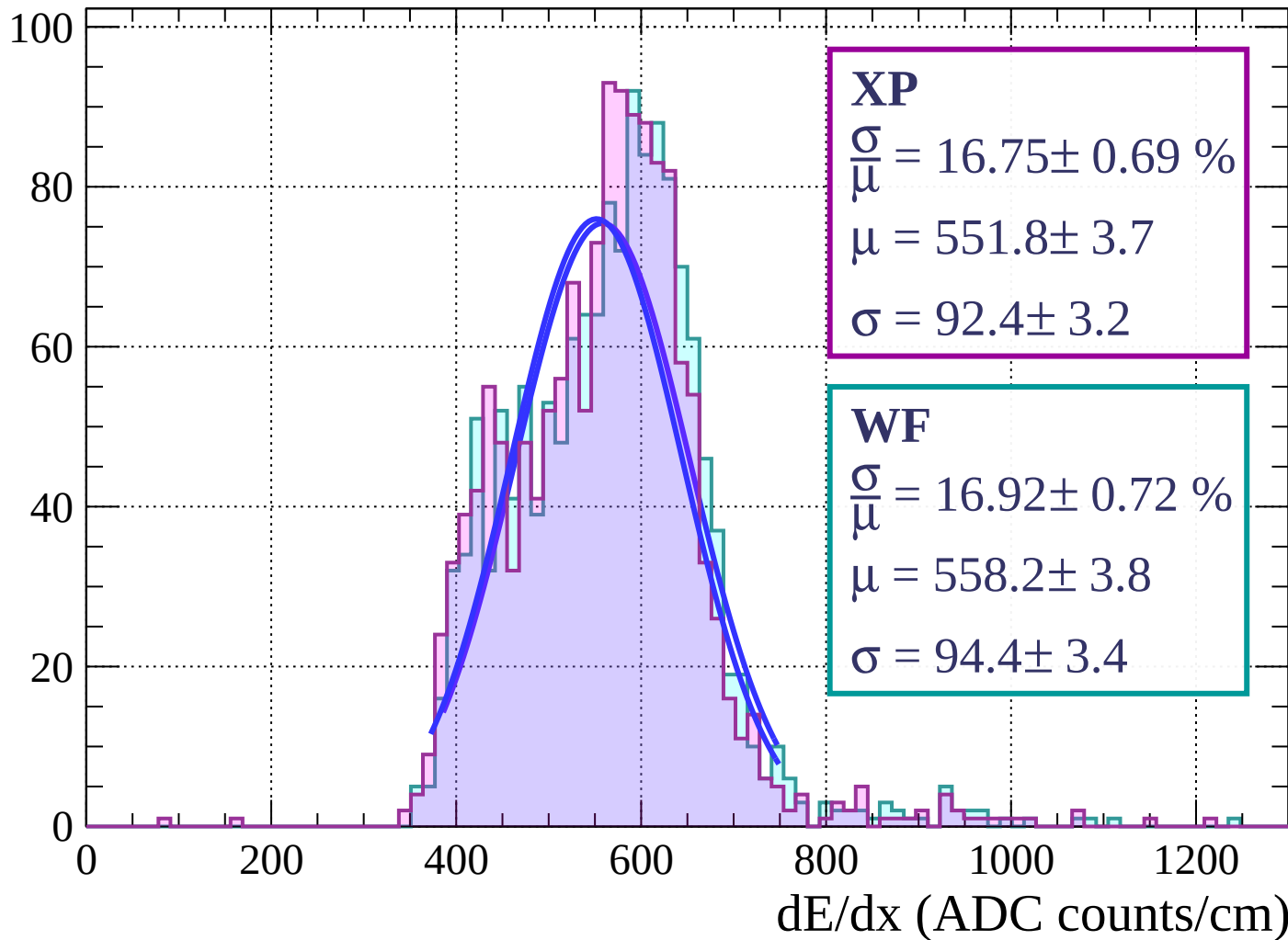
Energy loss | $9 < \phi < 12$

Count



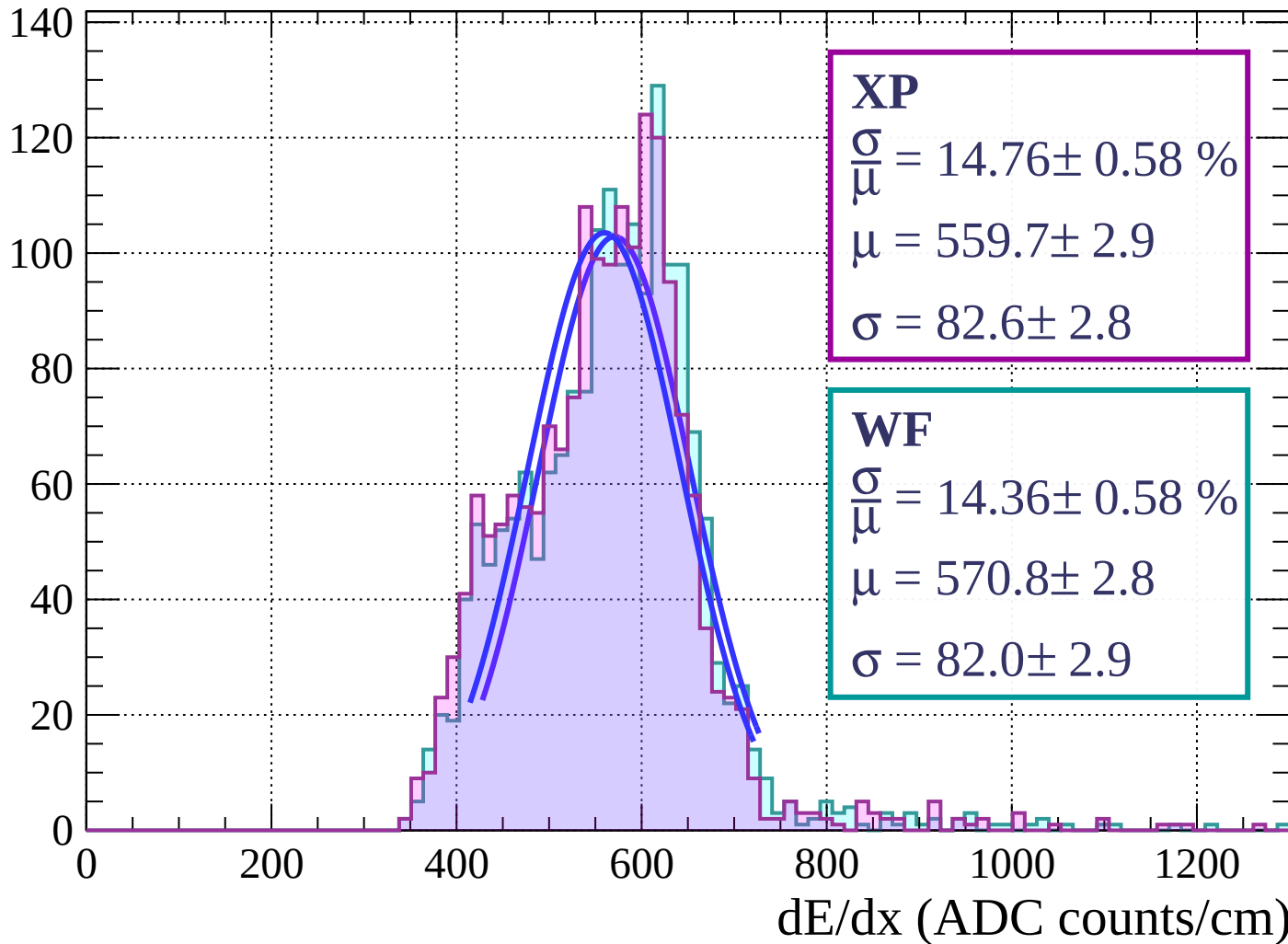
Energy loss | $12 < \phi < 15$

Count

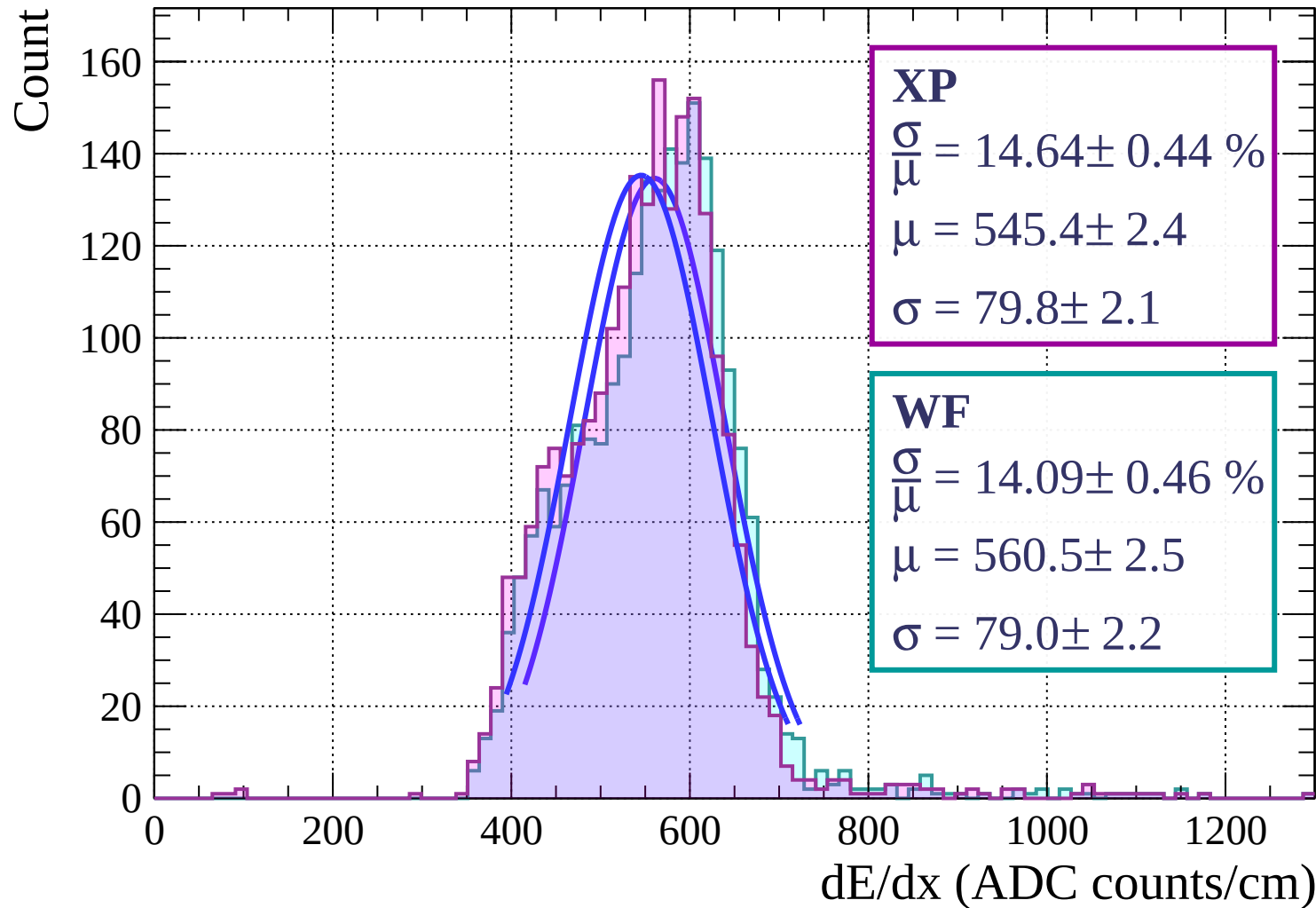


Energy loss | $15 < \phi < 18$

Count

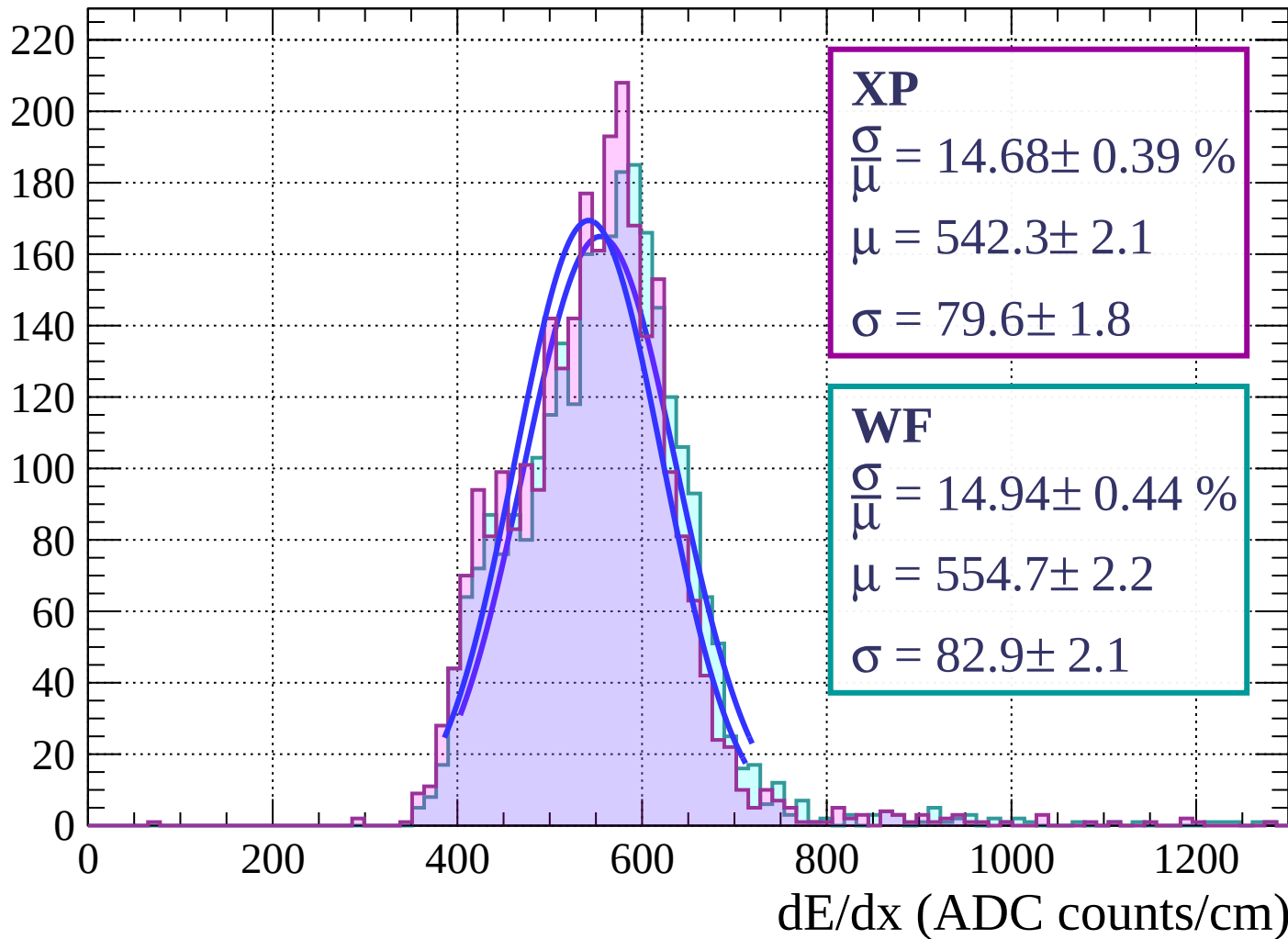


Energy loss | $18 < \phi < 21$

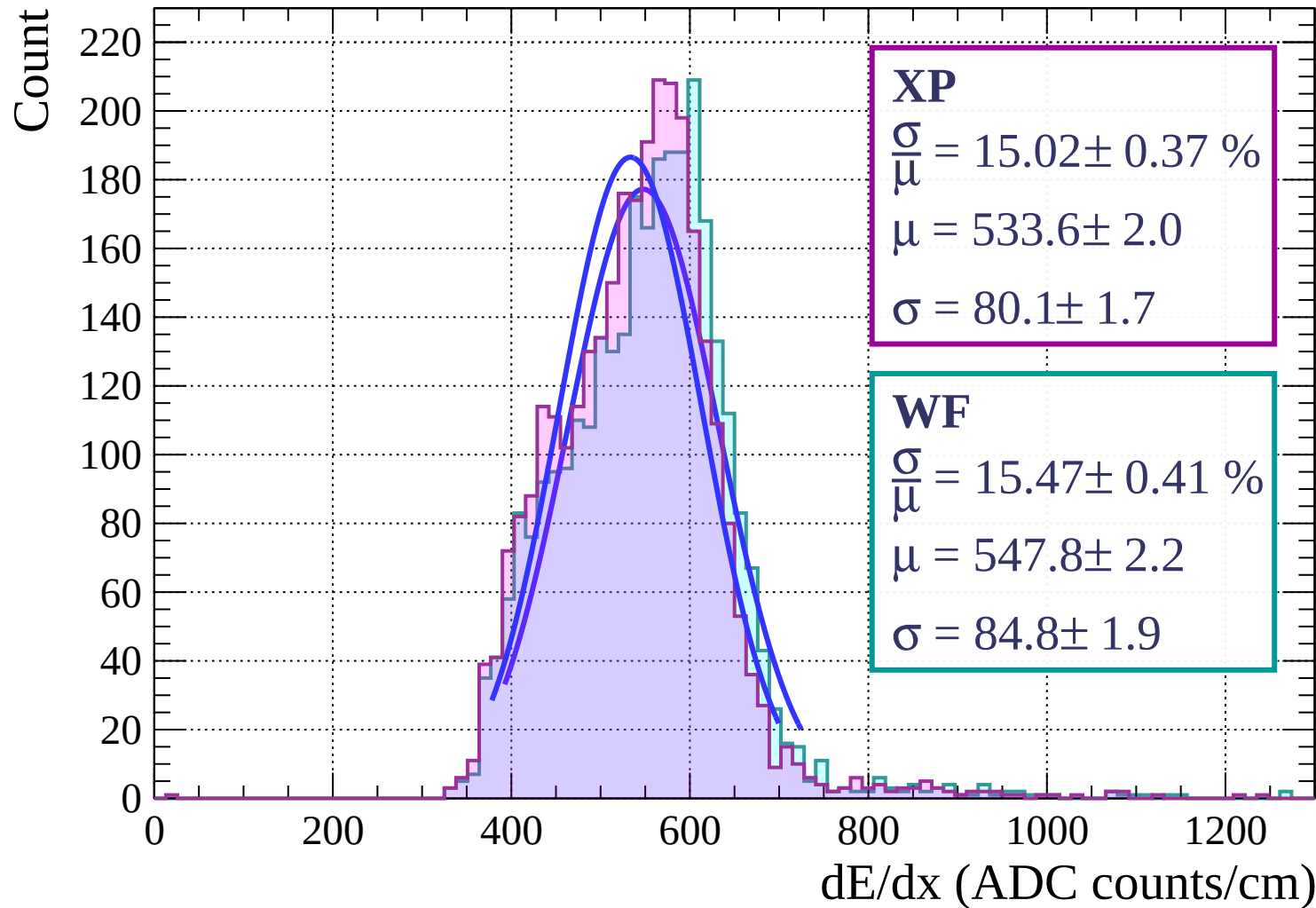


Energy loss | $21 < \phi < 24$

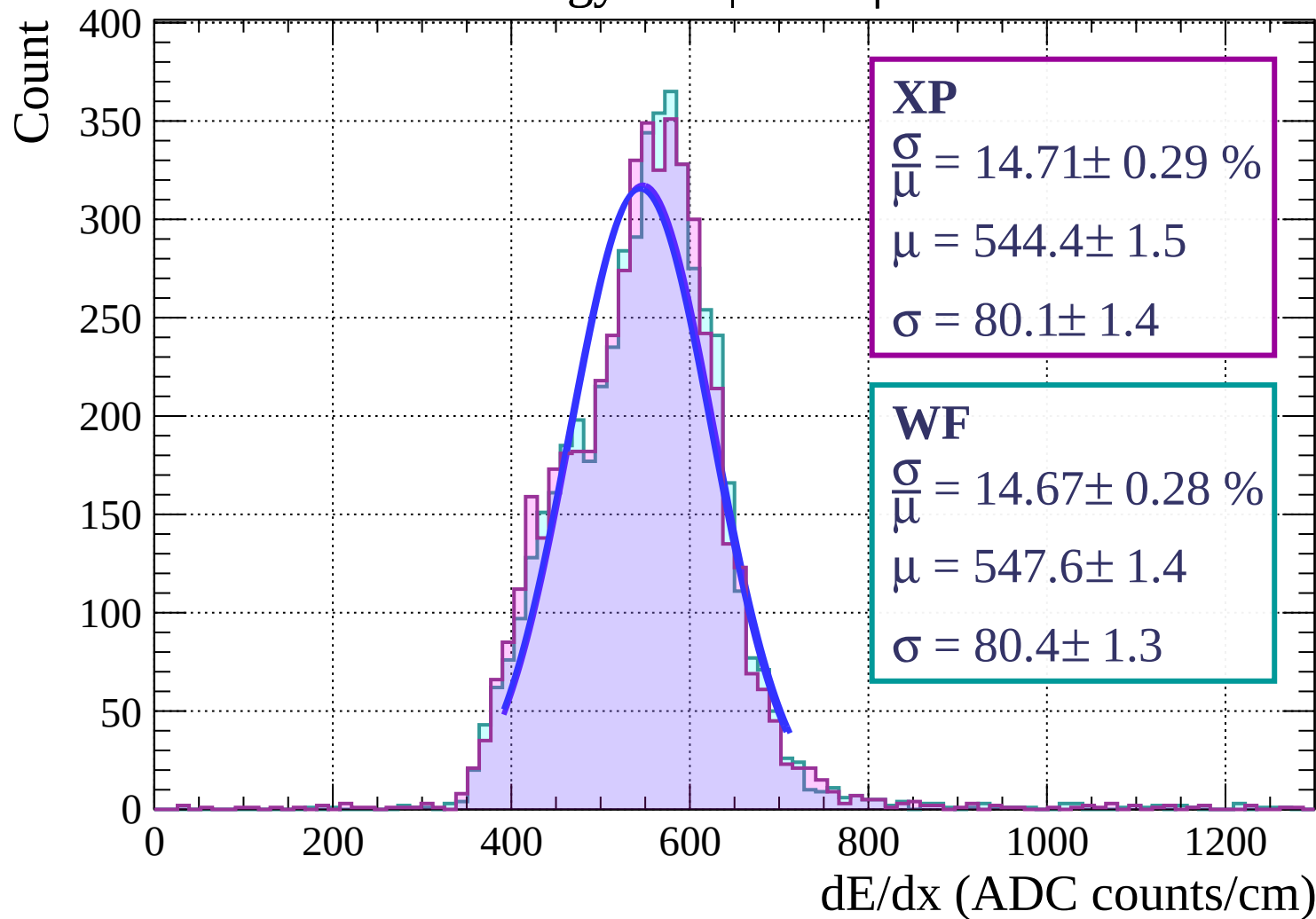
Count



Energy loss | $24 < \phi < 27$

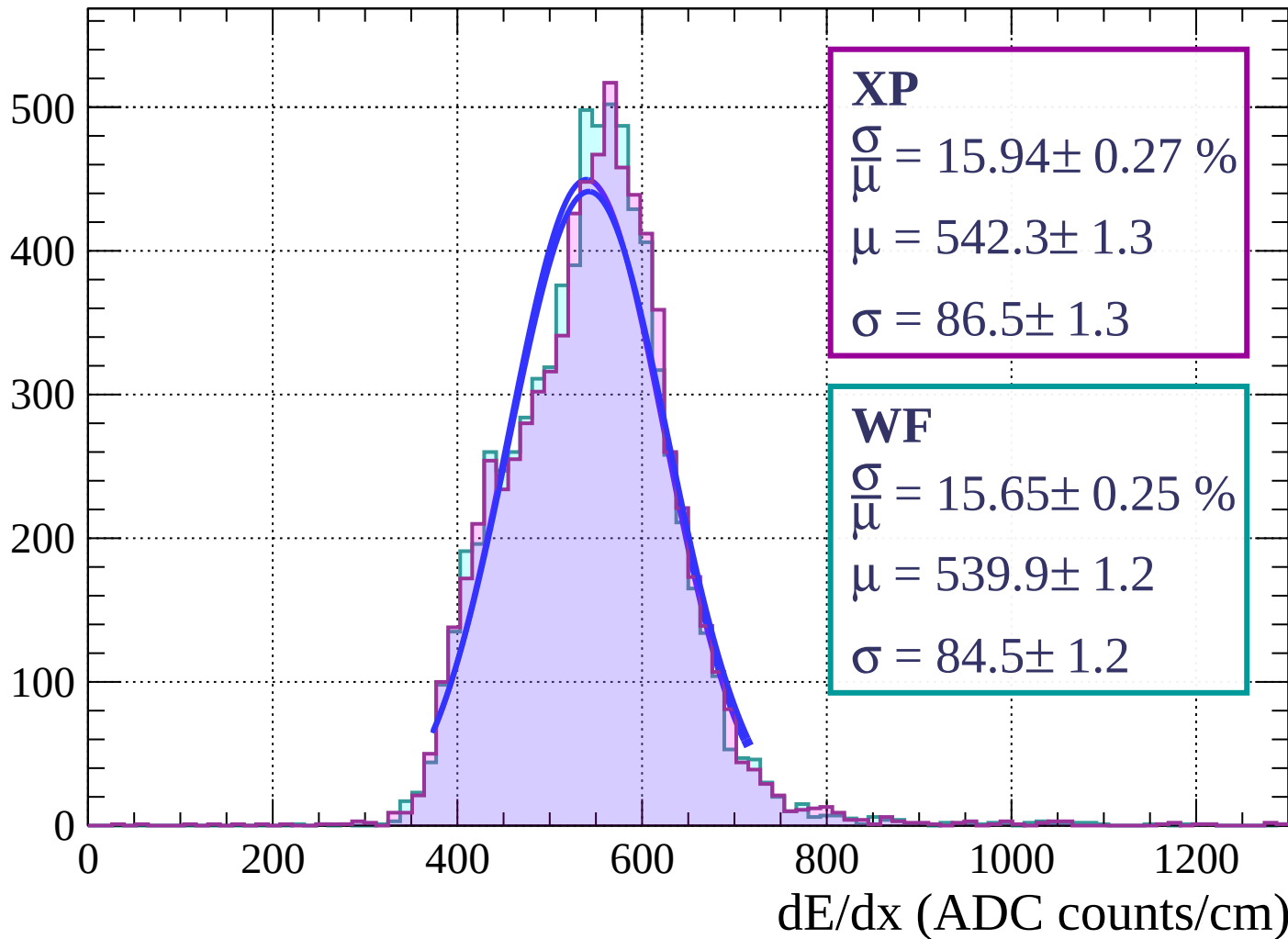


Energy loss | $27 < \phi < 30$



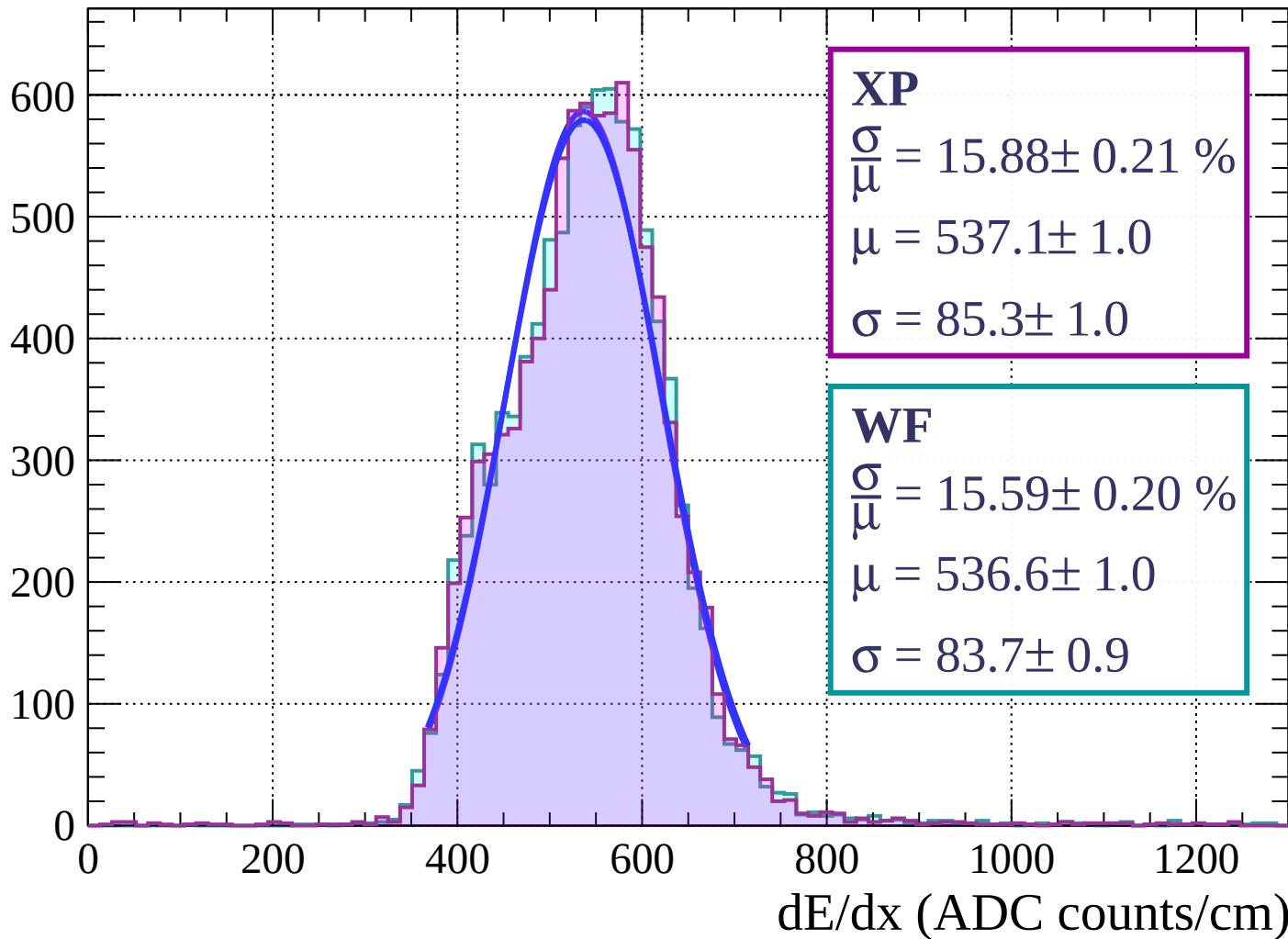
Energy loss | $30 < \phi < 33$

Count

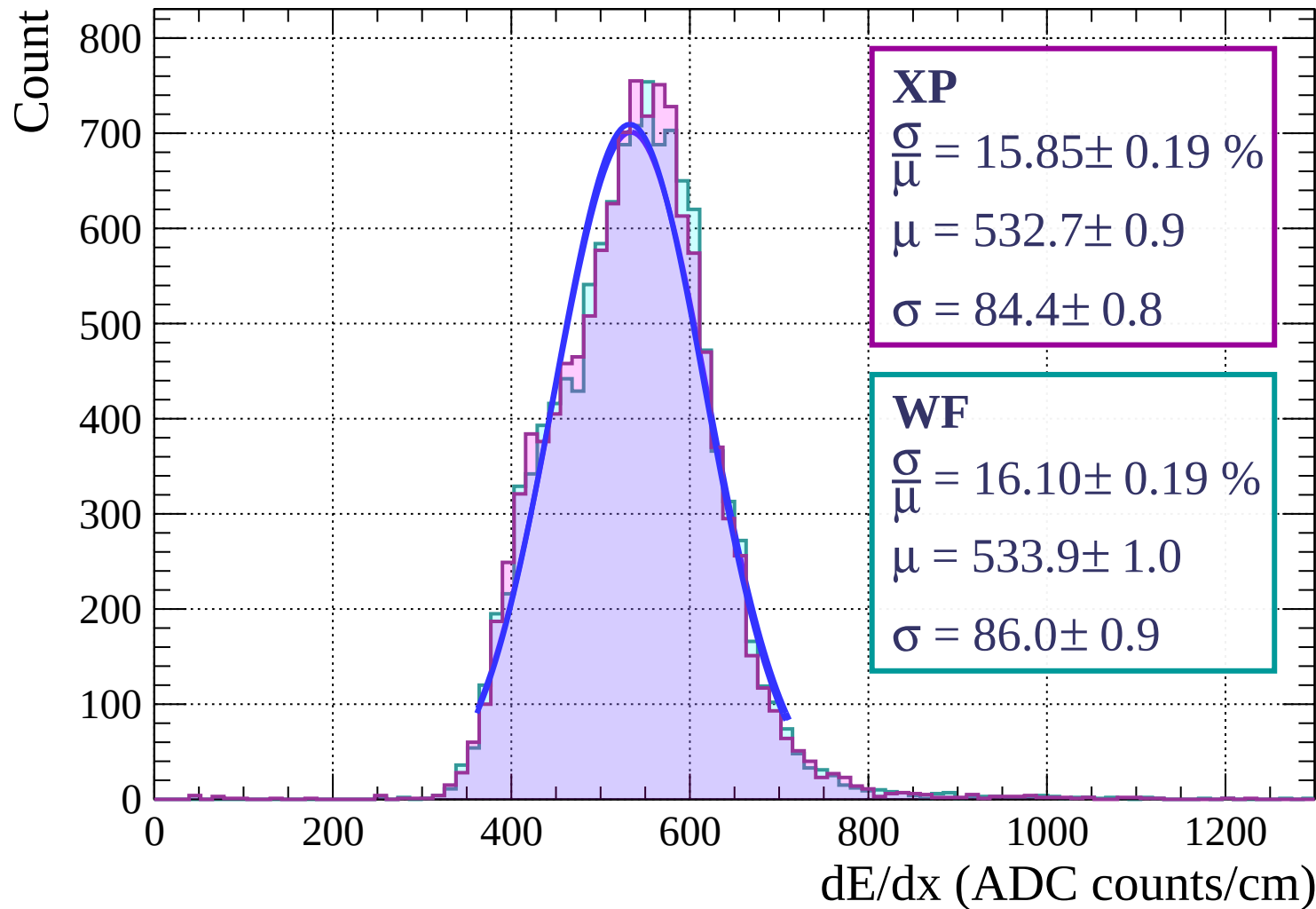


Energy loss | $33 < \phi < 36$

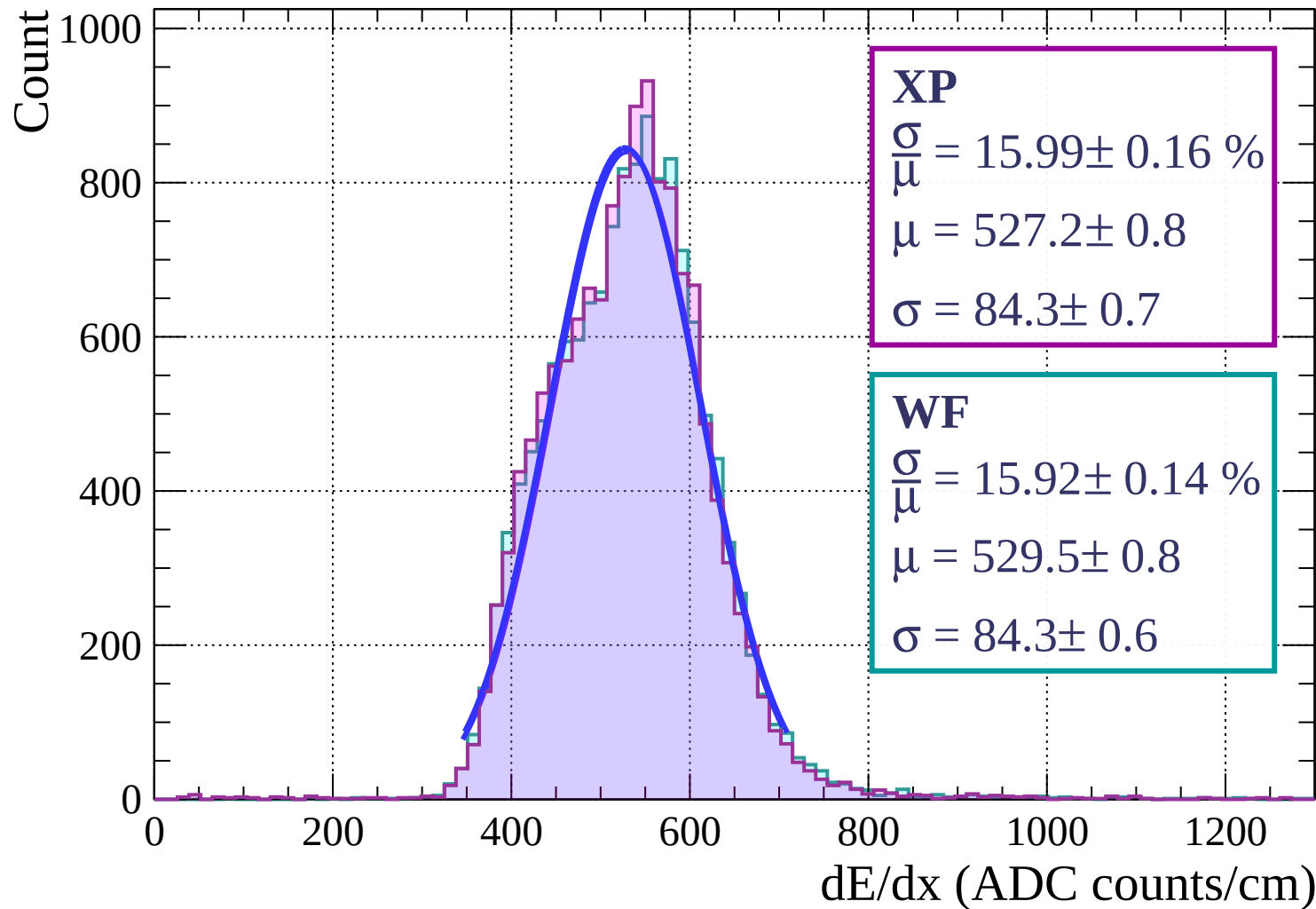
Count



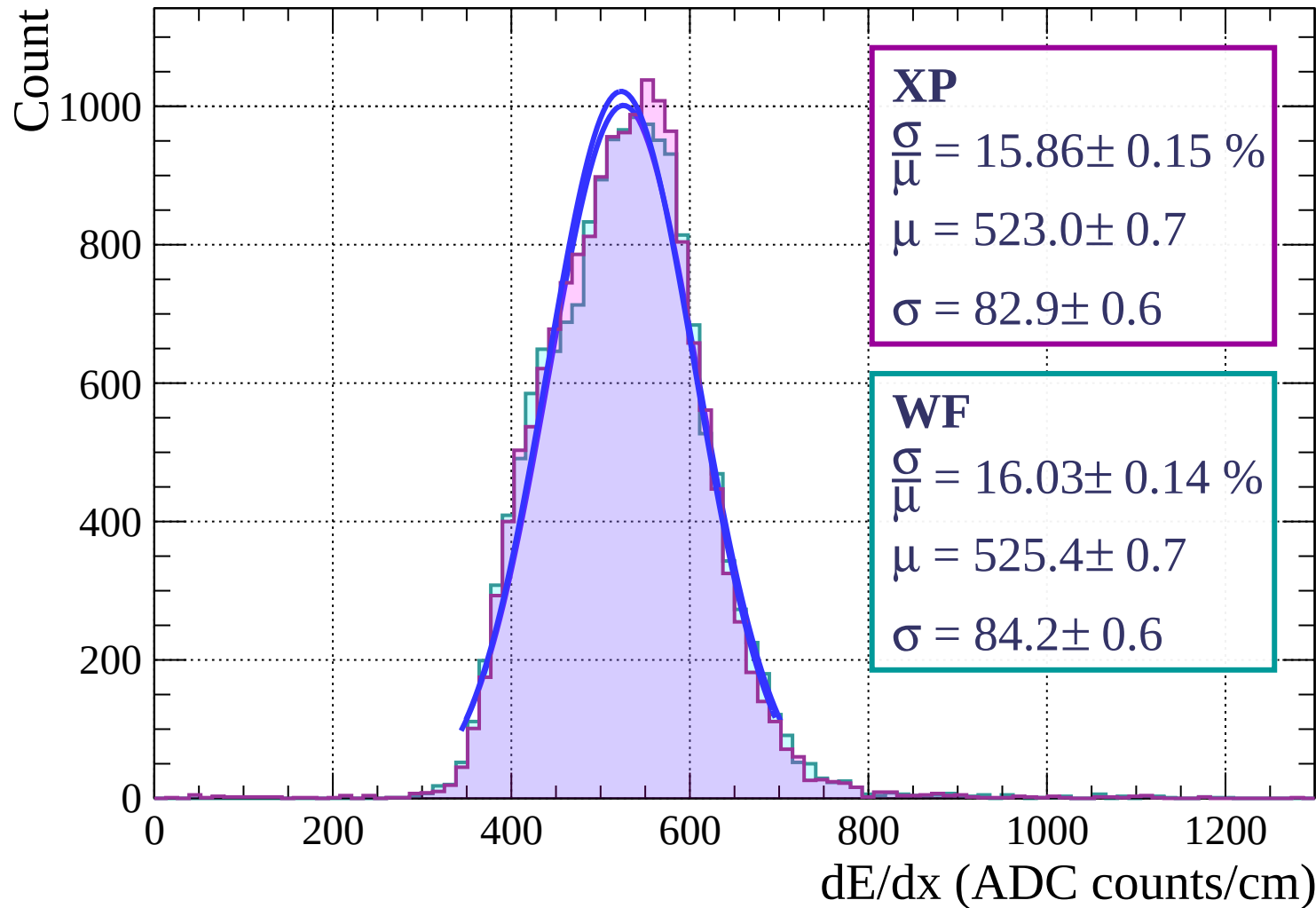
Energy loss | $36 < \phi < 39$



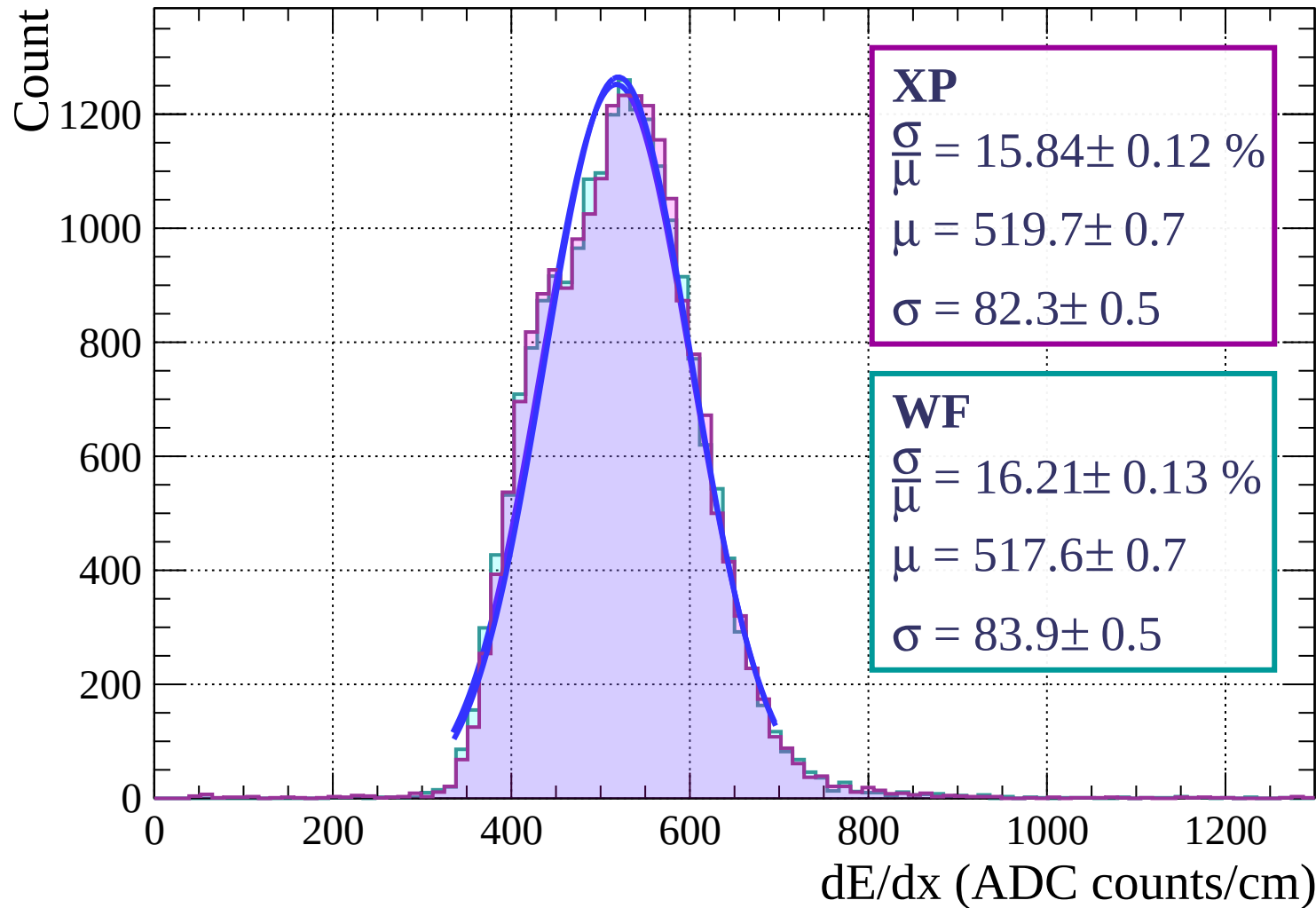
Energy loss | $39 < \phi < 42$



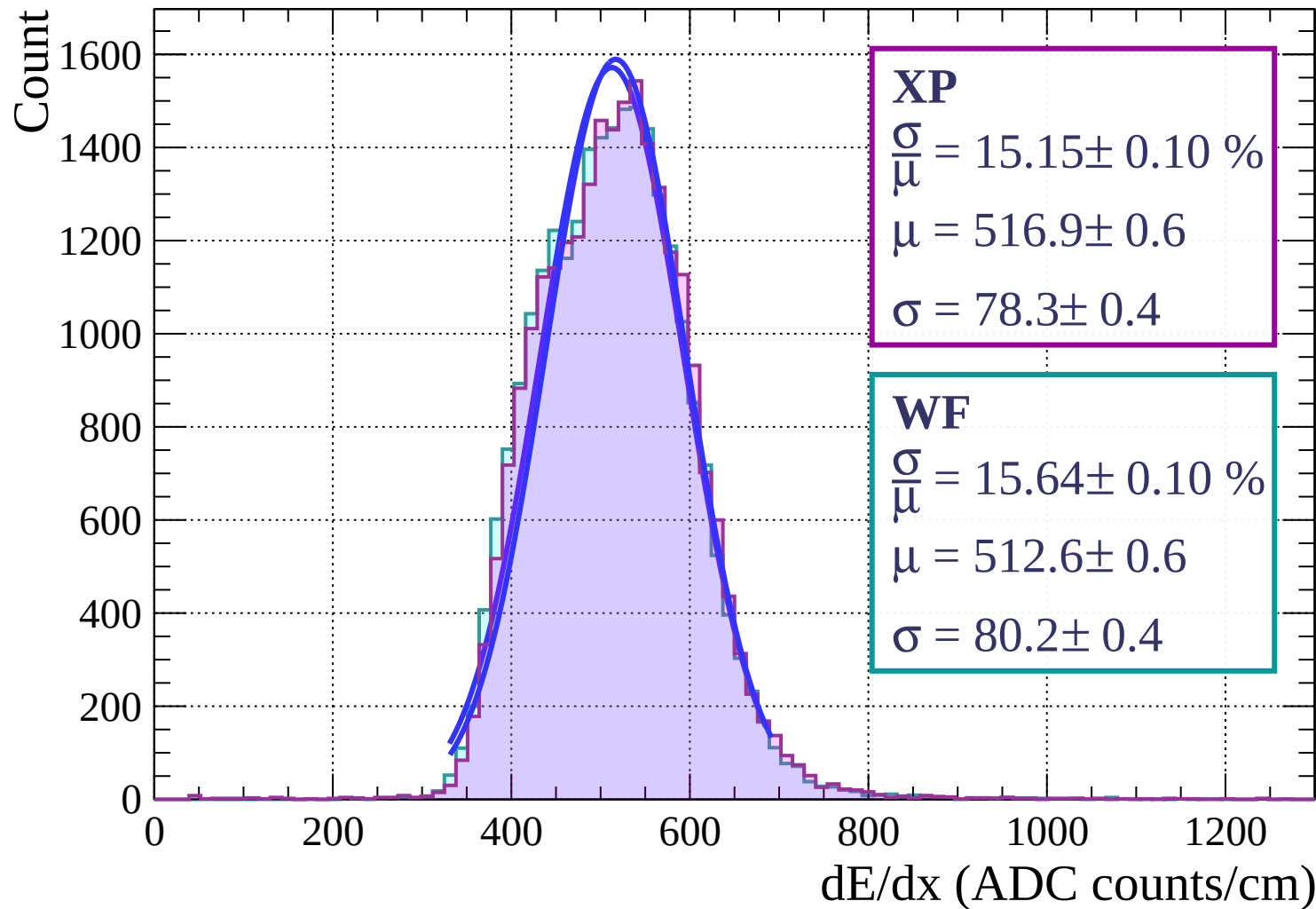
Energy loss | $42 < \phi < 45$



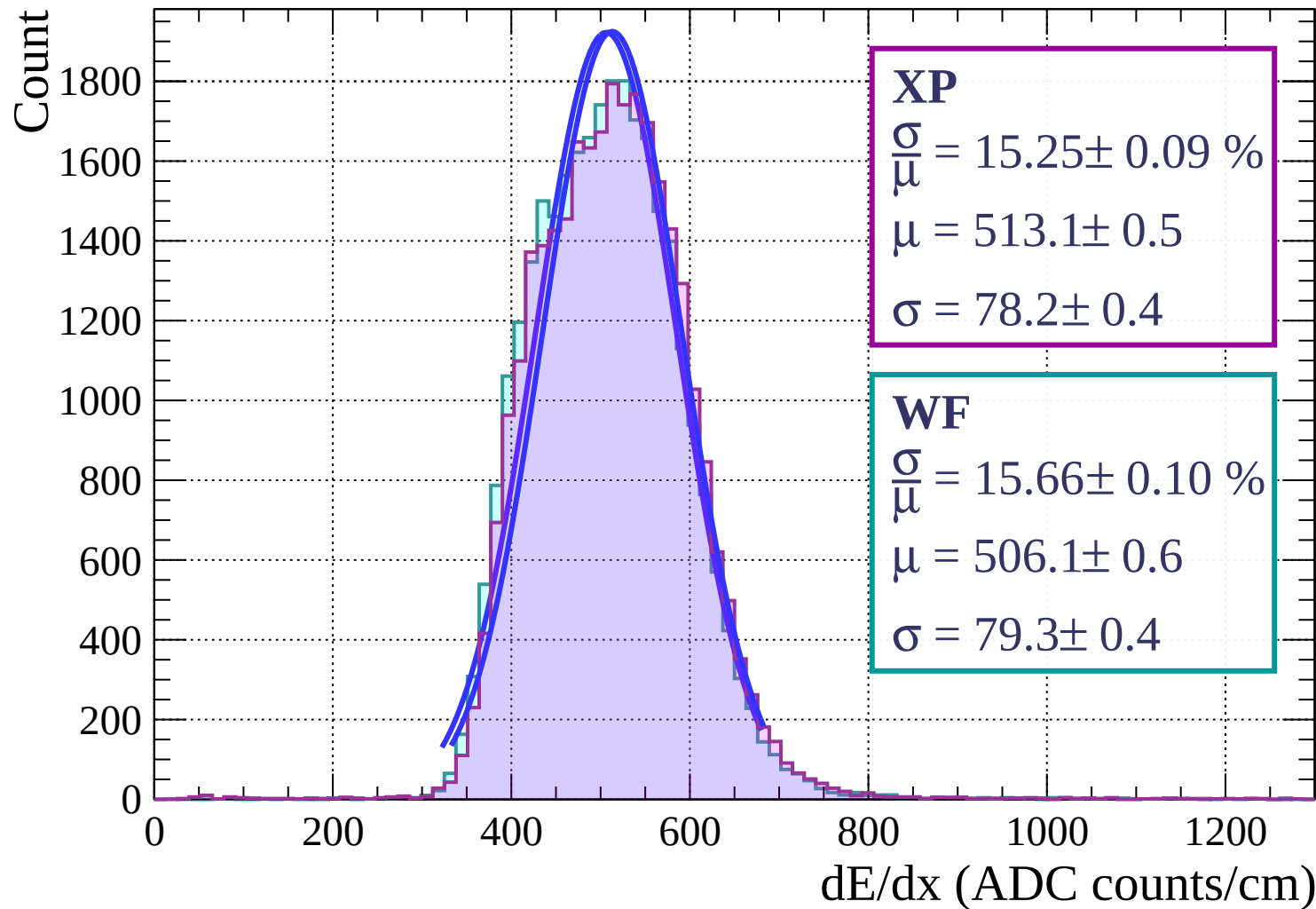
Energy loss | $45 < \phi < 48$



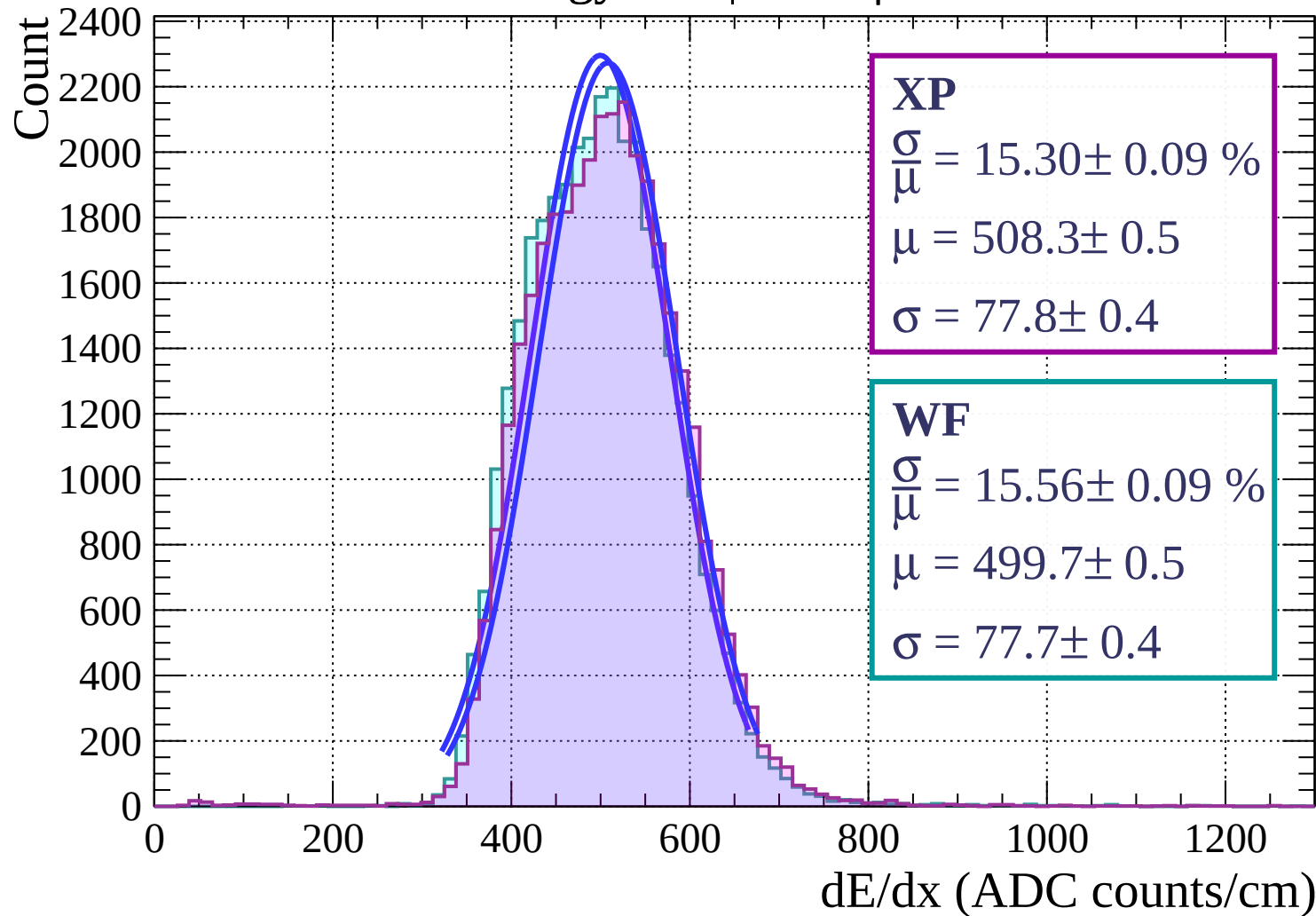
Energy loss | $48 < \phi < 51$



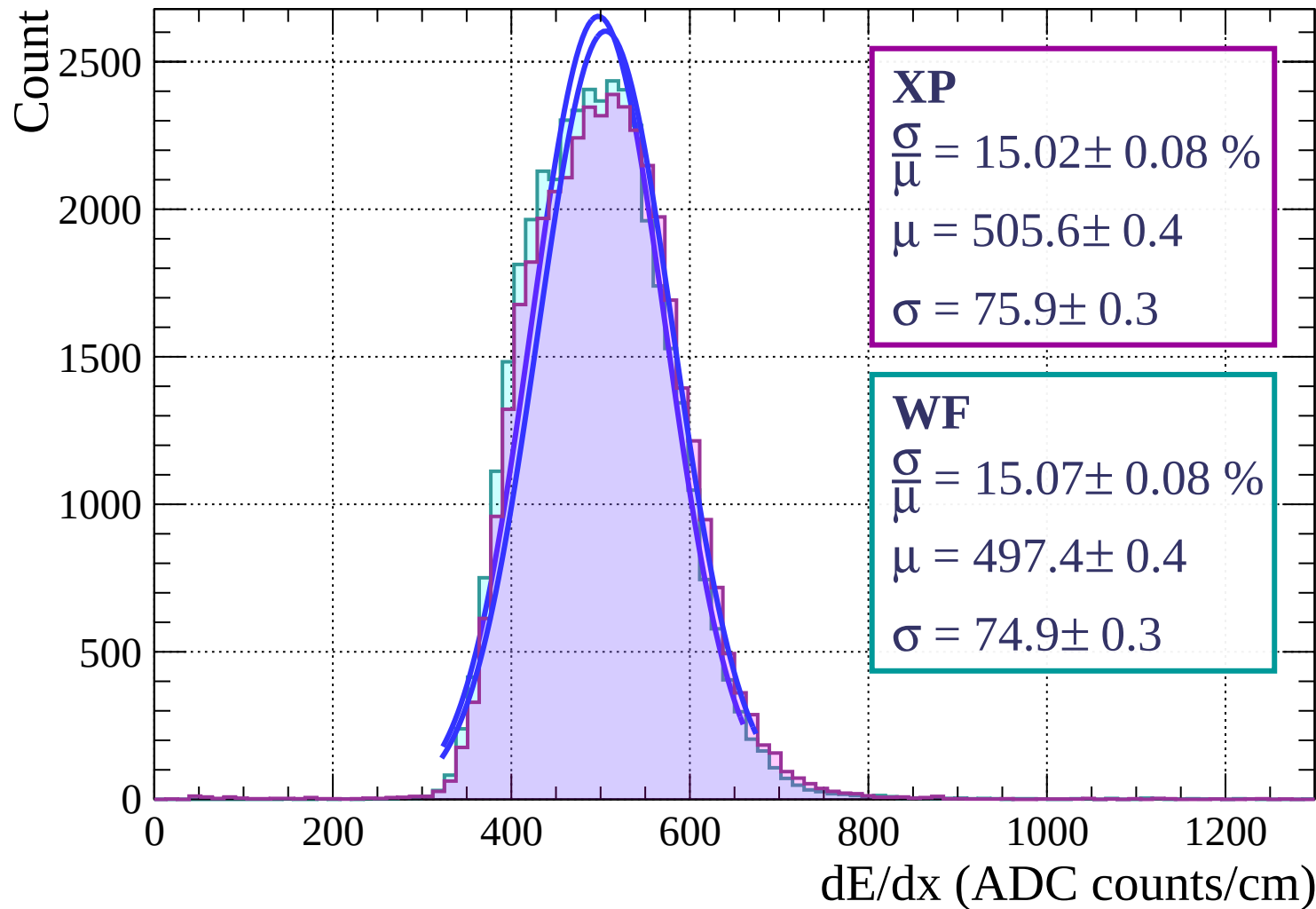
Energy loss | $51 < \phi < 54$



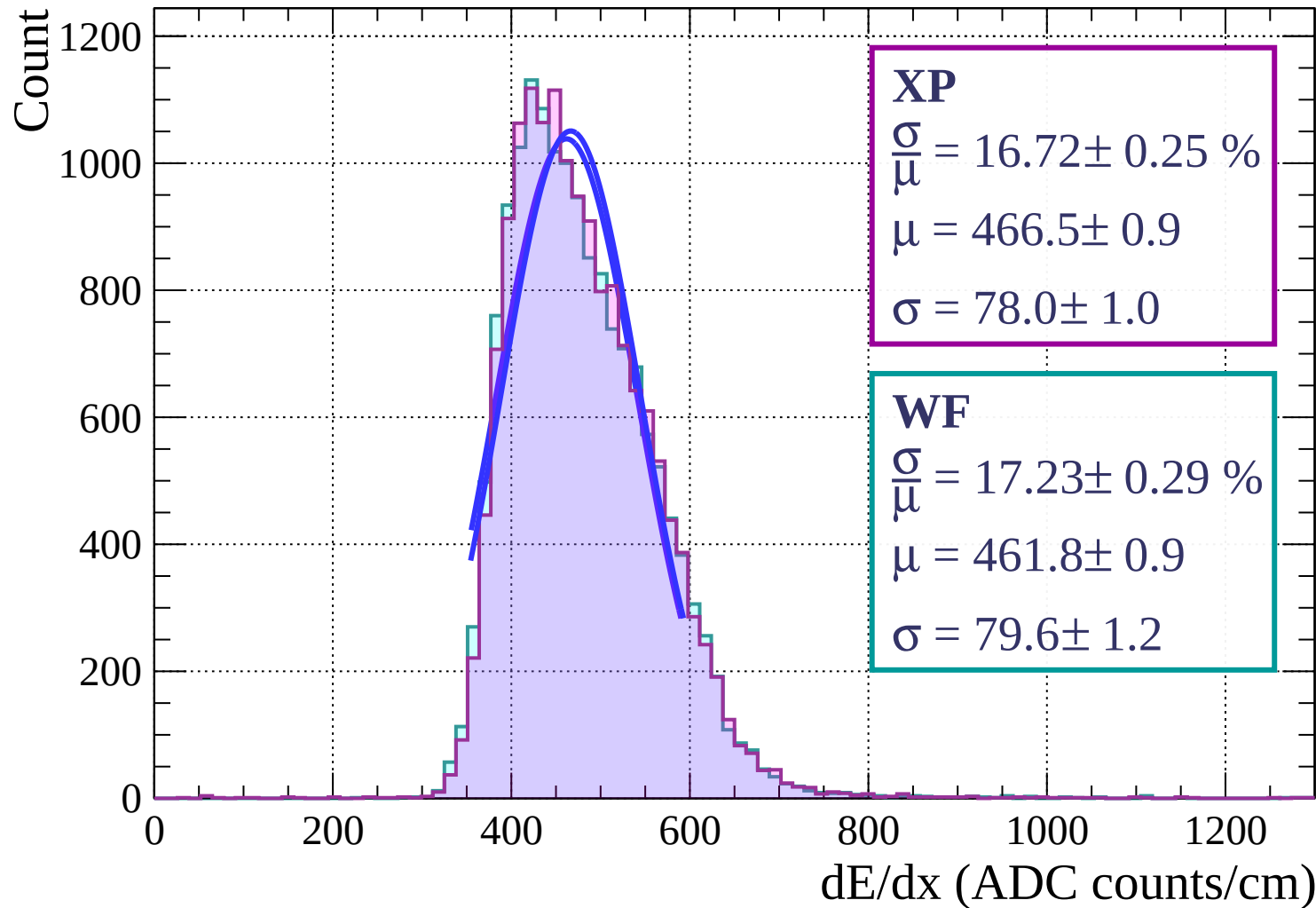
Energy loss | $54 < \phi < 57$



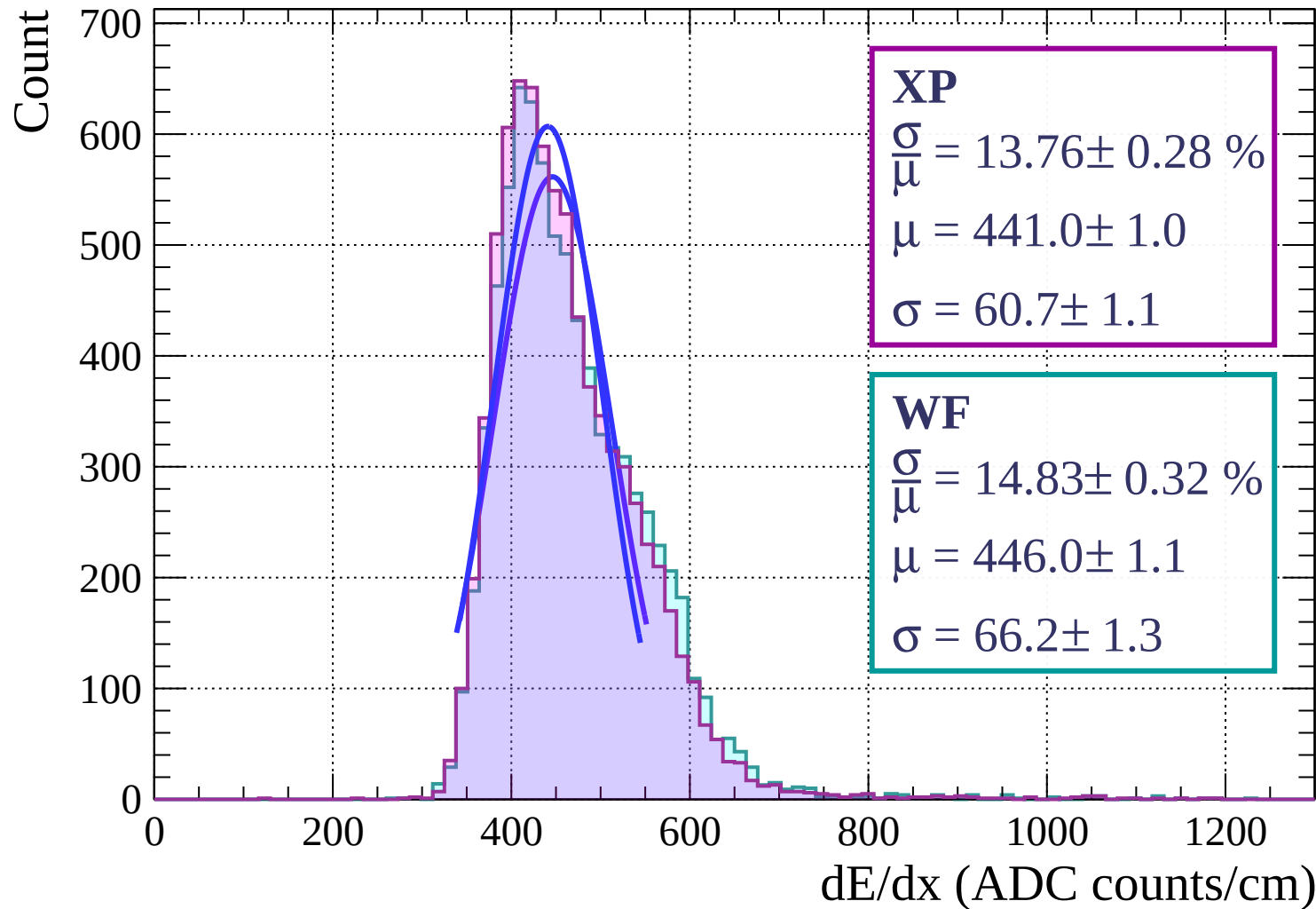
Energy loss | $57 < \phi < 60$



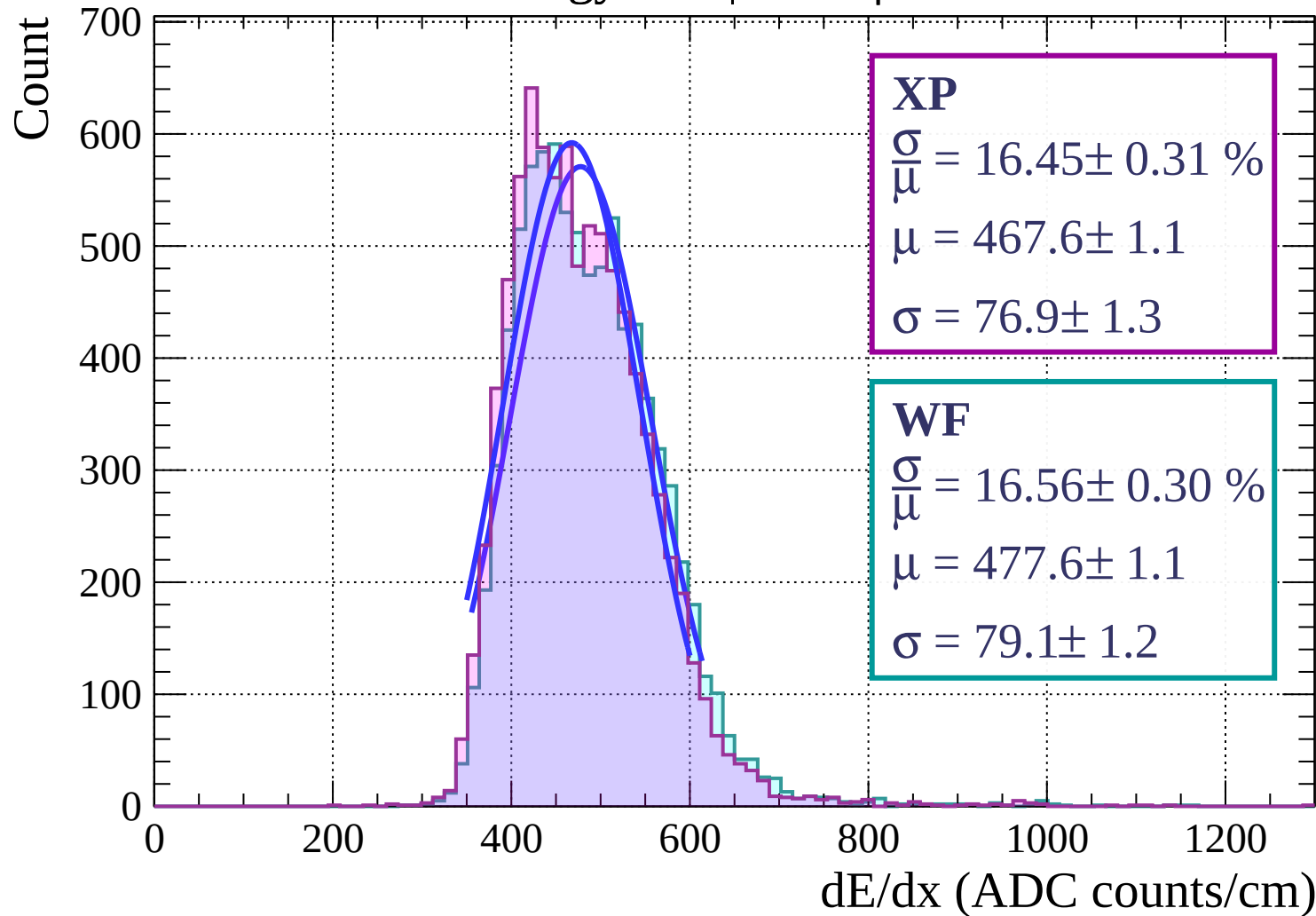
Energy loss | $60 < \phi < 63$



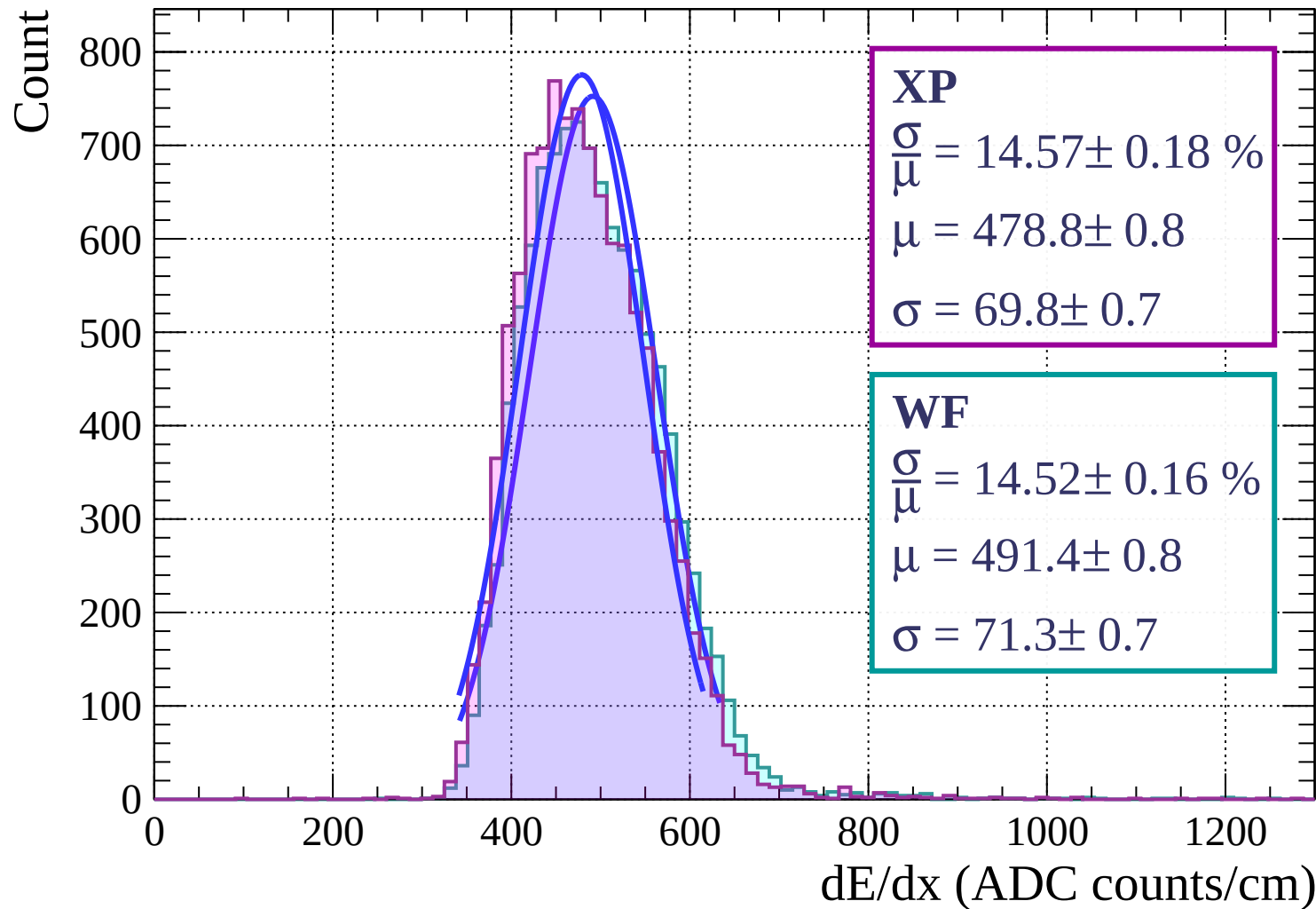
Energy loss | $63 < \phi < 66$



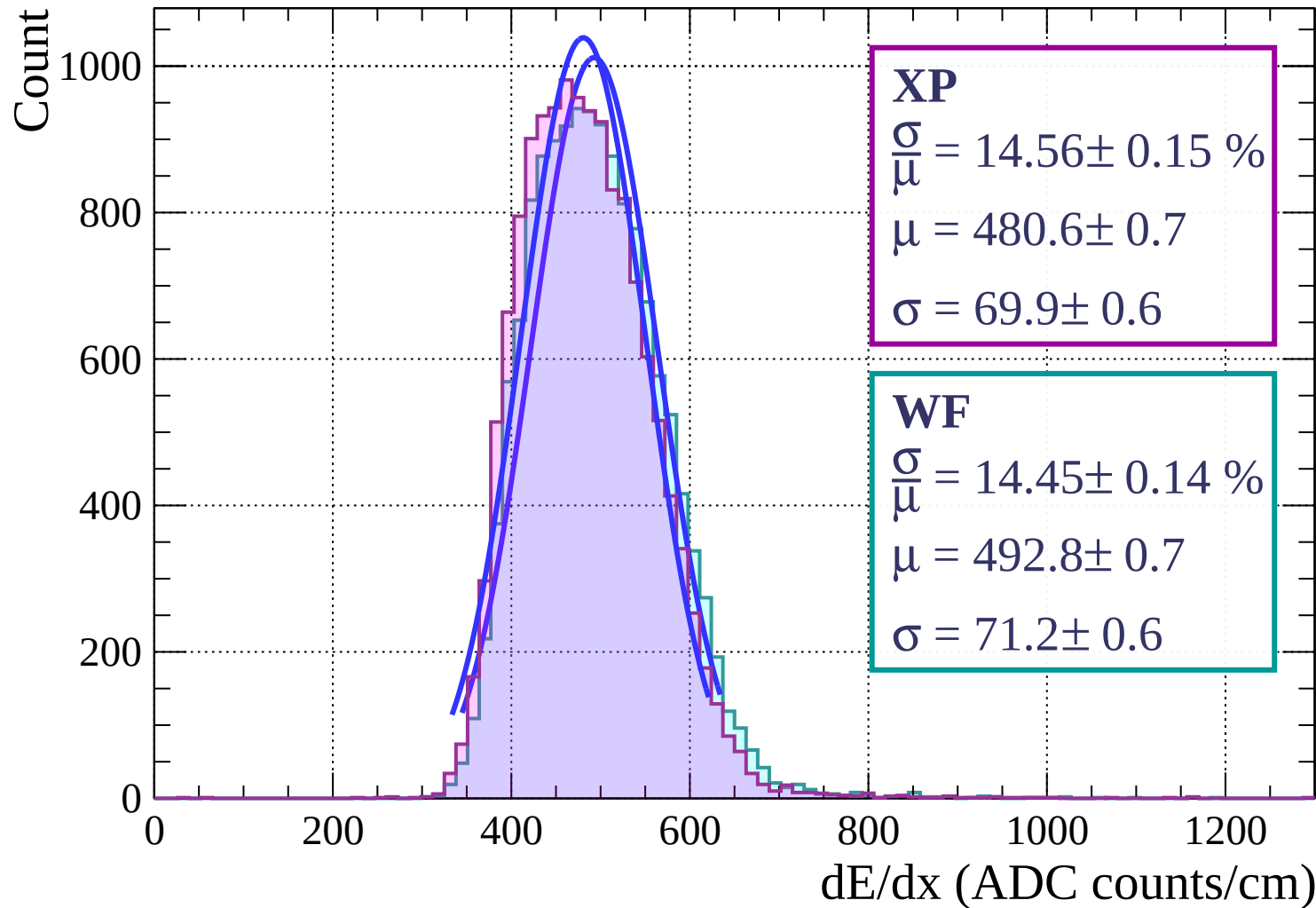
Energy loss | $66 < \phi < 69$



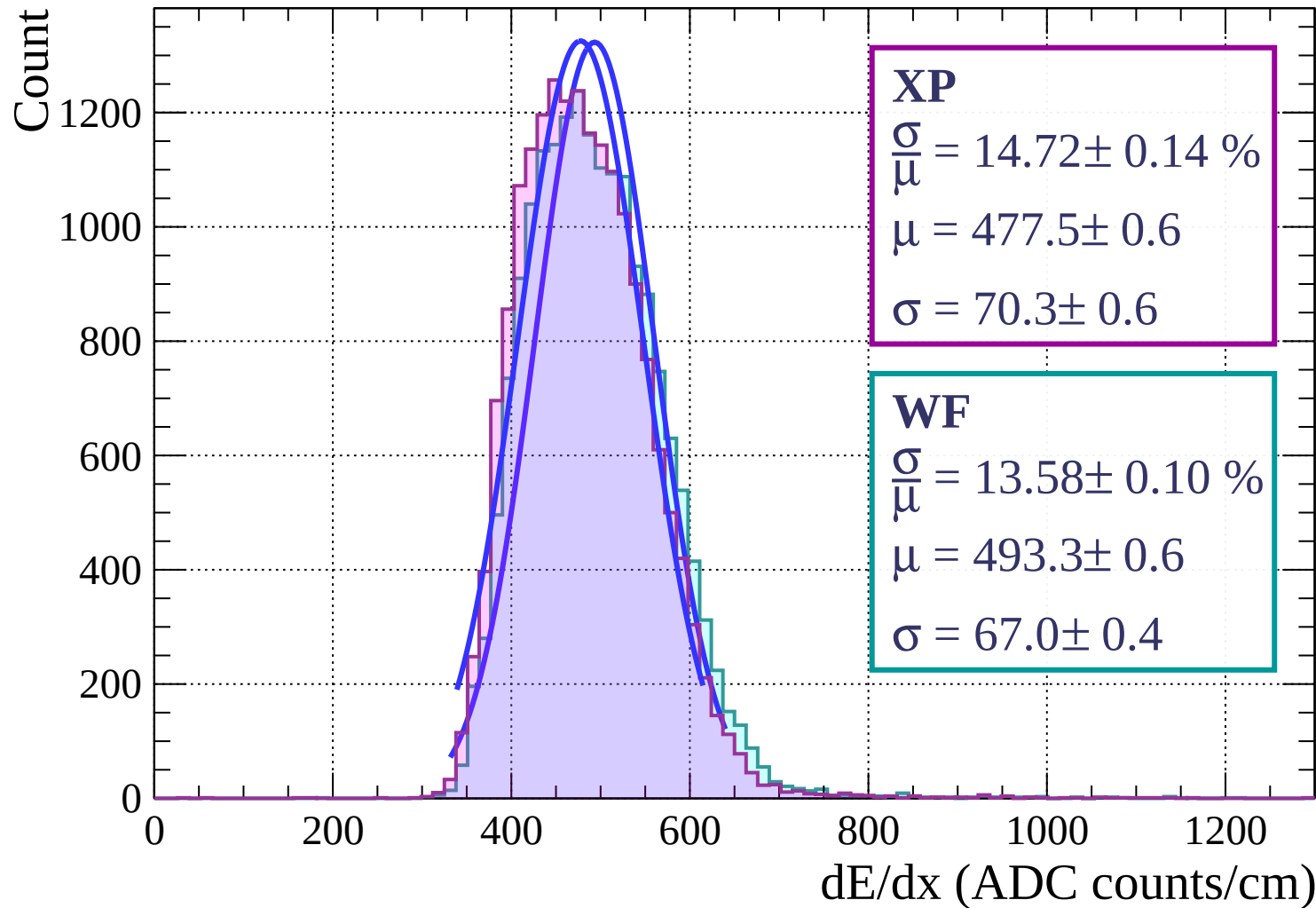
Energy loss | $69 < \phi < 72$



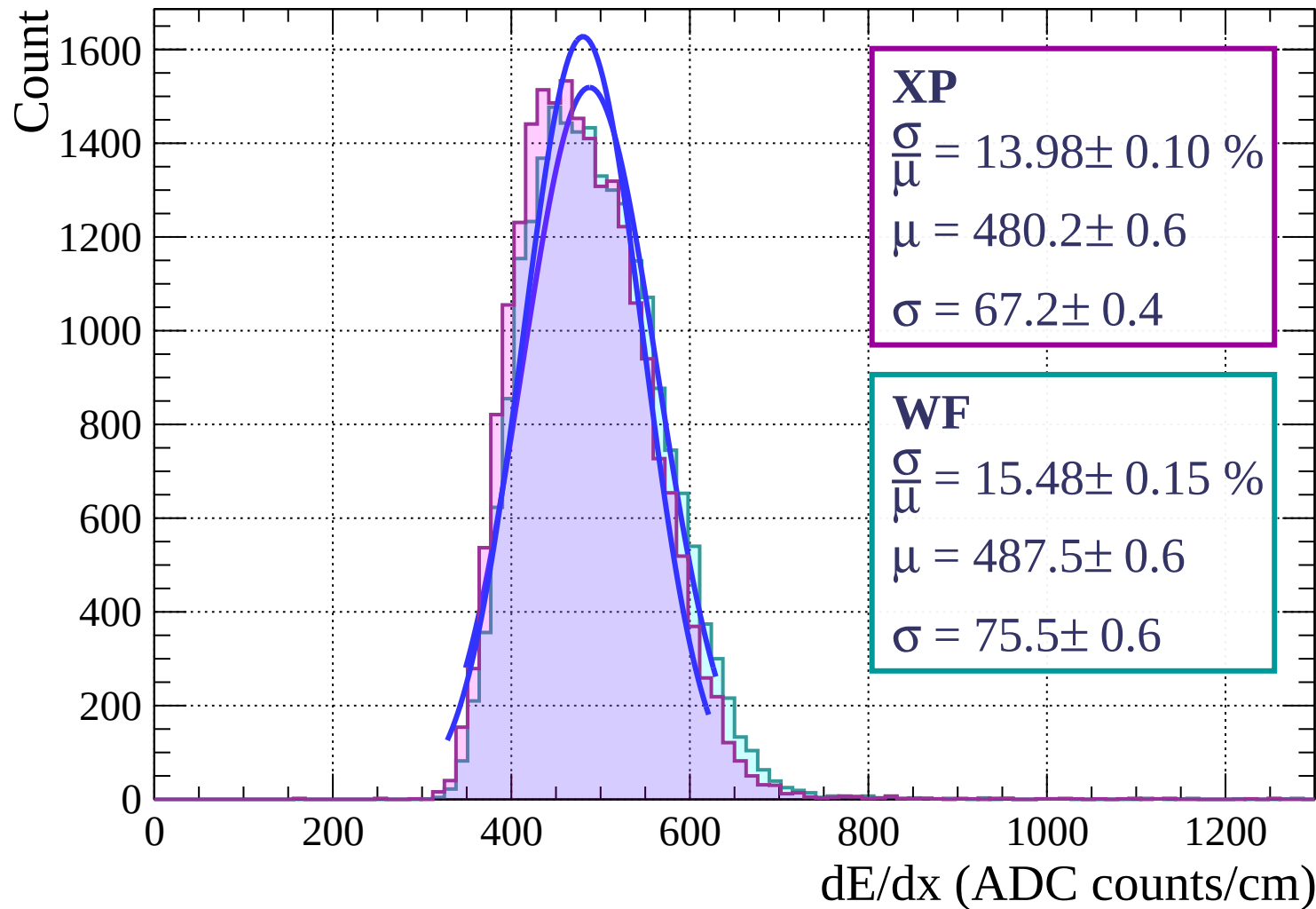
Energy loss | $72 < \phi < 75$



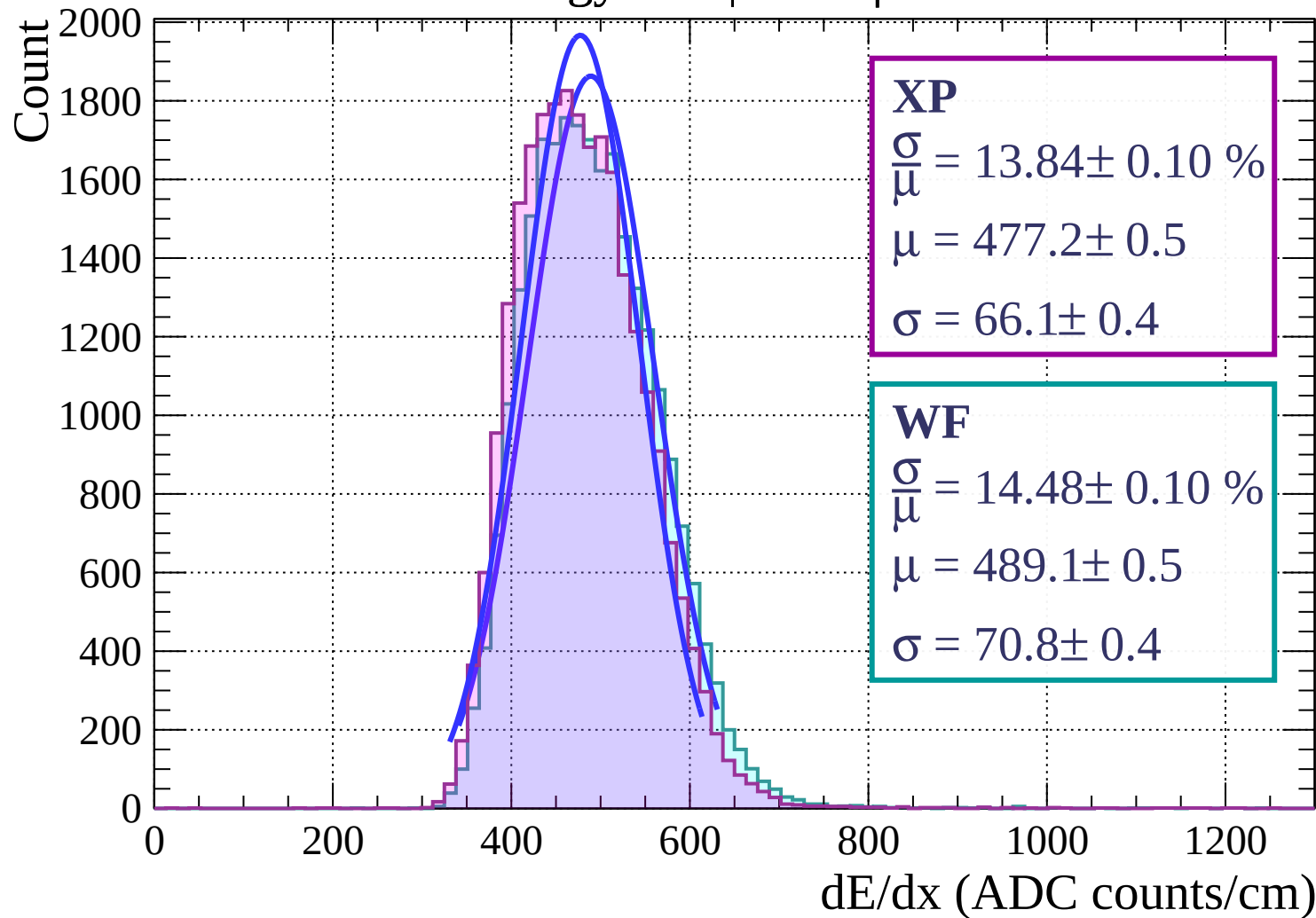
Energy loss | $75 < \phi < 78$



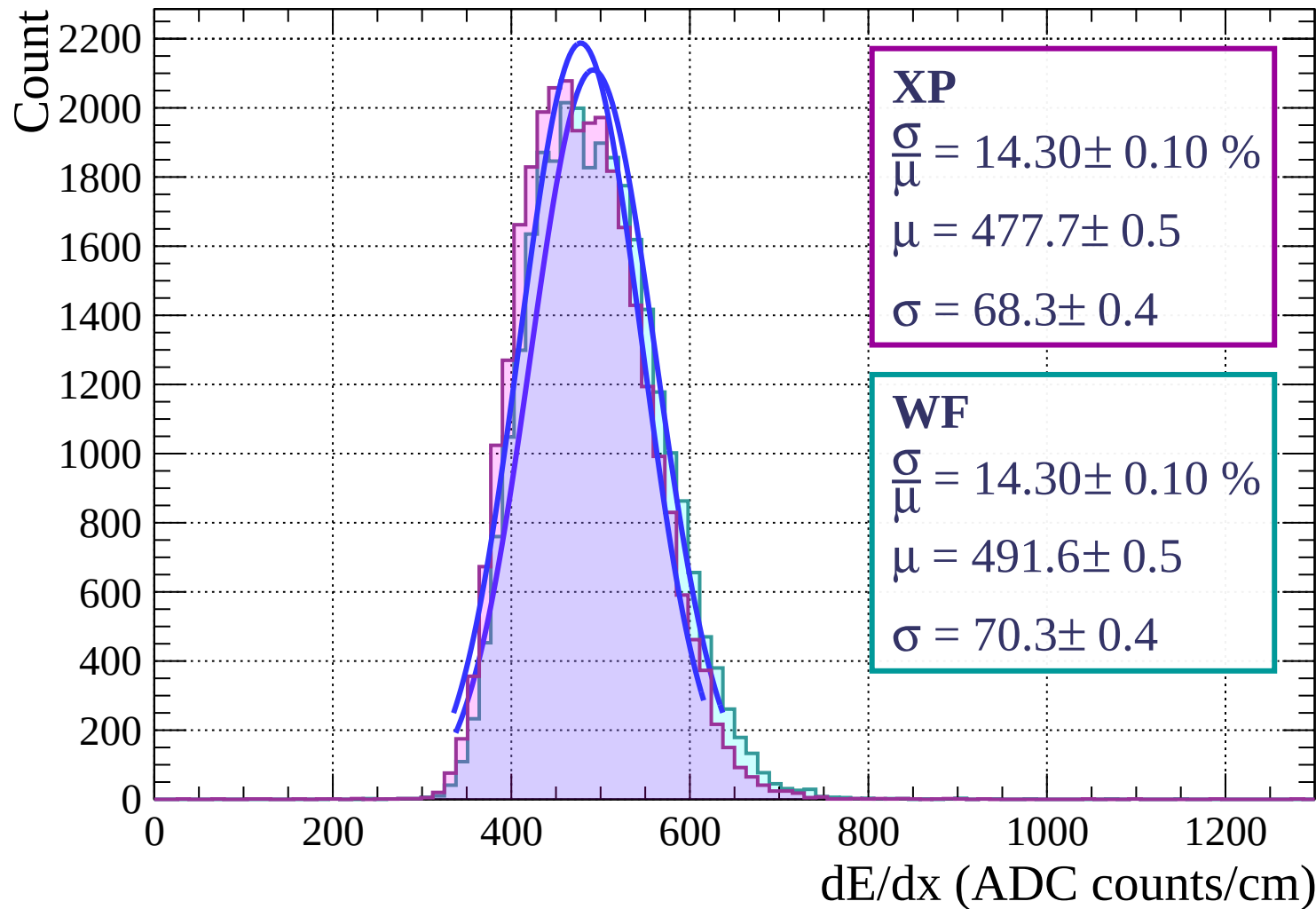
Energy loss | $78 < \phi < 81$



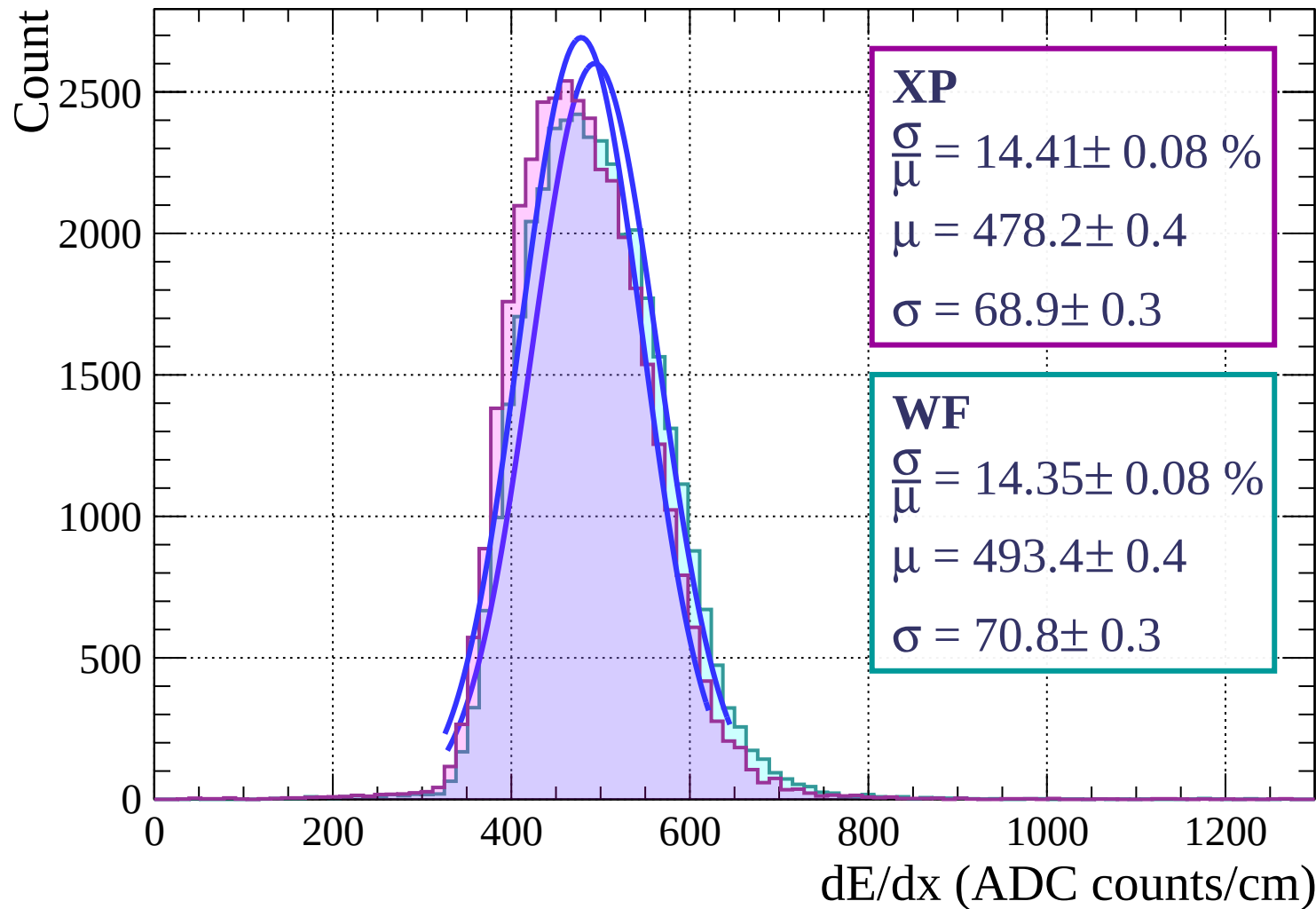
Energy loss | $81 < \phi < 84$



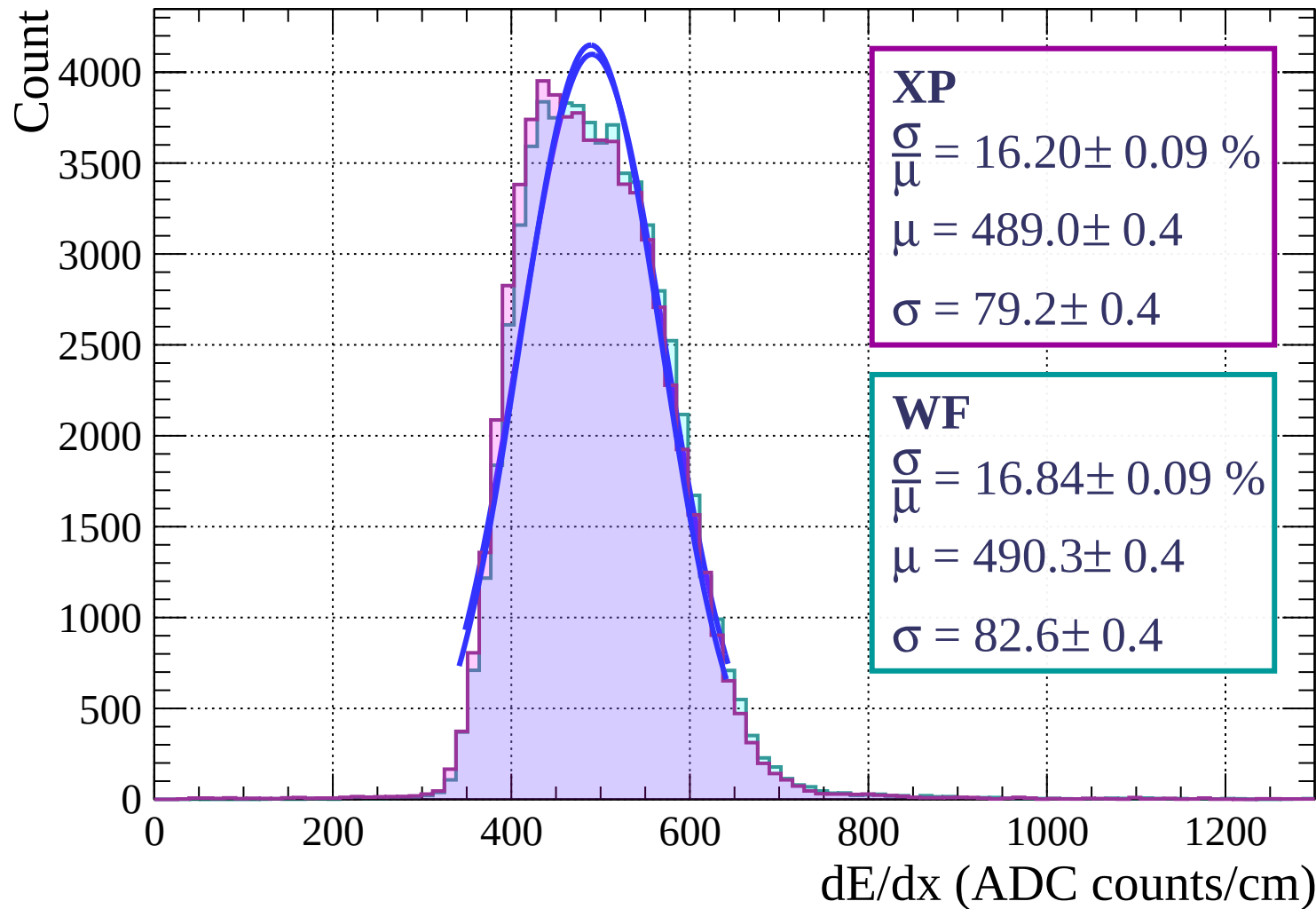
Energy loss | $84 < \phi < 87$



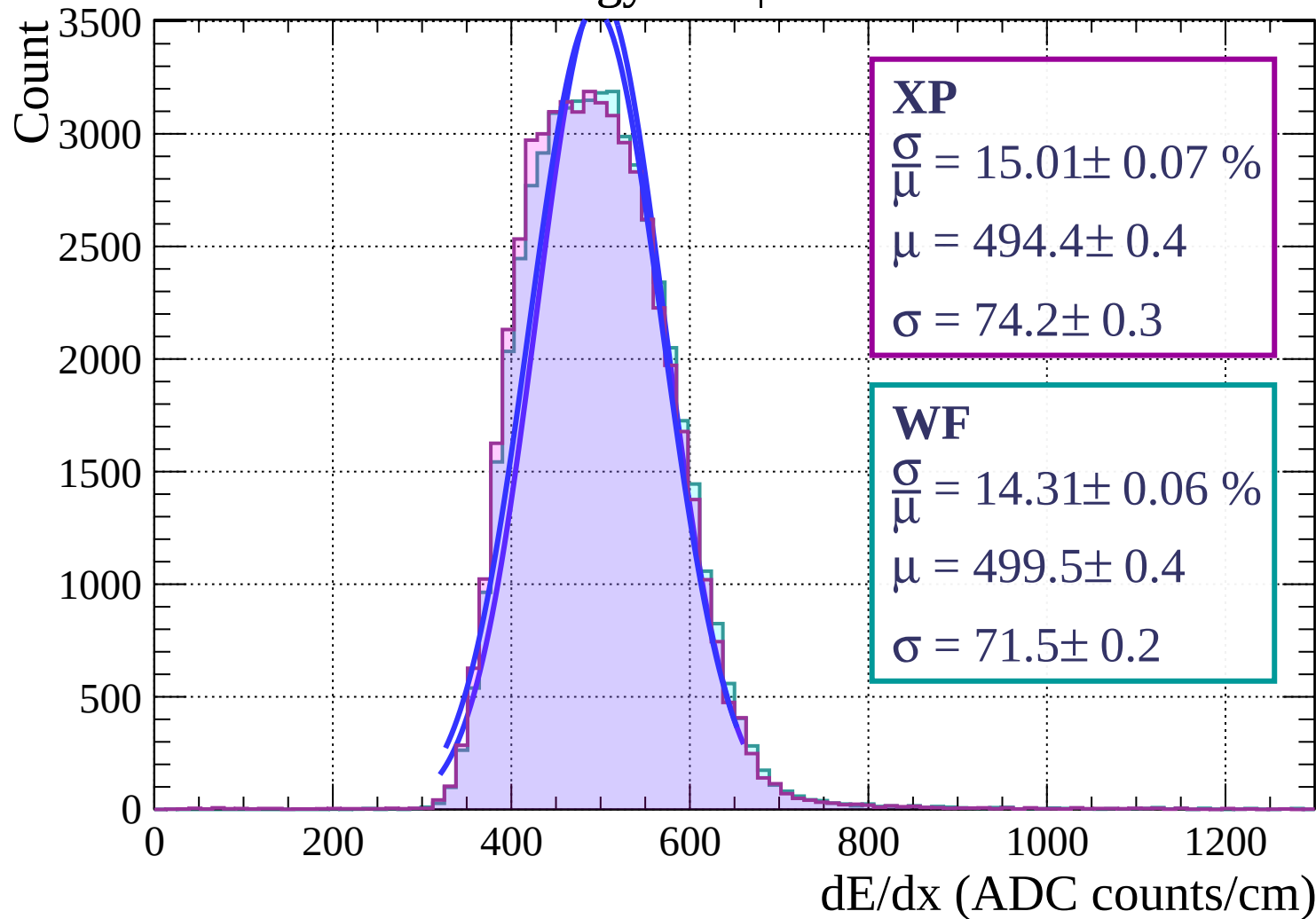
Energy loss | $87 < \phi < 90$



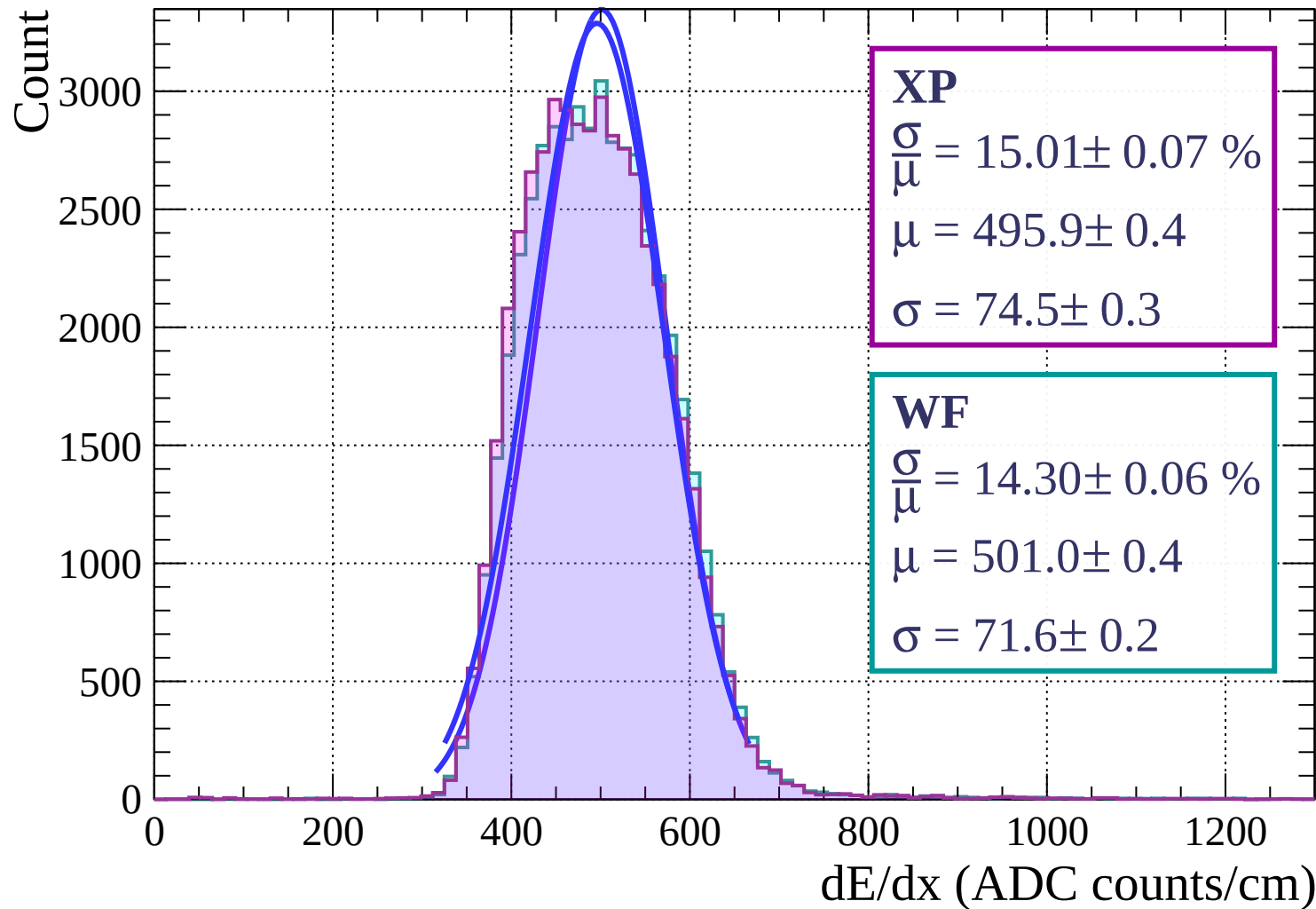
Energy loss | $0 < \theta < 3$



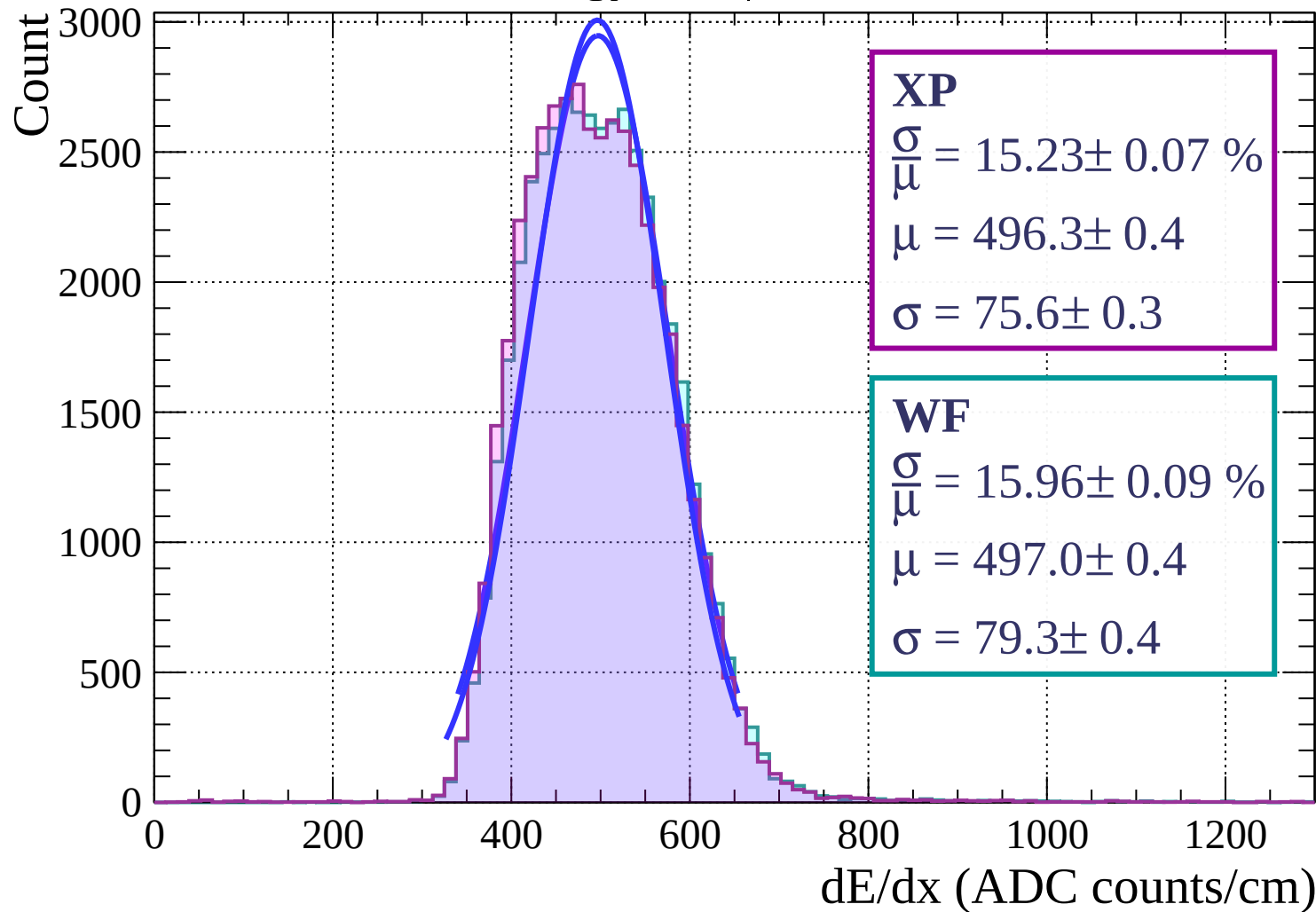
Energy loss | $3 < \theta < 6$



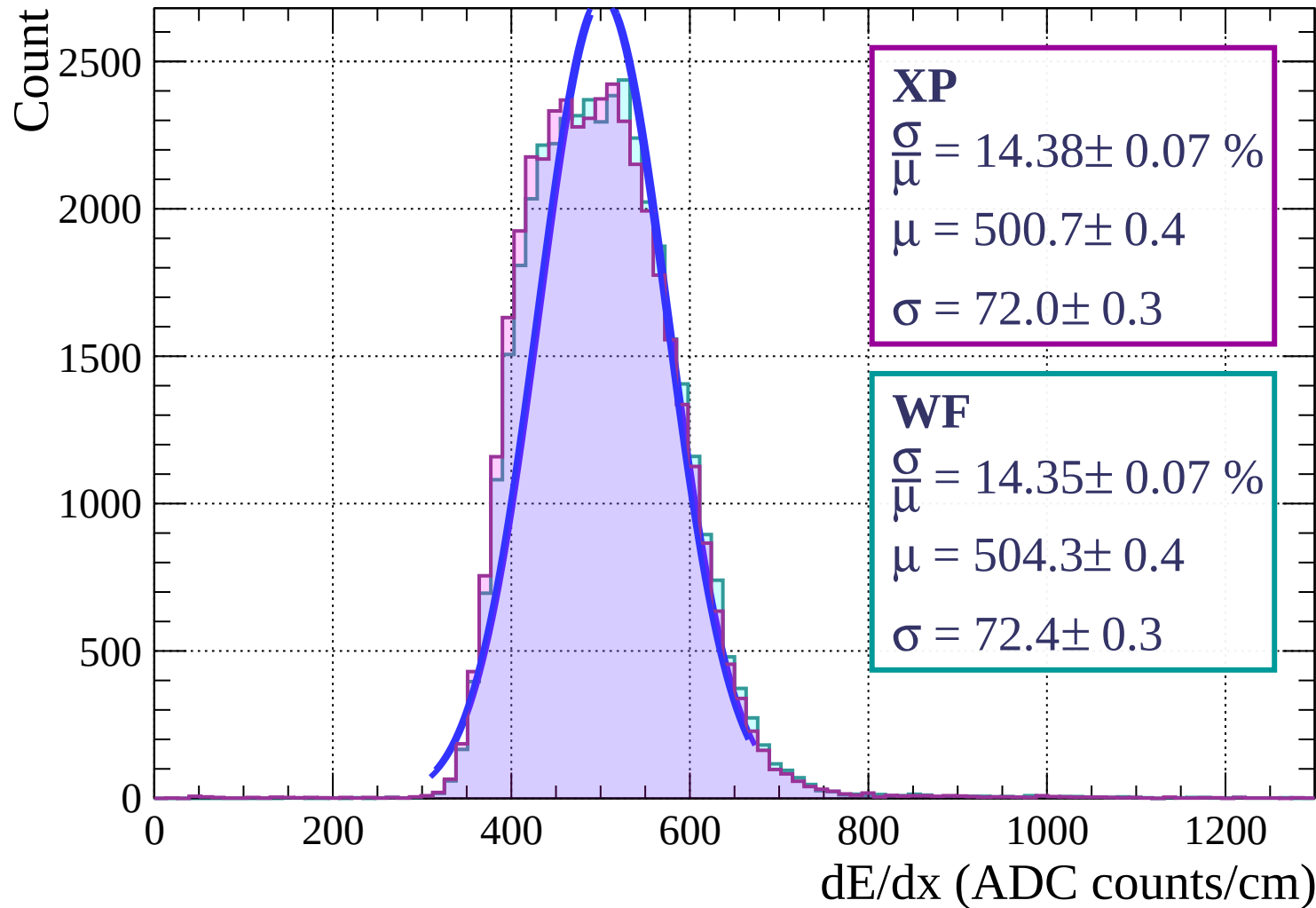
Energy loss | $6 < \theta < 9$



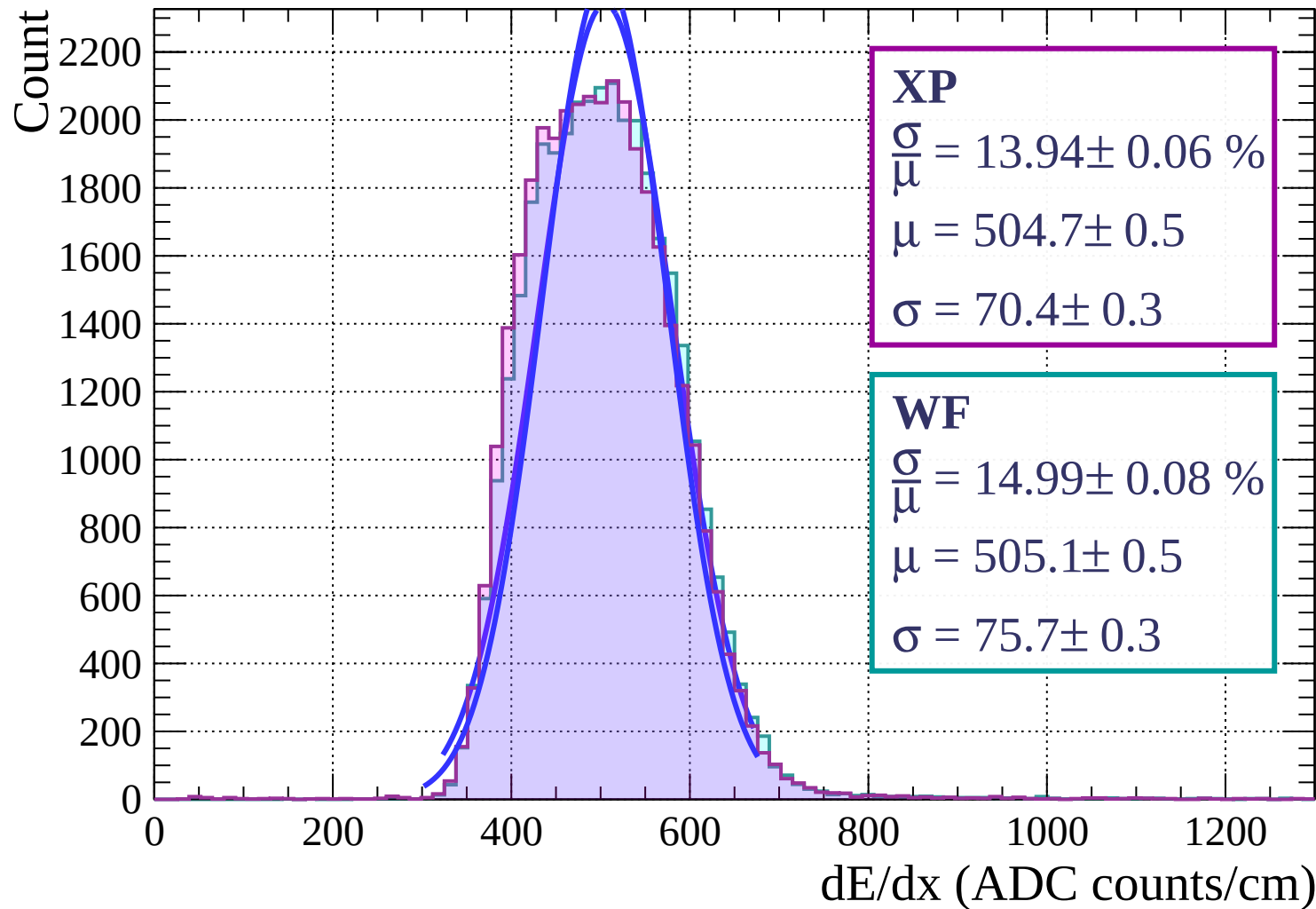
Energy loss | $9 < \theta < 12$



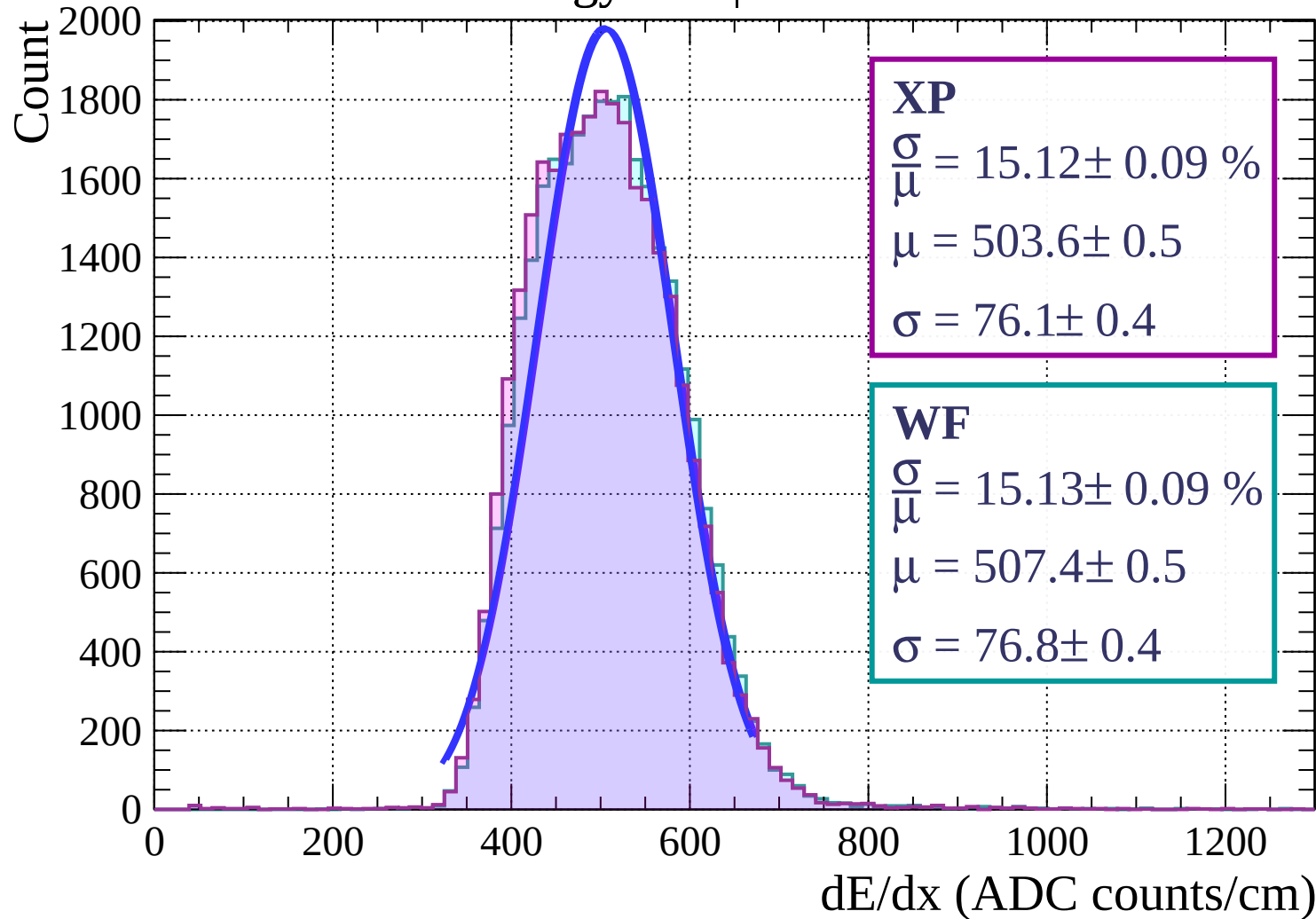
Energy loss | $12 < \theta < 15$



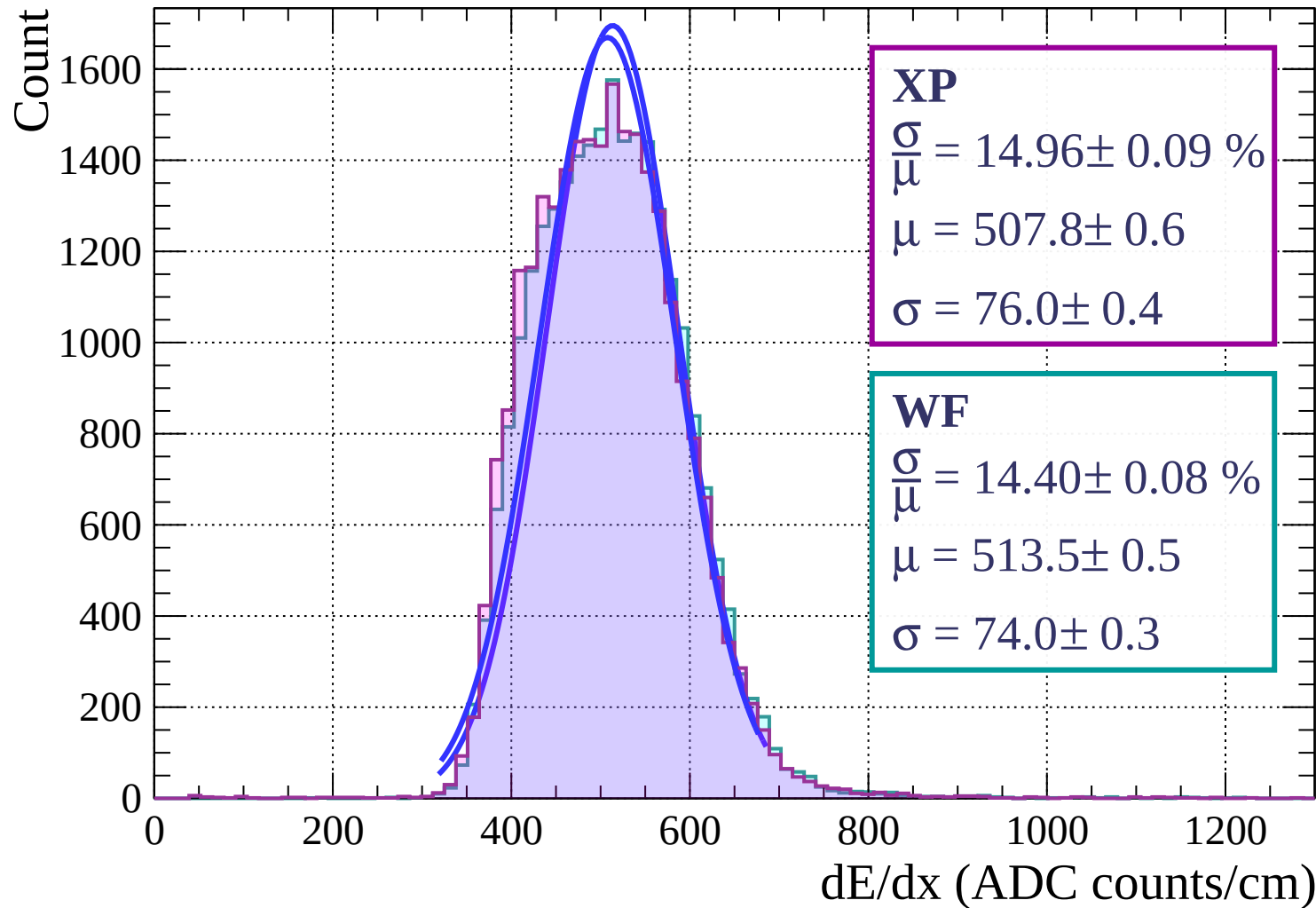
Energy loss | $15 < \theta < 18$



Energy loss | $18 < \theta < 21$

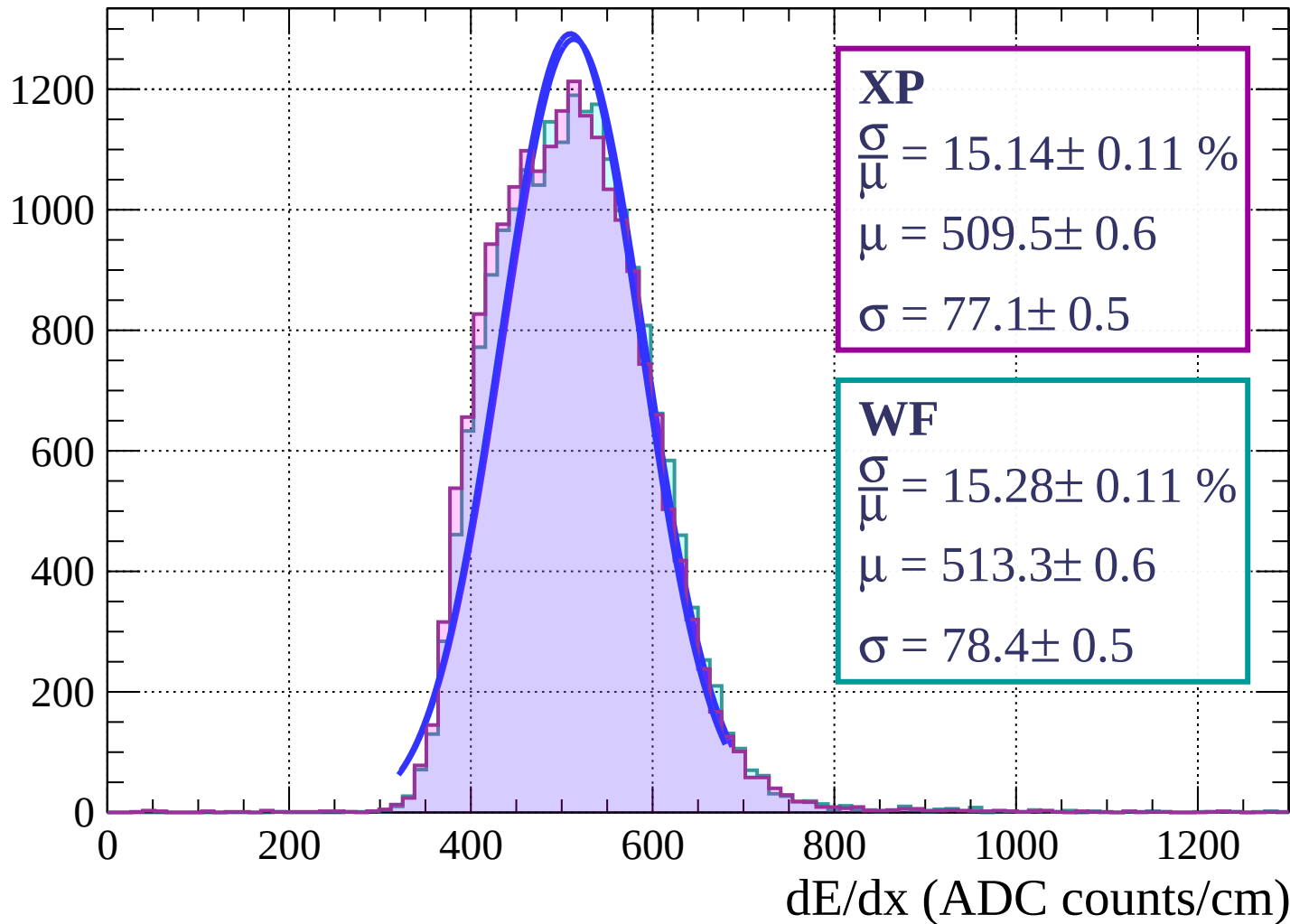


Energy loss | $21 < \theta < 24$

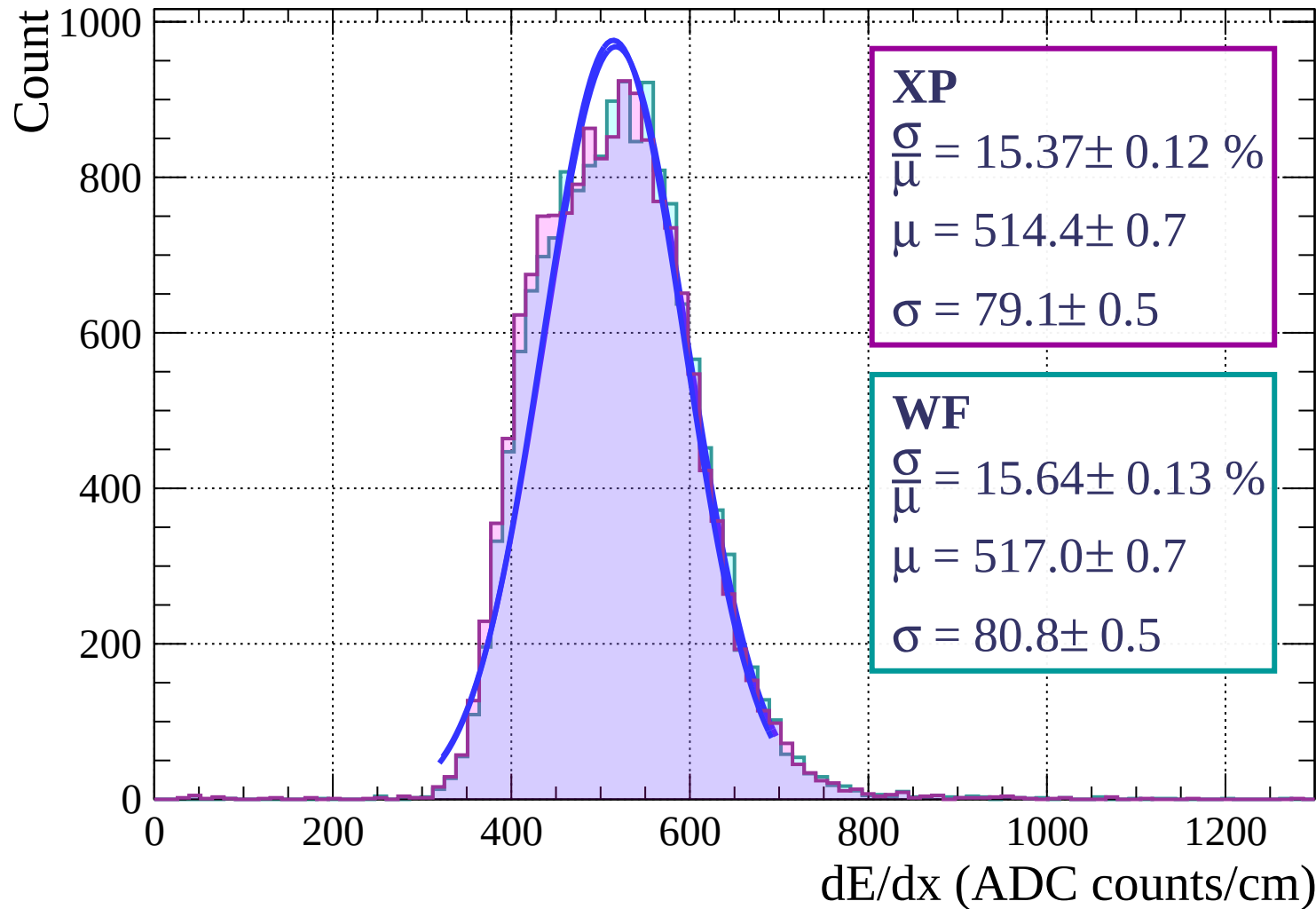


Energy loss | $24 < \theta < 27$

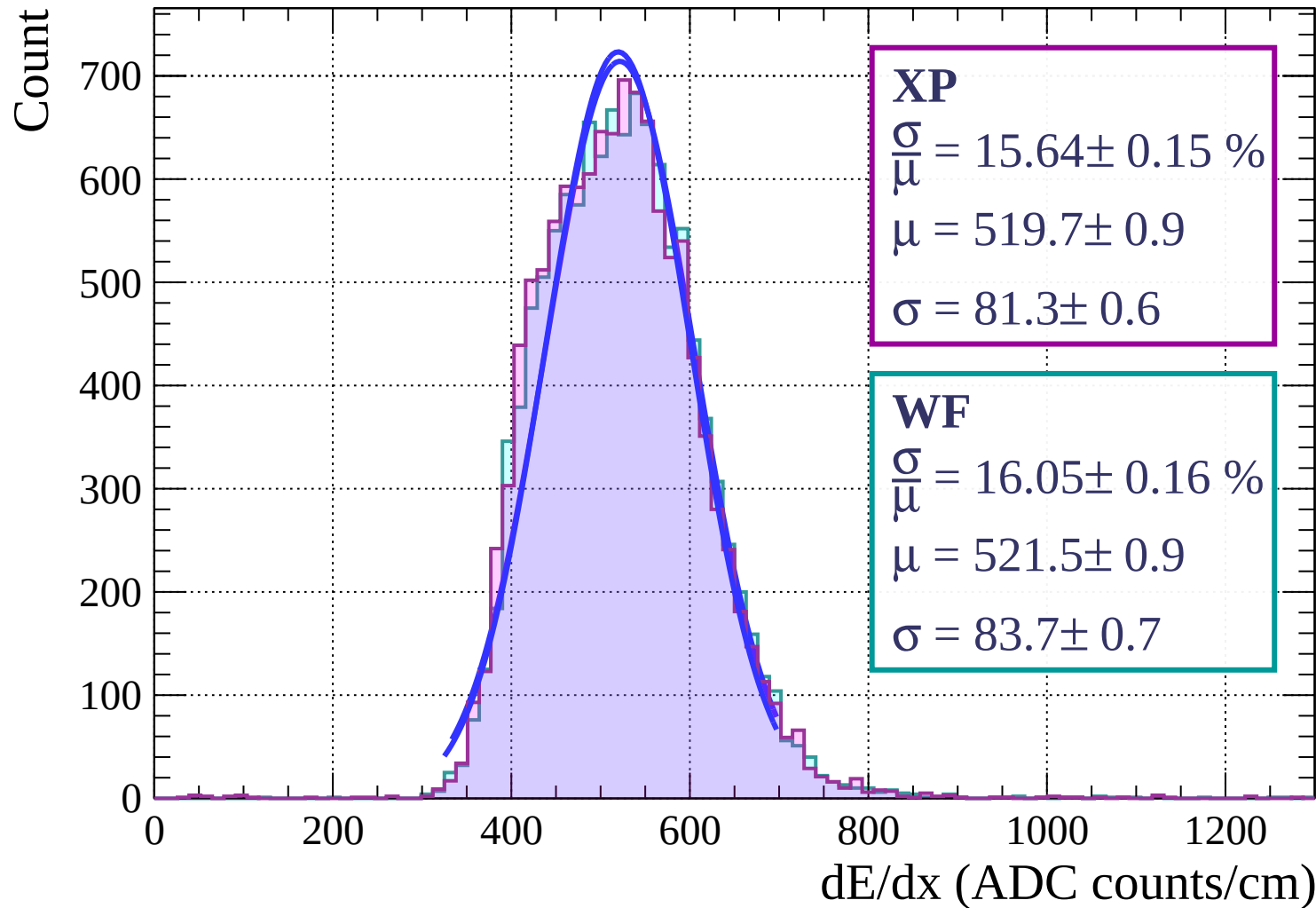
Count



Energy loss | $27 < \theta < 30$

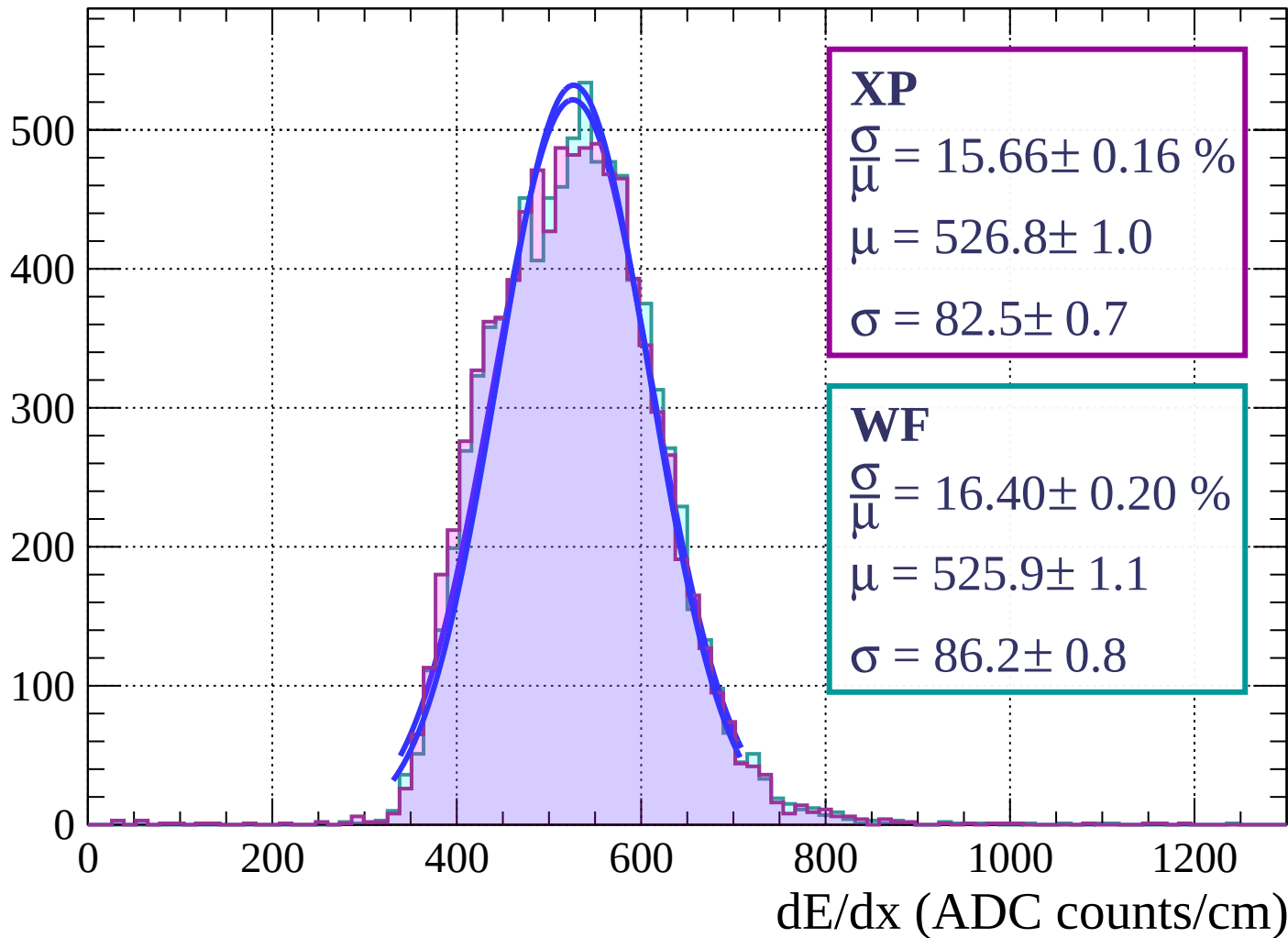


Energy loss | $30 < \theta < 33$



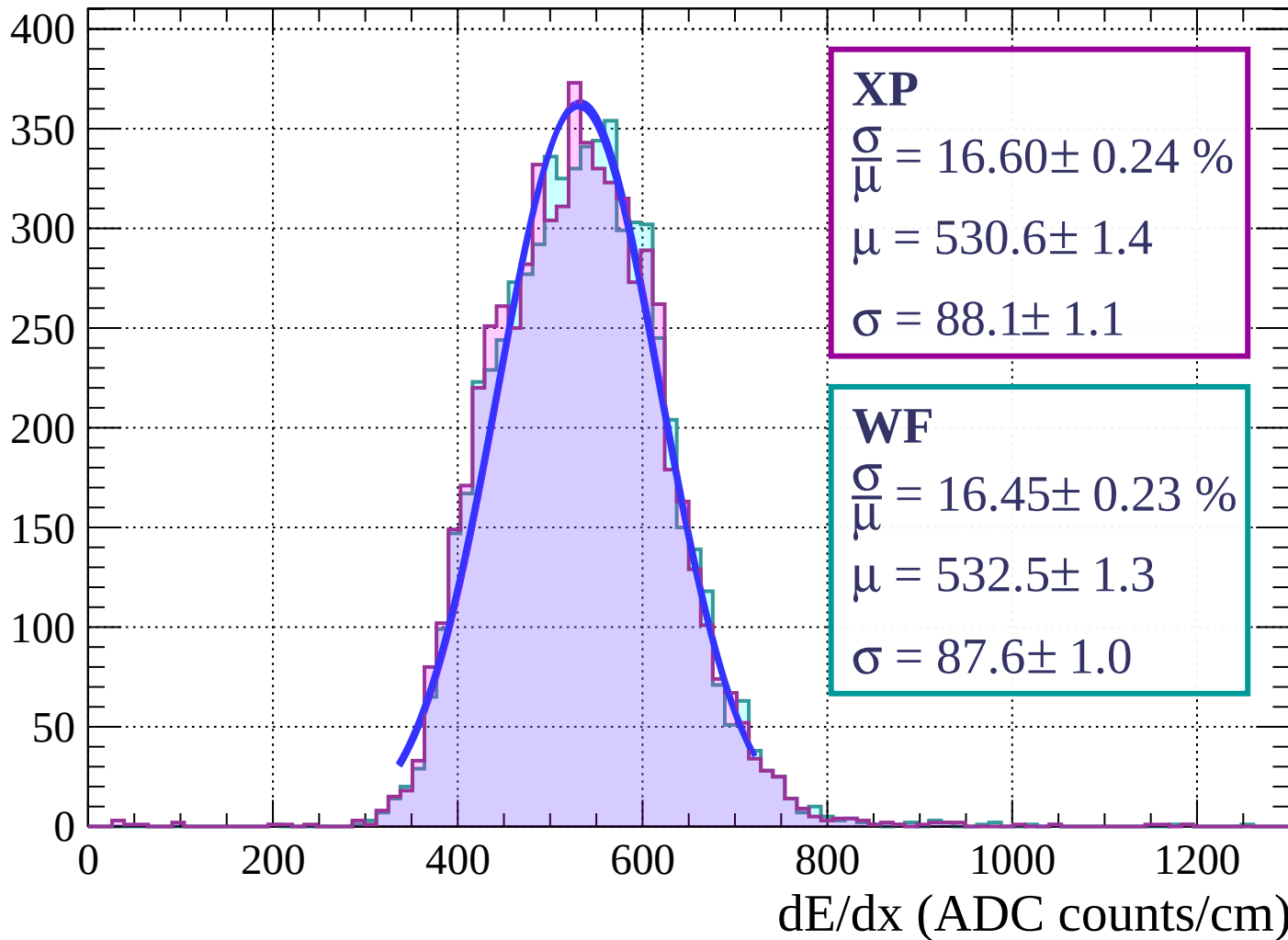
Energy loss | $33 < \theta < 36$

Count

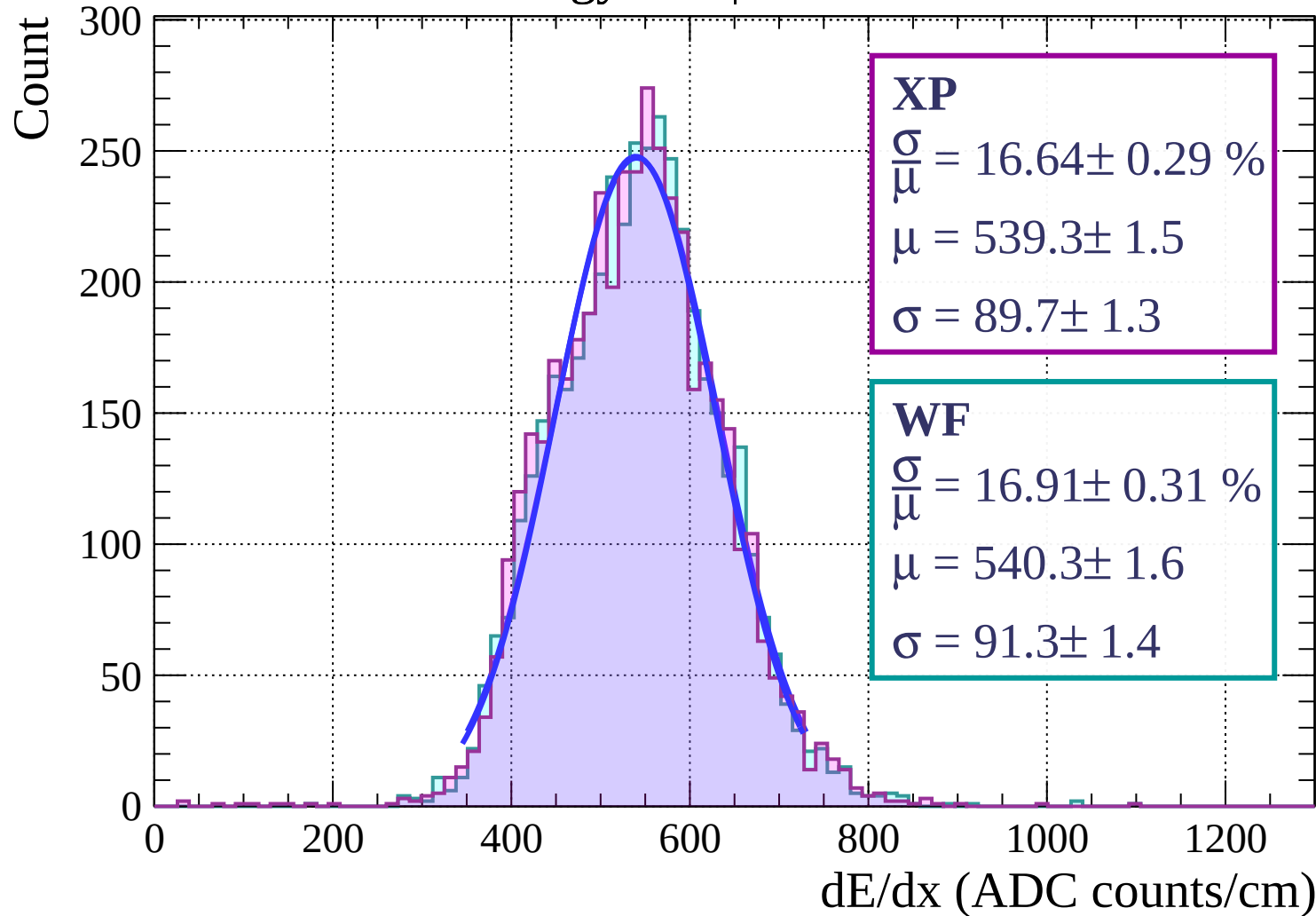


Energy loss | $36 < \theta < 39$

Count

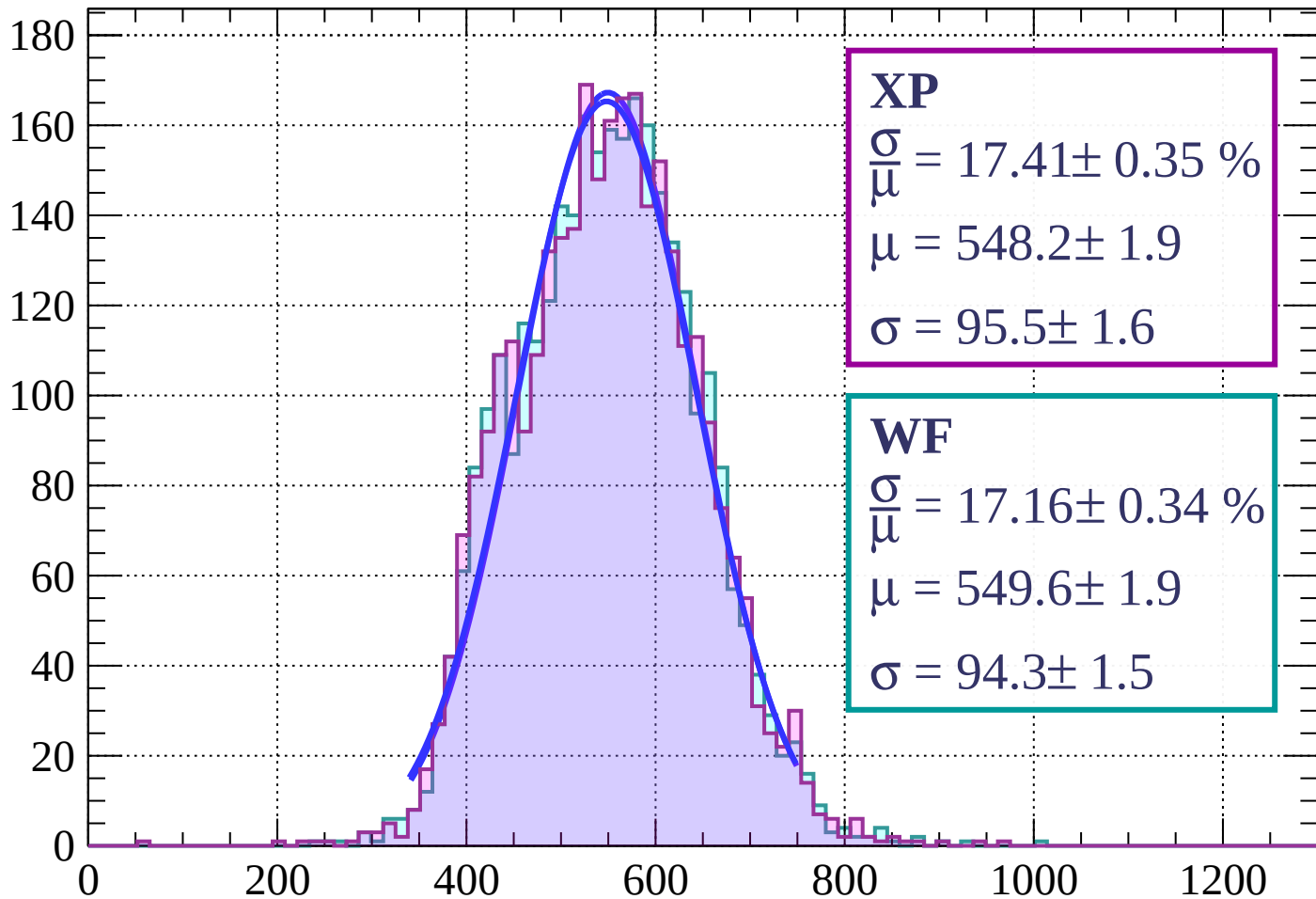


Energy loss | $39 < \theta < 42$



Energy loss | $42 < \theta < 45$

Count



XP

$$\frac{\sigma}{\mu} = 17.41 \pm 0.35 \%$$

$$\mu = 548.2 \pm 1.9$$

$$\sigma = 95.5 \pm 1.6$$

WF

$$\frac{\sigma}{\mu} = 17.16 \pm 0.34 \%$$

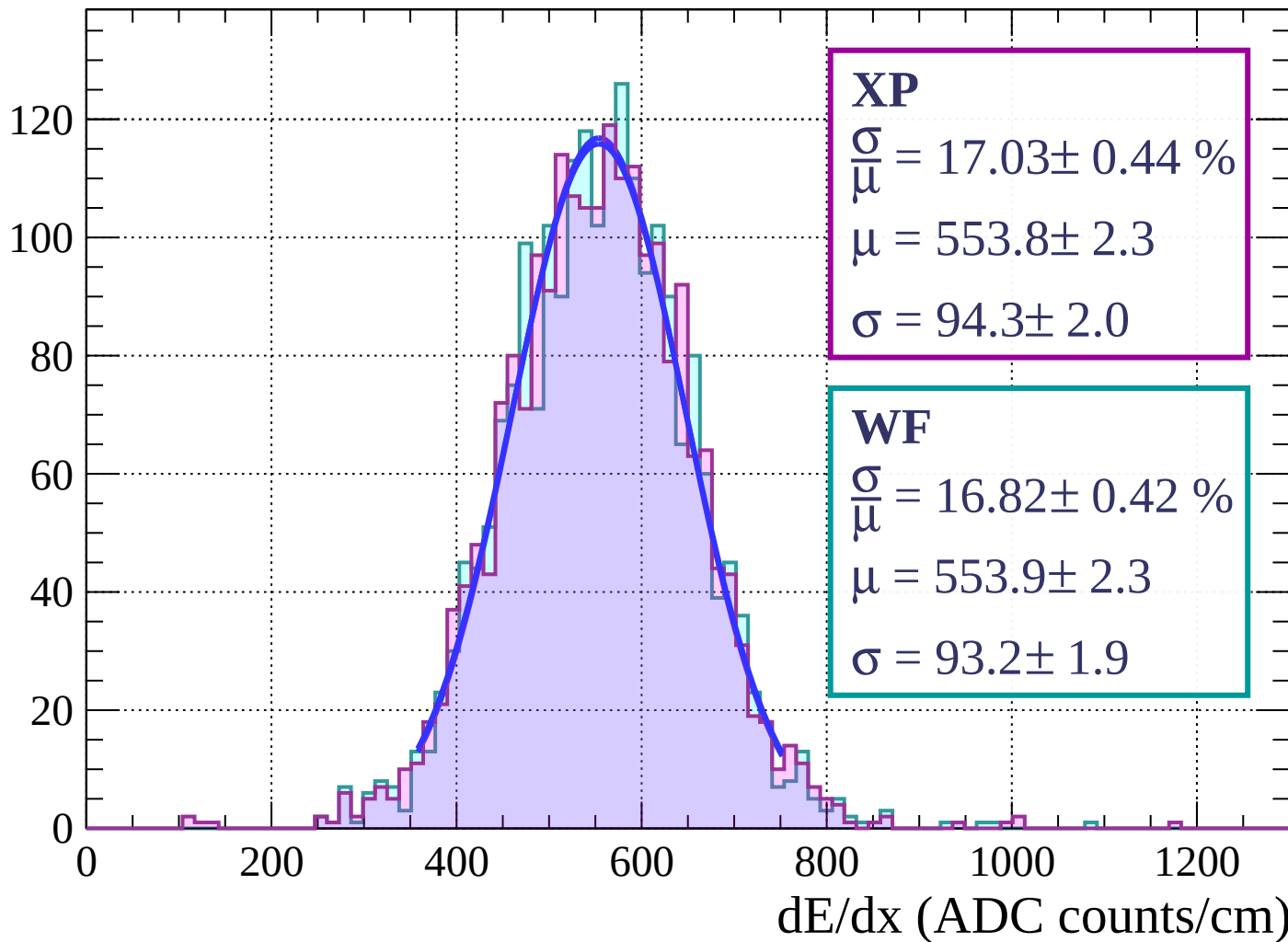
$$\mu = 549.6 \pm 1.9$$

$$\sigma = 94.3 \pm 1.5$$

dE/dx (ADC counts/cm)

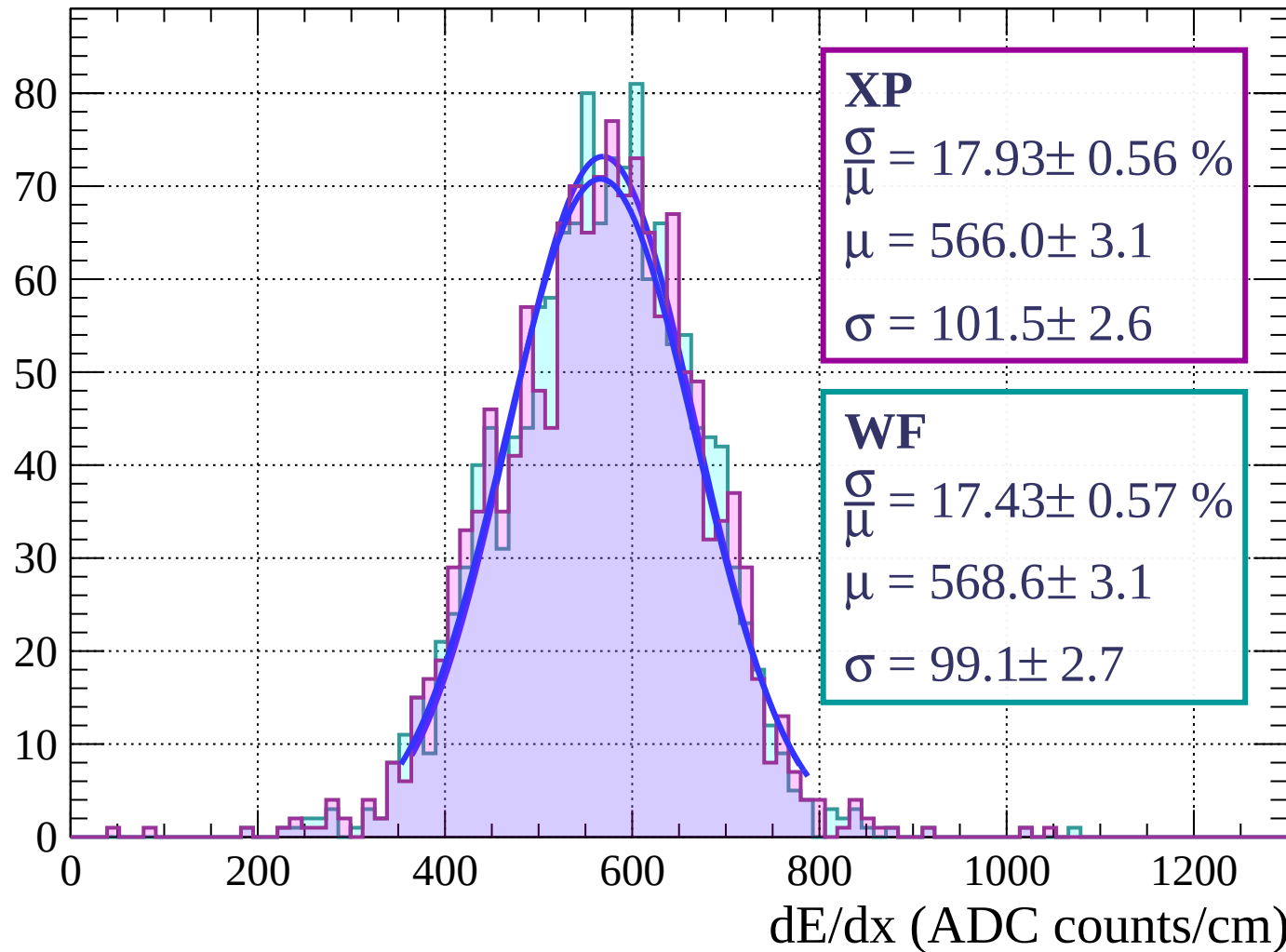
Energy loss | $45 < \theta < 48$

Count



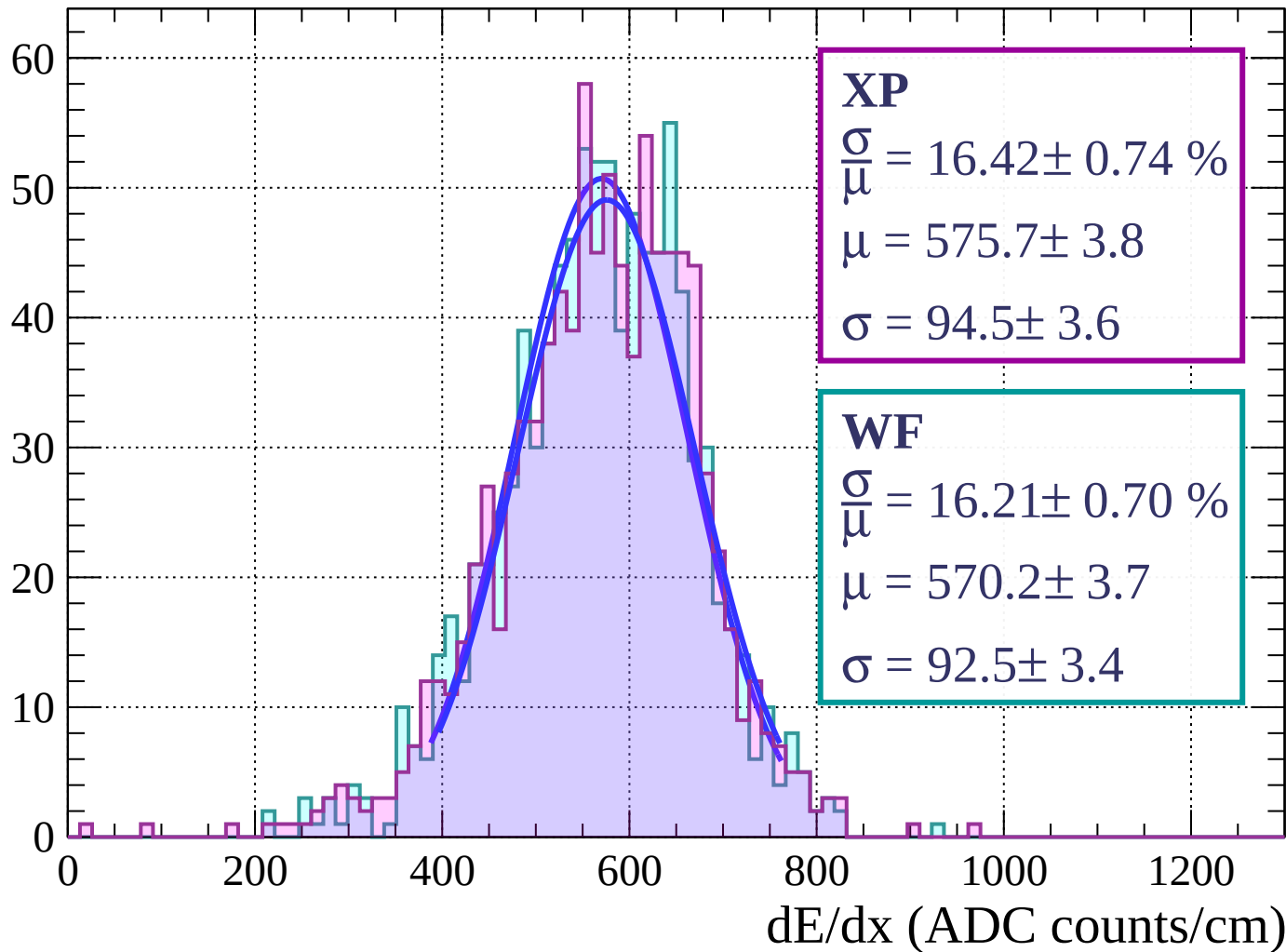
Energy loss | $48 < \theta < 51$

Count



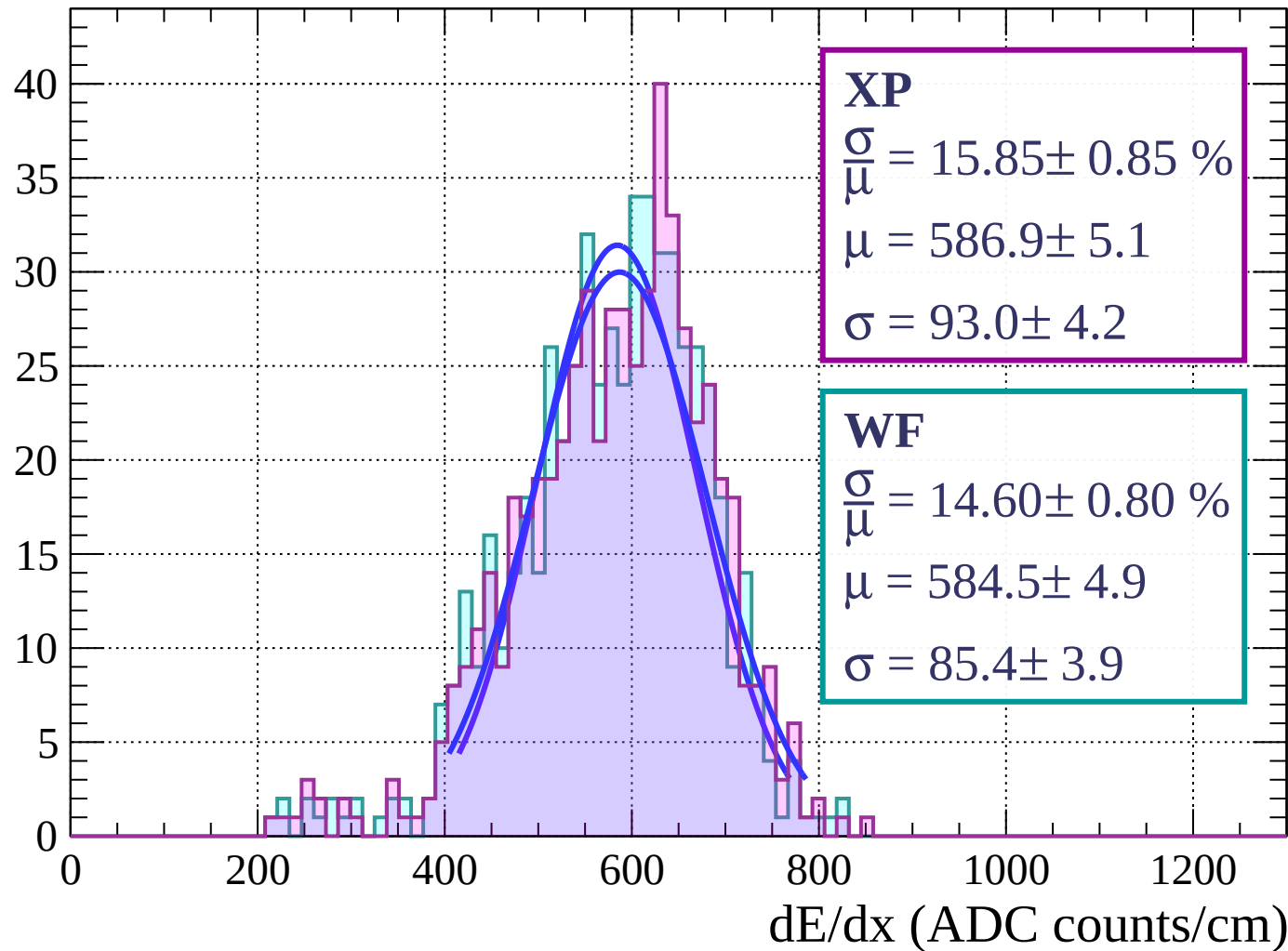
Energy loss | $51 < \theta < 54$

Count



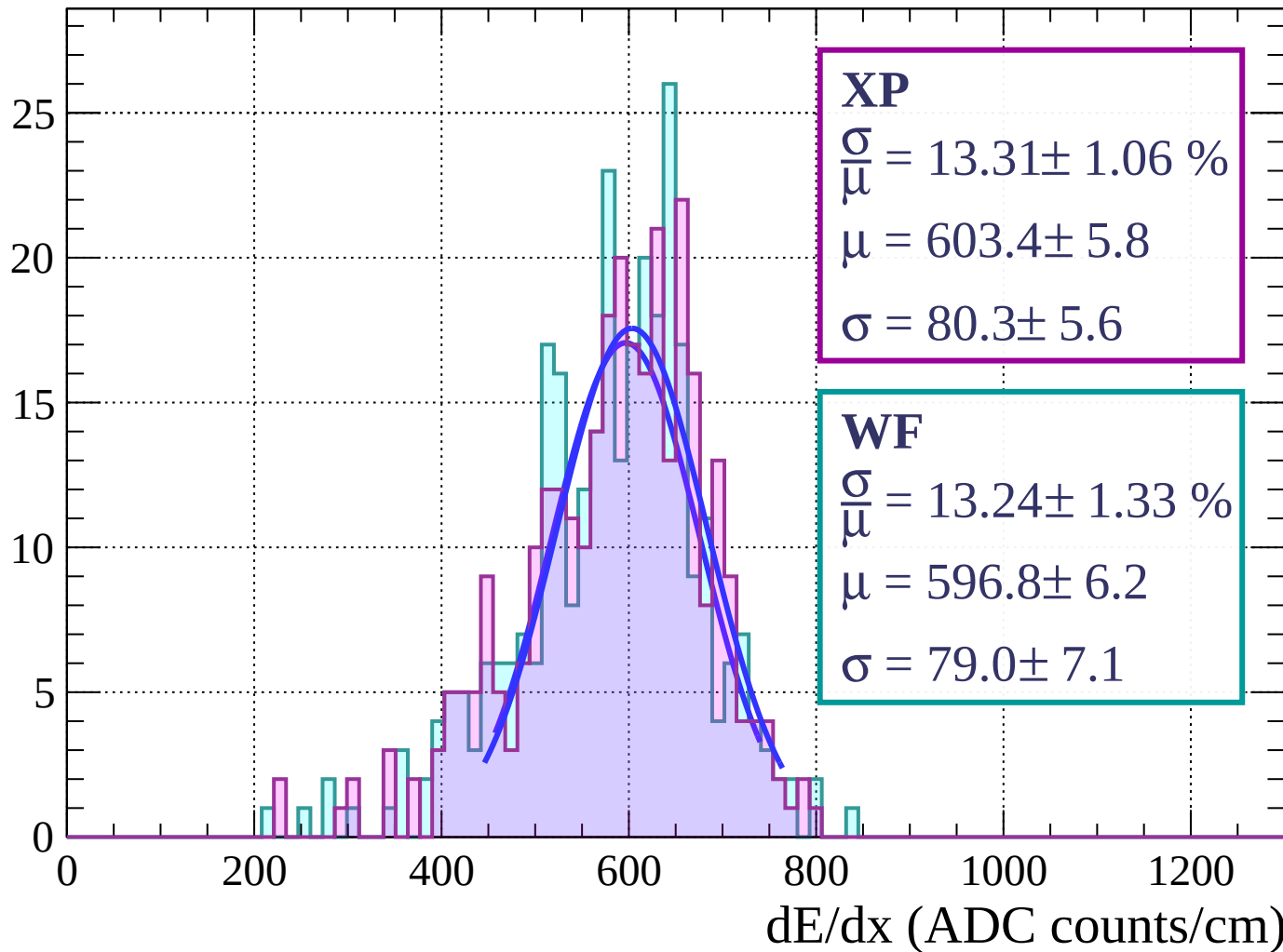
Energy loss | $54 < \theta < 57$

Count



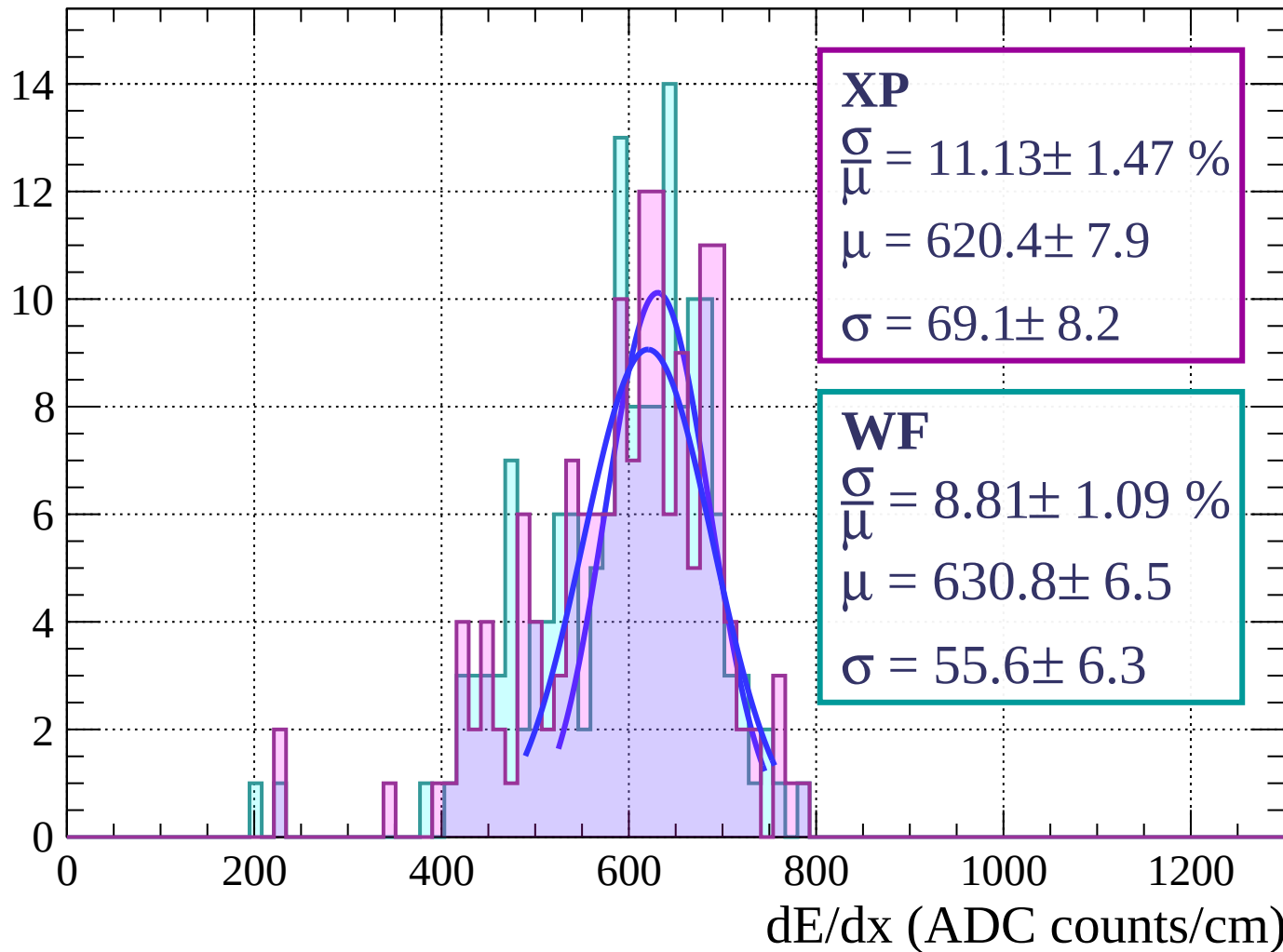
Energy loss | $57 < \theta < 60$

Count



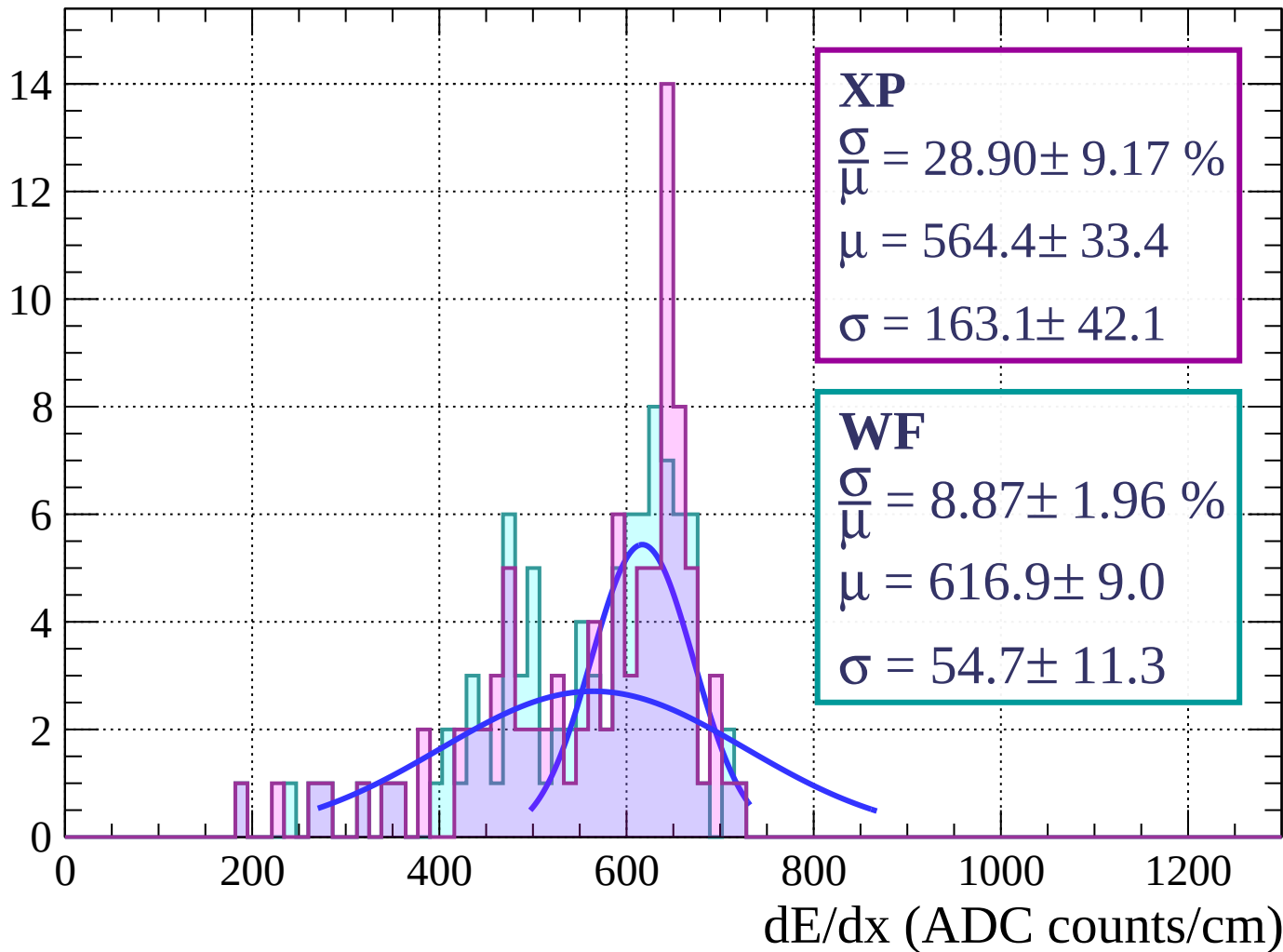
Energy loss | $60 < \theta < 63$

Count



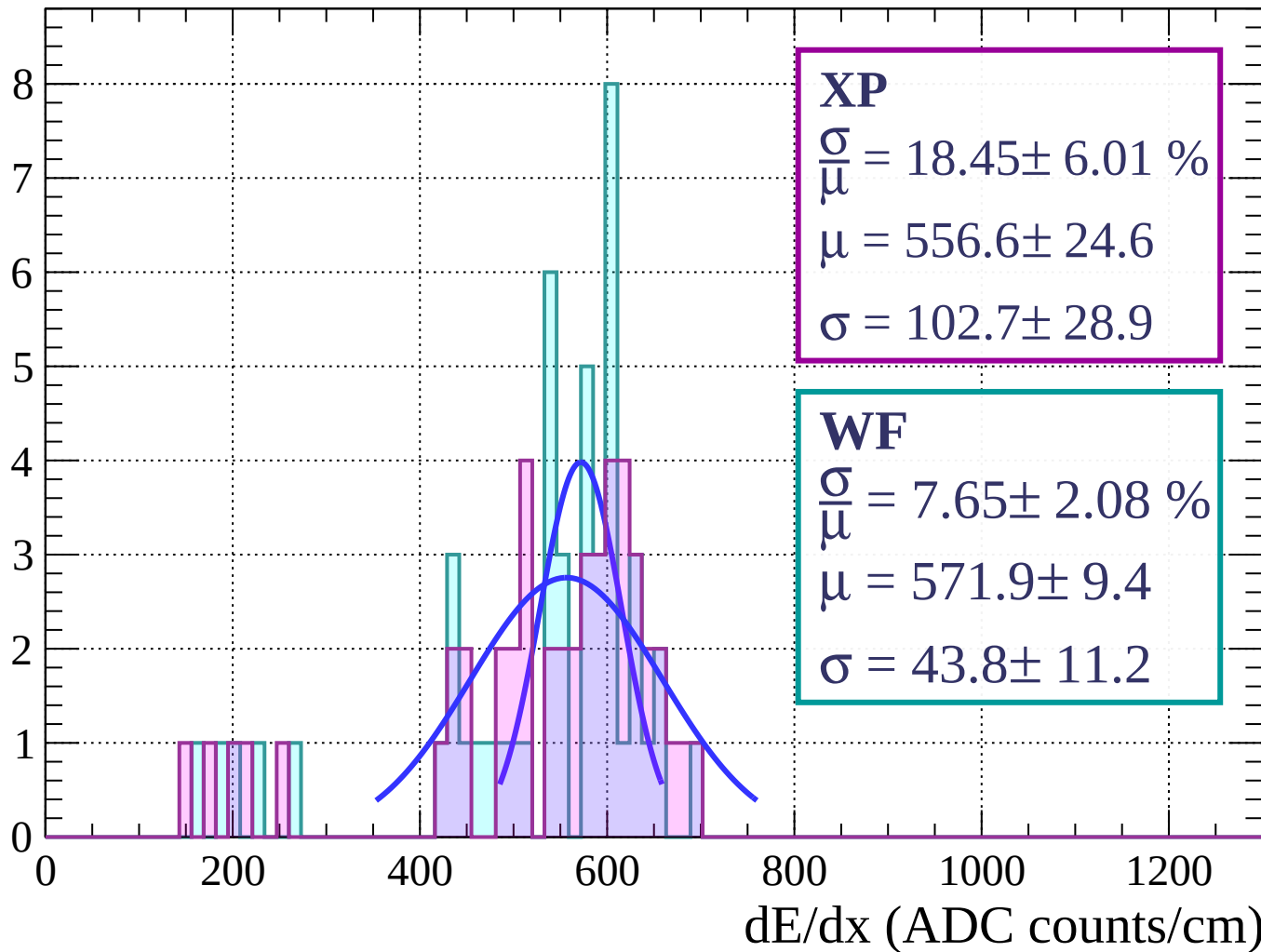
Energy loss | $63 < \theta < 66$

Count



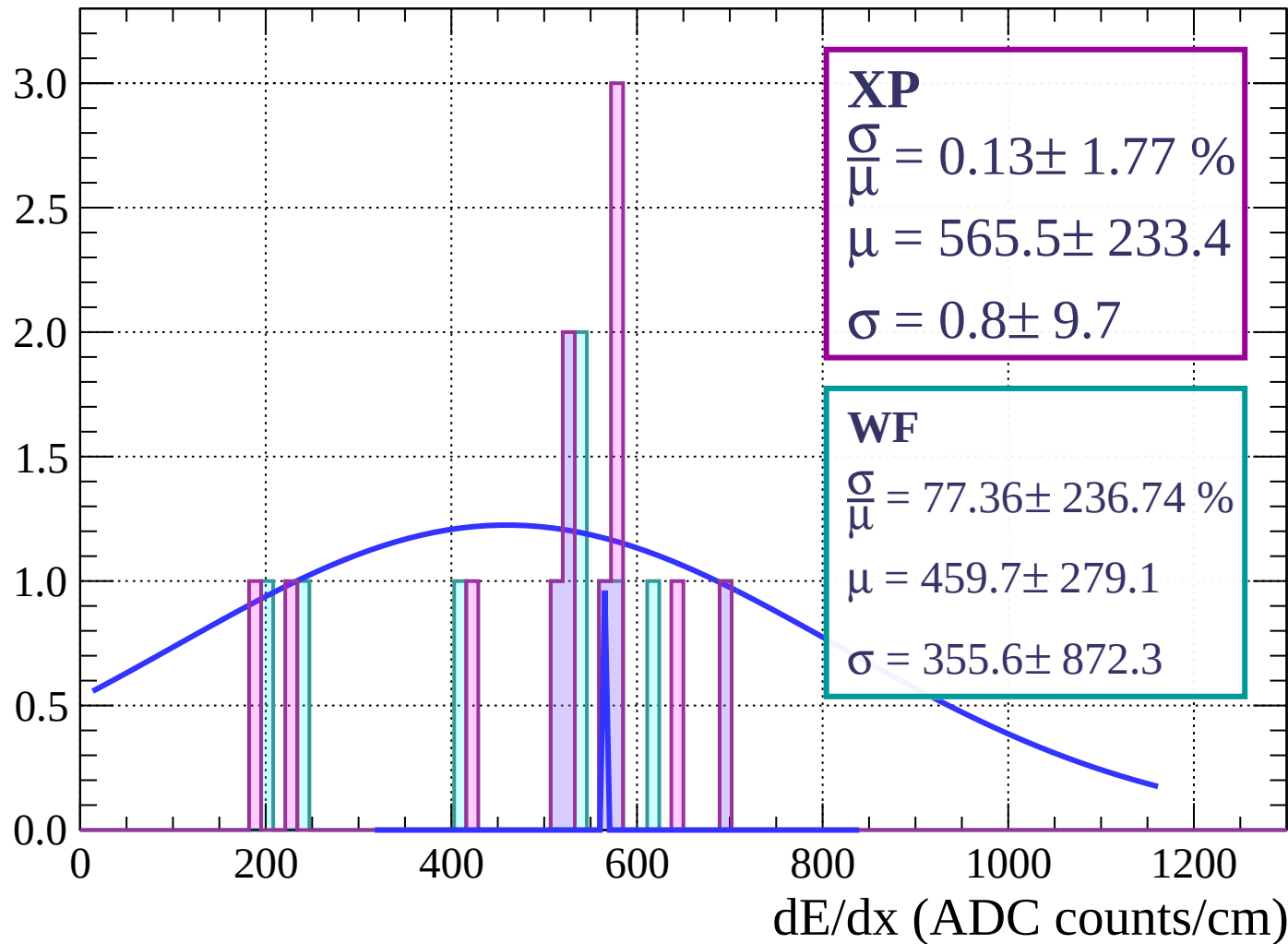
Energy loss | $66 < \theta < 69$

Count



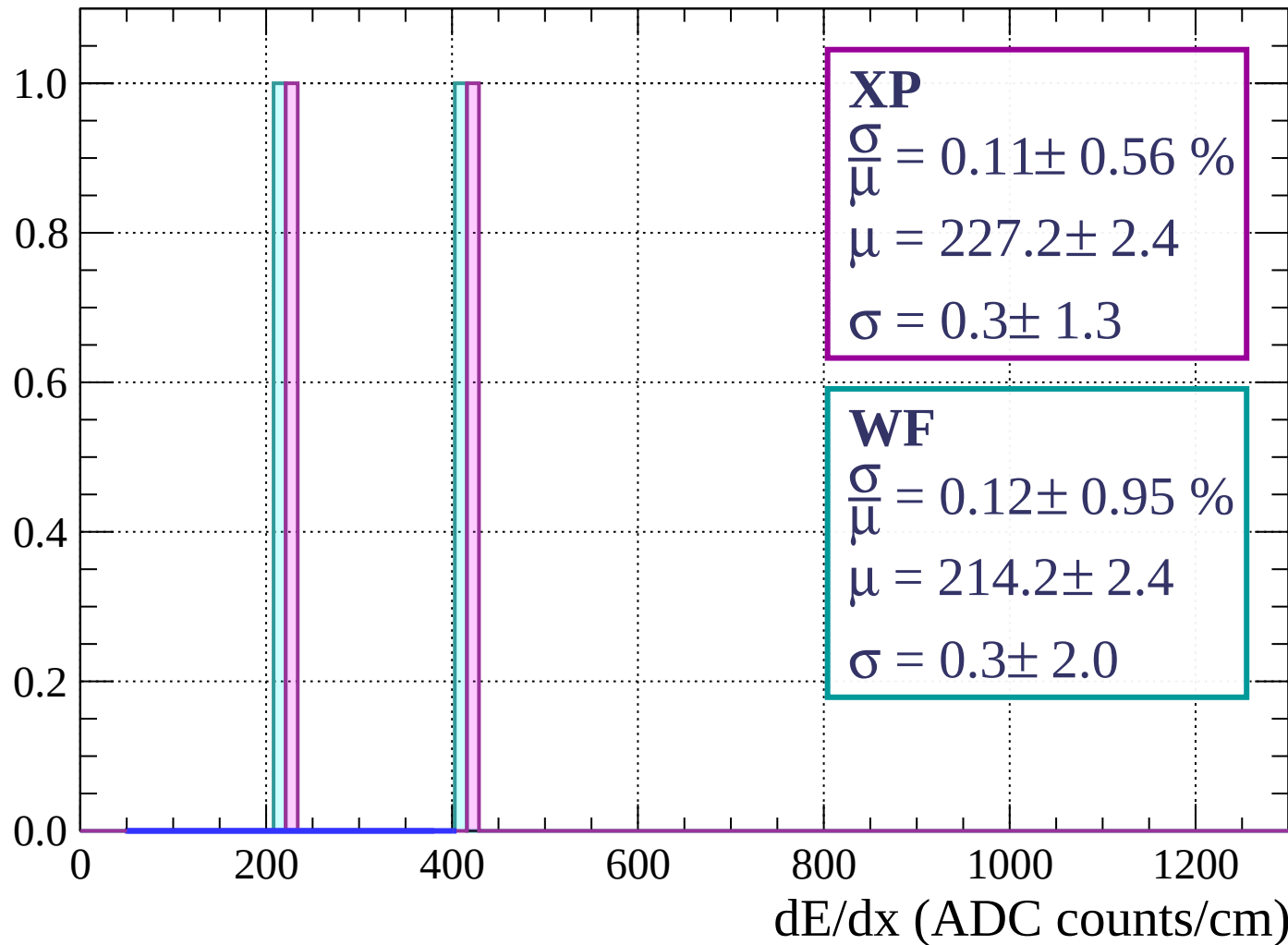
Energy loss | $69 < \theta < 72$

Count



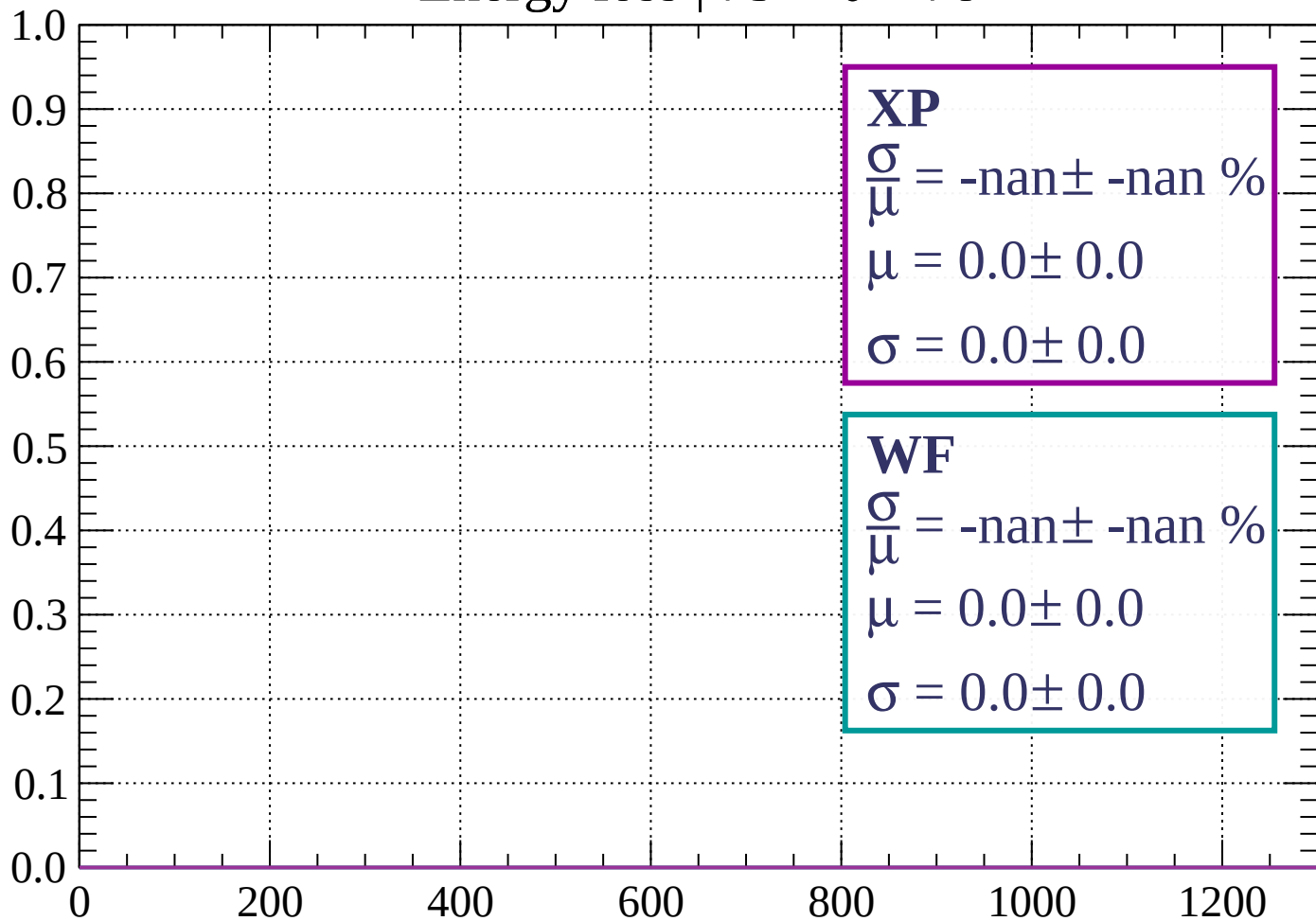
Energy loss | $72 < \theta < 75$

Count



Energy loss | $75 < \theta < 78$

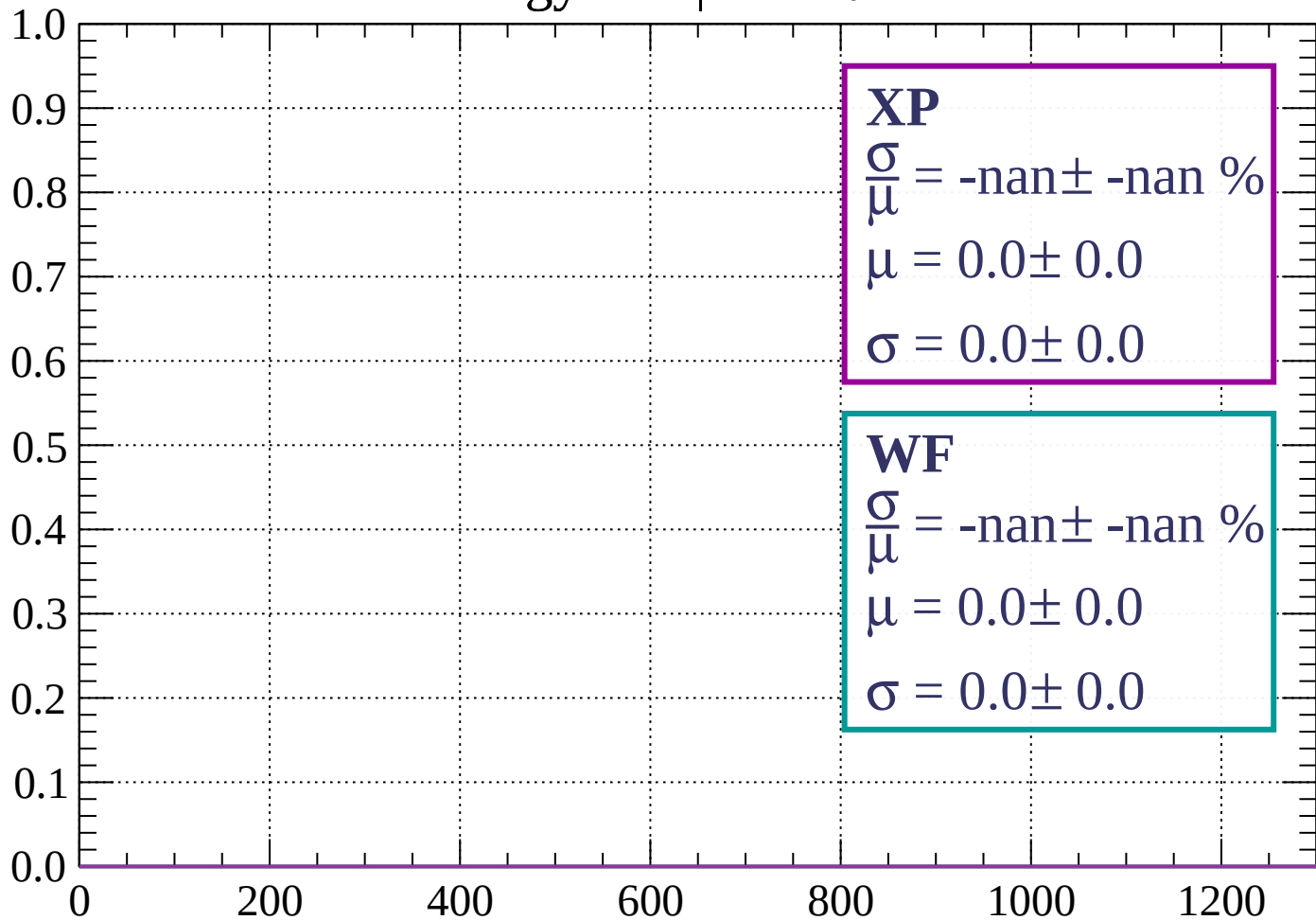
Count



dE/dx (ADC counts/cm)

Energy loss | $78 < \theta < 81$

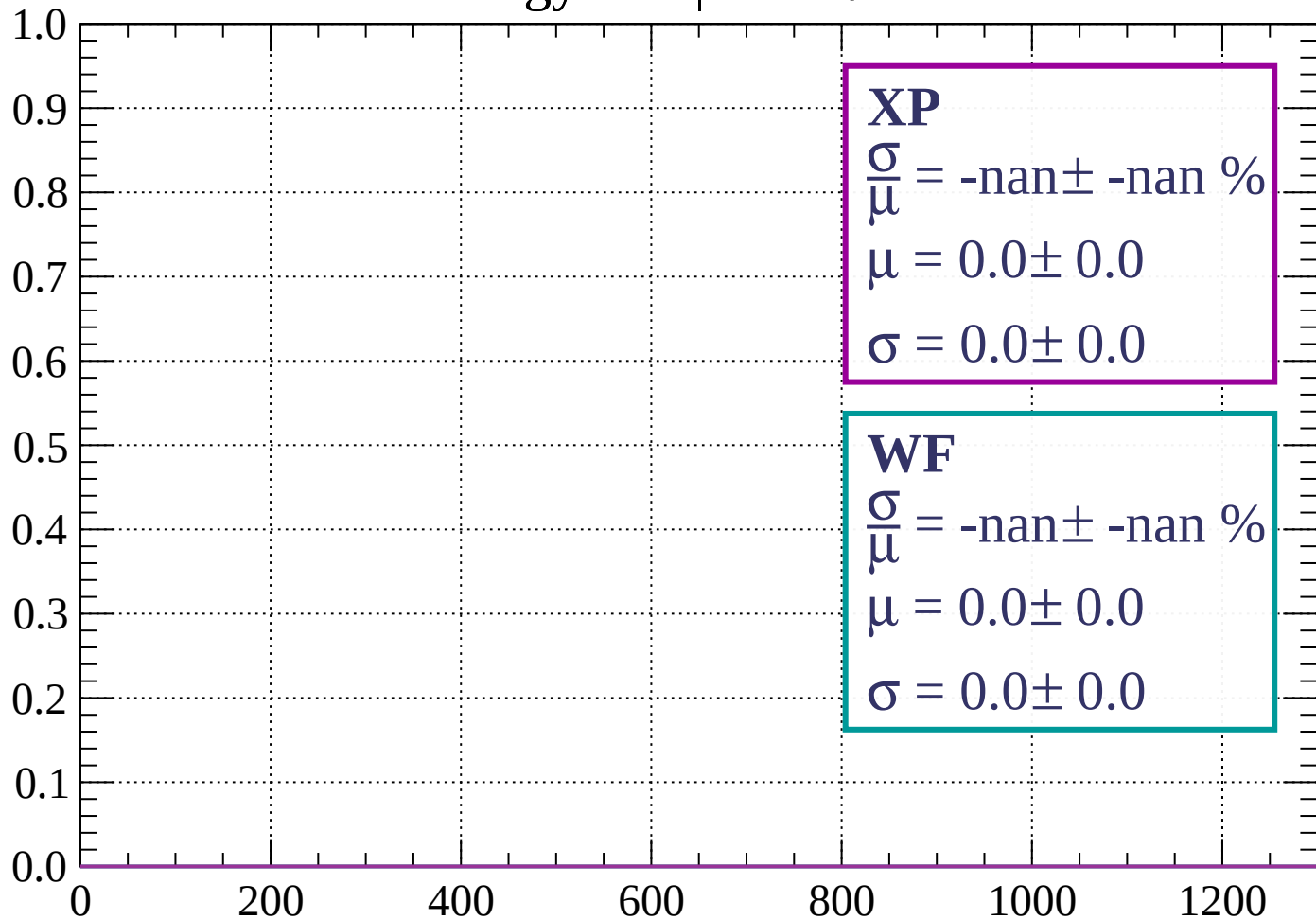
Count



dE/dx (ADC counts/cm)

Energy loss | $81 < \theta < 84$

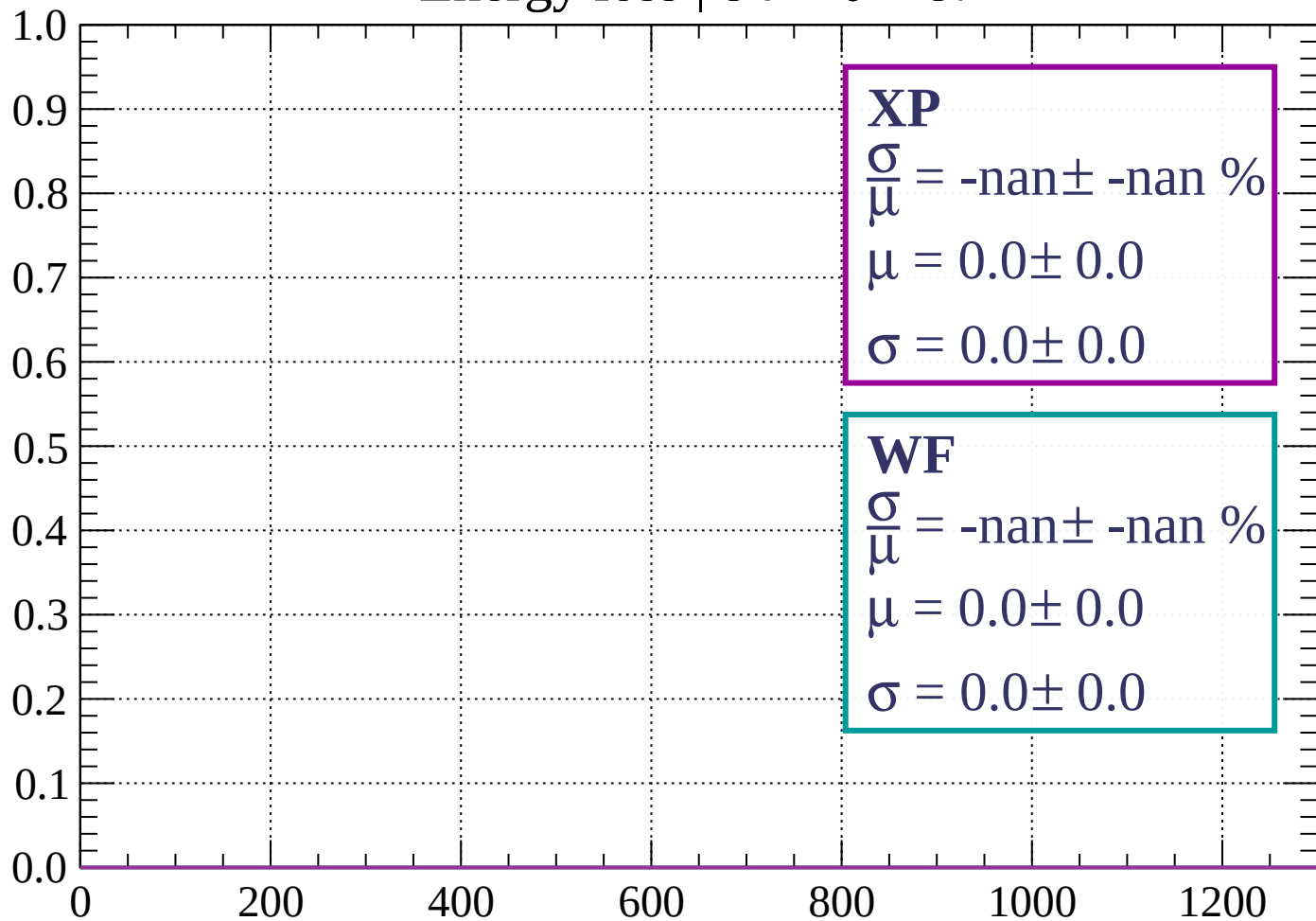
Count



dE/dx (ADC counts/cm)

Energy loss | $84 < \theta < 87$

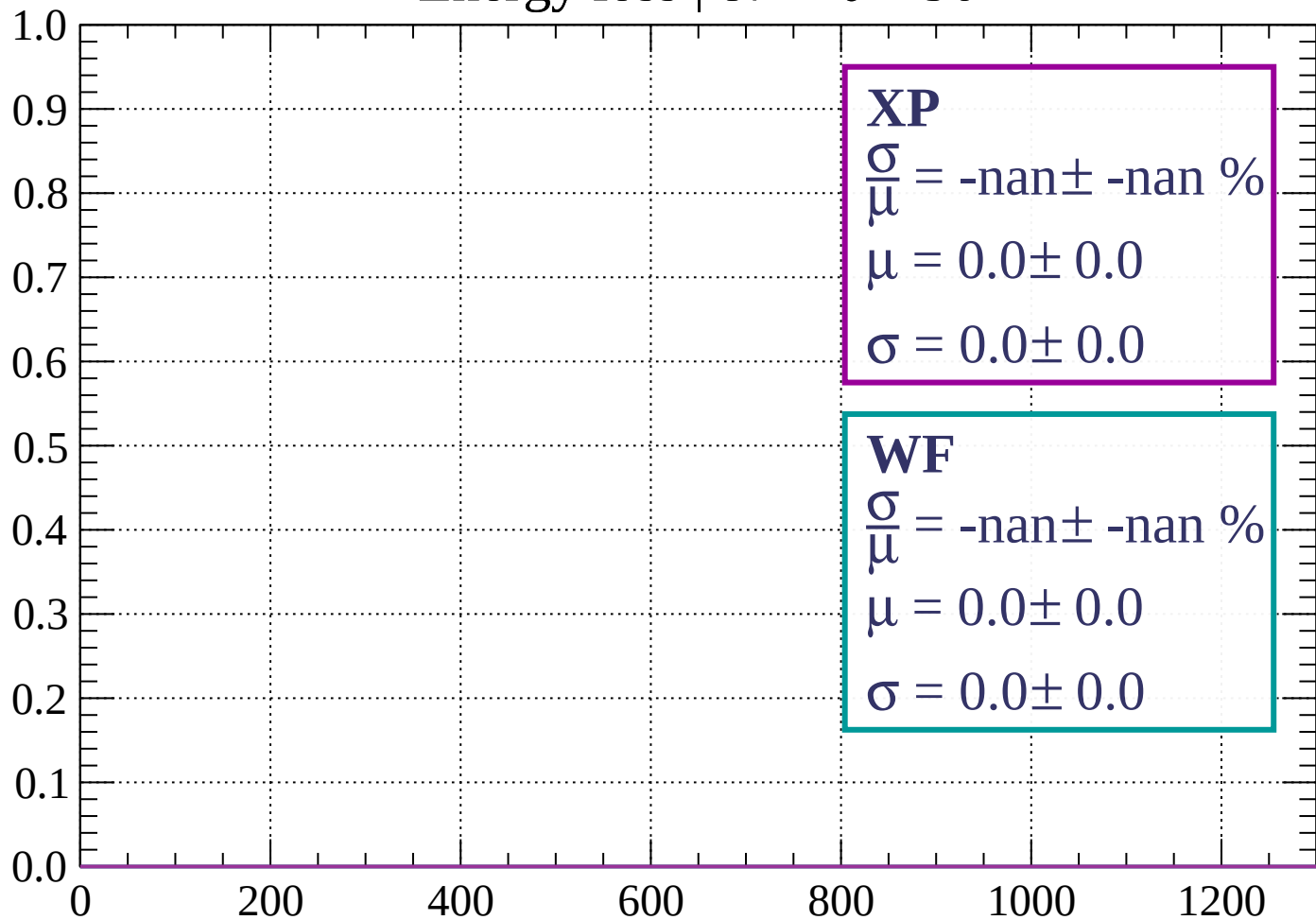
Count



dE/dx (ADC counts/cm)

Energy loss | $87 < \theta < 90$

Count



dE/dx (ADC counts/cm)

